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(202111.2) SONiC Command Line Interface Guide by Edgecore

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Introduction

SONiC is an open source network operating system based on Linux that runs on switches from multiple vendors and ASICs. SONiC offers a full-suite of network functionality, like BGP and RDMA, that has been production-hardened in the data centers of some of the largest cloud-service providers. It offers teams the flexibility to create the network solutions they need while leveraging the collective strength of a large ecosystem and community.

SONiC software shall be loaded in these [supported devices](#) and this CLI guide shall be used to configure the devices as well as to display the configuration, state and status.

Follow the [Quick Start Guide](#) to boot the device in ONIE mode, install the SONiC software using the steps specified in the document and login to the device using the default username and password.

After logging into the device, SONiC software can be configured in following three methods.

1. Command Line Interface (CLI)
2. [config_db.json](#)
3. [miniagraph.xml](#)

This document explains the first method and gives the complete list of commands that are supported in SONiC 201904 version (build#19). All the configuration commands need root privileges to execute them. Note that show commands can be executed by all users without the root privileges. Root privileges can be obtained either by using "sudo" keyword in front of all config commands, or by going to root prompt using "sudo -i". Note that all commands are case sensitive.

Example

```
admin@sonic:~$ sudo config aaa authentication login tacacs+
OR
admin@sonic:~$ sudo -i
root@sonic:~# config aaa authentication login tacacs+
```

Note that the command list given in this document is just a subset of all possible configurations in SONiC. Please follow config_db.json based configuration for the complete list of configuration options.

Scope of this Document

It is assumed that all configuration commands start with the keyword "config" as prefix. Any other scripts/utilities/commands that need user configuration control are wrapped as sub-commands under the "config" command. The direct scripts/utilities/commands (examples given below) that are not wrapped under the "config" command are not in the scope of this document.

1. `acl_loader` - This script is already wrapped inside "config acl" command; i.e. any ACL configuration that user is allowed to do is already part of "config acl" command; users are not expected to use the `acl_loader` script directly and hence this document need not explain the "acl_loader" script.
2. `crm` - this command is not explained in this document.
3. `sonic-clear`, `sftutil`, etc., This document does not explain these scripts also.

Not Supported Commands in this Document

In this document, some CLI commands or parameters of specific features are marked as "Not support". I may caused by varies reasons. These non-supported commands and parameters are still listed in this document to notify the user these commands, although existed in the Sonic CLI command Line, but it is not recommended to use these commands or parameters. Errors or unexpected results may occur if these commands or parameters are issued.

These commands existed in Community Sonic version, however, these commands/parameters are not working well due to the following reasons.

- The corresponding operations are not supported in the hardware platform.
- The corresponding operations are not fully implemented in Sonic platform or SAI driver.
- The corresponding operations are obsoleted compared with current Sonic implementation but the command is still kept.

- The corresponding operations has defects and under fixing.

Document History

Version	Modification Date	Details
v5	Aug-18-2021	Add new commands (autonego and advertised-speed) for interface speed auto-negotiation in 202012.
v4	Oct-17-2019	Unify usage statements and other formatting; Replace tabs with spaces; Modify heading sizes; Fix spelling, grammar and other errors; Fix organization of new commands
v3	Jun-26-2019	Update based on 201904 (build#19) release, "config interface" command changes related to interfacename order, FRR/Quagga show command changes, platform specific changes, ACL show changes and few formatting changes
v2	Apr-22-2019	CLI Guide for SONiC 201811 version (build#32) with complete "config" command set
v1	Mar-23-2019	Initial version of CLI Guide with minimal command set

Content Index of Articles

(202111.2) Getting Started with SONiC

This section covers the basic configurations related to the following:

- [SSH Login](#)
- [Configuring the Management Interface](#)

SSH Login

All SONiC devices support both serial console-based login and SSH-based login by default. The default credentials (if not modified at image build time) for login is admin/YourPaSsWoRd. In the case of SSH login, users can log in to the management interface (eth0) IP address after configuring the same using the serial console. Refer to the following section for configuring the IP address for the management interface.

Example:

```
At Console:
Debian GNU/Linux 9 sonic ttyS1
sonic login: admin
Password: YourPaSsWoRd
SSH from any remote server to sonic can be done by connecting to SONiC IP
user@debug:~$ ssh admin@sonic_ip_address(or SONIC DNS Name)
admin@sonic's password:
```

By default, login takes the user to the default prompt from which all the show commands can be executed.

Configuring the Management Interface

The management interface (eth0) in SONiC is configured (by default) to use DHCP client to get an IP address from a DHCP server. Connect the management interface to the same network in which your DHCP server is connected and get the IP address from DHCP server. The IP address received from the DHCP server can be verified using the `/sbin/ifconfig eth0` Linux command. SONiC provides a CLI to configure a static IP for the management interface. There are a few ways by which a static IP address can be configured for the management interface.

Use the `config interface ip add eth0` command.

Example

```
admin@sonic:~$ sudo config interface ip add eth0 20.11.12.13/24 20.11.12.254
```

Use `config_db.json` and configure the `MGMT_INTERFACE` key with the appropriate values. Refer to this [page](#).

Use `minigraph.xml` and configure "ManagementIPInterfaces" tag inside "DpgDesc" tag, as given on this [page](#).

Once the IP address is configured, it can be verified using either the `show management_interface address` command or the `/sbin/ifconfig eth0` Linux command. Users can use SSH to log in to this management interface IP address from their management network.

Example

```
admin@sonic:~$ /sbin/ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.11.11.13 netmask 255.255.255.0 broadcast 10.11.12.255
```

(202111.2) Getting Help

- [Help for Config Commands](#)
 - [config --help](#)
- [Help for Show Commands](#)
 - [show --help](#)

Help for Config Commands

All commands have in-built help that aids the user in understanding the command as well as the possible sub-commands and options. "--help" can be used at any level of the command; i.e. it can be used at the command level, or sub-command level or at argument level. The in-built help will display the available possibilities corresponding to that particular command/sub-command.

config --help

This command lists all the possible configuration commands at the top level.

Usage
config (-? -h --help)

Example

```
admin@sonic:~$ config --help
Usage: config [OPTIONS] COMMAND [ARGS]...

SONiC command line - 'config' command

Options:
  -?, -h, --help  Show this message and exit.

Commands:
  aaa          AAA command line
  acl          ACL-related configuration tasks
  bgp          BGP-related configuration tasks
  container    Modify configuration of containers
  dropcounters Drop counter related configuration tasks
  ecn          ECN-related configuration tasks
  feature      Configure status of feature
  hostname     Change device hostname without impacting the...
  interface    Interface-related configuration tasks
  interface_naming_mode Modify interface naming mode for interacting with...
  is-loopback-name-valid Loopback name validation
  kdump        Configure kdump
  load         Import a previous saved config DB dump file.
  load_mgmt_config Reconfigure hostname and mgmt interface based on...
  load_minigraph Reconfigure based on minigraph.
  loopback     Loopback-related configuration tasks
  mclag        Multi-Chassis Link Aggregation Group-related...
  mirror_session
  nat          NAT-related configuration tasks
  neigh_suppress Neighbour Suppress VLAN-related configuration
  ntp          NTP server configuration tasks
  pfcwd        Configure pfc watchdog
  platform     Platform-related configuration tasks
  portchannel
  pvlan        Private VLAN-related configuration tasks
  qos          QoS-related configuration tasks
  reload       Clear current configuration and import a previous...
  route        route-related configuration tasks
  sag          Static Anycast Gateway
  save         Export current config DB to a file on disk.
  sflow        sFlow-related configuration tasks
  snmpagentaddress SNMP agent listening IP address, port, vrf...
  snmptrap     SNMP Trap server configuration to send traps
  syslog       Syslog server configuration tasks
  tacacs       TACACS+ server configuration
  vlan         VLAN-related configuration tasks
  vrf          VRF-related configuration tasks
  vxlan
  warm_restart warm_restart-related configuration tasks
  watermark    Configure watermark
  ztp          Configure Zero Touch Provisioning
```

Help for Show Commands

show --help

This command displays the full list of show commands available in the software; the output of each of those show commands can be used to analyze, debug or troubleshoot the network node.

Usage

show (-?|-h|--help)

Example

```
admin@sonic:~$ show --help
Usage: show [OPTIONS] COMMAND [ARGS]...
```

SONiC command line - 'show' command

Options:

-?, -h, --help Show this message and exit.

Commands:

aaa	Show AAA configuration
acl	Show ACL related information
arp	Show IP ARP table
boot	Show boot configuration
buffer_pool	Show details of the buffer pools
clock	Show date and time
container	Show container
dropcounters	Show drop counter related information
ecn	Show ECN configuration
environment	Show environmentals (voltages, fans, temps)
features	Show status of optional features
interfaces	Show details of the network interfaces
ip	Show IP (IPv4) commands
ipv6	Show IPv6 commands
kdump	Show kdump configuration, status and information
line	Show all /dev/ttyUSB lines and their info
lldp	LLDP (Link Layer Discovery Protocol) information
logging	Show system log
mac	Show MAC (FDB) entries
management_interface	Show management interface parameters
mclag	Show MCLAG related information
mgmt-vrf	Show management VRF attributes
mirror_session	Show existing everflow sessions
mmu	Show mmu configuration
nat	Show details of the nat
ndp	Show IPv6 Neighbour table
neigh-suppress	show neigh_suppress
ntp	Show NTP information
pfc	Show details of the priority-flow-control (pfc)
pfcwd	Show details of the pfc watchdog
pim	Show details of the pims
platform	Show platform-specific hardware info
policer	Show existing policers
priority-group	Show details of the PGs
processes	Display process information
pvlan	Show Private VLAN related information
queue	Show details of the queues
reboot-cause	Show cause of most recent reboot
route-map	show route-map
runningconfiguration	Show current running configuration information
sag	show sag information
services	Show all daemon services
sflow	Show sFlow related information
snmpagentaddress	Show SNMP agent listening IP address configuration
snmptrap	Show SNMP agent Trap server configuration
startupconfiguration	Show startup configuration information
subinterfaces	Show details of the sub port interfaces
system-memory	Show memory information

tacacs	Show TACACS+ configuration
techsupport	Gather information for troubleshooting
uptime	Show system uptime
users	Show users
version	Show version information
vlan	Show VLAN information
vnet	Show vnet related information
vrf	Show vrf config
vxlan	Show VXLAN information
warm_restart	Show warm restart configuration and state
watermark	Show details of watermark
ztp	Show Zero Touch Provisioning status

The same syntax applies to all subgroups of **show** which themselves contain subcommands, and subcommands which accept options/arguments.

Example

```
admin@sonic:~$ show interface -?
Usage: show interface [OPTIONS] COMMAND [ARGS]...

    Show details of the network interfaces

Options:
  -?, -h, --help  Show this message and exit.

Commands:
  alias          Show Interface Name/Alias Mapping
  breakout      Show Breakout Mode information by interfaces
  counters       Show interface counters
  description    Show interface status, protocol and description
  naming_mode    Show interface naming_mode status
  neighbor       Show neighbor related information
  portchannel    Show PortChannel information
  status         Show Interface status information
  transceiver    Show SFP Transceiver information
```


(202111.2) Basic Show Commands

- Show Version
- Show System Status
 - show clock
 - show boot
 - show environment
 - show system status
 - show reboot-cause
 - show uptime
 - show logging
 - show users
- Show Hardware Platform
 - show platform fan
 - show platform firmware status
 - show platform firmware update status(not supported)
 - show platform firmware updates(not supported)
 - show platform firmware version
 - show platform pcieinfo
 - show platform psustatus
 - show platform ssdhealth
 - show platform summary
 - show platform syseeprom
 - show platform temperature
- Transceivers
 - show interfaces transceiver
- Show PIM Status
 - show pim status

Show Version

This command displays software component versions of the currently running SONiC image. This includes the SONiC image version as well as Docker image versions. This command displays relevant information as the SONiC and Linux kernel version being utilized, as well as the ID of the commit used to build the SONiC image. The second section of the output displays the various docker images and their associated IDs.

Usage
<code>show version</code>

Example

```
admin@as9716-32d-1:~$ show version
```

```
SONiC Software Version: SONiC.Edgecore-SONiC_20201120_145138_ec202006_72
Distribution: Debian 10.6
Kernel: 4.19.0-6-2-amd64
Build commit: 8ab86b25
Build date: Fri Nov 20 17:04:20 UTC 2020
Built by: ubuntu@ip-10-5-1-110
```

```
Platform: x86_64-accton_as9716_32d-r0
HwSKU: Accton-AS9716-32D-100G
ASIC: broadcom
Serial Number: 971632D1913030
Uptime: 03:43:54 up 2:51, 1 user, load average: 0.42, 0.48, 0.37
```

```
Docker images:
REPOSITORY                                TAG                                IMAGE ID                                SIZE
docker-sonic-mgmt-framework              Edgecore-SONiC_20201120_145138_ec202006_72  96af75b82358                          602MB
docker-sonic-mgmt-framework              latest                              96af75b82358                          602MB
docker-lldp                              Edgecore-SONiC_20201120_145138_ec202006_72  093fa2e59379                          380MB
docker-lldp                              latest                              093fa2e59379                          380MB
docker-sonic-telemetry                   Edgecore-SONiC_20201120_145138_ec202006_72  0268e4e30d64                          434MB
docker-sonic-telemetry                   latest                              0268e4e30d64                          434MB
docker-snmp                              Edgecore-SONiC_20201120_145138_ec202006_72  0160394139fc                          392MB
docker-snmp                              latest                              0160394139fc                          392MB
docker-router-advertiser                 Edgecore-SONiC_20201120_145138_ec202006_72  bc94b87e2483                          353MB
docker-router-advertiser                 latest                              bc94b87e2483                          353MB
docker-database                         Edgecore-SONiC_20201120_145138_ec202006_72  ff978d3189ed                          353MB
docker-database                         latest                              ff978d3189ed                          353MB
docker-dhcp-relay                       Edgecore-SONiC_20201120_145138_ec202006_72  0cd99667b957                          360MB
docker-dhcp-relay                       latest                              0cd99667b957                          360MB
docker-teamd                             Edgecore-SONiC_20201120_145138_ec202006_72  6a5900dc0ab3                          314MB
docker-teamd                             latest                              6a5900dc0ab3                          314MB
docker-sflow                             Edgecore-SONiC_20201120_145138_ec202006_72  00689631bb20                          318MB
docker-sflow                             latest                              00689631bb20                          318MB
docker-fpm-frr                          Edgecore-SONiC_20201120_145138_ec202006_72  50dec9c90285                          362MB
docker-fpm-frr                          latest                              50dec9c90285                          362MB
docker-orchagent                        Edgecore-SONiC_20201120_145138_ec202006_72  a93e4cf5c511                          334MB
docker-orchagent                        latest                              a93e4cf5c511                          334MB
docker-nat                              Edgecore-SONiC_20201120_145138_ec202006_72  235bdb8e81068                         316MB
docker-nat                              latest                              235bdb8e81068                         316MB
docker-iccpd                             Edgecore-SONiC_20201120_145138_ec202006_72  0b63060c53e5                          315MB
docker-iccpd                             latest                              0b63060c53e5                          315MB
docker-syncd-brcm                        Edgecore-SONiC_20201120_145138_ec202006_72  00fad1e61fcb                          478MB
docker-syncd-brcm                        latest                              00fad1e61fcb                          478MB
docker-platform-monitor                 Edgecore-SONiC_20201120_145138_ec202006_72  6dc4652fe01e                          356MB
docker-platform-monitor                 latest                              6dc4652fe01e                          356MB
```

Show System Status

This section explains some command sets that are used to display the status of various parameters pertaining to the physical state of the network node.

show clock

This command displays the current date and time configured on the system.

Usage

```
show clock
```

Example

```
admin@sonic:~$ show clock
Mon Mar 25 20:25:16 UTC 2019
```

show boot

This command displays the current OS image, the image to be loaded on next reboot, and lists all the available images installed on the device.

Usage

```
show boot
```

Example

```
admin@sonic:~$ show boot
Current: SONiC-OS-20181130.31
Next: SONiC-OS-20181130.31
Available:
SONiC-OS-20181130.31
```

show environment

This command displays the platform environmentals, such as voltages, temperatures and fan speeds.

Usage

```
show environment
```

Example

```
admin@sonic:~$ show environment
coretemp-isa-0000
Adapter: ISA adapter
Core 0:      +28.0 C (high = +98.0 C, crit = +98.0 C)
Core 1:      +28.0 C (high = +98.0 C, crit = +98.0 C)
Core 2:      +28.0 C (high = +98.0 C, crit = +98.0 C)
Core 3:      +28.0 C (high = +98.0 C, crit = +98.0 C)
SMF_Z9100_ON-isa-0000
Adapter: ISA adapter
CPU XP3R3V_EARLY:      +3.22 V
<... few more things ...>

Onboard Temperature Sensors:
CPU:                  30 C
BCM56960 (PSU side):  35 C
<... few more things ...>

Onboard Voltage Sensors:
CPU XP3R3V_EARLY      3.22 V
<... few more things ...>

Fan Trays:
Fan Tray 1:
Fan1 Speed:    6192 RPM
Fan2 Speed:    6362 RPM
Fan1 State:    Normal
Fan2 State:    Normal
Air Flow:      F2B
<... few more things ...>

PSUs:
PSU 1:
Input:         AC
<... few more things ...>
```

NOTE: The show output includes a lot of information; only a sample output is shown in the above example. Though the displayed output slightly differs from one platform to another platform, the overall content will be similar to the example above.

show system status

This command displays if the SONiC system has finished booting or not.

You may use this command to inspect the progress state of the SONiC system bring-up. It might show the message in Example 2 (or 3) in the middle of the bring-up. If it is seen after the system is up, there could be some configuration issues in the configuration file.

Usage

```
show system status
```

The result of executing this command could be:

- SONiC is ready, i.e. port provision is done and services are activated. (refer to Example1)
- Port configuration is not fully provisioned (refer to Example2)
- Not all SONiC services are ready (Example3, the "teamd" is not in service)

Example1 (SONiC is ready)

```
admin@sonic:~$ show system status
System is ready
```

Example2 (port provisioning incomplete)

```
admin@sonic:~$ show system status
System is not ready - Ports are not up
```

Example3 (SONiC system is not ready)

```
admin@sonic:~$ show system status
System is not ready - core service is down
Container and Inside Process    Status
-----
database                        Up
- redis                        Up
swss                            Up
- buffermgrd                   Up
- fdbsyncd                     Up
- intfmgrd                     Up
- intfsyncd                    Up
- nbrmgrd                      Up
- neighsyncd                   Up
- orchagent                    Up
- portmgrd                     Up
- portsyncd                    Up
- vlanmgrd                     Up
- vrfmgrd                      Up
- vxlanmgrd                    Up
syncd                          Up
- syncd                        Up
bgp                             Up
- bgpconfd                    Up
- bgpd                         Up
- fpm-syncd                    Up
- staticd                     Up
- zebra                        Up
mgmt-framework                 Up
pmon                           Up
- psud                         Up
- xcvr                         Up
teamd                          Down
- teammgrd                     N/A
- teamsyncd                    N/A
lldp                           Up
- lldp-syncd                   Up
- lldpd                        Up
- lldpMgrd                     Up
```

Status:

- **Up**: container or process (inside the container) is active/running.
- **Down**: container or process (inside the container) is not active/running.
 - In this example, **teamd** is the container.
- **N/A**: daemons (inside of the container) are not available when the container is **Down**.

◦ In this example, **teammgrd** or **teamsyncd** is the daemon.

show reboot-cause

This command displays the cause of the previous reboot.

Usage

```
show reboot-cause
```

Example

```
admin@sonic:~$ show reboot-cause
User issued reboot command [User: admin, Time: Mon Mar 25 01:02:03 UTC 2019]
```

show uptime

This command displays the current system uptime.

Usage

```
show uptime
```

Example

```
admin@sonic:~$ show uptime
up 2 days, 21 hours, 30 minutes
```

show logging

This command displays all the currently stored log messages. All the latest processes and corresponding transactions are stored in the "syslog" file. This file is saved in the path **/var/log** and can be viewed using the command **sudo cat syslog** as this requires root login.

Usage

```
show logging [(<process_name> [-l|--lines <number_of_lines>]) | (-f|--follow)]
```

Example

```
admin@sonic:~$ show logging
```

It can be useful to pipe the output from **show logging** to the command **more** in order to examine one screenful of log messages at a time.

Example

```
admin@sonic:~$ show logging | more
```

Optionally, you can specify a process name in order to display only log messages mentioning that process.

Example

```
admin@sonic:~$ show logging sensord
```

Optionally, you can specify a number of lines to display using the **-l** or **--lines** option. Only the most recent N lines will be displayed. Also, note that this option can be combined with a process name.

Example

```
admin@sonic:~$ show logging --lines 50
admin@sonic:~$ show logging sensord --lines 50
```

Optionally, you can follow the log live as entries are written to it by specifying the **-f** or **--follow** flag.

Example

```
admin@sonic:~$ show logging --follow
```

show users

This command displays a list of users currently logged in to the device.

Usage

```
show users
```

Example

```
admin@sonic:~$ show users
admin pts/9 Mar 25 20:31 (100.127.20.23)
admin ttyS1 2019-03-25 20:31
```

Show Hardware Platform

The information displayed in this set of commands partially overlaps with the one generated by the "show environment" instruction. In this case though, the information is presented in a more succinct fashion. In the future, these two CLI stanzas may end up being combined.

show platform fan

This command displays the fans status and information.

Usage

```
show platform fan
```

Example

```
admin@sonic:~$ show platform fan
```

Drawer	LED	FAN	Speed	Direction	Presence	Status	Timestamp
FanTray1	green	FAN-1F	40%	exhaust	Present	OK	20220527 07:21:02
FanTray1	green	FAN-1R	40%	exhaust	Present	OK	20220527 07:21:02
FanTray2	green	FAN-2F	40%	exhaust	Present	OK	20220527 07:21:02
FanTray2	green	FAN-2R	40%	exhaust	Present	OK	20220527 07:21:02
FanTray3	green	FAN-3F	40%	exhaust	Present	OK	20220527 07:21:02
FanTray3	green	FAN-3R	40%	exhaust	Present	OK	20220527 07:21:02
FanTray4	green	FAN-4F	40%	exhaust	Present	OK	20220527 07:21:02
FanTray4	green	FAN-4R	40%	exhaust	Present	OK	20220527 07:21:02
FanTray5	green	FAN-5F	40%	exhaust	Present	OK	20220527 07:21:02
FanTray5	green	FAN-5R	40%	exhaust	Present	OK	20220527 07:21:02
N/A	green	PSU-1 FAN-1	12%	exhaust	Present	OK	20220527 07:21:19
N/A	green	PSU-2 FAN-1	13%	exhaust	Present	OK	20220527 07:21:20

show platform firmware status

This command displays the platform components status.

Usage

```
show platform firmware status
```

Example

```
admin@sonic:~$ show platform firmware status
```

Chassis	Module	Component	Version	Description
5835-54X-0-AC-F	N/A	CPLD1	1	CPLD 1
		CPLD2	1	CPLD 2
		CPLD3	1	CPLD 3
		BIOS	v47.01.01.00	Basic Input/Output System

show platform firmware update status(not supported)

This command displays the platform components auto_update status

Usage

```
show platform firmware update status
```

show platform firmware updates(not supported)

This command displays the available updates.

Usage

```
show platform firmware updates
```

show platform firmware version

This command displays the utility version.

Usage

```
show platform firmware version
```

Example

```
admin@sonic:~$ show platform firmware version  
fwutil version 2.0.0.0
```

show platform pcieinfo

This command displays the platform pcie information .

Usage

```
show platform pcieinfo
```

Example

```
admin@as5835-54x:~$ show platform pcieinfo
Failed to load platform PCIe module. Error : No module named 'sonic_platform.pcie', fallback to load PCIe
common utility.
=====Display PCIe Device=====
bus:dev.fn 00:00.0 - dev_id=0x1980, Host bridge: Intel Corporation Atom Processor C3000 Series System Agent
(rev 11)
bus:dev.fn 00:04.0 - dev_id=0x19a1, Host bridge: Intel Corporation Atom Processor C3000 Series Error
Registers (rev 11)
bus:dev.fn 00:05.0 - dev_id=0x19a2, Generic system peripheral [0807]: Intel Corporation Atom Processor C3000
Series Root Complex Event Collector (rev 11)
bus:dev.fn 00:06.0 - dev_id=0x19a3, PCI bridge: Intel Corporation Atom Processor C3000 Series Integrated QAT
Root Port (rev 11)
bus:dev.fn 00:09.0 - dev_id=0x19a4, PCI bridge: Intel Corporation Atom Processor C3000 Series PCI Express
Root Port #0 (rev 11)
bus:dev.fn 00:0b.0 - dev_id=0x19a6, PCI bridge: Intel Corporation Atom Processor C3000 Series PCI Express
Root Port #2 (rev 11)
bus:dev.fn 00:0e.0 - dev_id=0x19a8, PCI bridge: Intel Corporation Atom Processor C3000 Series PCI Express
Root Port #4 (rev 11)
bus:dev.fn 00:10.0 - dev_id=0x19aa, PCI bridge: Intel Corporation Atom Processor C3000 Series PCI Express
Root Port #6 (rev 11)
bus:dev.fn 00:12.0 - dev_id=0x19ac, System peripheral: Intel Corporation Atom Processor C3000 Series SMBus
Controller - Host (rev 11)
bus:dev.fn 00:13.0 - dev_id=0x19b2, SATA controller: Intel Corporation Atom Processor C3000 Series SATA
Controller 0 (rev 11)
bus:dev.fn 00:15.0 - dev_id=0x19d0, USB controller: Intel Corporation Atom Processor C3000 Series USB 3.0
xHCI Controller (rev 11)
bus:dev.fn 00:16.0 - dev_id=0x19d1, PCI bridge: Intel Corporation Atom Processor C3000 Series Integrated LAN
Root Port #0 (rev 11)
bus:dev.fn 00:17.0 - dev_id=0x19d2, PCI bridge: Intel Corporation Atom Processor C3000 Series Integrated LAN
Root Port #1 (rev 11)
bus:dev.fn 00:18.0 - dev_id=0x19d3, Communication controller: Intel Corporation Atom Processor C3000 Series
ME HECI 1 (rev 11)
bus:dev.fn 00:1a.0 - dev_id=0x19d8, Serial controller: Intel Corporation Atom Processor C3000 Series HUART
Controller (rev 11)
bus:dev.fn 00:1a.1 - dev_id=0x19d8, Serial controller: Intel Corporation Atom Processor C3000 Series HUART
Controller (rev 11)
bus:dev.fn 00:1a.2 - dev_id=0x19d8, Serial controller: Intel Corporation Atom Processor C3000 Series HUART
Controller (rev 11)
bus:dev.fn 00:1f.0 - dev_id=0x19dc, ISA bridge: Intel Corporation Atom Processor C3000 Series LPC or eSPI
(rev 11)
bus:dev.fn 00:1f.1 - dev_id=0x19dd, Memory controller: Intel Corporation Atom Processor C3000 Series Primary
to Side Band (P2SB) Bridge (rev 11)
bus:dev.fn 00:1f.2 - dev_id=0x19de, Memory controller: Intel Corporation Atom Processor C3000 Series Power
Management Controller (rev 11)
bus:dev.fn 00:1f.4 - dev_id=0x19df, SMBus: Intel Corporation Atom Processor C3000 Series SMBus controller
(rev 11)
bus:dev.fn 00:1f.5 - dev_id=0x19e0, Serial bus controller [0c80]: Intel Corporation Atom Processor C3000
Series SPI Controller (rev 11)
bus:dev.fn 01:00.0 - dev_id=0x19e2, Co-processor: Intel Corporation Atom Processor C3000 Series QuickAssist
Technology (rev 11)
bus:dev.fn 02:00.0 - dev_id=0xb771, Ethernet controller: Broadcom Inc. and subsidiaries Device b771 (rev 01)
bus:dev.fn 05:00.0 - dev_id=0x1533, Ethernet controller: Intel Corporation I210 Gigabit Network Connection
(rev 03)
bus:dev.fn 06:00.0 - dev_id=0x15c2, Ethernet controller: Intel Corporation Ethernet Connection X553
Backplane (rev 11)
bus:dev.fn 08:00.0 - dev_id=0x15e5, Ethernet controller: Intel Corporation Ethernet Connection X553 1GbE
(rev 11)
```

show platform psustatus

This command displays the status of the device's power supply units.

Usage

```
show platform psustatus
```

Example

```
admin@sonic:~$ show platform psustatus
```

PSU	Model	Serial	HW Rev	Voltage (V)	Current (A)	Power (W)	Status	LED
PSU 1	YM-2651Y	N/A	N/A	12.06	4.62	56.00	OK	green
PSU 2	YM-2651Y	N/A	N/A	12.23	5.82	71.00	OK	green

show platform ssdhealth

This command displays health parameters of the device's SSD.

Usage

```
show platform ssdhealth [--vendor]
```

Example

```
admin@as9716-32d-1:~$ show platform ssdhealth
Device Model : M.2 (S42) 3ME4
Health       : 98.633%
Temperature  : 40C
```

Note: This command needs root privilege.

show platform summary

This command displays a summary of the device's hardware platform.

Usage

```
show platform summary
```

Example

```
admin@as9716-32d-1:~$ show platform summary
Platform: x86_64-accton_as9716_32d-r0
HwSKU: Accton-AS9716-32D-100G
ASIC: broadcom
```

show platform syseeprom

This command displays information stored on the system EEPROM. Note that the output of this command is not the same for all vendor's platforms. A couple of example outputs are given below.

Usage

```
show platform syseeprom
```

Example

```
admin@sonic:~$ show platform syseeprom
TlvInfo Header:
  Id String:    TlvInfo
  Version:      1
  Total Length: 117
TLV Name      Code      Len  Value
-----
Product Name   0x21      15  5835-54X-0-AC-F
Part Number    0x22      13  FP1ZZ56540B7A
Serial Number   0x23      14  583554X1218123
Base MAC Address 0x24       6  80:A2:35:D2:4C:B5
Platform Name   0x28      27  x86_64-accton_as5835_54x-r0
MAC Addresses   0x2A       2  256
Manufacturer    0x2B       6  Accton
Manufacture Country 0x2C       2  TW
Vendor Name     0x2D       8  Edgecore
CRC-32          0xFE       4  0xA334B01A

(checksum valid)
```

show platform temperature

The command displays device temperature information.

Usage

```
show platform temperature
```

Example

```
admin@sonic:~$ show platform temperature
Sensor      Temperature  High TH  Low TH  Crit High TH  Crit Low TH  Warning
Timestamp
-----
PSU-1 temp sensor 1      32      80      N/A      N/A      N/A      False 20220527
07:51:05
PSU-2 temp sensor 2      32      80      N/A      N/A      N/A      False 20220527
07:51:06
Temp sensor 1      35.5      80      N/A      N/A      N/A      False 20220527
07:51:04
Temp sensor 2      26.5      80      N/A      N/A      N/A      False 20220527
07:51:04
Temp sensor 3      28.5      80      N/A      N/A      N/A      False 20220527
07:51:04
Temp sensor 4      28      80      N/A      N/A      N/A      False 20220527
07:51:04
```

Transceivers

Displays diagnostic monitoring information of the transceivers.

show interfaces transceiver

This command displays transceiver information for all interfaces, or for a specific interface when the option **interface_name** is specified.

Usage

```
show interfaces transceiver (eeprom [-d|--dom] | lpmode | presence) [<interface_name>]
```

Example (Decode and display information stored on the EEPROM of SFP transceiver connected to Ethernet0)

```
admin@sonic:~$ show interfaces transceiver eeprom --dom Ethernet0
Ethernet0: SFP detected
  Connector : No separable connector
  Encoding : Unspecified
  Extended Identifier : Unknown
  Extended RateSelect Compliance : QSFP+ Rate Select Version 1
  Identifier : QSFP+
  Length Cable Assembly(m) : 1
  Specification compliance :
    10/40G Ethernet Compliance Code : 40GBASE-CR4
    Fibre Channel Speed : 1200 Mbytes/Sec
    Fibre Channel link length/Transmitter Technology : Electrical inter-enclosure (EL)
    Fibre Channel transmission media : Twin Axial Pair (TW)
  Vendor Date Code(YYYY-MM-DD Lot) : 2015-10-31
  Vendor Name : XXXXX
  Vendor OUI : XX-XX-XX
  Vendor PN : 1111111111
  Vendor Rev :
  Vendor SN : 1111111111
  ChannelMonitorValues:
    RX1Power: -1.1936dBm
    RX2Power: -1.1793dBm
    RX3Power: -0.9388dBm
    RX4Power: -1.0729dBm
    TX1Bias: 4.0140mA
    TX2Bias: 4.0140mA
    TX3Bias: 4.0140mA
    TX4Bias: 4.0140mA
  ModuleMonitorValues :
    Temperature : 1.1111C
    Vcc : 0.0000Volts
```

Example (Display status of low-power mode of SFP transceiver connected to Ethernet100):

```
admin@sonic:~$ show interfaces transceiver lpmode Ethernet100
Port          Low-power Mode
-----
Ethernet100  On
```

Example (Display presence of SFP transceiver connected to Ethernet100):

```
admin@sonic:~$ show interfaces transceiver presence Ethernet100
Port          Presence
-----
Ethernet100   Present
```

Show PIM Status

Displays the status of the PIM cards.

show pim status

This command displays information for all PIM cards.

Example (Display status for all PIM card):

```
admin@sonic:~$ show pim status
Slot Index    Operation Status    Type    HW Status
-----
slot 2        up                  PIM-16Q status code: 0b1
slot 3        up                  PIM-16Q status code: 0b1
slot 4        up                  PIM-16Q status code: 0b1
slot 5        up                  PIM-16Q status code: 0b1
slot 6        up                  PIM-16Q status code: 0b1
slot 7        up                  PIM-16Q status code: 0b1
slot 8        up                  PIM-16Q status code: 0b1
slot 9        up                  PIM-16Q status code: 0b0
```

Note: This command only works for platforms that support PIM hotswap, for example the AS8000.

(202111.2) Authentication, Authorization, and Accounting

This section captures the various show and configuration commands that are applicable to the AAA (Authentication, Authorization, and Accounting) module. You can configure the type of authentication (local or remote TACACS based) required for the users and also the authentication failthrough and fallback options.

(202111.2) AAA

- AAA show commands
 - show aaa
- AAA config commands
 - aaa authentication failthrough
 - aaa authentication fallback
 - aaa authentication login
 - aaa authentication login default
 - aaa authorization
 - aaa authorization default
 - aaa accounting
 - aaa accounting disable

AAA show commands

This command is used to view the Authentication, Authorization & Accounting settings that are configured in the network node.

show aaa

This command displays the AAA settings currently present in the network node.

Usage

```
show aaa
```

Example

```
admin@sonic:~$ show aaa
AAA authentication login local (default)
AAA authentication failthrough False (default)
AAA authorization login local (default)
AAA accounting login disable (default)
```

AAA config commands

This sub-section explains all the possible CLI based configuration options for the AAA module. The list of commands/sub-commands possible for AAA is given below.

Commands:

aaa authentication sub-commands:

- aaa authentication failthrough
- aaa authentication fallback
- aaa authentication login

aaa authentication failthrough

This command is used to either enable or disable the failthrough option. This command is useful when a user has configured more than one authentication servers or methods. When an authentication request to the first server fails, this configuration allows it to continue the request to the next server or method. When this configuration is enabled, the authentication process continues through all servers or methods configured. When disabled and if the authentication request fails on first server, the authentication process will stop and the login will be disallowed.

Usage

```
config aaa authentication failthrough (enable | disable | default)
```

Parameters:

- enable: This allows the AAA module to process with local authentication if remote authentication fails.
- disable: This disallows the AAA module to proceed further if remote authentication fails.
- default: This re-configures the default value, which is "disable".

The below example enables the failthrough option.

Example

```
admin@sonic:~$ sudo config aaa authentication failthrough enable
```

aaa authentication fallback

When the servers used in TACACS+ authentication are unreachable, it falls back to local authentication by default.

The fallback function is always enabled in the current design, which means this command takes no effect.

Usage

```
config aaa authentication fallback (enable | disable | default)
```

The below example enables the fallback option.

Example

```
admin@sonic:~$ sudo config aaa authentication fallback enable
```

NOTE: In the SONiC current design, the authentication fallback is always enabled. It makes this command as a malfunction one.

aaa authentication login

This command is used to configure the sequence of the authentication methods must be performed.

Usage

```
config aaa authentication login {radius | tacacs+ | local} [radius | tacacs+ | local]...
```

Parameters:

- radius: Enables remote authentication based on RADIUS.
- tacacs+: Enables remote authentication based on TACACS+.
- local: Uses local authentication.
- default: Resets back to the default value, which is only "local" authentication.

The command below configures AAA to use a remote TACACS+ database for user authentication.

Example

```
admin@sonic:~$ sudo config aaa authentication login tacacs+ local
```

aaa authentication login default

This command is used to restore the aaa authentication login configuration.

Usage

```
config aaa authentication login default
```

Parameters:

- default: Resets back to the default value, which is only "local" authentication.

aaa authorization

This command is used to enable authorization and define the sequence of the authorization methods must be performed.

Usage

```
config aaa authorization {tacacs+ | local} [local]
```

Parameters:

- tacacs+: Enables TACACS+ authorization.
- local: Enables local authorization.
- default: Resets back to the default value, which is only "local" authorization.

Example

```
admin@sonic:~$ sudo config aaa authorization tacacs+
```

aaa authorization default

This command is used to restore the aaa authorization configuration.

Usage

```
config aaa authorization default
```

Parameters:

- default: Resets back to the default value, which is only "local" authorization.

aaa accounting

This command is used to enable or disable accounting. If accounting enabled, all the methods will be performed.

Usage

```
config aaa accounting {tacacs+ | local} [tacacs+ | local]
```

Parameters:

- tacacs+: Enable TACACS+ accounting.
- local: Enable local accounting in syslog.
- disable: Disable accounting.

Example

```
admin@sonic:~$ sudo config aaa accounting tacacs+
```

aaa accounting disable

This command is used to disable accounting.

Usage

```
config aaa accounting disable
```

Parameters:

- disable: Disable accounting.

(202111.2) TACACS+

- TACACS+ show commands
 - show tacacs
- TACACS+ config commands
 - config tacacs add
 - config tacacs delete
 - config tacacs authtype
 - config tacacs default authtype
 - config tacacs passkey
 - config tacacs default passkey
 - config tacacs timeout
 - config tacacs default timeout
 - config tacacs sourceip

TACACS+ show commands

show tacacs

This command displays the global configuration fields and the list of all TACACS+ servers and their corresponding configurations.

Usage

```
show tacacs
```

The command below shows the TACACS+ configuration.

Example

```
admin@sonic:~$ show tacacs
TACPLUS global auth_type pap (default)
TACPLUS global timeout 99
TACPLUS global passkey <EMPTY_STRING> (default)
TACPLUS_SERVER address 10.11.12.14
priority 9
tcp_port 50
auth_type mschap
timeout 10
passkey testing789
TACPLUS_SERVER address 10.0.0.9
priority 1
tcp_port 49
```

TACACS+ config commands

config tacacs add

This command is used to add a TACACS+ server to the TACACS server list. Note that more than one TACACS+ server (maximum of eight) can be added to the device. When a user tries to log in, the TACACS client will contact the servers one by one. When any server times out, the device will try the other servers one by one based on the priority value configured for that server. When this command is executed, the configured TACACS+ server addresses are updated in the `/etc/pam.d/common-auth-sonic` configuration file that is being used by the TACACS service.

Usage

```
config tacacs add <ip_address> [-t|--timeout <seconds>] [-k|--key <secret>] [-a|--type <type>] [-o|--port <port>] [-p|--pri <priority>] [-m|--use-mgmt-vrf]
```

Parameters:

- **ip_address**: TACACS+ server IP address.
- **timeout**: transmission timeout interval in seconds.
 - range 1 to 60,
 - default 5
- **key**: shared secret key for authentication.
- **type**, "chap" or "pap" or "mschap" or "login".
 - default is "pap". port: TCP
 - "mschap" is not supported now and "pap" is used instead if authtype is configured as "mschap".
- **port**: port range is 1 to 65535.
 - default 49
- **pri**: priority.
 - range 1 to 64,
 - default 1.
- **use-mgmt-vrf**: indicate that the server is part of the Management VRF.
 - default is "no vrf"

The command below adds a new TACACS+ server with the parameters.

Example

```
admin@sonic:~$ sudo config tacacs add 10.11.12.13 -t 10 -k testing789 -a pap -o 50 -p 9
```

config tacacs delete

This command is used to delete the TACACS+ servers configured.

Usage

```
config tacacs delete <ip_address>
```

Parameters:

- **ip_address**: TACACS+ server IP address.

The example below deletes a TACACS+ server by IP address.

Example

```
admin@sonic:~$ sudo config tacacs delete 10.11.12.13
```

config tacacs authtype

This command is used to specify the global auth-type of TACACS+ servers.

Usage

```
config tacacs authtype {chap|pap|mschap|login}
```

Parameters:

- **chap**: Challenge-handshake authentication protocol.
- **pap**: Password authentication protocol.
- **mschap**: Microsoft version (version 2) of the challenge-handshake authentication protocol.
- **login**: Clear text.

Example

```
sudo config tacacs authtype mschap
```

config tacacs default authtype

This command is used to restore the global auth-type of TACACS+ servers.

Usage

```
config tacacs default authtype
```

Parameters:

none

The default is **pap**.

Example

```
sudo config tacacs default authtype
```

config tacacs passkey

This command is used to specify the global passkey of TACACS+ servers.

Usage

```
config tacacs passkey <passkey>
```

Parameters:

- **passkey**: shared secret key (Valid chars: ASCII printable except SPACE, '#', and COMMA).

Example

```
admin@sonic:~$ sudo config tacacs passkey testing123
```

config tacacs default passkey

This command is used to restore the global passkey of TACACS+ servers.

Usage

```
config tacacs default passkey
```

Parameters:

none

The default is blank string.

Example

```
sudo config tacacs default passkey
```

config tacacs timeout

This command is used to specify the global timeout of TACACS+ servers.

Usage

```
config tacacs timeout <timeout>
```

Parameters:

- **timeout**: duration in seconds to wait for before determining that the server has failed to respond. Minimum: 1. Maximum: 60.

Example (Configure non-default case)

```
sudo config tacacs timeout 60
```

config tacacs default timeout

This command is used to restore the global timeout of TACACS+ servers.

Usage

```
config tacacs default timeout
```

Parameters:

none

The default is 5 seconds.

Example

```
sudo config tacacs default timeout
```

config tacacs sourceip

This command is used to specify the global source address of TACACS+ servers.

Usage

```
config tacacs sourceip <source-ip>
```

Parameters:

- source-ip: source IP address (IPv4 or IPv6) for the outgoing TCP/IP dgram.

Example

```
sudo config tacacs sourceip 192.168.3.45
```


(202111.2) RADIUS

- [RADIUS](#)
 - [RADIUS Show Commands](#)
 - [show radius](#)
 - [RADIUS Configuration Commands](#)
 - [config radius timeout](#)
 - [config radius default timeout](#)
 - [config radius retransmit](#)
 - [config radius default retransmit](#)
 - [config radius authtype](#)
 - [config radius default authtype](#)
 - [config radius passkey](#)
 - [config radius default passkey](#)
 - [config radius sourceip](#)
 - [config radius default sourceip](#)
 - [config radius nasip](#)
 - [config radius default nasip](#)
 - [config radius statistics](#)
 - [config radius add](#)
 - [config radius delete](#)

RADIUS

The document shows several commands related to RADIUS service as a RADIUS client (NAS).

RADIUS Show Commands

show radius

This command is used to show RADIUS configuration.

Usage
<code>show radius</code>

Example
<pre>admin@sonic:~\$ show radius RADIUS global auth_type pap (default) RADIUS global retransmit 3 (default) RADIUS global timeout 5 (default) RADIUS global passkey <EMPTY_STRING> (default) RADIUS_SERVER address 192.168.3.45 auth_port 1812 priority 1</pre>

RADIUS Configuration Commands

config radius timeout

This command is used to specify the global timeout of RADIUS servers.

Usage

```
config radius timeout <timeout>
```

Parameters:

- timeout: duration in seconds to wait for before determining that the server has failed to respond. Minimum: 1. Maximum: 60.

Example

```
sudo config radius time 5
```

config radius default timeout

This command is used to restore the global timeout of RADIUS servers.

Usage

```
config radius default timeout
```

Parameters:

none

The default is 5 seconds.

Example

```
sudo config radius default timeout
```

config radius retransmit

This command is used to specify the global retry attempts of RADIUS servers.

Usage

```
config radius retransmit <retries>
```

Parameters:

- retries: number of times to re-send a packet, if there is no response. Minimum: 0. Maximum: 10.

Example

```
config radius retransmit 5
```

config radius default retransmit

This command is used to restore the global retry attempts of RADIUS servers.

Usage

```
config radius default retransmit
```

Parameters:

none

The default is 3 times.

Example

```
sudo config radius default retransmit
```

config radius authtype

This command is used to specify the global auth-type of RADIUS servers.

Usage

```
config radius authtype {chap|pap|mschapv2}
```

Parameters:

- **chap**: Challenge-handshake authentication protocol.
- **pap**: Password authentication protocol.
- **mschapv2**: Microsoft version (version 2) of the challenge-handshake authentication protocol.

Example

```
sudo config radius authtype chap
```

config radius default authtype

This command is used to restore the global auth-type of RADIUS servers.

Usage

```
config radius default authtype
```

Parameters:

none

The default is **pap**.

Example

```
sudo config radius default authtype
```

config radius passkey

This command is used to specify the global passkey of RADIUS servers.

Usage

```
config radius passkey <passkey>
```

Parameters:

- passkey: shared secret key (Valid chars: ASCII printable except SPACE, '#', and COMMA).

Example

```
sudo config radius passkey PASSKEY
```

config radius default passkey

This command is used to restore the global passkey of RADIUS servers.

Usage

```
config radius default passkey
```

Parameters:

none

The default is blank string.

Example

```
sudo config radius default passkey
```

config radius sourceip

This command is used to specify the global source address of RADIUS servers.

Usage

```
config radius sourceip <source-ip>
```

Parameters:

- source-ip: source IP address (IPv4 or IPv6) used in the IP header of the outgoing UDP/IP dgram.

Example

```
sudo config radius sourceip 192.168.3.45
```

config radius default sourceip

This command is use to restore the global source address of RADIUS servers.

Usage

```
config radius default sourceip
```

Parameters:

none

The default ip address is determined by routing stack.

Example

```
sudo config radius default sourceip
```

config radius nasip

This command is used to specify the global NAS-IP address of RADIUS servers.

Usage

```
config radius nasip <nas-ip>
```

Parameters:

- nas-ip: NAS-IP (IPv4 Address, Type4) or IPV6-Address (Type 95) attribute in the outgoing RADIUS PDU. If the nas-ip is not configured, one of the management IP addresses or the source IP address will be used as NAS-IP.

Example

```
sudo config radius nasip 192.168.3.45
```

config radius default nasip

This command is used to remove the global NAS-IP address of RADIUS servers.

Usage

```
config radius default nasip
```

Parameters:

none

Example

```
sudo config radius default nasip
```

config radius statistics

This command is used to specify the global statistics RADIUS servers.

Usage

```
config radius statistics {enable|disable|default}
```

Parameters:

- enable: enable statistics collection.
- disable: disable statistics collection.
- default: reset to disable.

Example

```
sudo config radius statistics default
```

config radius add

Usage

```
config radius add <ip_address_or_domain_name> [--retransmit <retransmit>] [--timeout <timeout>] [--key <key>] [--auth_type {chap | pap | mschapv2}] [--auth-port <auth-port>] [--pri <priority>] [--user-mgmt-vrf] [--source-interface <source-interface>]
```

Parameters:

- <ip_address_or_domain_name>: RADIUS server's DNS name, IPv4 or IPv6 address.
- <retransmit>: number of times to re-send a packet if there is no response.
- <timeout>: duration in seconds to wait for before determining that the server has failed to respond.
- <key>: shared secret key (Valid chars: ASCII printable except SPACE, '#', and COMMA).
- <auth_type>: authentication type. Valid values are:
 - **chap**: Challenge-handshake authentication protocol.
 - **pap**: Password authentication protocol.
 - **mschapv2**: Microsoft version (version 2) of the challenge-handshake authentication protocol.
- <auth-port>: authentication port number.

- <priority>: specify RADIUS server's priority.
- use-mgmt-vrf: use default vrf if not specified.
- <source-interface>: eg: eth0, Loopback0... Use the first IP address retrieved for the interface for the outgoing UDP/IP dgram. The default interface is determined by the routing stack.

Example

```
sudo config radius add 192.168.3.45 --key test --auth-port 10240
```

config radius delete

This command is used to delete a RADIUS server.

Usage

```
config radius delete <ip_address_or_domain_name>
```

Parameters:

- <ip_address_or_domain_name>: IP address or domain name of RADIUS server.

Example

```
sudo config radius delete 192.168.3.45
```

(202111.2) ACL

- [ACL Show Commands](#)
 - `show acl table`
 - `show acl rule`
 - `aclshow`
 - `counterpoll show`
- [ACL Configuration Commands](#)
 - `config acl add table`
 - `config acl remove table`
 - `config acl update full`
 - `config acl update incremental`
 - `counterpoll acl enable`
 - `counterpoll acl disable`
 - `counterpoll acl interval`

This section explains the various show and configuration commands available for users.

ACL Show Commands

`show acl table`

Displays a specific table or all.

Usage

```
show acl table [<table_name>]
```

Parameter

- `<table_name>`, the table to be displayed.

This command shows the table's name, type, bound interface(s), and applied stage.

Example

```
admin@sonic:~$ show acl table
```

Name	Type	Binding	Description	Stage
EVERFLOW	MIRROR	Ethernet16 Ethernet96 Ethernet108 Ethernet112 PortChannel0001 PortChannel0002	EVERFLOW	ingress
SNMP_ACL	CTRLPLANE	SNMP	SNMP_ACL	ingress
DT_ACL_T1	L3	Ethernet0 Ethernet4 Ethernet112 Ethernet116	DATA_ACL_TABLE_1	egress
SSH_ONLY	CTRLPLANE	SSH	SSH_ONLY	ingress

`show acl rule`

Displays the rules in a specific ACL table or all tables, or a specific rule in the specified table.

Usage

```
show acl rule [<table_name> [<rule_name>]]
```

Parameter

- <table_name>, the table to be displayed
- <rule_name>, the rule name to be searched

Notes:

- All ACL rules shown in this command are the rules in the configuration file. It doesn't indicate they are successfully configured into the hardware. To examine which rules are configured into the hardware, use **aclshow** command.

Example

```
admin@sonic:~$ show acl rule
Table      Rule      Priority  Action      Match
-----
SNMP_ACL   RULE_1     9999     ACCEPT      IP_PROTOCOL: 17
SSH_ONLY   RULE_2     9998     ACCEPT      SRC_IP: 1.1.1.1/32
EVERFLOW   RULE_3     9997     MIRROR INGRESS: everflow0 SRC_IP: 1.1.1.1/32
EVERFLOW   RULE_4     9996     MIRROR EGRESS : everflow1 SRC_IP: 20.0.0.2/32
DATAACL    RULE_5     9995     REDIRECT: Ethernet8      L4_SRC_PORT: 4621
DATAACL    RULE_6     9994     FORWARD     IP_PROTOCOL: 126
DATAACL    RULE_7     9993     FORWARD     L4_SRC_PORT: 179
SNMP_ACL   DEFAULT_RULE 1        DROP         L4_DST_PORT: 179
SSH_ONLY   DEFAULT_RULE 1        DROP         ETHER_TYPE: 2048
SSH_ONLY   DEFAULT_RULE 1        DROP         ETHER_TYPE: 2048
```

Information in the table(s):

- Table, ACL table name
- Rule, ACL rule name in the table
- Priority, priority of the rule
- Action
 - action to be executed when a packet matches the rule. One of the following:
 - ACCEPT
 - for control plane ACL
 - DROP
 - FORWARD
 - MIRROR INGRESS <mirror section>
 - <mirror section> refer to [\(202111.2\) Mirroring#\(202012\)Mirroring-showmirror_session](#)
 - MIRROR EGRESS <mirror section>
 - <mirror section> refer to [\(202111.2\) Mirroring#\(202012\)Mirroring-showmirror_session](#)
 - REDIRECT <redirect-object>
 - <redirect-object> One of the following::
 - interface; e.g., physical interface (Ethernet10) or port channel (PortChannel0002)
 - next-hop IP address, e.g., "10.0.0.1"
 - next-hop group of IP address(s), e.g., "10.0.0.1, 10.0.0.3"
- Match, the matching pattern of the frame header. The rule is executed when the ingress frame matches the pattern.

aclshow

Using **aclshow** to display ACL rules and counters.

Usage

```
aclshow [{-c | --clear}] [{-a | --all}] [{-t | --tables} TABLES] [{-r | --rules} RULES]
```

Parameters:

- clear, clear ACL counters' statistics.
- all, show all ACL counters.
- rules <RULES>, list of ACL rules. For example: "Rule_1,Rule_2".
- table <TABLES>, list of ACL tables. For example: "Table_1,Table_2".

Notes:

- If the **PACKETS COUNT** and **BYTES COUNT** are shown as **N/A**, either the rule hasn't been successfully written into the hardware or the counter value hasn't be read by the flex-counter. Check the syslog for the specified error message, or use **counterpoll** command to change the pool interval.

Example

```
admin@sonic:~$ aclshow -a
RULE NAME      TABLE NAME    PRIO    PACKETS COUNT    BYTES COUNT
-----
RULE_1         L3_TABLE       9999      0                0
DEFAULT_RULE   L3_TABLE       1         0                0
```

counterpoll show

Show each counter configuration.

Usage

```
counterpoll show
```

Example

```
admin@sonic:~$ counterpoll show
Type              Interval (in ms)    Status
-----
QUEUE_STAT        1000                enable
PORT_STAT         default (1000)      enable
PORT_BUFFER_DROP  default (60000)     enable
RIF_STAT          default (1000)      enable
QUEUE_WATERMARK_STAT 10000              enable
PG_WATERMARK_STAT  10000               enable
PG_DROP_STAT      default (10000)     enable
BUFFER_POOL_WATERMARK_STAT 10000             enable
ACL               default (10000)     enable
```

ACL Configuration Commands

This sub-section explains the list of configuration options available for the ACL module.

config acl add table

This command applies an ACL to the data plane or control plane. When it's applied to the data plane, the rules are applied to the frame in chip. When it's applied to the control plane, they are applied to the frame trapped to the CPU from the chip.

When applying an ACL to the data plane, you need to use <type> to specify the IP header type or mirroring, the interface, and the stage at which the ACL rules of the table are applied for filtering or forwarding packets.

Usage for data plane

```
config acl add table <table_name> <type> [{-d | --description} <description>] [{-p | --ports} <interfaces>]
[{-s | --stage} {ingress | egress}]
```

Parameter

- <table_name>, the ACL table to be bound on the interface(s)
- <type>, one of the following:
 - **L3** - inspect IPv4 header
 - **L3V6** - inspect IPv6 header
 - **MIRROR** - inspect IPv4 header for mirroring
 - **MIRRORV6** - inspect IPv6 header for mirroring
 - **MIRROR_DSCP** - inspect DSCP for mirroring
- <description>, a description to remind users of the goal. The default is the ACL table name.
- <interfaces>, the interface(s) that are bound to this ACL.
 - The interface(s) can be Ethernet, PortChannel, or VLAN. You may use commas ',' to separate multiple interfaces in the list (for example, "Ethernet01, Ethernet02, PortChannel0001"), but VLANs cannot be mix-configured with Ethernet or PortChannel interfaces.
 - The default option is to apply the ACL to all front ports and PortChannels.
- <stage>, the stage this ACL table will be applied to, either ingress or egress. (The default is ingress)

When applying to the control plane, use the keyword CTRLPLANE and list the service protocols to be examined. You cannot use <interface> since it is a management frame and the destination is the CPU.

Usage for control plane

```
config acl add table <table_name> CTRLPLANE [{-d | --description} <description>] {-v | --services}
<services_string>
```

Parameters:

- <table_name>, the ACL table to be applied to management frames.
- <description>, a description to remind users of the goal. The default is the ACL table name.
- <services_string>, a list of services to be matched, which are separated by commas. Valid values are:
 - **SNMP**: SNMP protocol.
 - **NTP**: NTP protocol.
 - **SSH**: SSH protocol.

Some examples follow:

The ACL table "DATA_L3" is applied to interfaces "Ethernet0, Ethernet4" and inspects the IPv4 header in the ingress direction.

Example

```
admin@sonic:~$ sudo config acl add table DATA_L3 L3 --description "L3 table" --ports "Ethernet0,Ethernet4" --stage "ingress"
```

The ACL table "DATA_L3V6" is applied to interfaces "PortChannel0001, PortChannel0002" and inspects the IPv6 header in the ingress direction.

Example

```
admin@sonic:~$ sudo config acl add table DATA_L3V6 L3V6 --description "L3V6 table" --ports "PortChannel0001, PortChannel0002" --stage "ingress"
```

The ACL table "DATA_CTRL" is applied to frames that are trapped to the CPU and whose services are SSH or SNMP.

Example

```
admin@sonic:~$ sudo config acl add table DATA_CTRL CTRLPLANE --description "CTRLPLANE table" --services "SSH, SNMP"
```

Note:

The ACL table may not be provisioned if the chip is out of capacity, even when the ACL table exists in the configuration file. You would see "sonic ERR swss#orchagent: :- addAclTable: Failed to create ACL table" in syslog when an ACL table provisioning has failed.

config acl remove table

This command is used to remove the binding of an ACL table on an interface. That is, the frames going through the interface are not inspected by the rule(s).

Usage

```
config acl remove table <table_name>
```

Parameters:

- <table_name>, the ACL table to be removed from binding.

Example

```
admin@sonic:~$ sudo config acl remove table DATA_L3
admin@sonic:~$ sudo config acl remove table DATA_L3V6
```

config acl update full

This command is used to update the rules in all tables, or one specific table that already exists. That is, this command does not create an ACL table if it has not been created. You can use the **show acl rule** command to examine the updated ACL rules.

Usage

```
config acl update full [--table_name <table_name>] [--session_name <session_name>] <acl_configuration_file>
```

Parameters:

- <table_name>, the name of the ACL table to be loaded from the configuration file. Example: **config acl update full "--table_name DT_ACL_T1 /etc/sonic/acl_table_1.json"**
- <session_name>, the mirror session to be loaded from the configuration file. Example: **config acl update full "--session_name mirror_ses1 /etc/sonic/acl_table_1.json"**
 - this command fails if the <session_name> does not exist in the the configuration file when it is specified.
- <acl_configuration_file>, a json file that contains the ACL configurations. Refer to section [ACL Rule Configuration File](#) for details.

Note:

- The ACL rules may not be provisioned if the chip is out of capacity (even when the rules exist in the configuration file) or the value of some parameters are invalid. You would see "sonic ERR swss#orchagent: :- Failed to create ACL rule. Rule configuration is invalid" in syslog when an ACL rule provisioning has failed.
- Each ACL table contains a default rule that matches everything if none of the higher priority rules are matched. The default rule is used to deny any unmatched traffic.

The following examples show how to 1. replace all ACL tables, 2. replace the ACL table "DATA_L3" only, 3. replace the mirroring session "flow_local_l3" only.

Example

```
admin@sonic:~$ sudo config acl update full acl_rule_L3_scrip_dstip.json
admin@sonic:~$ sudo config acl update full --table_name DATA_L3 acl_rule_L3_scrip_dstip.json
admin@sonic:~$ sudo config acl update full --session_name flow_local_l3 acl_rule_L3_mirror_session.json
```

The configuration file "acl_rule_L3_scrip_dstip.json" has one rule in the "DATA_L3" table: forward (ACCEPT) the IPv4 packet whose destination IP address is "4.4.4.4" and source IP address is "3.3.3.3".

acl_rule_L3_scrip_dstip.json

```
{
  "acl":
  {
    "acl-sets":
    {
      "acl-set":
      {
        "DATA_L3":
        {
          "acl-entries":
          {
            "acl-entry":
            {
              "1":
              {
                "ip":
                {
                  "config":
                  {
                    "destination-ip-address": "4.4.4.4/32",
                    "source-ip-address": "3.3.3.3/32"
                  }
                },
                "config":
                {
                  "sequence-id": 1
                },
                "actions":
                {
                  "config":
                  {
                    "forwarding-action": "ACCEPT"
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
```

Note: This command removes all the rules in ACL tables and inserts the rules in the input file, i.e., it replaces the rule(s) with those in the configuration file.

The following example shows how to create a rule to match any packets and forward them.

Example

```
admin@sonic:~$ sudo config acl update full acl_rule_match_any_forward.json
```

The configuration file "acl_rule_match_any_forward.json" has one rule in the "DATA_L3" table: forward (ACCEPT) any packets.

acl_rule_match_any_forward.json

```
cat > acl_rule.json << EEE
{
  "acl":
  {
    "acl-sets":
    {
      "acl-set":
      {
        "DATA_L3":
        {
          "acl-entries":
          {
            "acl-entry":
            {
              "1":
              {
                "config":
                {
                  "sequence-id": 1
                },
                "actions":
                {
                  "config":
                  {
                    "forwarding-action": "ACCEPT"
                  }
                }
              }
            }
          }
        }
      }
    }
  }
}
EEE
```

config acl update incremental

Compared to **config acl update full**, this command performs incremental updates, which checks and modifies the corresponding rule(s), or updates the mirror session if <session name> is specified and configured.

Usage

```
config acl update incremental [--session_name <session_name>] <acl_json_file>
```

Parameters

- <session_name>, the mirror session to be loaded from the configuration file. Example: **config acl update incremental "--session_name mirror_ses1 /etc/sonic/acl_table_1.json"**
- <acl_json_file>, a json file that contains the ACL configurations. Refer to section [ACL Rule Configuration File](#) for details.

Note:

- The ACL rules may not be provisioned if the chip is out of capacity (even when the rules exist in the configuration file) or the value of some parameters are invalid. You would see "sonic ERR swss#orchagent: :- Failed to create ACL rule. Rule configuration is invalid" in syslog when an ACL table provisioning has failed.
- Each ACL table contains a default rule that matches everything if none of the higher priority rules are matched. The default rule is used to deny any unmatched traffic.

Example

```
admin@sonic:~$ sudo config acl update incremental acl_rule_L3_scrip_dstip.json
admin@sonic:~$ sudo config acl update incremental --session_name flow_local_l3 acl_rule_L3_mirror_session.json
```

counterpoll acl enable

Activate the flex counter to monitor the number and the size (in bytes) of packets that match the ACL rule.

Usage

```
counterpoll acl enable
```

Example

```
admin@sonic:~$ counterpoll acl enable
```

counterpoll acl disable

Deactivate the flex counter to monitor the number of packets and bytes that match the ACL rule.

Usage

```
counterpoll acl disable
```

Example

```
admin@sonic:~$ counterpoll acl disable
```

counterpoll acl interval

Set the polling interval of flex counter.

Usage

```
counterpoll acl interval <poll_interval>
```

Parameters

- <poll_interval>: the interval in milliseconds to poll the ACL rule matched count. Minimum: 1000. Maximum: 30000.

Configure the polling interval to 2 seconds.

Example

```
admin@sonic:~$ counterpoll acl interval 2000
```


(202111.2) ARP

- [ARP show commands](#)
 - [show arp](#)

ARP show commands

show arp

This command displays the ARP entries in the device with following options.

1. Display the entire table.
2. Display the ARP entries learnt on a specific interface.
3. Display the ARP of a specific IP address.

Usage

```
show arp [-if|--iface <interface_name>] [<ip_address>]
```

Details:

- show arp: Displays all entries.
- show arp -if <interface_name>: Displays the entries that are associated with the specified interface <interface_name>.
- show arp <ip_address>: Displays the entries that are associated with the specified address <ip_address>.

Example

```
admin@sonic:~$ show arp
Address          MacAddress      Iface          Vlan
-----
192.168.1.183    88:5a:92:fb:bf:41 Ethernet44     -
192.168.1.175    88:5a:92:fc:95:81 Ethernet28     -
192.168.1.181    e4:c7:22:c1:07:7c Ethernet40     -
192.168.1.179    88:5a:92:de:a8:bc Ethernet36     -
192.168.1.118    00:1c:73:3c:de:43 Ethernet64     -
192.168.1.11     00:1c:73:3c:e1:38 Ethernet88     -
192.168.1.161    24:e9:b3:71:3a:01 Ethernet0      -
192.168.1.189    24:e9:b3:9d:57:41 Ethernet56     -
192.168.1.187    74:26:ac:8b:8f:c1 Ethernet52     -
192.168.1.165    88:5a:92:de:a0:7c Ethernet8      -

Total number of entries 10
```

Optionally, you can specify the interface in order to display the ARPs learnt on that particular interface.

Example

```
admin@sonic:~$ show arp -if Ethernet40
Address          MacAddress      Iface          Vlan
-----
192.168.1.181    e4:c7:22:c1:07:7c Ethernet40     -

Total number of entries 1
```

Optionally, you can specify an IP address in order to display only that particular entry.

Example

```
admin@sonic:~$ show arp 192.168.1.181
Address      MacAddress      Iface      Vlan
-----
192.168.1.181  e4:c7:22:c1:07:7c  Ethernet40  -
Total number of entries 1
```

(202111.2) BFD (Basic)

Bidirectional Forwarding Detection (BFD) uses the FRR vtysh shell CLI. The following commands are available in the FRR shell. Use the command **vytysh** to enter FRR mode.

Example

```
admin@sonic:~$ vtysh
```

```
Hello, this is FRRouting (version 7.2.1-sonic).
Copyright 1996-2005 Kunihiro Ishiguro, et al.
```

```
sonic#
```

Refer to [FRR BFDd Commands](#) and [FRR BGP Commands](#) for more related FRR commands.

- [BFD Show Commands](#)
 - [show bfd peers](#)
 - [show bfd peer](#)
 - [show bfd peer\(s\) counters](#)
- [BFD Peer Specific Commands](#)
 - [peer](#)
 - [receive-interval](#)
 - [transmit-interval](#)
 - [echo-interval](#)
 - [echo-mode](#)
- [BGP BFD Commands](#)
 - [bfd](#)
 - [ebgp-multihop](#)
 - [bfd info on bgp neighbors](#)

BFD Show Commands

This sub-section explains how to display BFD information.

show bfd peers

Shows all configured BFD peer information and the current status.

Usage

```
show bfd [vrf <vrf_name>] peers [json]
```

Example

```
sonic# show bfd peers
BFD Peers:
  peer 192.169.3.33 vrf default
    ID: 1766365608
    Remote ID: 455730325
    Status: up
    Uptime: 9 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Local timers:
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo transmission interval: 50ms
    Remote timers:
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo transmission interval: 50ms

  peer 192.169.1.11 vrf default
    ID: 1991080464
    Remote ID: 3120344281
    Status: up
    Uptime: 2 hour(s), 5 minute(s), 11 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Local timers:
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo transmission interval: 50ms
    Remote timers:
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo transmission interval: 50ms
```

Example (json)

```
sonic# show bfd peers json
[{"multihop":false,"peer":"192.169.3.33","vrf":"default","id":1766365608,"remote-id":455730325,"status":"up","uptime":355,"diagnostic":"ok","remote-diagnostic":"ok","receive-interval":300,"transmit-interval":300,"echo-interval":0,"remote-receive-interval":300,"remote-transmit-interval":300,"remote-echo-interval":50}, {"multihop":false,"peer":"192.169.1.11","vrf":"default","id":1991080464,"remote-id":3120344281,"status":"up","uptime":7857,"diagnostic":"ok","remote-diagnostic":"ok","receive-interval":300,"transmit-interval":300,"echo-interval":0,"remote-receive-interval":300,"remote-transmit-interval":300,"remote-echo-interval":50}]
```

show bfd peer

Shows specified BFD peer information and the current status.

Usage

```
show bfd [vrf <vrf_name>] peer [ipv4_addr|peer_label|ipv6_addr]
```

Example

```
sonic# show bfd vrf default peer 192.169.1.11
BFD Peer:
  peer 192.169.1.11 vrf default
    ID: 1991080464
    Remote ID: 3120344281
    Status: up
    Uptime: 2 hour(s), 42 minute(s), 25 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Local timers:
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo transmission interval: 50ms
    Remote timers:
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo transmission interval: 50ms
```

Example (json)

```
sonic# show bfd peer 192.169.1.11 json
{"multihop":false,"peer":"192.169.1.11","vrf":"default","id":1991080464,"remote-id":3120344281,"status":"up","uptime":10273,"diagnostic":"ok","remote-diagnostic":"ok","receive-interval":300,"transmit-interval":300,"echo-interval":0,"remote-receive-interval":300,"remote-transmit-interval":300,"remote-echo-interval":50}
```

show bfd peer(s) counters

Shows BFD counters for a particular BFD session.

Usage

```
show bfd peer counters
show bfd peers counters
```

Example (all peers)

```
sonic# show bfd peers counters
BFD Peers:
  peer 192.169.3.33 vrf default
    Control packet input: 15996 packets
    Control packet output: 16008 packets
    Echo packet input: 0 packets
    Echo packet output: 0 packets
    Session up events: 1
    Session down events: 0
    Zebra notifications: 1

  peer 192.169.1.11 vrf default
    Control packet input: 44536 packets
    Control packet output: 44542 packets
    Echo packet input: 0 packets
    Echo packet output: 0 packets
    Session up events: 1
    Session down events: 0
    Zebra notifications: 1
```

Example (specified peer)

```
sonic# show bfd peer 192.169.3.33 counters
peer 192.169.3.33 vrf default
Control packet input: 16032 packets
Control packet output: 16044 packets
Echo packet input: 0 packets
Echo packet output: 0 packets
Session up events: 1
Session down events: 0
Zebra notifications: 1
```

BFD Peer Specific Commands

This sub-section explains how to configure a BFD peer and its settings.

All commands in this sub-section are entered in BFD mode. Use the **bfd** command to open the BFD daemon configuration mode.

Usage

```
bfd
```

Example

```
sonic# config
sonic(config)# bfd
sonic(config-bfd)#
```

peer

Creates and configures a new BFD peer to listen and talk to.

Usage

```
peer <ipv4_addr|ipv6_addr>
```

Example

```
sonic(config-bfd)# peer 192.169.1.11
sonic(config-bfd-peer)#
```

receive-interval

Configures the minimum interval that this system is capable of receiving control packets. The default value is 300 milliseconds.

Usage

```
receive-interval <value_in_ms>
```

Example

```
sonic(config-bfd)# peer 192.169.1.11
sonic(config-bfd-peer)# receive-interval 1000
sonic(config-bfd-peer)# end
sonic# show bfd peer 192.169.1.11
BFD Peer:
    peer 192.169.1.11 vrf default
        ID: 1991080464
        Remote ID: 3120344281
        Status: up
        Uptime: 4 hour(s), 11 minute(s), 22 second(s)
        Diagnostics: ok
        Remote diagnostics: ok
        Local timers:
            Receive interval: 1000ms
            Transmission interval: 300ms
            Echo transmission interval: 50ms
        Remote timers:
            Receive interval: 300ms
            Transmission interval: 300ms
            Echo transmission interval: 50ms
```

transmit-interval

Configures the minimum transmission interval that this system wants to use to send BFD control packets. The default value is 300 milliseconds.

Usage

```
transmit-interval <value_in_ms>
```

Example

```
sonic(config-bfd)# peer 192.169.1.11
sonic(config-bfd-peer)# transmit-interval 1000
sonic(config-bfd-peer)# end
sonic# show bfd peer 192.169.1.11
BFD Peer:
    peer 192.169.1.11 vrf default
        ID: 1991080464
        Remote ID: 3120344281
        Status: up
        Uptime: 4 hour(s), 13 minute(s), 36 second(s)
        Diagnostics: ok
        Remote diagnostics: ok
        Local timers:
            Receive interval: 1000ms
            Transmission interval: 1000ms
            Echo transmission interval: 50ms
        Remote timers:
            Receive interval: 300ms
            Transmission interval: 300ms
            Echo transmission interval: 50ms
```

echo-interval

Configures the minimum echo receive transmission interval. The default value is 50 milliseconds.

Usage

```
echo-interval <value_in_ms>
```

Example

```
sonic(config-bfd)# peer 192.169.1.11
sonic(config-bfd-peer)# echo-interval 100
sonic(config-bfd-peer)# end
sonic# show bfd peer 192.169.1.11
BFD Peer:
    peer 192.169.1.11 vrf default
        ID: 1991080464
        Remote ID: 3120344281
        Status: up
        Uptime: 4 hour(s), 29 minute(s), 12 second(s)
        Diagnostics: ok
        Remote diagnostics: ok
        Local timers:
            Receive interval: 1000ms
            Transmission interval: 1000ms
            Echo transmission interval: 100ms
        Remote timers:
            Receive interval: 300ms
            Transmission interval: 300ms
            Echo transmission interval: 50ms
```

echo-mode

Enables/disables the echo transmission mode. This mode is disabled by default.

It is recommended that the transmission interval of control packets to be increased after enabling echo-mode to reduce bandwidth usage. For example: transmit-interval 2000.

When using multi-hop mode, **echo-mode** will not work (see [RFC 5883](#) section 3).

Usage

```
[no] echo-mode
```

Example

```
sonic(config-bfd)# peer 192.169.1.11
sonic(config-bfd-peer)# echo-mode
sonic(config-bfd-peer)# no echo-mode
```

BGP BFD Commands

This sub-section explains how to configure or display BFD related settings on BGP neighbors.

The following commands are available in BGP configuration mode.

Usage

```
router bgp <as_id>
```


Example

```
sonic# config
sonic(config)# router bgp 65101
sonic(config-router)#
```

bfd

Enables/disables BFD on BGP neighbors.

Usage

```
[no] neighbor <ipv4_addr|ipv6_addr|peer_label> bfd
```

Example

```
sonic(config-router)# neighbor 192.169.0.2 remote-as 65101
sonic(config-router)# neighbor 192.169.0.2 bfd
```

ebgp-multihop

Specifying **ebgp-multihop** allows sessions with eBGP neighbors to establish when they are multiple hops away.

Usage

```
[no] neighbor <ipv4_addr|neighbor_interface|ipv6_addr> ebgp-multihop
```

Example

```
sonic(config-router)# neighbor 192.169.0.2 ebgp-multihop
```

bfd info on bgp neighbors

When BGP neighbors use BFD to detect if the next hop router is alive, the BFD related information is shown in **show bgp neighbors**.

Usage

```
show bgp neighbors [ipv4_addr|neighbor_interface|ipv6_addr]
```

Example (single hop)

```
sonic# config
sonic(config)# router bgp 65101
sonic(config-router)# neighbor 192.169.0.2 remote-as 65101
sonic(config-router)# neighbor 192.169.0.2 bfd
sonic(config-router)# end
sonic# show bgp neighbors
BGP neighbor is 192.169.0.2, remote AS 65101, local AS 65101, internal link
Hostname: sonic
  BGP version 4, remote router ID 192.169.0.2, local router ID 192.169.0.1
  BGP state = Established, up for 00:01:18
  Last read 00:00:18, Last write 00:00:18
  Hold time is 180, keepalive interval is 60 seconds
  Neighbor capabilities:
    4 Byte AS: advertised and received
    AddPath:
      IPv4 Unicast: RX advertised IPv4 Unicast and received
    Route refresh: advertised and received(old & new)
    Address Family IPv4 Unicast: advertised and received
    Hostname Capability: advertised (name: sonic, domain name: n/a) received (name: sonic, domain name: n/a)
    Graceful Restart Capabilty: advertised and received
      Remote Restart timer is 120 seconds
    Address families by peer:
      none
  Graceful restart information:
    End-of-RIB send: IPv4 Unicast
    End-of-RIB received: IPv4 Unicast
  Message statistics:
    Inq depth is 0
    Outq depth is 0

      Sent      Rcvd
Opens:          1          1
Notifications:  0          0
Updates:        1          1
Keepalives:     2          2
Route Refresh:  0          0
Capability:     0          0
Total:          4          4
  Minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
  Update group 1, subgroup 1
  Packet Queue length 0
  Community attribute sent to this neighbor(all)
  0 accepted prefixes

Connections established 1; dropped 0
  Last reset 00:01:36, No AFI/SAFI activated for peer
Local host: 192.169.0.1, Local port: 179
Foreign host: 192.169.0.2, Foreign port: 53912
Nexthop: 192.169.0.1
Nexthop global: fe80::6f8:f8ff:fee5:c2ba
Nexthop local: fe80::6f8:f8ff:fee5:c2ba
BGP connection: shared network
BGP Connect Retry Timer in Seconds: 120
Read thread: on Write thread: on FD used: 27

BFD: Type: single hop
  Detect Multiplier: 3, Min Rx interval: 300, Min Tx interval: 300
  Status: Up, Last update: 0:00:00:05
```

Example (multi hop)

```
sonic# config
sonic(config)# router bgp 65101
sonic(config-router)# neighbor 192.169.2.2 remote-as 65102
sonic(config-router)# neighbor 192.169.2.2 ebgp-multihop
sonic(config-router)# neighbor 192.169.2.2 bfd
sonic# show bgp neighbors
BGP neighbor is 192.169.2.2, remote AS 65102, local AS 65101, external link
Hostname: sonic
  BGP version 4, remote router ID 192.169.2.2, local router ID 192.169.1.1
  BGP state = Established, up for 00:00:02
  Last read 00:00:01, Last write 00:00:01
  Hold time is 180, keepalive interval is 60 seconds
  Neighbor capabilities:
    4 Byte AS: advertised and received
    AddPath:
      IPv4 Unicast: RX advertised IPv4 Unicast and received
    Route refresh: advertised and received(old & new)
    Address Family IPv4 Unicast: advertised and received
    Hostname Capability: advertised (name: sonic, domain name: n/a) received (name: sonic, domain name: n/a)
    Graceful Restart Capabilty: advertised and received
      Remote Restart timer is 120 seconds
    Address families by peer:
      none
  Graceful restart information:
    End-of-RIB send: IPv4 Unicast
    End-of-RIB received: IPv4 Unicast
  Message statistics:
    Inq depth is 0
    Outq depth is 0

      Sent      Rcvd
Opens:          2          2
Notifications:  0          0
Updates:        2          2
Keepalives:     5          5
Route Refresh:  0          0
Capability:     0          0
Total:          9          9
  Minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
  Update group 2, subgroup 2
  Packet Queue length 0
  Community attribute sent to this neighbor(all)
  0 accepted prefixes

Connections established 2; dropped 1
Last reset 00:01:59, Waiting for NHT
  External BGP neighbor may be up to 255 hops away.
Local host: 192.169.1.1, Local port: 51462
Foreign host: 192.169.2.2, Foreign port: 179
Nexthop: 192.169.1.1
Nexthop global: fe80::6f8:f8ff:fee5:c2ba
Nexthop local: fe80::6f8:f8ff:fee5:c2ba
BGP connection: non shared network
BGP Connect Retry Timer in Seconds: 120
Estimated round trip time: 877 ms
Read thread: on Write thread: on FD used: 27

BFD: Type: multi hop
  Detect Multiplier: 3, Min Rx interval: 300, Min Tx interval: 300
  Status: Up, Last update: 0:00:00:02
```


(202111.2) BGP

This section explains all the BGP show commands and BGP configuration commands in both "Quagga" and "FRR" routing software that are supported in SONiC. In 201811 and older versions "Quagga" was enabled by default. In current version "FRR" is enabled by default.

Most of the FRR show commands start with "show bgp" (201904 - 201911) or "show ip bgp" (after 202006). Similar commands in Quagga starts with "show ip bgp". All sub-options supported in all these show commands are common for FRR and Quagga. Detailed show command examples for Quagga are provided at the end of this document. This section covers only the commands supported by FRR.

- BGP show commands
 - show bgp summary (FRR, SONiC versions 201904 - 201911)
 - show ip bgp summary (Quagga, SONiC versions before 201811 or FRR, SONiC versions after 202006)
 - show bgp neighbors (FRR, SONiC version after 201904 and before 201911)
 - show ip bgp neighbors (Quagga, SONiC version before 201811 or FRR, SONiC version after 202006)
 - show ip bgp network
 - show bgp ipv6 summary (FRR, SONiC versions after 201904 and before 201911)
 - show ipv6 bgp summary (Quagga, SONiC versions before 201811 or FRR, SONiC versions after 202006)
 - show bgp ipv6 neighbors (FRR, SONiC versions after 201904 and before 201911)
 - show ipv6 bgp neighbors (Quagga, SONiC versions before 201811 or FRR, SONiC versions after 202006)
 - show ipv6 bgp network
 - show route-map
- BGP config commands
 - config bgp shutdown all
 - config bgp shutdown neighbor
 - config bgp startup all
 - config bgp startup neighbor
 - config bgp remove neighbor
- Notes

BGP show commands

show bgp summary (FRR, SONiC versions 201904 - 201911)

This command displays the summary of all IPv4 and IPv6 BGP neighbors that are configured and the corresponding states.

(FRR, SONiC versions $\geq$ 201904 and $\leq$ 201911)

Usage (FRR, version after 201904 and before 201911)

show bgp summary

Example (FRR, bgp)

```
admin@sonic:~$ show bgp summary
```

IPv4 Unicast Summary:

BGP router identifier 10.1.0.32, local AS number 65100 vrf-id 0

BGP table version 6465

RIB entries 12807, using 2001 KiB of memory

Peers 4, using 83 KiB of memory

Peer groups 2, using 128 bytes of memory

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.0.0.57	4	64600	3995	4001	0	0	0	00:39:32	6400
10.0.0.59	4	64600	3995	3998	0	0	0	00:39:32	6400
10.0.0.61	4	64600	3995	4001	0	0	0	00:39:32	6400
10.0.0.63	4	64600	3995	3998	0	0	0	00:39:32	6400

Total number of neighbors 4

IPv6 Unicast Summary:

BGP router identifier 10.1.0.32, local AS number 65100 vrf-id 0

BGP table version 12803

RIB entries 12805, using 2001 KiB of memory

Peers 4, using 83 KiB of memory

Peer groups 2, using 128 bytes of memory

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
fc00::72	4	64600	3995	5208	0	0	0	00:39:30	6400
fc00::76	4	64600	3994	5208	0	0	0	00:39:30	6400
fc00::7a	4	64600	3993	5208	0	0	0	00:39:30	6400
fc00::7e	4	64600	3993	5208	0	0	0	00:39:30	6400

Total number of neighbors 4

show ip bgp summary (Quagga, SONiC versions before 201811 or FRR, SONiC versions after 202006)

This command displays a summary of all IPv4 and IPv6 BGP neighbors that are configured and the corresponding states.

(Quagga, SONiC version <= 201811 or FRR, SONiC version >= 202006)

Usage (Quagga, version before 201811 or FRR, version after 202006)

```
show ip bgp summary
```

Example (Quagga/FRR after 202006, ip bgp)

```
admin@sonic:~$ show ip bgp summary
```

IPv4 Unicast Summary:

BGP router identifier 10.1.0.32, local AS number 65100 vrf-id 0

BGP table version 12770

RIB entries 12807, using 2301 KiB of memory

Peers 4, using 82 KiB of memory

Peer groups 6, using 384 bytes of memory

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd	NeighborName
10.0.0.57	4	64600	3346	4494	0	0	0	00:07:09	6400	ARISTA01T1
10.0.0.59	4	64600	3346	4494	0	0	0	00:07:09	6400	ARISTA02T1
10.0.0.61	4	64600	3348	4493	0	0	0	00:07:09	6400	ARISTA03T1
10.0.0.63	4	64600	3347	4493	0	0	0	00:07:09	6400	ARISTA04T1

Total number of neighbors 4

show bgp neighbors (FRR, SONiC version after 201904 and before 201911)

This command displays all the details of IPv4 and IPv6 BGP neighbors when no optional argument is specified. When the optional argument **IPv4_address** is specified, it displays the detailed neighbor information about that specific IPv4 neighbor. The command has additional optional arguments to display only the advertised routes, or the received routes, or all routes. In order to see details for an IPv6 neighbor, use the **show bgp ipv6 neighbor <ipv6_address>** command.

Usage (FRR, after 201904 and before 201911)

```
show bgp neighbors [<ipv4-address> [advertised-routes | received-routes | routes]]
```

Example (FRR, after 201904 and before 201911)

```
admin@sonic:~$ show bgp neighbors
BGP neighbor is 10.0.0.57, remote AS 64600, local AS 65100, external link
Description: ARISTA01T1
BGP version 4, remote router ID 100.1.0.29, local router ID 10.1.0.32
BGP state = Established, up for 00:42:15
Last read 00:00:00, Last write 00:00:03
Hold time is 10, keepalive interval is 3 seconds
Configured hold time is 10, keepalive interval is 3 seconds
Neighbor capabilities:
  4 Byte AS: advertised and received
  AddPath:
    IPv4 Unicast: RX advertised IPv4 Unicast and received
  Route refresh: advertised and received(new)
  Address Family IPv4 Unicast: advertised and received
  Hostname Capability: advertised (name: sonic-z9264f-9251, domain name: n/a) not received
  Graceful Restart Capability: advertised and received
    Remote Restart timer is 300 seconds
  Address families by peer:
    none
Graceful restart information:
  End-of-RIB send: IPv4 Unicast
  End-of-RIB received: IPv4 Unicast
Message statistics:
  Inq depth is 0
  Outq depth is 0

```

	Sent	Rcvd
Opens:	2	1
Notifications:	2	0
Updates:	3206	3202
Keepalives:	845	847
Route Refresh:	0	0
Capability:	0	0
Total:	4055	4050

```

Minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
Update group 1, subgroup 1
Packet Queue length 0
Inbound soft reconfiguration allowed
Community attribute sent to this neighbor(all)
6400 accepted prefixes

Connections established 1; dropped 0
Last reset 00:42:37, due to NOTIFICATION sent (Cease/Connection collision resolution)
Local host: 10.0.0.56, Local port: 179
Foreign host: 10.0.0.57, Foreign port: 46419
Nexthop: 10.0.0.56
Nexthop global: fc00::71
Nexthop local: fe80::2204:fff:fe36:9449
BGP connection: shared network
BGP Connect Retry Timer in Seconds: 120
Read thread: on Write thread: on

```

Optionally, you can specify an IP address in order to display only that particular neighbor. In this mode, you can optionally specify whether you want to display all routes advertised to the specified neighbor, all routes received from the specified neighbor, or all routes (received and accepted) from the specified neighbor.

Example (FRR, specific neighbor's information)

```
admin@sonic:~$ show bgp neighbors 10.0.0.57
BGP neighbor is 10.0.0.57, remote AS 64600, local AS 65100, external link
```



```

Description: ARISTA01T1
Member of peer-group PEER_V4 for session parameters
BGP version 4, remote router ID 100.1.0.29, local router ID 10.1.0.32
BGP state = Established, up for 00:04:24
Last read 00:00:00, Last write 00:00:01
Hold time is 10, keepalive interval is 3 seconds
Configured hold time is 10, keepalive interval is 3 seconds
Neighbor capabilities:
  4 Byte AS: advertised and received
  AddPath:
    IPv4 Unicast: RX advertised IPv4 Unicast and received
  Route refresh: advertised and received(new)
  Address Family IPv4 Unicast: advertised and received
  Hostname Capability: advertised (name: as7726-32x-1, domain name: n/a) not received
  Graceful Restart Capability: advertised and received
    Remote Restart timer is 300 seconds
  Address families by peer:
    none
Graceful restart information:
  End-of-RIB send: IPv4 Unicast
  End-of-RIB received: IPv4 Unicast
Message statistics:
  Inq depth is 0
  Outq depth is 0

      Sent      Rcvd
Opens:          2         1
Notifications:  2         0
Updates:       4228      3202
Keepalives:     89        88
Route Refresh:   0         0
Capability:      0         0
Total:         4321      3291
Minimum time between advertisement runs is 0 seconds

For address family: IPv4 Unicast
PEER_V4 peer-group member
Update group 1, subgroup 1
Packet Queue length 0
Inbound soft reconfiguration allowed
Community attribute sent to this neighbor(all)
Inbound path policy configured
Outbound path policy configured
Route map for incoming advertisements is *FROM_BGP_PEER_V4
Route map for outgoing advertisements is *TO_BGP_PEER_V4
6400 accepted prefixes

Connections established 1; dropped 0
Last reset 00:04:36, No AFI/SAFI activated for peer
Local host: 10.0.0.56, Local port: 179
Foreign host: 10.0.0.57, Foreign port: 57108
Nexthop: 10.0.0.56
Nexthop global: fc00::71
Nexthop local: fe80::82a2:35ff:fe46:999
BGP connection: shared network
BGP Connect Retry Timer in Seconds: 120
Read thread: on Write thread: on FD used: 32

admin@sonic:~$ show bgp neighbors 10.0.0.57 advertised-routes
BGP table version is 11920, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

      Network      Next Hop      Metric LocPrf Weight Path
*> 0.0.0.0/0      10.0.0.59          0 64600 65534 6666 6667 i

```

```

*> 10.1.0.32/32      0.0.0.0          0          32768 i
*> 100.1.0.29/32    10.0.0.57        0 64600 i
*> 100.1.0.30/32    10.0.0.59        0 64600 i

...

*> 193.11.248.0/25  10.0.0.61        0 64600 65534 64799 65515 i
*> 193.11.248.128/25
    10.0.0.61      0 64600 65534 64799 65515 i

Total number of prefixes 6405
admin@sonic:~$ show bgp neighbors 10.0.0.57 received-routes
BGP table version is 0, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

    Network          Next Hop          Metric LocPrf Weight Path
*> 0.0.0.0/0         10.0.0.57          0 64600 65534 6666 6667 i
*> 100.1.0.29/32     10.0.0.57          0 64600 i
*> 192.168.8.0/25    10.0.0.57          0 64600 65501 i
*> 192.168.8.128/25 10.0.0.57          0 64600 65501 i
*> 192.168.16.0/25   10.0.0.57          0 64600 65502 i

...

*> 193.11.248.0/25  10.0.0.57          0 64600 65534 64799 65515 i
*> 193.11.248.128/25
    10.0.0.57      0 64600 65534 64799 65515 i

Total number of prefixes 6400
admin@sonic:~$ show bgp neighbors 10.0.0.57 routes
BGP table version is 11920, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

    Network          Next Hop          Metric LocPrf Weight Path
*= 0.0.0.0/0         10.0.0.57          0 64600 65534 6666 6667 i
*> 100.1.0.29/32     10.0.0.57          0 64600 i
*= 192.168.8.0/25    10.0.0.57          0 64600 65501 i
*= 192.168.8.128/25 10.0.0.57          0 64600 65501 i
*= 192.168.16.0/25   10.0.0.57          0 64600 65502 i
*= 192.168.16.128/25
    10.0.0.57      0 64600 65502 i

...

*= 193.11.240.0/25  10.0.0.57          0 64600 65534 64799 65514 i
*= 193.11.240.128/25
    10.0.0.57      0 64600 65534 64799 65514 i
*= 193.11.248.0/25  10.0.0.57          0 64600 65534 64799 65515 i
*= 193.11.248.128/25
    10.0.0.57      0 64600 65534 64799 65515 i

Displayed 6400 routes and 25602 total paths

```

show ip bgp neighbors (Quagga, SONiC version before 201811 or FRR, SONiC version after 202006)

This command displays all the details of IPv4 and IPv6 BGP neighbors when no optional argument is specified. When the optional argument **IPv4_address** is specified, it displays the detailed neighbor information about that specific IPv4 neighbor. The command has additional optional arguments to display only the advertised routes, or the received routes, or all routes. In order to see details for an IPv6 neighbor, use the **show bgp ipv6 neighbor <ipv6_address>** command.

Usage (Quagga, before 201811 or FRR, after 202006)

```
show ip bgp neighbors [<ipv4-address> [advertised-routes | received-routes | routes]]
```

For an example of **show ip bgp neighbors**, you can refer to **show bgp neighbors** with the same options.

show ip bgp network

This command displays all the details of IPv4 Border Gateway Protocol (BGP) prefixes.

Usage

```
show ip bgp network [[<ipv4-address>|<ipv4-prefix>] [(bestpath | multipath | longer-prefixes | json)]]
```

Example

```
admin@sonic:~$ show ip bgp network
BGP table version is 10839, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*= 0.0.0.0/0        10.0.0.63          0 64600 65534 6666 6667 i
*=                  10.0.0.61          0 64600 65534 6666 6667 i
*>                  10.0.0.57          0 64600 65534 6666 6667 i
*=                  10.0.0.59          0 64600 65534 6666 6667 i

...

*= 193.11.248.128/25
                  10.0.0.63          0 64600 65534 64799 65515 i
*=                  10.0.0.61          0 64600 65534 64799 65515 i
*=                  10.0.0.59          0 64600 65534 64799 65515 i
*>                  10.0.0.57          0 64600 65534 64799 65515 i

Displayed 6405 routes and 25602 total paths
admin@sonic:~$ show ip bgp network 192.168.8.130 bestpath
BGP routing table entry for 192.168.8.128/25
Paths: (4 available, best #3, table default)
  Advertised to non peer-group peers:
  10.0.0.57 10.0.0.59 10.0.0.61 10.0.0.63
  64600 65501
  10.0.0.57 from 10.0.0.57 (100.1.0.29)
    Origin IGP, valid, external, multipath, best (Router ID)
    Last update: Tue Nov 24 07:23:34 2020

admin@sonic:~$ show ip bgp network 192.168.8.130 multipath
BGP routing table entry for 192.168.8.128/25
Paths: (4 available, best #3, table default)
  Advertised to non peer-group peers:
```

```
10.0.0.57 10.0.0.59 10.0.0.61 10.0.0.63
64600 65501
  10.0.0.63 from 10.0.0.63 (100.1.0.32)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 07:23:34 2020

64600 65501
  10.0.0.61 from 10.0.0.61 (100.1.0.31)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 07:23:34 2020

64600 65501
  10.0.0.57 from 10.0.0.57 (100.1.0.29)
    Origin IGP, valid, external, multipath, best (Router ID)
    Last update: Tue Nov 24 07:23:34 2020

64600 65501
  10.0.0.59 from 10.0.0.59 (100.1.0.30)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 07:23:34 2020
```

```
admin@sonic:~$ show ip bgp network 10.1.0.32 json
```

```
{
  "prefix": "10.1.0.32/32",
  "advertisedTo": {
    "10.0.0.57": {
    },
    "10.0.0.59": {
    },
    "10.0.0.61": {
    },
    "10.0.0.63": {
    }
  },
  "paths": [
    {
      "aspath": {
        "string": "Local",
        "segments": [
        ],
        "length": 0
      },
      "origin": "IGP",
      "med": 0,
      "metric": 0,
      "weight": 32768,
      "valid": true,
      "sourced": true,
      "local": true,
      "bestpath": {
        "overall": true,
        "selectionReason": "First path received"
      },
      "lastUpdate": {
        "epoch": 1606202576,
        "string": "Tue Nov 24 07:22:56 2020\n"
      },
      "nexthops": [
        {
          "ip": "0.0.0.0",
          "afi": "ipv4",
          "metric": 0,
          "accessible": true,
          "used": true
        }
      ],
      "peer": {
```

```

        "peerId":"0.0.0.0",
        "routerId":"10.1.0.32"
    }
}
]
}
admin@sonic:~$ show ip bgp network 10.1.0.32/32 bestpath
BGP routing table entry for 10.1.0.32/32
Paths: (1 available, best #1, table default)
  Advertised to non peer-group peers:
    10.0.0.57 10.0.0.59 10.0.0.61 10.0.0.63
  Local
    0.0.0.0 from 0.0.0.0 (10.1.0.32)
      Origin IGP, metric 0, weight 32768, valid, sourced, local, best (First path received)
      Last update: Tue Nov 24 07:22:56 2020

admin@sonic:~$ show ip bgp network 10.1.0.32/32 multipath
BGP routing table entry for 10.1.0.32/32
Paths: (1 available, best #1, table default)
  Advertised to non peer-group peers:
    10.0.0.57 10.0.0.59 10.0.0.61 10.0.0.63
  Local
    0.0.0.0 from 0.0.0.0 (10.1.0.32)
      Origin IGP, metric 0, weight 32768, valid, sourced, local, best (First path received)
      Last update: Tue Nov 24 07:22:56 2020

admin@sonic:~$ show ip bgp network 10.1.0.32/32 json
{
  "prefix":"10.1.0.32\32",
  "advertisedTo":{
    "10.0.0.57":{
    },
    "10.0.0.59":{
    },
    "10.0.0.61":{
    },
    "10.0.0.63":{
    }
  },
  "paths":[
    {
      "aspath":{
        "string":"Local",
        "segments":[
        ],
        "length":0
      },
      "origin":"IGP",
      "med":0,
      "metric":0,
      "weight":32768,
      "valid":true,
      "sourced":true,
      "local":true,
      "bestpath":{
        "overall":true,
        "selectionReason":"First path received"
      },
      "lastUpdate":{
        "epoch":1606202576,
        "string":"Tue Nov 24 07:22:56 2020\n"
      },
      "nexthops":[
        {
          "ip":"0.0.0.0",
          "afi":"ipv4",
          "metric":0,

```

```

        "accessible":true,
        "used":true
    }
],
"peer":{
    "peerId":"0.0.0.0",
    "routerId":"10.1.0.32"
}
}
]
}
admin@sonic:~$ show ip bgp network 10.1.0.32/32 longer-prefixes
BGP table version is 10839, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop           Metric LocPrf Weight Path
*> 10.1.0.32/32     0.0.0.0              0         32768 i

Displayed 1 routes and 25602 total paths

```

Note: The **longer-prefixes** option is only available when a network prefix with a "/" notation is used.

show bgp ipv6 summary (FRR, SONiC versions after 201904 and before 201911)

This command displays the summary of all IPv6 BGP neighbors that are configured and the corresponding states.

Usage (FRR, after 201904 and before 201911)

```
show bgp ipv6 summary
```

Example (FRR, after 201904 and before 201911)

```

admin@sonic:~$ show bgp ipv6 summary
BGP router identifier 10.1.0.32, local AS number 65100 vrf-id 0
BGP table version 12803
RIB entries 12805, using 2001 KiB of memory
Peers 4, using 83 KiB of memory
Peer groups 2, using 128 bytes of memory

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ OutQ  Up/Down State/PfxRcd NeighborName
fc00::72      4      64600   3995   5208        0    0    0 00:39:30        6400 Lab-T1-01
fc00::76      4      64600   3994   5208        0    0    0 00:39:30        6400 Lab-T1-02
fc00::7a      4      64600   3993   5208        0    0    0 00:39:30        6400 Lab-T1-03
fc00::7e      4      64600   3993   5208        0    0    0 00:39:30        6400 Lab-T1-04

Total number of neighbors 4

```

show ipv6 bgp summary (Quagga, SONiC versions before 201811 or FRR, SONiC versions after 202006)

This command displays the summary of all IPv6 BGP neighbors that are configured and the corresponding states.

Usage (Quagga, before 201811 or FRR, after 202006)

```
show ipv6 bgp summary
```

Example (Quagga, before 201811 or FRR, after 202006)

```
admin@sonic:~$ show ipv6 bgp summary
```

IPv6 Unicast Summary:

BGP router identifier 10.1.0.32, local AS number 65100 vrf-id 0

BGP table version 11661

RIB entries 12807, using 2301 KiB of memory

Peers 4, using 82 KiB of memory

Peer groups 6, using 384 bytes of memory

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd	NeighborName
fc00::72	4	64600	4341	6224	0	0	0	00:56:55	6400	ARISTA01T1
fc00::76	4	64600	4341	6224	0	0	0	00:56:55	6400	ARISTA02T1
fc00::7a	4	64600	4340	6224	0	0	0	00:56:55	6400	ARISTA03T1
fc00::7e	4	64600	4339	6224	0	0	0	00:56:55	6400	ARISTA04T1

Total number of neighbors 4

show bgp ipv6 neighbors (FRR, SONiC versions after 201904 and before 201911)

This command displays all the details of one particular IPv6 Border Gateway Protocol (BGP) neighbor. An option is also available to display only the advertised routes, or the received routes, or all routes.

Usage (FRR, after 201904 and before 201911)

```
show bgp ipv6 neighbors [<ipv6-address> [(advertised-routes | received-routes | routes)]]
```

Example (FRR, after 201904 and before 201911)

```
admin@sonic:~$ show bgp ipv6 neighbors
```

BGP neighbor is fc00::72, remote AS 64600, local AS 65100, external link

Description: ARISTA01T1

Member of peer-group PEER_V6 for session parameters

BGP version 4, remote router ID 100.1.0.29, local router ID 10.1.0.32

BGP state = Established, up for 01:18:47

Last read 00:00:02, Last write 00:00:02

Hold time is 10, keepalive interval is 3 seconds

Configured hold time is 10, keepalive interval is 3 seconds

Neighbor capabilities:

4 Byte AS: advertised and received

AddPath:

IPv6 Unicast: RX advertised IPv6 Unicast and received

Route refresh: advertised and received(new)

Address Family IPv6 Unicast: advertised and received

Hostname Capability: advertised (name: as5812-54x, domain name: n/a) not received

Graceful Restart Capabilty: advertised and received

Remote Restart timer is 300 seconds

Address families by peer:

none

Graceful restart information:

```

End-of-RIB send: IPv6 Unicast
End-of-RIB received: IPv6 Unicast
Message statistics:
  Inq depth is 0
  Outq depth is 0

                Sent      Rcvd
Opens:           2         1
Notifications:   2         0
Updates:        5081      3202
Keepalives:     1576     1575
Route Refresh:    0         0
Capability:       0         0
Total:          6661     4778
Minimum time between advertisement runs is 0 seconds

```

```

For address family: IPv6 Unicast
PEER_V6 peer-group member
Update group 2, subgroup 2
Packet Queue length 0
Inbound soft reconfiguration allowed
Community attribute sent to this neighbor(all)
Inbound path policy configured
Outbound path policy configured
Route map for incoming advertisements is *FROM_BGP_PEER_V6
Route map for outgoing advertisements is *TO_BGP_PEER_V6
6400 accepted prefixes

```

```

Connections established 1; dropped 0
Last reset 01:19:13, No AFI/SAFI activated for peer
Local host: fc00::71, Local port: 179
Foreign host: fc00::72, Foreign port: 43581
Nexthop: 10.0.0.56
Nexthop global: fc00::71
Nexthop local: fe80::3e2c:99ff:fe15:aa60
BGP connection: shared network
BGP Connect Retry Timer in Seconds: 120
Read thread: on Write thread: on FD used: 22

```

...

```

admin@sonic:~$ show bgp ipv6 neighbors fc00::72
BGP neighbor is fc00::72, remote AS 64600, local AS 65100, external link
Description: ARISTA01T1
Member of peer-group PEER_V6 for session parameters
BGP version 4, remote router ID 100.1.0.29, local router ID 10.1.0.32
BGP state = Established, up for 01:18:47
Last read 00:00:02, Last write 00:00:02
Hold time is 10, keepalive interval is 3 seconds
Configured hold time is 10, keepalive interval is 3 seconds
Neighbor capabilities:
  4 Byte AS: advertised and received
  AddPath:
    IPv6 Unicast: RX advertised IPv6 Unicast and received
    Route refresh: advertised and received(new)
    Address Family IPv6 Unicast: advertised and received
    Hostname Capability: advertised (name: as5812-54x, domain name: n/a) not received
    Graceful Restart Capability: advertised and received
    Remote Restart timer is 300 seconds
    Address families by peer:
      none
Graceful restart information:
  End-of-RIB send: IPv6 Unicast
  End-of-RIB received: IPv6 Unicast
Message statistics:
  Inq depth is 0
  Outq depth is 0

```

```

                Sent      Rcvd

```



```

Opens:                2          1
Notifications:        2          0
Updates:              5081       3202
Keepalives:           1576       1575
Route Refresh:         0          0
Capability:            0          0
Total:                6661       4778

```

Minimum time between advertisement runs is 0 seconds

```

For address family: IPv6 Unicast
PEER_V6 peer-group member
Update group 2, subgroup 2
Packet Queue length 0
Inbound soft reconfiguration allowed
Community attribute sent to this neighbor(all)
Inbound path policy configured
Outbound path policy configured
Route map for incoming advertisements is *FROM_BGP_PEER_V6
Route map for outgoing advertisements is *TO_BGP_PEER_V6
6400 accepted prefixes

```

Connections established 1; dropped 0

Last reset 01:19:13, No AFI/SAFI activated for peer

Local host: fc00::71, Local port: 179

Foreign host: fc00::72, Foreign port: 43581

Nexthop: 10.0.0.56

Nexthop global: fc00::71

Nexthop local: fe80::3e2c:99ff:fe15:aa60

BGP connection: shared network

BGP Connect Retry Timer in Seconds: 120

Read thread: on Write thread: on FD used: 22

admin@sonic:~\$ show bgp ipv6 neighbors fc00::72 advertised-routes

BGP table version is 11661, local router ID is 10.1.0.32, vrf id 0

Default local pref 100, local AS 65100

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, r RIB-failure, S Stale, R Removed

Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self

Origin codes: i - IGP, e - EGP, ? - incomplete

Network	Next Hop	Metric	LocPrf	Weight	Path
*> ::/0	fc00::72	0	64600	65534	6666 6667 i
*> 2064:100::1d/128	fc00::72	0	64600		i
*> 2064:100::1e/128	fc00::76	0	64600		i
*> 2064:100::1f/128	fc00::7a	0	64600		i
*> 2064:100::20/128	fc00::7e	0	64600		i
...					
*> 20c1:bf8::/64	fc00::72	0	64600	65534	64799 65515 i
*> 20c1:bf8:0:80::/64	fc00::72	0	64600	65534	64799 65515 i
*> fc00:1::/64	::	0	32768		i
*> fc02:1000::/64	::	0	32768		i

Total number of prefixes 6405

admin@sonic:~\$ show bgp ipv6 neighbors fc00::72 received-routes

BGP table version is 0, local router ID is 10.1.0.32, vrf id 0

Default local pref 100, local AS 65100

Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, r RIB-failure, S Stale, R Removed

Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self

Origin codes: i - IGP, e - EGP, ? - incomplete

Network	Next Hop	Metric	LocPrf	Weight	Path
*> ::/0	fc00::72	0	64600	65534	6666 6667 i

```
*> 2064:100::1d/128 fc00::72          0 64600 i
*> 20c0:a808::/64 fc00::72          0 64600 65501 i
*> 20c0:a808:0:80::/64
    fc00::72                          0 64600 65501 i
...

*> 20c1:bf8::/64 fc00::72          0 64600 65534 64799 65515 i
*> 20c1:bf8:0:80::/64
    fc00::72                          0 64600 65534 64799 65515 i

Total number of prefixes 6400

admin@sonic:~$ show bgp ipv6 neighbors fc00::72 routes
BGP table version is 11661, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> ::/0             fc00::72          0 64600 65534 6666 6667 i
*> 2064:100::1d/128 fc00::72          0 64600 i
*> 20c0:a808::/64   fc00::72          0 64600 65501 i
*> 20c0:a808:0:80::/64
    fc00::72          0 64600 65501 i
...

*> 20c1:bf8::/64   fc00::72          0 64600 65534 64799 65515 i
*> 20c1:bf8:0:80::/64
    fc00::72          0 64600 65534 64799 65515 i

Displayed 6400 routes and 25602 total paths
```

show ipv6 bgp neighbors (Quagga, SONiC versions before 201811 or FRR, SONiC versions after 202006)

This command displays all the details of one particular IPv6 Border Gateway Protocol (BGP) neighbor. An option is also available to display only the advertised routes, or the received routes, or all routes.

Usage (Quagga, before 201811 or FRR, after 202006)

```
show ipv6 bgp neighbors [<ipv6-address> [(advertised-routes | received-routes | routes)]]
```

For an example of **show ipv6 bgp neighbors**, you can refer to **show bgp ipv6 neighbors** with the same options.

show ipv6 bgp network

This command displays all the details of IPv6 Border Gateway Protocol (BGP) prefixes.

Usage (Quagga or FRR after 202006)

```
show ipv6 bgp network [[<ipv6-address>|<ipv6-prefix>] [(bestpath | multipath | longer-prefixes | json)]]
```

Example (Quagga or FRR after 202006)

```
admin@sonic:~$ show ipv6 bgp network
BGP table version is 11661, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*= ::/0             fc00::7a              0 64600 65534 6666 6667 i
*=                  fc00::7e              0 64600 65534 6666 6667 i
*>                  fc00::72              0 64600 65534 6666 6667 i
*=                  fc00::76              0 64600 65534 6666 6667 i
*> 2064:100::1d/128 fc00::72              0 64600 i
*> 2064:100::1e/128 fc00::76              0 64600 i
*> 2064:100::1f/128 fc00::7a              0 64600 i

...

*= 20c1:bf8:0:80::/64
                  fc00::7e              0 64600 65534 64799 65515 i
*=                  fc00::7a              0 64600 65534 64799 65515 i
*=                  fc00::76              0 64600 65534 64799 65515 i
*>                  fc00::72              0 64600 65534 64799 65515 i
*> fc00:1::/64      ::              0          32768 i
*> fc02:1000::/64   ::              0          32768 i

Displayed 6405 routes and 25602 total paths

admin@sonic:~$ show ipv6 bgp network 2064:100::1d bestpath
BGP routing table entry for 2064:100::1d/128
Paths: (1 available, best #1, table default)
  Advertised to non peer-group peers:
    fc00::72 fc00::76 fc00::7a fc00::7e
    64600
    fc00::72 from fc00::72 (100.1.0.29)
    (fe80::5054:ff:fe02:b909) (prefer-global)
    Origin IGP, valid, external, best (First path received)
    Last update: Tue Nov 24 08:56:51 2020

admin@sonic:~$ show ipv6 bgp network 20c0:a810::/64 multipath
BGP routing table entry for 20c0:a810::/64
Paths: (4 available, best #4, table default)
  Advertised to non peer-group peers:
    fc00::72 fc00::76 fc00::7a fc00::7e
    64600 65502
    fc00::76 from fc00::76 (100.1.0.30)
    (fe80::5054:ff:fe25:758e) (prefer-global)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 08:56:51 2020

    64600 65502
    fc00::7e from fc00::7e (100.1.0.32)
    (fe80::5054:ff:fe51:5f51) (prefer-global)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 08:56:51 2020

    64600 65502
    fc00::72 from fc00::72 (100.1.0.29)
    (fe80::5054:ff:fe02:b909) (prefer-global)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 08:56:51 2020

    64600 65502
```

```
fc00::7a from fc00::7a (100.1.0.31)
(fe80::5054:ff:fe2:88ce) (prefer-global)
  Origin IGP, valid, external, multipath, best (Older Path)
  Last update: Tue Nov 24 08:56:51 2020
```

```
admin@sonic:~$ show ipv6 bgp network 2064:100::1d json
```

```
{
  "prefix":"2064:100::1d\128",
  "advertisedTo":{
    "fc00::72":{
    },
    "fc00::76":{
    },
    "fc00::7a":{
    },
    "fc00::7e":{
    }
  },
  "paths":[
    {
      "aspath":{
        "string":"64600",
        "segments":[
          {
            "type":"as-sequence",
            "list":[
              64600
            ]
          }
        ],
        "length":1
      },
      "origin":"IGP",
      "valid":true,
      "bestpath":{
        "overall":true,
        "selectionReason":"First path received"
      },
      "lastUpdate":{
        "epoch":1606208211,
        "string":"Tue Nov 24 08:56:51 2020\n"
      },
      "nexthops":[
        {
          "ip":"fc00::72",
          "afi":"ipv6",
          "scope":"global",
          "metric":0,
          "accessible":true,
          "used":true
        },
        {
          "ip":"fe80::5054:ff:fe02:b909",
          "afi":"ipv6",
          "scope":"link-local",
          "accessible":true
        }
      ],
      "peer":{
        "peerId":"fc00::72",
        "routerId":"100.1.0.29",
        "type":"external"
      }
    }
  ]
}
```

```
admin@sonic:~$ show ipv6 bgp network 2064:100::1d/128 bestpath
BGP routing table entry for 2064:100::1d/128
Paths: (1 available, best #1, table default)
  Advertised to non peer-group peers:
    fc00::72 fc00::76 fc00::7a fc00::7e
  64600
    fc00::72 from fc00::72 (100.1.0.29)
    (fe80::5054:ff:fe02:b909) (prefer-global)
    Origin IGP, valid, external, best (First path received)
    Last update: Tue Nov 24 08:56:51 2020

admin@sonic:~$ show ipv6 bgp network 20c0:a810::/64 multipath
BGP routing table entry for 20c0:a810::/64
Paths: (4 available, best #4, table default)
  Advertised to non peer-group peers:
    fc00::72 fc00::76 fc00::7a fc00::7e
  64600 65502
    fc00::76 from fc00::76 (100.1.0.30)
    (fe80::5054:ff:fe25:758e) (prefer-global)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 08:56:51 2020

  64600 65502
    fc00::7e from fc00::7e (100.1.0.32)
    (fe80::5054:ff:fe51:5f51) (prefer-global)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 08:56:51 2020

  64600 65502
    fc00::72 from fc00::72 (100.1.0.29)
    (fe80::5054:ff:fe02:b909) (prefer-global)
    Origin IGP, valid, external, multipath
    Last update: Tue Nov 24 08:56:51 2020

  64600 65502
    fc00::7a from fc00::7a (100.1.0.31)
    (fe80::5054:ff:fef2:88ce) (prefer-global)
    Origin IGP, valid, external, multipath, best (Older Path)
    Last update: Tue Nov 24 08:56:51 2020

admin@sonic:~$ show ipv6 bgp network 2064:100::1d/128 json
{
  "prefix":"2064:100::1d\128",
  "advertisedTo":{
    "fc00::72":{
    },
    "fc00::76":{
    },
    "fc00::7a":{
    },
    "fc00::7e":{
    }
  },
  "paths":[
    {
      "aspath":{
        "string":"64600",
        "segments":[
          {
            "type":"as-sequence",
            "list":[
              64600
            ]
          }
        ],
        "length":1
      },
    },
  ],
}
```

```

    "origin": "IGP",
    "valid": true,
    "bestpath": {
      "overall": true,
      "selectionReason": "First path received"
    },
    "lastUpdate": {
      "epoch": 1606208211,
      "string": "Tue Nov 24 08:56:51 2020\n"
    },
    "nexthops": [
      {
        "ip": "fc00::72",
        "afi": "ipv6",
        "scope": "global",
        "metric": 0,
        "accessible": true,
        "used": true
      },
      {
        "ip": "fe80::5054:ff:fe02:b909",
        "afi": "ipv6",
        "scope": "link-local",
        "accessible": true
      }
    ],
    "peer": {
      "peerId": "fc00::72",
      "routerId": "100.1.0.29",
      "type": "external"
    }
  }
}

```

```

admin@sonic:~$ show ipv6 bgp network 2064:100::1d/128 longer-prefixes
BGP table version is 12770, local router ID is 10.1.0.32, vrf id 0
Default local pref 100, local AS 65100
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes:  i - IGP, e - EGP, ? - incomplete

   Network          Next Hop           Metric LocPrf Weight Path
*> 2064:100::1d/128 fc00::72              0 64600 i

Displayed 1 routes and 25602 total paths

```

Note: The **longer-prefixes** option is only available when a network prefix with a "/" notation is used.

show route-map

This command displays the routing policy that takes precedence over the other route processes that are configured.

Usage

```
show route-map
```

Example

```
admin@sonic:~$ show route-map
ZEBRA:
route-map RM_SET_SRC, permit, sequence 10
  Match clauses:
  Set clauses:
    src 10.12.0.102
  Call clause:
  Action:
    Exit routemap
ZEBRA:
route-map RM_SET_SRC6, permit, sequence 10
  Match clauses:
  Set clauses:
    src fc00:1::102
  Call clause:
  Action:
    Exit routemap
BGP:
route-map FROM_BGP_SPEAKER_V4, permit, sequence 10
  Match clauses:
  Set clauses:
  Call clause:
  Action:
    Exit routemap
BGP:
route-map TO_BGP_SPEAKER_V4, deny, sequence 10
  Match clauses:
  Set clauses:
  Call clause:
  Action:
    Exit routemap
BGP:
route-map ISOLATE, permit, sequence 10
  Match clauses:
  Set clauses:
    as-path prepend 65000
  Call clause:
  Action:
    Exit routemap
```

BGP config commands

This sub-section explains the list of configuration options available for the BGP module, both for IPv4 and IPv6 BGP neighbors.

config bgp shutdown all

This command is used to shutdown all the BGP IPv4 and IPv6 sessions. When a session is shutdown using this command, the BGP state in **show ip bgp summary** is displayed as "Idle (Admin)"

Usage

```
config bgp shutdown all
```

Example

```
admin@sonic:~$ sudo config bgp shutdown all
Shutting down BGP session with neighbor 10.0.0.57...
Shutting down BGP session with neighbor 10.0.0.59...
Shutting down BGP session with neighbor 10.0.0.61...
Shutting down BGP session with neighbor 10.0.0.63...
admin@sonic:~$ show ip bgp summary

IPv4 Unicast Summary:
BGP router identifier 10.1.0.32, local AS number 65100 vrf-id 0
BGP table version 21863
RIB entries 3, using 552 bytes of memory
Peers 4, using 82 KiB of memory
Peer groups 6, using 384 bytes of memory
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd	NeighborName
10.0.0.57	4	64600	4437	5982	0	0	0	00:06:02	Idle (Admin)	ARISTA01T1
10.0.0.59	4	64600	4434	4682	0	0	0	00:06:11	Idle (Admin)	ARISTA02T1
10.0.0.61	4	64600	4433	4649	0	0	0	00:06:14	Idle (Admin)	ARISTA03T1
10.0.0.63	4	64600	4431	4603	0	0	0	00:06:16	Idle (Admin)	ARISTA04T1

```
Total number of neighbors 4
```

config bgp shutdown neighbor

This command shuts down a BGP session with a neighbor by specifying the IP address or hostname.

Usage

```
sudo config bgp shutdown neighbor (<ip_address> | <hostname>)
```

Example

```
admin@sonic:~$ sudo config bgp shutdown neighbor 10.0.0.57
Shutting down BGP session with neighbor 10.0.0.57...
admin@sonic:~$ sudo config bgp shutdown neighbor ARISTA01T1
Shutting down BGP session with neighbor 10.0.0.57...
Shutting down BGP session with neighbor fc00::72...
```

config bgp startup all

This command starts up all the IPv4 and IPv6 BGP neighbors

Usage

```
config bgp startup all
```


Example

```
admin@sonic:~$ sudo config bgp startup all
Starting up BGP session with neighbor 10.0.0.57...
Starting up BGP session with neighbor 10.0.0.59...
Starting up BGP session with neighbor 10.0.0.61...
Starting up BGP session with neighbor 10.0.0.63...
```

config bgp startup neighbor

This command starts up a particular IPv4 or IPv6 BGP neighbor using either the IP address or hostname.

Usage

```
config bgp startup neighbor (<ip-address> | <hostname>)
```

Example

```
admin@sonic:~$ sudo config bgp startup neighbor 10.0.0.57
Starting up BGP session with neighbor 10.0.0.57...
admin@sonic:~$ sudo config bgp startup neighbor ARISTA01T1
Starting up BGP session with neighbor 10.0.0.57...
Starting up BGP session with neighbor fc00::72...
```

config bgp remove neighbor

This command removes a particular IPv4 or IPv6 BGP neighbor configuration using either the IP address or hostname.

Usage

```
config bgp remove neighbor <neighbor_ip_or_hostname>
```

Example

```
admin@sonic:~$ sudo config bgp remove neighbor 10.0.0.57
Removed configuration of BGP neighbor 10.0.0.57
admin@sonic:~$ sudo config bgp remove neighbor fc00::72
Removed configuration of BGP neighbor fc00::72
admin@sonic:~$ sudo config bgp remove neighbor ARISTA01T1
Removed configuration of BGP neighbor 10.0.0.57
Removed configuration of BGP neighbor fc00::72
```

Notes

- These SONiC bgp commands (**shutdown/startup/remove**) will only apply to the neighbor entries included in config_db.json (redis db). Otherwise, it will show error message about it could not locate the neighbor and take no effect.

An example entry in config_db.json

```
"BGP_NEIGHBOR": {
  "10.0.0.57": {
    "admin_status": "up",
    "asn": "64600",
    "holdtime": "10",
    "keepalive": "3",
    "local_addr": "10.0.0.56",
    "name": "ARISTA01T1",
    "nhopself": "0",
    "rrclient": "0"
  },

```

Error

```
admin@sonic:~$ sudo config bgp shutdown neighbor 10.1.1.1
Usage: config bgp shutdown neighbor [OPTIONS] <ipaddr_or_hostname>
Try "config bgp shutdown neighbor -h" for help.

Error: Could not locate neighbor '10.1.1.1'
```

For FRR 'split' mode, please use these commands in FRR VTY shell instead:

Example

```
admin@sonic:~$ vtysh

Hello, this is FRRouting (version 8.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

sonic# config
sonic(config)# router bgp 65100
sonic(config-router)# neighbor 10.0.0.61 shutdown (shutdown)
sonic(config-router)# no neighbor 10.0.0.61 shutdown (startup)
sonic(config-router)# no neighbor 10.0.0.61 (remove)
sonic(config-router)# bgp shutdown (shutdown all)
sonic(config-router)# no bgp shutdown (startup all)
```

(202111.2) Configuration File Management

- Loading the Configuration from a JSON File
 - config load
- Loading the Configuration from a Minigraph (XML) File
 - config load_minigraph
- Reloading the Configuration
 - config reload
- Loading the Management Configuration
 - config load_mgmt_config
- Saving the Configuration to a File for Persistence
 - config save

This section explains the commands that are used to load the configuration from either ConfigDB or from minigraph.

Loading the Configuration from a JSON File

config load

This command loads the configuration from a JSON file, such as the file to which SONiC saves its configuration, `/etc/sonic/config_db.json`.

This command loads the configuration from the input file (if an optional filename is specified, that input file is used, otherwise the default `/etc/sonic/config_db.json` file is used) into CONFIG_DB. The configuration present in the input file is applied on top of the already running configuration.

This command does not flush CONFIG_DB before loading the new configuration (i.e., if the configuration present in the input file is same as the current running configuration, nothing happens). If the config present in the input file is not present in the running configuration, it will be added. If the config present in the input file differs (when key matches) from that of the running configuration, it will be modified as per the new values for those keys.

Usage

```
config load [-y|--yes] [<filename>]
```

Parameters:

- "-y" or "--yes", forces the loading without prompting the user for confirmation.
 - If this argument is not specified, it prompts the user to confirm whether the user really wants to load this configuration file.
- <filename>, loads the configuration from the input file (<filename>) and stores into CONFIG_DB; otherwise, it uses `/etc/sonic/config_db.json` as the default input file.

Note: This command does not flush CONFIG_DB before loading the new configuration. That is, if there is a new configuration in <filename>, it will be added into CONFIG_DB (running configuration); if there is any configuration changed in <filename>, it will update CONFIG_DB with those entries.

Example

```
admin@sonic:~$ sudo config load
Load config from the file /etc/sonic/config_db.json? [y/N]: y
Running command: /usr/local/bin/sonic-cfggen -j /etc/sonic/config_db.json --write-to-db
```

Loading the Configuration from a Minigraph (XML) File

Minigraph is used in commands defined before 2017 Sep.

config load_minigraph

This command is used to load the configuration from `/etc/sonic/minigraph.xml`, when users do not want to use the configuration in `config_db.json`.

You can copy the minigraph.xml configuration file to the device and load it using this command. This command restarts various services running in the device and it takes some time to complete the command.

Usage

```
config load_minigraph [-y|--yes] [-n|--no-service-restart]
```

Parameter:

- "-y" or "--yes", forces the loading without prompting the user for confirmation.
 - If the argument is not specified, it prompts the user to confirm whether user really wants to load this configuration file.
- "-n" or "--no-service-restart", loads the configuration without restarting dependent services running on the device.
 - One use case for this option is during boot time when config-setup service loads the minigraph configuration and there are no services running on the device.

Note:

- If you are logged in using SSH, you might get disconnected and some configuration failures could happen, from which it might be hard to recover. You need to reconnect the SSH session after configuring the management IP address. It is recommended to execute this command from the console port.
- The management interface IP address and default route (or specific route) may require reconfiguration in case those parameters are not part of minigraph.xml.

Example

```
admin@sonic:~$ sudo config load_minigraph
Reload config from minigraph? [y/N]: y
Running command: /usr/local/bin/sonic-cfggen -j /etc/sonic/config_db.json --write-to-db
```

Reloading the Configuration

config reload

Clears the current configuration and imports a new configuration from an input file, or from `/etc/sonic/config_db.json`.

Usage

```
config reload [-y|--yes] [-l|--load-sysinfo] [<filename>] [-n|--no-service-restart]
```

Parameters:

- "-y" or "--yes", forces the loading without prompting the user for confirmation.
 - If the argument is not specified, it prompts the user to confirm whether user really wants to load this configuration file.
- "-n" or "--no-service-restart", clears and loads the configuration without restarting dependent services running on the device.
 - One use case for this option is during boot time when config-setup service loads the existing old configuration and there are no services running on the device.

Notes:

- Stops all services before clearing the configuration and then restarts those services after configuration is reloaded.
- It takes some time to complete the command since it needs to restart various running services in the device.
- If users are logged in using SSH, they *might be disconnected* depending upon the new management IP address. Users will need to reconnect their SSH sessions. In general, it is recommended to execute this command from the console port after disconnecting all SSH sessions to the device.
- **config reload** may change the management IP address in the input file (<filename>), this introduces the following possible scenario:
 1. **mgmtIP** (management IP address) is not changed. The SSH session may not be affected if the SSH session has not timed out, but it needs to resume the connection after the interface is re-configured and up.
 2. **mgmtIP** is changed in new configuration file. Users will lose their SSH connections.
 3. There is not any **mgmtIP** in the new configuration file. Users will lose their SSH connections.

- The management interface IP address and default route (or specific route) may require reconfiguration if those parameters are not part of minigraph.xml.

Example

```
admin@sonic:~$ sudo config reload
Clear current config and reload config from the file /etc/sonic/config_db.json? \[y/N\]: y
Running command: systemctl stop dhcp_relay
Running command: systemctl stop swss
Running command: systemctl stop snmp
Warning: Stopping snmp.service, but it can still be activated by:
  snmp.timer
Running command: systemctl stop lldp
Running command: systemctl stop pmon
Running command: systemctl stop bgp
Running command: systemctl stop teamd
Running command: /usr/local/bin/sonic-cfggen -H -k Force10-Z9100-C32 --write-to-db
Running command: /usr/local/bin/sonic-cfggen -j /etc/sonic/config_db.json --write-to-db
Running command: systemctl restart hostname-config
Running command: systemctl restart interfaces-config
Timeout, server 10.11.162.42 not responding.
```

Loading the Management Configuration

config load_mgmt_config

This command is used to reconfigure the hostname and mgmt interface based on the device description file.

Usage

```
config load_mgmt_config [-y|--yes] [<filename>]
```

Parameters:

- "-y" or "--yes", forces the loading without prompting the user for confirmation.
 - If the argument is not specified, it prompts the user to confirm whether user really wants to load this configuration file.

Note: This command either uses the optional file specified in <filename> or looks for the file `/etc/sonic/device_desc.xml`. The execution fails If the file does not exist or does not have valid fields for **hostname** and **ManagementAddress**.

Example

```
admin@sonic:~$ sudo config load_mgmt_config
Reload config from minigraph? [y/N]: y
Running command: /usr/local/bin/sonic-cfggen -M /etc/sonic/device_desc.xml --write-to-db
```

Saving the Configuration to a File for Persistence

config save

This command saves CONFIG_DB (the running configuration) into the user-specified filename or the default `/etc/sonic/config_db.json`.

Usage

```
config save [-y|--yes] [<filename>]
```

Note: This command retains the configuration in storage after rebooting. The file can be transferred to remote machines for debugging. You can use **config load** to load the configuration file at any point and generate a new CONFIG_DB (running configuration) or copy to **/etc/sonic/config_db.json** if you want this newly generated file to be used during reboot.

Example (boot configuration file)

```
admin@sonic:~$ sudo config save -y
```

or

Example (specific file)

```
admin@sonic:~$ sudo config save -y /etc/sonic/config2.json
```

(202111.2) Container Auto-restart

SONiC includes a feature in which Docker containers can be automatically shut down and restarted if one of the critical processes running in the container exits unexpectedly. Restarting the entire container ensures that the configuration is reloaded and all processes in the container are restarted, thus increasing the likelihood of entering a healthy state.

- [Container Auto-restart Show Commands](#)
 - [show feature autorestart](#)
- [Container Auto-restart Config Command](#)
 - [config feature autorestart <feature_name> <autorestart_status>](#)

Container Auto-restart Show Commands

show feature autorestart

This command displays the status of the auto-restart feature for containers.

Usage

```
show feature autorestart [<feature_name>]
```

Example

```
admin@sonic:~$ show feature autorestart
Container Name      Status
-----
database            enabled
syncd               enabled
teamd               disabled
dhcp_relay          enabled
lldp                enabled
pmon                enabled
bgp                 enabled
swss                disabled
telemetry            enabled
sflow               enabled
snmp                enabled
radv                disabled
```

Optionally, you can specify a container name in order to display the auto-restart feature status for that container only.

Container Auto-restart Config Command

config feature autorestart <feature_name> <autorestart_status>

This command configures the status of the auto-restart feature for a specific container.

Usage

```
config feature autorestart <feature_name> (enabled | disabled)
```

Example

```
admin@sonic:~$ sudo config feature autorestart database disabled
```


(202111.2) CoPP

- [Schema](#)
- [Example](#)

CoPP (Control Plane Policing) allows users to manage and rate limit the control plane packets.

The CoPP configuration file is composed of one or more trap groups. For each trap group, it includes one or more predefined trap IDs that are mapped to known protocols (STP, LACP, EAPOL, etc.), and a meter (if required) to decide how to process specific packets (forward, trap to CPU, etc.). For each predefined trap ID, it represents one or more ACL rules to match the expected traffic. For example, one ACL rule is used for the trap ID **stp** to match the traffic whose destination MAC address is **01:80:c2:00:00:00**.

Schema

The following shows the schema of a CoPP configuration file:

Schema

```
{
  "COPP_GROUP": {
    "<group-name>": {
      "queue": "Queue number",
      "trap_action": "Packet trap action",
      "trap_priority": "Priority",
      "meter_type": "Meter type",
      "mode": "Meter mode",
      "color": "Color mode",
      "cbs": "Number",
      "cir": "Number",
      "pbs": "Number",
      "pir": "Number",
      "green_action": "Packet action",
      "yellow_action": "Packet action",
      "red_action": "Packet action",
      "genetlink_name": "Genetlink name",
      "genetlink_mcgrp_name": "Genetlink multicast group name"
    }
  },
  "COPP_TRAP": {
    "<trap-name>": {
      "trap_ids": "List of trap IDs",
      "trap_group": "CoPP group name"
    }
  }
}
```

Parameters:

- **<group-name>**: group name.
 - **queue**: Specifies the CPU queue number (0..47) the packets will enqueue into for processing when **trap_action** is **trap** or **copy**.
 - **trap_action**: Specifies the action for the trap group.
 - **trap**: trap the packets to CPU for process but not forward the packets.
 - **copy**: trap the packets to CPU for process and forward the packets.
 - **drop**: drop the packets.
 - **trap_priority**: Specifies the priority (0..65535) of the trap group. A higher number means higher priority.
 - **meter_type**: Specifies the meter type.
 - *packets*
 - *bytes*
 - **mode**: Specifies the meter mode.
 - *sr_tcm*: srTCM (Single Rate Three Color Marker).
 - *tr_tcm*: trTCM (Triple Rate Three Color Marker).
 - *storm*: storm control.
 - **color**: Specifies the color mode.
 - *aware*: color-aware mode.
 - *blind*: color-blind mode.
 - **cbs**: Committed Burst Size. Packets or bytes depending on the **meter_type** value.

- **cir** : Committed Information Rate. Packets or bytes depending on the **meter_type** value.
- **pbs**: Peak Burst Size. Packets or bytes depending on the **meter_type** value.
- **pir**: Peak Information Rate. Packets or bytes depending on the **meter_type** value.
- **green_action**: specifies the action (??) for packets marked as green color by meter. Valid valuse are same as **trap_action**.
- **yellow_action**: specifies the action (??) for packets marked as yellow color by meter. Valid valuse are same as **trap_action**.
- **red_action**: specifies the action (??) for packets marked as red color by meter. Valid valuse are same as **trap_action**.
- **genetlink_name**: genetlink name. Ignore (Dont add this parameter) if sFlow disabled. The value only support "**psample**" which is used for sFlow only.
- **genetlink_mcgrp_name**: genetlink multicast group name. Ignore (Dont add this parameter) if sFlow disabled. The value only support "**packets**" which is used for sFlow only.
- **<trap-name>**: Trap name to identify different trap configuration.
 - **trap_ids**: Specifies the trap ID in the trap group. It will associate the corresponding packet type with specified trap group. The supported trap IDs are listed in the below.
 - **trap_group**: Specifies the trap group to which the specified Trap ID associates with.

The following shows the supported trap IDs:

- stp
- lacp
- eapol
- lldp
- pvrst
- igmp_query
- sample_packet
- udld
- switch_cust_range
- arp_req
- arp_resp
- dhcp
- ospf
- pim
- dhcpv6
- neigh_discovery
- src_nat_miss
- dest_nat_miss
- isis
- arp_suppression
- nd_suppression
- ip2me
- ssh
- snmp
- bgp
- bgpv6
- bfd
- bfdv6
- l3_mtu_error
- ttl_error
- vxlan_arp

Note: The trap ID **vxlan_arp** is only supported in AS4630-54PE, AS5835-54X, AS5835-54T, AS7326-56X, and AS7726-32X models.

The following shows the predefined trap groups and IDs added by default:

Trap Group:

- Group **default**
 - queue: 0
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: 600
 - cbs: 600
 - red_action: drop
- Group **queue4_group1**
 - trap_action: trap
 - trap_priority: 4
 - queue: 4
- Group **queue4_group2**
 - trap_action: copy
 - trap_priority: 4
 - queue: 4
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: 600
 - cbs: 600
 - red_action: drop
- Group **queue4_group3**
 - trap_action: trap

- trap_priority: 4
- queue: 4
- Group **queue1_group1**
 - trap_action: **trap**
 - trap_priority: 1
 - queue: 1
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: **6000**
 - cbs: **6000**
 - red_action: **drop**
- Group **queue1_group2**
 - trap_action: **trap**
 - trap_priority: 1
 - queue: 1
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: **600**
 - cbs: **600**
 - red_action: **drop**
- Group **queue2_group1**
 - trap_action: **trap**
 - trap_priority: 1
 - queue: 2
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: **1000**
 - cbs: **1000**
 - red_action: **drop**
 - genetlink_name: **psample**
 - genetlink_mcgrp_name: **packets**
- Group **queue4_group4**
 - trap_action: **trap**
 - trap_priority: 4
 - queue: 4
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: **1000**
 - cbs: **1000**
 - red_action: **drop**
- Group **queue4_group5**
 - trap_action: **trap**
 - trap_priority: 4
 - queue: 4
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: **1000**
 - cbs: **1000**
 - red_action: **drop**
- Group **queue4_group6**
 - trap_action: **trap**
 - trap_priority: 5
 - queue: 4
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: **600**
 - cbs: **600**
 - red_action: **drop**
- Group **queue4_group7**
 - trap_action: **copy**
 - trap_priority: 5
 - queue: 4
 - meter_type: **packets**
 - mode: **sr_tcm**
 - cir: **600**
 - cbs: **600**
 - red_action: **drop**

Trap ID:

- Trap **bgp**
 - trap_ids: **bgp,bgpv6**
 - trap_group: **queue4_group1**
- Trap **lACP**
 - trap_ids: **lACP**
 - trap_group: **queue4_group1**
- Trap **arp**
 - trap_ids: **arp_req,arp_resp,neigh_discovery**
 - trap_group: **queue4_group2**

- Trap **lldp**
 - trap_ids: **lldp**
 - trap_group: **queue4_group3**
- Trap **dhcp**
 - trap_ids: **dhcp,dhcpv6**
 - trap_group: **queue4_group3**
- Trap **udld**
 - trap_ids: **udld**
 - trap_group: **queue4_group3**
- Trap **ip2me**
 - trap_ids: **ip2me**
 - trap_group: **queue1_group1**
- Trap **nat**
 - trap_ids: **src_nat_miss,dest_nat_miss**
 - trap_group: **queue1_group2**
- Trap **sflow**
 - trap_ids: **sample_packet**
 - trap_group: **queue2_group1**
- Trap **ospf**
 - trap_ids: **ospf**
 - trap_group: **queue4_group4**
- Trap **isis**
 - trap_ids: **isis**
 - trap_group: **queue4_group5**
- Trap **evpn**
 - trap_ids: **arp_suppression,nd_suppression**
 - trap_group: **queue4_group6**
- Trap **evpn_vxlan**
 - trap_ids: **vxlan_arp**
 - trap_group: **queue4_group7**

Note:

- The trap group **default** is created to trap the IPv4/IPv6 packets whose TTL fields in the IP headers are 0 to CPU.
- The trap group **queue4_group7** is only created for AS4630-54PE, AS5835-54X, AS5835-54T, AS7326-56X, and AS7726-32X models.

Example

The following example shows a trap group named **queue1_group2** is created that and a Trap ID **nat** is also created which includes two NAT related trap IDs (**src_nat_miss** and **dest_nat_miss**) and associate with the **trap group "queue1_group2"** which define the meter:

Example of trap group

```
{
  "COPP_GROUP": {
    "queue1_group2": {
      "trap_action": "trap",
      "trap_priority": "1",
      "queue": "1",
      "meter_type": "packets",
      "mode": "sr_tcm",
      "cir": "600",
      "cbs": "600",
      "red_action": "drop"
    }
  },
  "COPP_TRAP": {
    "nat": {
      "trap_ids": "src_nat_miss,dest_nat_miss",
      "trap_group": "queue1_group2"
    }
  }
}
```

Users can update the user configuration file **/etc/sonic/config_db.json** (or update **Config_DB** and then save it to the **config_db.json** file) to apply changes and to the default config, which is loaded from the file **/etc/sonic/copp_cfg** at system startup. After the file **/etc/sonic/config_db.json** is changed, it is required to reboot the device to make the changes taking effect. The **config_db.json** COPP configuration will override the default configuration if the COPP group name or Trap ID name is same.

Example

```
admin@sonic:~$ vi /etc/sonic/config_db.json
admin@sonic:~$ config reload -y
```

(202111.2) Critical Resource Monitoring

- CRM Show Commands
 - `crm show`
 - `crm show summary`
 - `crm show resources`
 - `crm show resources all`
 - `crm show resources acl`
 - `crm show resources acl group`
 - `crm show resources acl table`
 - `crm show resources dn timer`
 - `crm show resources fdb`
 - `crm show resources ipmc`
 - `crm show resources ipv4`
 - `crm show resources ipv4 neighbor`
 - `crm show resources ipv4 nexthop`
 - `crm show resources ipv4 route`
 - `crm show resources ipv6`
 - `crm show resources ipv6 neighbor`
 - `crm show resources ipv6 nexthop`
 - `crm show resources ipv6 route`
 - `crm show resources mpls`
 - `crm show resources mpls inseg`
 - `crm show resources mpls nexthop`
 - `crm show resources nexthop group`
 - `crm show resources nexthop group member`
 - `crm show resources nexthop group object`
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- CRM Config Commands
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 - `crm config thresholds acl table lowhigh <value>`
 - `crm config thresholds dn timer lowhigh <value>`
 - `crm config thresholds dn timer type <value>`
 - `crm config thresholds fdb type <value>`
 - `crm config thresholds fdb lowhigh <value>`
 - `crm config thresholds ipmc lowhigh <value>`
 - `crm config thresholds ipmc type`
 - `crm config thresholds ipv4 neighbor type <value>`
 - `crm config thresholds ipv4 neighbor lowhigh <value>`
 - `crm config thresholds ipv4 nexthop type <value>`
 - `crm config thresholds ipv4 nexthop lowhigh <value>`
 - `crm config thresholds ipv4 route type <value>`
 - `crm config thresholds ipv4 route lowhigh <value>`
 - `crm config thresholds ipv6 neighbor type <value>`
 - `crm config thresholds ipv6 neighbor lowhigh <value>`
 - `crm config thresholds ipv6 nexthop type <value>`
 - `crm config thresholds ipv6 nexthop lowhigh <value>`
 - `crm config thresholds ipv6 route type <value>`
 - `crm config thresholds ipv6 route lowhigh <value>`
 - `crm config thresholds mpls inseg lowhigh <value>`
 - `crm config thresholds mpls inseg type <value>`
 - `crm config thresholds mpls nexthop lowhigh <value>`
 - `crm config thresholds mpls nexthop type <value>`
 - `crm config thresholds nexthop group member type <value>`
 - `crm config thresholds nexthop group member lowhigh <value>`
 - `crm config thresholds nexthop group object type <value>`
 - `crm config thresholds nexthop group object lowhigh <value>`
 - `crm config thresholds snat lowhigh <value>`
 - `crm config thresholds snat type <value>`
 - `rm config thresholds srv6-my-sid-entry lowhigh <value>`
 - `crm config thresholds srv6-my-sid-entry type <value>`
 - `crm config thresholds srv6-nexthop lowhigh <value>`
 - `crm config thresholds srv6-nexthop type <value>`

CRM Show Commands

`crm show`

Displays the usage of the show command.

Usage

```
admin@sonic:~$ crm show ?
Usage: crm show [OPTIONS] COMMAND [ARGS]...

  Show CRM related information

Options:
  --help  Show this message and exit.

Commands:
  resources  Show CRM resources information
  summary    Show CRM general information
  thresholds Show CRM thresholds information
```

crm show summary

Displays the polling interval setting.

Example

```
admin@sonic:~$ crm show summary

Polling Interval: 300 second(s)
```

crm show resources

Displays the usage of the **show resources** command.

Usage

```
admin@sonic:~$ crm show resources
Usage: crm show resources [OPTIONS] COMMAND [ARGS]...

  Show CRM resources information

Options:
  --help  Show this message and exit.

Commands:
  acl      Show CRM information for acl resource
  all      Show CRM information for all resources
  fdb      Show CRM information for fdb resource
  ipv4     CRM resource IPv4 address family
  ipv6     CRM resource IPv6 address family
  nexthop  Show CRM information for nexthop resource
```

crm show resources all

Displays all statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources all
```

Resource Name	Used Count	Available Count
-----	-----	-----
ipv4_route	66	98238
ipv6_route	68	16316
ipv4_nexthop	0	49150
ipv6_nexthop	0	49150
ipv4_neighbor	0	8192
ipv6_neighbor	0	4096
nexthop_group_member	0	16384
nexthop_group	0	128
fdb_entry	0	32768

Stage	Bind Point	Resource Name	Used Count	Available Count
-----	-----	-----	-----	-----
INGRESS	PORT	acl_group	32	224
INGRESS	PORT	acl_table	3	2
INGRESS	LAG	acl_group	0	224
INGRESS	LAG	acl_table	0	2
INGRESS	VLAN	acl_group	0	224
INGRESS	VLAN	acl_table	0	5
INGRESS	RIF	acl_group	0	224
INGRESS	RIF	acl_table	0	5
INGRESS	SWITCH	acl_group	0	224
INGRESS	SWITCH	acl_table	0	5
EGRESS	PORT	acl_group	0	224
EGRESS	PORT	acl_table	0	2
EGRESS	LAG	acl_group	0	224
EGRESS	LAG	acl_table	0	2
EGRESS	VLAN	acl_group	0	224
EGRESS	VLAN	acl_table	0	2
EGRESS	RIF	acl_group	0	224
EGRESS	RIF	acl_table	0	2
EGRESS	SWITCH	acl_group	0	224
EGRESS	SWITCH	acl_table	0	2

Table ID	Resource Name	Used Count	Available Count
-----	-----	-----	-----
0x7000000000a20	acl_entry	2	6142
0x7000000000a20	acl_counter	2	6142

crm show resources acl

Displays the usage of the ACL resources show command.

Usage

```
admin@sonic:~$ crm show resources acl
Usage: crm show resources acl [OPTIONS] COMMAND [ARGS]...

  Show CRM information for acl resource

Options:
  --help  Show this message and exit.

Commands:
  group  Show CRM information for acl group resource
  table  Show CRM information for acl table resource
```

crm show resources acl group

Displays the ACL group statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources acl group
```

Stage	Bind Point	Resource Name	Used Count	Available Count
INGRESS	PORT	acl_group	32	224
INGRESS	PORT	acl_table	2	3
INGRESS	LAG	acl_group	0	224
INGRESS	LAG	acl_table	0	3
INGRESS	VLAN	acl_group	0	224
INGRESS	VLAN	acl_table	0	6
INGRESS	RIF	acl_group	0	224
INGRESS	RIF	acl_table	0	6
INGRESS	SWITCH	acl_group	0	224
INGRESS	SWITCH	acl_table	0	6
EGRESS	PORT	acl_group	0	224
EGRESS	PORT	acl_table	0	2
EGRESS	LAG	acl_group	0	224
EGRESS	LAG	acl_table	0	2
EGRESS	VLAN	acl_group	0	224
EGRESS	VLAN	acl_table	0	2
EGRESS	RIF	acl_group	0	224
EGRESS	RIF	acl_table	0	2
EGRESS	SWITCH	acl_group	0	224
EGRESS	SWITCH	acl_table	0	2

crm show resources acl table

Displays the ACL table statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources acl table
```

Table ID	Resource Name	Used Count	Available Count
-----	-----	-----	-----
0x7000000000a20	acl_entry	2	6142
0x7000000000a20	acl_counter	2	6142

crm show resources dn timer

(Unsupported in Broadcom platform) Display the usage of DNAT resources show command

```
admin@sonic:~$ crm show resources dn timer
```

Resource Name	Used Count	Available Count
-----	-----	-----
dnat_entry	0	45056

crm show resources fdb

Displays FDB statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources fdb
```

Resource Name	Used Count	Available Count
-----	-----	-----
fdb_entry	0	32768

crm show resources ipmc

(Unsupported) Display the usage of IPMC resources show command

Example

```
admin@sonic:~$ crm show resources ipmc
```

Resource Name	Used Count	Available Count
-----	-----	-----
ipmc_entry	0	0

crm show resources ipv4

Displays the usage of the IPv4 resources show command.

Usage

```
admin@sonic:~$ crm show resources ipv4
Usage: crm show resources ipv4 [OPTIONS] COMMAND [ARGS]...

CRM resource IPv4 address family

Options:
  --help  Show this message and exit.

Commands:
  neighbor  Show CRM information for neighbor resource
  nexthop   Show CRM information for nexthop resource
  route     Show CRM information for route resource
```

crm show resources ipv4 neighbor

Displays the IPv4 neighbor statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources ipv4
Usage: crm show resources ipv4 [OPTIONS] COMMAND [ARGS]...

CRM resource IPv4 address family

Options:
  --help  Show this message and exit.

Commands:
  neighbor  Show CRM information for neighbor resource
  nexthop   Show CRM information for nexthop resource
  route     Show CRM information for route resource
admin@as5812-54x:~$ crm show resources ipv4 neighbor
```

Resource Name	Used Count	Available Count
ipv4_neighbor	0	8192

crm show resources ipv4 nexthop

Displays the IPv4 nexthop statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources ipv4 nexthop
```

Resource Name	Used Count	Available Count
ipv4_nexthop	0	49118

crm show resources ipv4 route

Displays the IPv4 route statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources ipv4 route
```

Resource Name	Used Count	Available Count
-----	-----	-----
ipv4_route	66	98238

crm show resources ipv6

Displays the usage of IPv6 resources show command.

Usage

```
admin@sonic:~$ crm show resources ipv6
```

```
Usage: crm show resources ipv6 [OPTIONS] COMMAND [ARGS]...
```

CRM resource IPv6 address family

Options:

--help Show this message and exit.

Commands:

neighbor Show CRM information for neighbor resource

nexthop Show CRM information for nexthop resource

route Show CRM information for route resource

crm show resources ipv6 neighbor

Displays the IPv6 neighbor statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources ipv6 neighbor
```

Resource Name	Used Count	Available Count
-----	-----	-----
ipv6_neighbor	33	4096

crm show resources ipv6 nexthop

Displays the IPv6 nexthop statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources ipv6 nexthop
```

Resource Name	Used Count	Available Count
-----	-----	-----
ipv6_nexthop	33	49117

crm show resources ipv6 route

Displays the IPv6 route statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources ipv6 route
```

Resource Name	Used Count	Available Count
-----	-----	-----
ipv6_route	68	16316

crm show resources mpls

(Unsupported in Broadcom platform) Display the usage of MPLS resources show command

Usage

```
admin@sonic:~$ crm show resources mpls
```

Usage: crm show resources mpls [OPTIONS] COMMAND [ARGS]...

CRM resource MPLS address family

Options:

--help Show this message and exit.

Commands:

inseg Show CRM information for in-segment resource
 nexthop Show CRM information for nexthop resource

crm show resources mpls inseg

(Unsupported in Broadcom platform) Display MPLS inseg statistic which include used by all components and available resources in the ASIC

Example

```
admin@sonic:~$ crm show resources mpls inseg
```

Resource Name	Used Count	Available Count
-----	-----	-----
mpls_inseg	0	0

crm show resources mpls nexthop

(Unsupported in Broadcom platform) Display MPLS nexthop statistic which include used by all components and available resources in the ASIC

Example

```
admin@sonic:~$ crm show resources mpls nexthop
```

Resource Name	Used Count	Available Count
mpls_nexthop	0	32765

crm show resources nexthop group

Displays the usage of the nexthop group resources show command.

Usage

```
admin@sonic:~$ crm show resources nexthop group
Usage: crm show resources nexthop group [OPTIONS] COMMAND [ARGS]...
```

Show CRM information for nexthop group resource

Options:

--help Show this message and exit.

Commands:

member Show CRM information for nexthop group member resource
 object Show CRM information for nexthop group resource

crm show resources nexthop group member

Displays the nexthop group member statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources nexthop group member
```

Resource Name	Used Count	Available Count
nexthop_group_member	0	16384

crm show resources nexthop group object

Displays the nexthop group object statistics, which includes those used by all components and available resources in the ASIC.

Example

```
admin@sonic:~$ crm show resources nexthop group object
```

Resource Name	Used Count	Available Count
nexthop_group	0	128

crm show resources snat

Display the usage of SNAT resources show command

Example

```
admin@sonic:~$ crm show resources snat
```

Resource Name	Used Count	Available Count
snat_entry	0	45056

crm show resources srv6-my-sid-entry

(Unsupported in Broadcom platform) Display the usage of srv6-my-sid-entry resources show command

Example

```
admin@sonic:~$ crm show resources srv6-my-sid-entry
```

Resource Name	Used Count	Available Count
srv6-my-sid-entry	0	45056

crm show resources srv6-nexthop

(Unsupported in Broadcom platform) Display the usage of srv6-nexthop resources show command

Example

```
admin@sonic:~$ crm show resources srv6-nexthop
```

Resource Name	Used Count	Available Count
srv6-nexthop	0	45056

CRM Config Commands

The config commands should be extended in order to add the "crm" alias to the CRM utility.

crm config

Displays the usage of the config command.

Usage

```
admin@sonic:~$ crm config
Usage: crm config [OPTIONS] COMMAND [ARGS]...
```

CRM related configuration

Options:

--help Show this message and exit.

Commands:

polling CRM polling configuration
thresholds CRM thresholds configuration

crm config polling interval <value>

Configures the polling interval for fetching available resources from the ASIC.

Usage

```
admin@sonic:~$ crm config polling interval
Usage: crm config polling interval [OPTIONS] INTERVAL
```

Example

```
admin@sonic:~$ crm config polling interval 400
```

crm config thresholds

Displays the usage of the **config thresholds** command.

Usage

```
admin@as5812-54x:~$ crm config thresholds
Usage: crm config thresholds [OPTIONS] COMMAND [ARGS]...
```

CRM thresholds configuration

Options:

--help Show this message and exit.

Commands:

acl CRM configuration for ACL resource
fdb CRM configuration for FDB resource
ipv4 CRM resource IPv4 address-family
ipv6 CRM resource IPv6 address-family
nexthop CRM configuration for nexthop resource

crm config thresholds acl group counter type <value>

Configures the threshold type of the ACL group.

Usage

```
admin@sonic:~$ crm config thresholds acl group type
Usage: crm config thresholds acl group type [OPTIONS] [percentage|used|free]
Try "crm config thresholds acl group type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds acl group type percentage
admin@sonic:~$ crm config thresholds acl group type used
admin@sonic:~$ crm config thresholds acl group type free
```

crm config thresholds acl group counter lowhigh <value>

Configures the lowhigh thresholds of the ACL group.

Usage

```
admin@sonic:~$ crm config thresholds acl group low
Usage: crm config thresholds acl group low [OPTIONS] VALUE
Try "crm config thresholds acl group low --help" for help.
```

```
admin@sonic:~$ crm config thresholds acl group high
Usage: crm config thresholds acl group high [OPTIONS] VALUE
Try "crm config thresholds acl group high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds acl group high 90
admin@sonic:~$ crm config thresholds acl group low 10
```

crm config thresholds acl table type <value>

Configures the threshold type of the ACL table.

Usage

```
admin@sonic:~$ crm config thresholds acl table type
Usage: crm config thresholds acl table type [OPTIONS] [percentage|used|free]
Try "crm config thresholds acl table type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds acl table type percentage
admin@sonic:~$ crm config thresholds acl table type used
admin@sonic:~$ crm config thresholds acl table type free
```

crm config thresholds acl table lowhigh <value>

Configures the lowhigh thresholds of the ACL table.

Usage

```
admin@sonic:~$ crm config thresholds acl table low
Usage: crm config thresholds acl table low [OPTIONS] VALUE
Try "crm config thresholds acl table low --help" for help.

admin@sonic:~$ crm config thresholds acl table high
Usage: crm config thresholds acl table high [OPTIONS] VALUE
Try "crm config thresholds acl table high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds acl table low 3

admin@sonic:~$ crm config thresholds acl table high 80
```

crm config thresholds dnat low|high <value>

(Unsupported in Broadcom platform) Configure low|high threshold of DNAT

Usage

```
admin@sonic:~$ crm config thresholds dnat low
Usage: crm config thresholds dnat low [OPTIONS] VALUE
Try "crm config thresholds dnat low --help" for help

admin@sonic:~$ crm config thresholds dnat high
Usage: crm config thresholds dnat high [OPTIONS] VALUE
Try "crm config thresholds dnat high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds dnat low 3

admin@sonic:~$ crm config thresholds dnat high 80
```

crm config thresholds dnat type <value>

(Unsupported in Broadcom platform) Configure threshold type of DNAT

Usage

```
admin@sonic:~$ crm config thresholds dnat type
Usage: crm config thresholds dnat type [OPTIONS] [percentage|used|free]
Try "crm config thresholds dnat type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds dnat type percentage
admin@sonic:~$ crm config thresholds dnat type used
admin@sonic:~$ crm config thresholds dnat type free
```

crm config thresholds fdb type <value>

Configures the threshold type of FDB.

Usage

```
admin@sonic:~$ crm config thresholds fdb type
Usage: crm config thresholds fdb type [OPTIONS] [percentage|used|free]
Try "crm config thresholds fdb type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds fdb type percentage
admin@sonic:~$ crm config thresholds fdb type used
admin@sonic:~$ crm config thresholds fdb type free
```

crm config thresholds fdb low|high <value>

Configures the low|high thresholds of FDB.

Usage

```
admin@sonic:~$ crm config thresholds fdb low
Usage: crm config thresholds fdb low [OPTIONS] VALUE
Try "crm config thresholds fdb low --help" for help.

admin@sonic:~$ crm config thresholds fdb high
Usage: crm config thresholds fdb high [OPTIONS] VALUE
Try "crm config thresholds fdb high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds fdb low 20
admin@sonic:~$ crm config thresholds fdb high 311111
```

crm config thresholds ipmc low|high <value>

(Unsupported) Configure low|high threshold of IPMC

Usage

```
admin@sonic:~$ crm config thresholds ipmc low
Usage: crm config thresholds ipmc low [OPTIONS] VALUE
Try "crm config thresholds ipmc low --help" for help.

admin@sonic:~$ crm config thresholds ipmc high
Usage: crm config thresholds ipmc high [OPTIONS] VALUE
Try "crm config thresholds ipmc high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipmc low 3
admin@sonic:~$ crm config thresholds ipmc high 80
```

crm config thresholds ipmc type

Configure threshold type of IPMC

Usage

```
admin@sonic:~$ crm config thresholds ipmc type
Usage: crm config thresholds ipmc type [OPTIONS] [percentage|used|free]
Try "crm config thresholds ipmc type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipmc type percentage
admin@sonic:~$ crm config thresholds ipmc type used
admin@sonic:~$ crm config thresholds ipmc type free
```

crm config thresholds ipv4 neighbor type <value>

Configures the threshold type of IPv4 neighbors.

Usage

```
admin@sonic:~$ crm config thresholds ipv4 neighbor type
Usage: crm config thresholds ipv4 neighbor type [OPTIONS]
[percentage|used|free]
Try "crm config thresholds ipv4 neighbor type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv4 neighbor type percentage
admin@sonic:~$ crm config thresholds ipv4 neighbor type used
admin@sonic:~$ crm config thresholds ipv4 neighbor type free
```

crm config thresholds ipv4 neighbor low|high <value>

Configures the low|high thresholds of IPv4 neighbors.

Usage

```
admin@sonic:~$ crm config thresholds ipv4 neighbor low
Usage: crm config thresholds ipv4 neighbor low [OPTIONS] VALUE
Try "crm config thresholds ipv4 neighbor low --help" for help.

admin@sonic:~$ crm config thresholds ipv4 neighbor high
Usage: crm config thresholds ipv4 neighbor high [OPTIONS] VALUE
Try "crm config thresholds ipv4 neighbor high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv4 neighbor low 2
admin@sonic:~$ crm config thresholds ipv4 neighbor high 5566
```

crm config thresholds ipv4 nexthop type <value>

Configures the threshold type of IPv4 nexthop.

Usage

```
admin@sonic:~$ crm config thresholds ipv4 nexthop type
Usage: crm config thresholds ipv4 nexthop type [OPTIONS]
                                     [percentage|used|free]
Try "crm config thresholds ipv4 nexthop type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv4 nexthop type percentage
admin@sonic:~$ crm config thresholds ipv4 nexthop type free
admin@sonic:~$ crm config thresholds ipv4 nexthop type used
```

crm config thresholds ipv4 nexthop low|high <value>

Configures the low|high thresholds of IPv4 nexthop.

Usage

```
admin@sonic:~$ crm config thresholds ipv4 nexthop low
Usage: crm config thresholds ipv4 nexthop low [OPTIONS] VALUE
Try "crm config thresholds ipv4 nexthop low --help" for help.

admin@sonic:~$ crm config thresholds ipv4 nexthop high
Usage: crm config thresholds ipv4 nexthop high [OPTIONS] VALUE
Try "crm config thresholds ipv4 nexthop high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv4 nexthop high 90
admin@sonic:~$ crm config thresholds ipv4 nexthop high 9000000
```

crm config thresholds ipv4 route type <value>

Configures the threshold type of IPv4 route.

Usage

```
admin@sonic:~$ crm config thresholds ipv4 route type
Usage: crm config thresholds ipv4 route type [OPTIONS] [percentage|used|free]
Try "crm config thresholds ipv4 route type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv4 route type percentage
admin@sonic:~$ crm config thresholds ipv4 route type used
admin@sonic:~$ crm config thresholds ipv4 route type free
```

crm config thresholds ipv4 route lowhigh <value>

Configures the lowhigh threshold of IPv4 route.

Usage

```
admin@sonic:~$ crm config thresholds ipv4 route low
Usage: crm config thresholds ipv4 route low [OPTIONS] VALUE
Try "crm config thresholds ipv4 route low --help" for help.

admin@sonic:~$ crm config thresholds ipv4 route high
Usage: crm config thresholds ipv4 route high [OPTIONS] VALUE
Try "crm config thresholds ipv4 route high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv4 route low 11
admin@sonic:~$ crm config thresholds ipv4 route high 800000
```

crm config thresholds ipv6 neighbor type <value>

Configures the threshold type of IPv6 neighbors.

Usage

```
admin@sonic:~$ crm config thresholds ipv6 neighbor type
Usage: crm config thresholds ipv6 neighbor type [OPTIONS]
[percentage|used|free]
Try "crm config thresholds ipv6 neighbor type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv6 neighbor type percentage
admin@sonic:~$ crm config thresholds ipv6 neighbor type used
admin@sonic:~$ crm config thresholds ipv6 neighbor type free
```

crm config thresholds ipv6 neighbor lowhigh <value>

Configures the lowhigh thresholds of IPv6 neighbors.

Usage

```
admin@sonic:~$ crm config thresholds ipv6 neighbor low
Usage: crm config thresholds ipv6 neighbor low [OPTIONS] VALUE
Try "crm config thresholds ipv6 neighbor low --help" for help.

admin@sonic:~$ crm config thresholds ipv6 neighbor high
Usage: crm config thresholds ipv6 neighbor high [OPTIONS] VALUE
Try "crm config thresholds ipv6 neighbor high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv6 neighbor low 100
admin@sonic:~$ crm config thresholds ipv6 neighbor high 50000
```

crm config thresholds ipv6 nexthop type <value>

Configures the threshold type of IPv6 nexthop.

Usage

```
admin@sonic:~$ crm config thresholds ipv6 nexthop type
Usage: crm config thresholds ipv6 nexthop type [OPTIONS]
                                     [percentage|used|free]
Try "crm config thresholds ipv6 nexthop type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv6 nexthop type percentage
admin@sonic:~$ crm config thresholds ipv6 nexthop type used
admin@sonic:~$ crm config thresholds ipv6 nexthop type free
```

crm config thresholds ipv6 nexthop low|high <value>

Configures the low|high thresholds of IPv6 nexthop.

Usage

```
admin@sonic:~$ crm config thresholds ipv6 nexthop low
Usage: crm config thresholds ipv6 nexthop low [OPTIONS] VALUE
Try "crm config thresholds ipv6 nexthop low --help" for help.

admin@sonic:~$ crm config thresholds ipv6 nexthop high
Usage: crm config thresholds ipv6 nexthop high [OPTIONS] VALUE
Try "crm config thresholds ipv6 nexthop high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv6 nexthop low 10
admin@sonic:~$ crm config thresholds ipv6 nexthop high 80000
```

crm config thresholds ipv6 route type <value>

Configures the threshold type of IPv6 route.

Usage

```
admin@sonic:~$ crm config thresholds ipv6 route type
Usage: crm config thresholds ipv6 route type [OPTIONS] [percentage|used|free]
Try "crm config thresholds ipv6 route type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv6 route type percentage
admin@sonic:~$ crm config thresholds ipv6 route type used
admin@sonic:~$ crm config thresholds ipv6 route type free
```

crm config thresholds ipv6 route lowhigh <value>

Configures the lowhigh thresholds of IPv6 route.

Usage

```
admin@sonic:~$ crm config thresholds ipv6 route low
Usage: crm config thresholds ipv6 route low [OPTIONS] VALUE
Try "crm config thresholds ipv6 route low --help" for help.

admin@sonic:~$ crm config thresholds ipv6 route high
Usage: crm config thresholds ipv6 route high [OPTIONS] VALUE
Try "crm config thresholds ipv6 route high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds ipv6 route low 100
admin@sonic:~$ crm config thresholds ipv6 route low 80000
```

crm config thresholds mpls inseg lowhigh <value>

(Unsupported in Broadcom platform) Configure lowhigh threshold of MPLS inseg

Usage

```
admin@sonic:~$ crm config thresholds mpls inseg low
Usage: crm config thresholds mpls inseg low [OPTIONS] VALUE
Try "crm config thresholds mpls inseg low --help" for help

admin@sonic:~$ crm config thresholds mpls inseg high
Usage: crm config thresholds mpls inseg high [OPTIONS] VALUE
Try "crm config thresholds mpls inseg high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds mpls inseg low 3
admin@sonic:~$ crm config thresholds mpls inseg high 80
```

crm config thresholds mpls inseg type <value>

(Unsupported in Broadcom platform) Configure threshold type of mpls inseg

Usage

```
admin@sonic:~$ crm config thresholds mpls inseg type
Usage: crm config thresholds mpls inseg type [OPTIONS] [percentage|used|free]
Try "crm config thresholds mpls inseg type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds mpls inseg type percentage
admin@sonic:~$ crm config thresholds mpls inseg type used
admin@sonic:~$ crm config thresholds mpls inseg type free
```

crm config thresholds mpls nexthop low|high <value>

(Unsupported in Broadcom platform) Configure low|high threshold of MPLS nexthop

Usage

```
admin@sonic:~$ crm config thresholds mpls nexthop low
Usage: crm config thresholds mpls nexthop low [OPTIONS] VALUE
Try "crm config thresholds mpls nexthop low --help" for help

admin@sonic:~$ crm config thresholds mpls nexthop high
Usage: crm config thresholds mpls nexthop high [OPTIONS] VALUE
Try "crm config thresholds mpls nexthop high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds mpls nexthop low 3
admin@sonic:~$ crm config thresholds mpls nexthop high 80
```

crm config thresholds mpls nexthop type <value>

(Unsupported in Broadcom platform) Configure threshold type of mpls nexthop

Usage

```
admin@sonic:~$ crm config thresholds mpls nexthop type
Usage: crm config thresholds mpls nexthop type [OPTIONS] [percentage|used|free]
Try "crm config thresholds mpls nexthop type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds mpls nexthop type percentage
admin@sonic:~$ crm config thresholds mpls nexthop type used
admin@sonic:~$ crm config thresholds mpls nexthop type free
```

crm config thresholds nexthop group member type <value>

Configures the threshold type of nexthop group members.

Usage

```
admin@sonic:~$ crm config thresholds nexthop group member type
Usage: crm config thresholds nexthop group member type [OPTIONS]
                                         [percentage|used|free]
Try "crm config thresholds nexthop group member type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds nexthop group member type percentage
admin@sonic:~$ crm config thresholds nexthop group member type used
admin@sonic:~$ crm config thresholds nexthop group member type free
```

crm config thresholds nexthop group member low|high <value>

Configures the low|high thresholds of nexthop group members.

Usage

```
admin@sonic:~$ crm config thresholds nexthop group member low
Usage: crm config thresholds nexthop group member low [OPTIONS] VALUE
Try "crm config thresholds nexthop group member low --help" for help.

admin@sonic:~$ crm config thresholds nexthop group member high
Usage: crm config thresholds nexthop group member high [OPTIONS] VALUE
Try "crm config thresholds nexthop group member high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds nexthop group member low 1000
admin@sonic:~$ crm config thresholds nexthop group member high 12345
```

crm config thresholds nexthop group object type <value>

Configures the threshold type of nexthop group objects.

Usage

```
admin@sonic:~$ crm config thresholds nexthop group object type
Usage: crm config thresholds nexthop group object type [OPTIONS]
                                [percentage|used|free]
Try "crm config thresholds nexthop group object type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds nexthop group object type percentage
admin@sonic:~$ crm config thresholds nexthop group object type used
admin@sonic:~$ crm config thresholds nexthop group object type free
```

crm config thresholds nexthop group object low|high <value>

Configures the threshold type of nexthop group objects.

Usage

```
admin@sonic:~$ crm config thresholds nexthop group object low
Usage: crm config thresholds nexthop group object low [OPTIONS] VALUE
Try "crm config thresholds nexthop group object low --help" for help.

admin@sonic:~$ crm config thresholds nexthop group object high
Usage: crm config thresholds nexthop group object high [OPTIONS] VALUE
Try "crm config thresholds nexthop group object high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds nexthop group object low 100
admin@sonic:~$ crm config thresholds nexthop group object high 2000
```

crm config thresholds snat low|high <value>

Configure low|high threshold of SNAT

Usage

```
admin@sonic:~$ crm config thresholds snat low
Usage: crm config thresholds snat low [OPTIONS] VALUE
Try "crm config thresholds snat low --help" for help.

admin@sonic:~$ crm config thresholds snat high
Usage: crm config thresholds snat high [OPTIONS] VALUE
Try "crm config thresholds snat high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds snat low 3
admin@sonic:~$ crm config thresholds snat high 80
```

crm config thresholds snat type <value>

Configure threshold type of SNAT

Usage

```
admin@sonic:~$ crm config thresholds snat type
Usage: crm config thresholds snat type [OPTIONS] [percentage|used|free]
Try "crm config thresholds snat type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds snat type percentage
admin@sonic:~$ crm config thresholds snat type used
admin@sonic:~$ crm config thresholds snat type free
```

rm config thresholds srv6-my-sid-entry low|high <value>

(Unsupported in Broadcom platform) Configure low|high threshold of srv6-my-sid-entry

Usage

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry low
Usage: crm config thresholds srv6-my-sid-entry low [OPTIONS] VALUE
Try "crm config thresholds srv6-my-sid-entry low --help" for help

admin@sonic:~$ crm config thresholds srv6-my-sid-entry high
Usage: crm config thresholds srv6-my-sid-entry high [OPTIONS] VALUE
Try "crm config thresholds srv6-my-sid-entry high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry low 3
admin@sonic:~$ crm config thresholds srv6-my-sid-entry high 80
```

crm config thresholds srv6-my-sid-entry type <value>

(Unsupported in Broadcom platform) Configure threshold type of srv6-my-sid-entry

Usage

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type
Usage: crm config thresholds srv6-my-sid-entry type [OPTIONS] [percentage|used|free]
Try "crm config thresholds srv6-my-sid-entry type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type percentage
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type used
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type free
```

crm config thresholds srv6-nexthop low|high <value>

Configure low|high threshold of srv6-nexthop

Usage

```
admin@sonic:~$ crm config thresholds srv6-nexthop low
Usage: crm config thresholds srv6-nexthop low [OPTIONS] VALUE
Try "crm config thresholds srv6-nexthop low --help" for help

admin@sonic:~$ crm config thresholds srv6-nexthop high
Usage: crm config thresholds srv6-nexthop high [OPTIONS] VALUE
Try "crm config thresholds srv6-nexthop high --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds srv6-nexthop low 3
admin@sonic:~$ crm config thresholds srv6-nexthop high 80
```

crm config thresholds srv6-nexthop type <value>

Configure threshold type of srv6-nexthop

Usage

```
admin@sonic:~$ crm config thresholds srv6-nexthop type
Usage: crm config thresholds srv6-nexthop type [OPTIONS] [percentage|used|free]
Try "crm config thresholds srv6-nexthop type --help" for help.
```

Example

```
admin@sonic:~$ crm config thresholds srv6-nexthop type percentage
admin@sonic:~$ crm config thresholds srv6-nexthop type used
admin@sonic:~$ crm config thresholds srv6-nexthop type free
```

(202111.2) CRM

CRM show commands

crm show

Display the usage of show command

- Usage:

```
admin@sonic:~$ crm show ?
Usage: crm show [OPTIONS] COMMAND [ARGS]...

Show CRM related information

Options:
  --help  Show this message and exit.

Commands:
  resources  Show CRM resources information
  summary    Show CRM general information
  thresholds Show CRM thresholds information
```

crm show summary

Display polling interval setting

- Example:

```
admin@sonic:~$ crm show summary

Polling Interval: 300 second(s)
```

crm show resources

Display the usage of show resources command

- Usage:

```
admin@sonic:~$ crm show resources
Usage: crm show resources [OPTIONS] COMMAND [ARGS]...

Show CRM resources information

Options:
  --help  Show this message and exit.

Commands:
  acl          Show CRM information for acl resource
  all          Show CRM information for all resources
  dn timer     Show CRM information for DNAT resource
  fdb          Show CRM information for fdb resource
  ipmc         Show CRM information for IPMC resource
  ipv4         Show CRM resource IPv4 address family
  ipv6         Show CRM resource IPv6 address family
  mpls         Show CRM resource MPLS address family
  nexthop      Show CRM information for nexthop resource
  snat         Show CRM information for SNAT resource
  srv6-my-sid-entry Show CRM information for SRV6 MY_SID entry
  srv6-nexthop Show CRM information for SRV6 Nexthop
```

crm show resources all

Display all statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources all
```

Resource Name	Used Count	Available Count
-----	-----	-----
ipv4_route	66	100286
ipv6_route	68	39279
ipv4_nextthop	0	32765
ipv6_nextthop	0	32765
ipv4_neighbor	0	32765
ipv6_neighbor	0	4062
nextthop_group_member	0	32767
nextthop_group	0	511
fdb_entry	0	16384
ipmc_entry	0	0
snat_entry	0	45056
dnat_entry	0	45056
mpls_inseg	0	0
mpls_nextthop	0	32765
srv6_nextthop	0	32765
srv6_my_sid_entry	0	0

Stage	Bind Point	Resource Name	Used Count	Available Count
-----	-----	-----	-----	-----
INGRESS	PORT	acl_group	32	224
INGRESS	PORT	acl_table	3	2
INGRESS	LAG	acl_group	0	224
INGRESS	LAG	acl_table	0	2
INGRESS	VLAN	acl_group	0	224
INGRESS	VLAN	acl_table	0	5
INGRESS	RIF	acl_group	0	224
INGRESS	RIF	acl_table	0	5
INGRESS	SWITCH	acl_group	0	224
INGRESS	SWITCH	acl_table	0	5
EGRESS	PORT	acl_group	0	224
EGRESS	PORT	acl_table	0	2
EGRESS	LAG	acl_group	0	224
EGRESS	LAG	acl_table	0	2
EGRESS	VLAN	acl_group	0	224
EGRESS	VLAN	acl_table	0	2
EGRESS	RIF	acl_group	0	224
EGRESS	RIF	acl_table	0	2
EGRESS	SWITCH	acl_group	0	224
EGRESS	SWITCH	acl_table	0	2

Table ID	Resource Name	Used Count	Available Count
-----	-----	-----	-----
0x7000000000a20	acl_entry	2	6142
0x7000000000a20	acl_counter	2	6142

crm show resources acl

Display the usage of ACL resources show command

- Usage:

```
admin@sonic:~$ crm show resources acl
Usage: crm show resources acl [OPTIONS] COMMAND [ARGS]...

Show CRM information for acl resource

Options:
  --help  Show this message and exit.

Commands:
  group  Show CRM information for acl group resource
  table  Show CRM information for acl table resource
```

crm show resources acl group

Display ACL group statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources acl group
```

Stage	Bind Point	Resource Name	Used Count	Available Count
INGRESS	PORT	acl_group	32	224
INGRESS	PORT	acl_table	2	3
INGRESS	LAG	acl_group	0	224
INGRESS	LAG	acl_table	0	3
INGRESS	VLAN	acl_group	0	224
INGRESS	VLAN	acl_table	0	6
INGRESS	RIF	acl_group	0	224
INGRESS	RIF	acl_table	0	6
INGRESS	SWITCH	acl_group	0	224
INGRESS	SWITCH	acl_table	0	6
EGRESS	PORT	acl_group	0	224
EGRESS	PORT	acl_table	0	2
EGRESS	LAG	acl_group	0	224
EGRESS	LAG	acl_table	0	2
EGRESS	VLAN	acl_group	0	224
EGRESS	VLAN	acl_table	0	2
EGRESS	RIF	acl_group	0	224
EGRESS	RIF	acl_table	0	2
EGRESS	SWITCH	acl_group	0	224
EGRESS	SWITCH	acl_table	0	2

crm show resources acl table

Display ACL table statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources acl table
```

Table ID	Resource Name	Used Count	Available Count
0x7000000000a20	acl_entry	2	6142
0x7000000000a20	acl_counter	2	6142

crm show resources dn timer

(Unsupported in Broadcom platform) Display the usage of DNAT resources show command

- Example:

```
admin@sonic:~$ crm show resources dnat
```

Resource Name	Used Count	Available Count
dnat_entry	0	45056

crm show resources fdb

Display FDB statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources fdb
```

Resource Name	Used Count	Available Count
fdb_entry	0	32768

crm show resources ipmc

(Unsupported) Display the usage of IPMC resources show command

- Example:

```
admin@sonic:~$ crm show resources ipmc
```

Resource Name	Used Count	Available Count
ipmc_entry	0	0

crm show resources ipv4

Display the usage of IPv4 resources show command

- Usage:

```
admin@sonic:~$ crm show resources ipv4
Usage: crm show resources ipv4 [OPTIONS] COMMAND [ARGS]...
```

CRM resource IPv4 address family

Options:

--help Show this message and exit.

Commands:

neighbor Show CRM information for neighbor resource
 nexthop Show CRM information for nexthop resource
 route Show CRM information for route resource

crm show resources ipv4 neighbor

Display IPv4 neighbor statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources ipv4
Usage: crm show resources ipv4 [OPTIONS] COMMAND [ARGS]...

CRM resource IPv4 address family

Options:
  --help  Show this message and exit.

Commands:
  neighbor  Show CRM information for neighbor resource
  nexthop   Show CRM information for nexthop resource
  route     Show CRM information for route resource
admin@as5812-54x:~$ crm show resources ipv4 neighbor
```

Resource Name	Used Count	Available Count
ipv4_neighbor	0	8192

crm show resources ipv4 nexthop

Display IPv4 nexthop statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources ipv4 nexthop
```

Resource Name	Used Count	Available Count
ipv4_nexthop	0	49118

crm show resources ipv4 route

Display IPv4 route statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources ipv4 route
```

Resource Name	Used Count	Available Count
ipv4_route	66	98238

crm show resources ipv6

Display the usage of IPv6 resources show command

- Usage:

```
admin@sonic:~$ crm show resources ipv6
Usage: crm show resources ipv6 [OPTIONS] COMMAND [ARGS]...
```

CRM resource IPv6 address family

Options:

--help Show this message and exit.

Commands:

neighbor Show CRM information for neighbor resource
nexthop Show CRM information for nexthop resource
route Show CRM information for route resource

crm show resources ipv6 neighbor

Display IPv6 neighbor statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources ipv6 neighbor
```

Resource Name	Used Count	Available Count
ipv6_neighbor	33	4096

crm show resources ipv6 nexthop

Display IPv6 nexthop statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources ipv6 nexthop
```

Resource Name	Used Count	Available Count
ipv6_nexthop	33	49117

crm show resources ipv6 route

Display IPv6 route statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources ipv6 route
```

Resource Name	Used Count	Available Count
ipv6_route	68	16316

crm show resources mpls

(Unsupported in Broadcom platform) Display the usage of MPLS resources show command

- Example:

```
admin@sonic:~$ crm show resources mpls
```

Usage: crm show resources mpls [OPTIONS] COMMAND [ARGS]...

CRM resource MPLS address family

Options:

--help Show this message and exit.

Commands:

inseg Show CRM information for in-segment resource
nexthop Show CRM information for nexthop resource

crm show resources mpls inseg

(Unsupported in Broadcom platform) Display MPLS inseg statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources mpls inseg
```

Resource Name	Used Count	Available Count
mpls_inseg	0	0

crm show resources mpls nexthop

(Unsupported in Broadcom platform) Display MPLS nexthop statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources mpls nexthop
```

Resource Name	Used Count	Available Count
mpls_nexthop	0	32765

crm show resources nexthop group

Display the usage of nexthop group resources show command

- Usage:

```
admin@sonic:~$ crm show resources nexthop group
```

Usage: crm show resources nexthop group [OPTIONS] COMMAND [ARGS]...

Show CRM information for nexthop group resource

Options:

--help Show this message and exit.

Commands:

member Show CRM information for nexthop group member resource
object Show CRM information for nexthop group resource

crm show resources nexthop group member

Display nexthop group member statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources nexthop group member
```

Resource Name	Used Count	Available Count
-----	-----	-----
nexthop_group_member	0	16384

crm show resources nexthop group object

Display nexthop group object statistic which include used by all components and available resources in the ASIC

- Example:

```
admin@sonic:~$ crm show resources nexthop group object
```

Resource Name	Used Count	Available Count
-----	-----	-----
nexthop_group	0	128

crm show resources snat

Display the usage of SNAT resources show command

- Example:

```
admin@sonic:~$ crm show resources snat
```

Resource Name	Used Count	Available Count
-----	-----	-----
snat_entry	0	45056

crm show resources srv6-my-sid-entry

(Unsupported in Broadcom platform) Display the usage of srv6-my-sid-entry resources show command

- Example:

```
admin@sonic:~$ crm show resources srv6-my-sid-entry
```

Resource Name	Used Count	Available Count
-----	-----	-----
srv6-my-sid-entry	0	45056

crm show resources srv6-nexthop

(Unsupported in Broadcom platform) Display the usage of srv6-nexthop resources show command

- Example:

```
admin@sonic:~$ crm show resources srv6-nexthop
```

Resource Name	Used Count	Available Count
-----	-----	-----
srv6-nexthop	0	45056

CRM config command

Config command should be extended in order to add "crm" alias to the CRM utility.

crm config

Display the usage of config command

- Usage:

```
admin@sonic:~$ crm config
Usage: crm config [OPTIONS] COMMAND [ARGS]...
```

CRM related configuration

Options:

--help Show this message and exit.

Commands:

polling CRM polling configuration
thresholds CRM thresholds configuration

crm config polling interval

Configure the polling interval to fetch available resources from ASIC

- Usage:

```
admin@sonic:~$ crm config polling interval
Usage: crm config polling interval [OPTIONS] INTERVAL
```

- Example:

```
admin@sonic:~$ crm config polling interval 400
```

crm config thresholds

Display the usage of config thresholds command

- Usage:

```
admin@as5812-54x:~$ crm config thresholds
Usage: crm config thresholds [OPTIONS] COMMAND [ARGS]...

CRM thresholds configuration

Options:
  --help  Show this message and exit.

Commands:
  acl          CRM configuration for ACL resource
  dn timer      CRM configuration for Destination NAT resource
  fdb          CRM configuration for FDB resource
  ipmc         CRM configuration for IPMC resource
  ipv4         CRM resource IPv4 address-family
  ipv6         CRM resource IPv6 address-family
  mpls         CRM resource MPLS address-family
  nexthop      CRM configuration for nexthop resource
  snat         CRM configuration for Source NAT resource
  srv6-my-sid-entry CRM configuration for SRV6 MY_SID resource
  srv6-nexthop CRM configuration for SRV6 Nexthop resource
```

crm config thresholds acl group counter type

Configure threshold type of ACL group

- Usage:

```
admin@sonic:~$ crm config thresholds acl group type
Usage: crm config thresholds acl group type [OPTIONS] [percentage|used|free]
Try "crm config thresholds acl group type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds acl group type percentage
admin@sonic:~$ crm config thresholds acl group type used
admin@sonic:~$ crm config thresholds acl group type free
```

crm config thresholds acl group counter lowhigh

Configure lowhigh threshold of ACL group

- Usage:

```
admin@sonic:~$ crm config thresholds acl group low
Usage: crm config thresholds acl group low [OPTIONS] VALUE
Try "crm config thresholds acl group low --help" for help.

admin@sonic:~$ crm config thresholds acl group high
Usage: crm config thresholds acl group high [OPTIONS] VALUE
Try "crm config thresholds acl group high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds acl group high 90

admin@sonic:~$ crm config thresholds acl group low 10
```

crm config thresholds acl table type

Configure threshold type of ACL table

- Usage:

```
admin@sonic:~$ crm config thresholds acl table type
Usage: crm config thresholds acl table type [OPTIONS] [percentage|used|free]
Try "crm config thresholds acl table type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds acl table type percentage
admin@sonic:~$ crm config thresholds acl table type used
admin@sonic:~$ crm config thresholds acl table type free
```

crm config thresholds acl table lowhigh

Configure lowhigh threshold of ACL table

- Usage:

```
admin@sonic:~$ crm config thresholds acl table low
Usage: crm config thresholds acl table low [OPTIONS] VALUE
Try "crm config thresholds acl table low --help" for help.

admin@sonic:~$ crm config thresholds acl table high
Usage: crm config thresholds acl table high [OPTIONS] VALUE
Try "crm config thresholds acl table high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds acl table low 3

admin@sonic:~$ crm config thresholds acl table high 80
```

crm config thresholds dnat lowhigh

(Unsupported in Broadcom platform) Configure lowhigh threshold of DNAT

- Usage:

```
admin@sonic:~$ crm config thresholds dnat low
Usage: crm config thresholds dnat low [OPTIONS] VALUE
Try "crm config thresholds dnat low --help" for help.

admin@sonic:~$ crm config thresholds dnat high
Usage: crm config thresholds dnat high [OPTIONS] VALUE
Try "crm config thresholds dnat high --help" for help.
```


- Example:

```
admin@sonic:~$ crm config thresholds dnat low 3
admin@sonic:~$ crm config thresholds dnat high 80
```

crm config thresholds dnat type

(Unsupported in Broadcom platform) Configure threshold type of DNAT

- Usage:

```
admin@sonic:~$ crm config thresholds dnat type
Usage: crm config thresholds dnat type [OPTIONS] [percentage|used|free]
Try "crm config thresholds dnat type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds dnat type percentage
admin@sonic:~$ crm config thresholds dnat type used
admin@sonic:~$ crm config thresholds dnat type free
```

crm config thresholds fdb type

Configure threshold type of FDB

- Usage:

```
dmin@sonic:~$ crm config thresholds fdb type
Usage: crm config thresholds fdb type [OPTIONS] [percentage|used|free]
Try "crm config thresholds fdb type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds fdb type percentage
admin@sonic:~$ crm config thresholds fdb type used
admin@sonic:~$ crm config thresholds fdb type free
```

crm config thresholds fdb low|high

Configure low|high threshold of FDB

- Usage:

```
admin@sonic:~$ crm config thresholds fdb low
Usage: crm config thresholds fdb low [OPTIONS] VALUE
Try "crm config thresholds fdb low --help" for help.

admin@sonic:~$ crm config thresholds fdb high
Usage: crm config thresholds fdb high [OPTIONS] VALUE
Try "crm config thresholds fdb high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds fdb low 20
admin@sonic:~$ crm config thresholds fdb high 311111
```

crm config thresholds ipmc lowhigh

(Unsupported) Configure lowhigh threshold of IPMC

- Usage:

```
admin@sonic:~$ crm config thresholds ipmc low
Usage: crm config thresholds ipmc low [OPTIONS] VALUE
Try "crm config thresholds ipmc low --help" for help.
```

```
admin@sonic:~$ crm config thresholds ipmc high
Usage: crm config thresholds ipmc high [OPTIONS] VALUE
Try "crm config thresholds ipmc high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipmc low 3
admin@sonic:~$ crm config thresholds ipmc high 80
```

crm config thresholds ipmc type

Configure threshold type of IPMC

- Usage:

```
admin@sonic:~$ crm config thresholds ipmc type
Usage: crm config thresholds ipmc type [OPTIONS] [percentage|used|free]
Try "crm config thresholds ipmc type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipmc type percentage
admin@sonic:~$ crm config thresholds ipmc type used
admin@sonic:~$ crm config thresholds ipmc type free
```

crm config thresholds ipv4 neighbor type

Configure threshold type of IPv4 neighbor

- Usage:

```
admin@sonic:~$ crm config thresholds ipv4 neighbor type
Usage: crm config thresholds ipv4 neighbor type [OPTIONS]
                                         [percentage|used|free]
Try "crm config thresholds ipv4 neighbor type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv4 neighbor type percentage
admin@sonic:~$ crm config thresholds ipv4 neighbor type used
admin@sonic:~$ crm config thresholds ipv4 neighbor type free
```

crm config thresholds ipv4 neighbor lowhigh

Configure lowhigh threshold of IPv4 neighbor

- Usage:

```
admin@sonic:~$ crm config thresholds ipv4 neighbor low
Usage: crm config thresholds ipv4 neighbor low [OPTIONS] VALUE
Try "crm config thresholds ipv4 neighbor low --help" for help.

admin@sonic:~$ crm config thresholds ipv4 neighbor high
Usage: crm config thresholds ipv4 neighbor high [OPTIONS] VALUE
Try "crm config thresholds ipv4 neighbor high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv4 neighbor low 2
admin@sonic:~$ crm config thresholds ipv4 neighbor high 5566
```

crm config thresholds ipv4 nexthop type

Configure threshold type of IPv4 nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds ipv4 nexthop type
Usage: crm config thresholds ipv4 nexthop type [OPTIONS]
                                [percentage|used|free]
Try "crm config thresholds ipv4 nexthop type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv4 nexthop type percentage
admin@sonic:~$ crm config thresholds ipv4 nexthop type free
admin@sonic:~$ crm config thresholds ipv4 nexthop type used
```

crm config thresholds ipv4 nexthop lowhigh

Configure lowhigh threshold of IPv4 nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds ipv4 nexthop low
Usage: crm config thresholds ipv4 nexthop low [OPTIONS] VALUE
Try "crm config thresholds ipv4 nexthop low --help" for help.

admin@sonic:~$ crm config thresholds ipv4 nexthop high
Usage: crm config thresholds ipv4 nexthop high [OPTIONS] VALUE
Try "crm config thresholds ipv4 nexthop high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv4 nexthop high 90
admin@sonic:~$ crm config thresholds ipv4 nexthop high 9000000
```

crm config thresholds ipv4 route type

Configure threshold type of IPv4 route

- Usage:

```
admin@sonic:~$ crm config thresholds ipv4 route type
Usage: crm config thresholds ipv4 route type [OPTIONS] [percentage|used|free]
Try "crm config thresholds ipv4 route type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv4 route type percentage
admin@sonic:~$ crm config thresholds ipv4 route type used
admin@sonic:~$ crm config thresholds ipv4 route type free
```

crm config thresholds ipv4 route lowhigh

Configure lowhigh threshold of IPv4 route

- Usage:

```
admin@sonic:~$ crm config thresholds ipv4 route low
Usage: crm config thresholds ipv4 route low [OPTIONS] VALUE
Try "crm config thresholds ipv4 route low --help" for help.

admin@sonic:~$ crm config thresholds ipv4 route high
Usage: crm config thresholds ipv4 route high [OPTIONS] VALUE
Try "crm config thresholds ipv4 route high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv4 route low 11
admin@sonic:~$ crm config thresholds ipv4 route high 800000
```

crm config thresholds ipv6 neighbor type

Configure threshold type of IPv6 neighbor

- Usage:

```
admin@sonic:~$ crm config thresholds ipv6 neighbor type
Usage: crm config thresholds ipv6 neighbor type [OPTIONS]
[percentage|used|free]
Try "crm config thresholds ipv6 neighbor type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv6 neighbor type percentage
admin@sonic:~$ crm config thresholds ipv6 neighbor type used
admin@sonic:~$ crm config thresholds ipv6 neighbor type free
```

crm config thresholds ipv6 neighbor lowhigh

Configure lowhigh threshold of IPv6 neighbor

- Usage:

```
admin@sonic:~$ crm config thresholds ipv6 neighbor low
Usage: crm config thresholds ipv6 neighbor low [OPTIONS] VALUE
Try "crm config thresholds ipv6 neighbor low --help" for help.

admin@sonic:~$ crm config thresholds ipv6 neighbor high
Usage: crm config thresholds ipv6 neighbor high [OPTIONS] VALUE
Try "crm config thresholds ipv6 neighbor high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv6 neighbor low 100
admin@sonic:~$ crm config thresholds ipv6 neighbor high 50000
```

crm config thresholds ipv6 nexthop type

Configure threshold type of IPv6 nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds ipv6 nexthop type
Usage: crm config thresholds ipv6 nexthop type [OPTIONS]
[percentage|used|free]
Try "crm config thresholds ipv6 nexthop type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv6 nexthop type percentage
admin@sonic:~$ crm config thresholds ipv6 nexthop type used
admin@sonic:~$ crm config thresholds ipv6 nexthop type free
```

crm config thresholds ipv6 nexthop lowhigh

Configure lowhigh threshold of IPv6 nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds ipv6 nexthop low
Usage: crm config thresholds ipv6 nexthop low [OPTIONS] VALUE
Try "crm config thresholds ipv6 nexthop low --help" for help.

admin@sonic:~$ crm config thresholds ipv6 nexthop high
Usage: crm config thresholds ipv6 nexthop high [OPTIONS] VALUE
Try "crm config thresholds ipv6 nexthop high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv6 nexthop low 10
admin@sonic:~$ crm config thresholds ipv6 nexthop high 80000
```

crm config thresholds ipv6 route type

Configure threshold type of IPv6 route

- Usage:

```
admin@sonic:~$ crm config thresholds ipv6 route type
Usage: crm config thresholds ipv6 route type [OPTIONS] [percentage|used|free]
Try "crm config thresholds ipv6 route type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv6 route type percentage
admin@sonic:~$ crm config thresholds ipv6 route type used
admin@sonic:~$ crm config thresholds ipv6 route type free
```

crm config thresholds ipv6 route lowhigh

Configure lowhigh threshold of IPv6 route

- Usage:

```
admin@sonic:~$ crm config thresholds ipv6 route low
Usage: crm config thresholds ipv6 route low [OPTIONS] VALUE
Try "crm config thresholds ipv6 route low --help" for help.

admin@sonic:~$ crm config thresholds ipv6 route high
Usage: crm config thresholds ipv6 route high [OPTIONS] VALUE
Try "crm config thresholds ipv6 route high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds ipv6 route low 100
admin@sonic:~$ crm config thresholds ipv6 route low 80000
```

crm config thresholds mpls inseg lowhigh

(Unsupported in Broadcom platform) Configure lowhigh threshold of MPLS inseg

- Usage:

```
admin@sonic:~$ crm config thresholds mpls inseg low
Usage: crm config thresholds mpls inseg low [OPTIONS] VALUE
Try "crm config thresholds mpls inseg low --help" for help.

admin@sonic:~$ crm config thresholds mpls inseg high
Usage: crm config thresholds mpls inseg high [OPTIONS] VALUE
Try "crm config thresholds mpls inseg high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds mpls inseg low 3
admin@sonic:~$ crm config thresholds mpls inseg high 80
```

crm config thresholds mpls inseg type

(Unsupported in Broadcom platform) Configure threshold type of mpls inseg

- Usage:

```
admin@sonic:~$ crm config thresholds mpls inseg type
Usage: crm config thresholds mpls inseg type [OPTIONS] [percentage|used|free]
Try "crm config thresholds mpls inseg type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds mpls inseg type percentage
admin@sonic:~$ crm config thresholds mpls inseg type used
admin@sonic:~$ crm config thresholds mpls inseg type free
```

crm config thresholds mpls nexthop low|high

(Unsupported in Broadcom platform) Configure low|high threshold of MPLS nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds mpls nexthop low
Usage: crm config thresholds mpls nexthop low [OPTIONS] VALUE
Try "crm config thresholds mpls nexthop low --help" for help

admin@sonic:~$ crm config thresholds mpls nexthop high
Usage: crm config thresholds mpls nexthop high [OPTIONS] VALUE
Try "crm config thresholds mpls nexthop high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds mpls nexthop low 3
admin@sonic:~$ crm config thresholds mpls nexthop high 80
```

crm config thresholds mpls nexthop type

(Unsupported in Broadcom platform) Configure threshold type of mpls nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds mpls nexthop type
Usage: crm config thresholds mpls nexthop type [OPTIONS] [percentage|used|free]
Try "crm config thresholds mpls nexthop type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds mpls nexthop type percentage
admin@sonic:~$ crm config thresholds mpls nexthop type used
admin@sonic:~$ crm config thresholds mpls nexthop type free
```

crm config thresholds nexthop group member type

Configure threshold type of nexthop group member

- Usage:

```
admin@sonic:~$ crm config thresholds nexthop group member type
Usage: crm config thresholds nexthop group member type [OPTIONS]
                                     [percentage|used|free]
Try "crm config thresholds nexthop group member type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds nexthop group member type percentage
admin@sonic:~$ crm config thresholds nexthop group member type used
admin@sonic:~$ crm config thresholds nexthop group member type free
```

crm config thresholds nexthop group member lowhigh

Configure lowhigh threshold of nexthop group member

- Usage:

```
admin@sonic:~$ crm config thresholds nexthop group member low
Usage: crm config thresholds nexthop group member low [OPTIONS] VALUE
Try "crm config thresholds nexthop group member low --help" for help.

admin@sonic:~$ crm config thresholds nexthop group member high
Usage: crm config thresholds nexthop group member high [OPTIONS] VALUE
Try "crm config thresholds nexthop group member high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds nexthop group member low 1000
admin@sonic:~$ crm config thresholds nexthop group member high 12345
```

crm config thresholds nexthop group object type

Configure threshold type of nexthop group object

- Usage:

```
admin@sonic:~$ crm config thresholds nexthop group object type
Usage: crm config thresholds nexthop group object type [OPTIONS]
                                     [percentage|used|free]
Try "crm config thresholds nexthop group object type --help" for help.
```

- Example:


```
admin@sonic:~$ crm config thresholds nexthop group object type percentage
admin@sonic:~$ crm config thresholds nexthop group object type used
admin@sonic:~$ crm config thresholds nexthop group object type free
```

crm config thresholds nexthop group object lowhigh

Configure threshold type of nexthop group object

- Usage:

```
admin@sonic:~$ crm config thresholds nexthop group object low
Usage: crm config thresholds nexthop group object low [OPTIONS] VALUE
Try "crm config thresholds nexthop group object low --help" for help.

admin@sonic:~$ crm config thresholds nexthop group object high
Usage: crm config thresholds nexthop group object high [OPTIONS] VALUE
Try "crm config thresholds nexthop group object high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds nexthop group object low 100
admin@sonic:~$ crm config thresholds nexthop group object high 2000
```

crm config thresholds snat lowhigh

Configure lowhigh threshold of SNAT

- Usage:

```
admin@sonic:~$ crm config thresholds snat low
Usage: crm config thresholds snat low [OPTIONS] VALUE
Try "crm config thresholds snat low --help" for help.

admin@sonic:~$ crm config thresholds snat high
Usage: crm config thresholds snat high [OPTIONS] VALUE
Try "crm config thresholds snat high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds snat low 3
admin@sonic:~$ crm config thresholds snat high 80
```

crm config thresholds snat type

Configure threshold type of SNAT

- Usage:

```
admin@sonic:~$ crm config thresholds snat type
Usage: crm config thresholds snat type [OPTIONS] [percentage|used|free]
Try "crm config thresholds snat type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds snat type percentage
admin@sonic:~$ crm config thresholds snat type used
admin@sonic:~$ crm config thresholds snat type free
```

crm config thresholds srv6-my-sid-entry low/high

(Unsupported in Broadcom platform) Configure low/high threshold of srv6-my-sid-entry

- Usage:

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry low
Usage: crm config thresholds srv6-my-sid-entry low [OPTIONS] VALUE
Try "crm config thresholds srv6-my-sid-entry low --help" for help

admin@sonic:~$ crm config thresholds srv6-my-sid-entry high
Usage: crm config thresholds srv6-my-sid-entry high [OPTIONS] VALUE
Try "crm config thresholds srv6-my-sid-entry high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry low 3
admin@sonic:~$ crm config thresholds srv6-my-sid-entry high 80
```

crm config thresholds srv6-my-sid-entry type

(Unsupported in Broadcom platform) Configure threshold type of srv6-my-sid-entry

- Usage:

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type
Usage: crm config thresholds srv6-my-sid-entry type [OPTIONS] [percentage|used|free]
Try "crm config thresholds srv6-my-sid-entry type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type percentage
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type used
admin@sonic:~$ crm config thresholds srv6-my-sid-entry type free
```

crm config thresholds srv6-nexthop low/high

Configure low/high threshold of srv6-nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds srv6-nexthop low
Usage: crm config thresholds srv6-nexthop low [OPTIONS] VALUE
Try "crm config thresholds srv6-nexthop low --help" for help

admin@sonic:~$ crm config thresholds srv6-nexthop high
Usage: crm config thresholds srv6-nexthop high [OPTIONS] VALUE
Try "crm config thresholds srv6-nexthop high --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds srv6-nexthop low 3
admin@sonic:~$ crm config thresholds srv6-nexthop high 80
```

crm config thresholds srv6-nexthop type

Configure threshold type of srv6-nexthop

- Usage:

```
admin@sonic:~$ crm config thresholds srv6-nexthop type
Usage: crm config thresholds srv6-nexthop type [OPTIONS] [percentage|used|free]
Try "crm config thresholds srv6-nexthop type --help" for help.
```

- Example:

```
admin@sonic:~$ crm config thresholds srv6-nexthop type percentage
admin@sonic:~$ crm config thresholds srv6-nexthop type used
admin@sonic:~$ crm config thresholds srv6-nexthop type free
```

(202111.2) Device Hostname

This section of commands is used to change the device hostname without traffic being impacted.

- [Update Device Hostname Configuration Commands](#)
 - [config hostname](#)

Update Device Hostname Configuration Commands

config hostname

This command is used to change the device hostname without traffic being impacted.

Usage

```
config hostname <new_hostname>
```

Example

```
admin@sonic:~$ sudo config hostname CSW06
```

Note: The configured setting will be lost after a system reboot. To preserve the setting, run **config save**.

(202111.2) DHCP Relay

- DHCP Relay show commands
 - `show vlan brief`
 - `show vlan brief --verbose`
- DHCP Relay config commands
 - `config vlan dhcp_relay add`
 - `config vlan dhcp_relay delete`
 - `config vlan dhcp_relay link_selection add`
 - `config vlan dhcp_relay link_selection del`
 - `config vlan dhcp_relay src_intf add`
 - `config vlan dhcp_relay src_intf del`

These commands are used to add or remove DHCP Relay destination IP addresses for VLAN interfaces.

When the DHCP Relay service is enabled, and the switch sees a DHCP client request from an attached host, it inserts its own IP address into the request so that the DHCP server will know the subnet where the client is located. Then, the switch forwards the packet to a DHCP server (destination IP) on another network. When the server receives the DHCP request, it allocates a free IP address for the DHCP client from its defined scope for the DHCP client's subnet, and sends a DHCP response back to the DHCP Relay agent (the switch). The switch then passes the DHCP response received from the server to the client.

DHCP Relay show commands

show vlan brief

show vlan brief --verbose

This command displays brief information about all the vlans configured in the device. It displays the vlan ID, IP address (if configured for the vlan), list of vlan member ports, whether the port is tagged or in untagged mode and the DHCP Helper Address.

Usage

```
show vlan brief
show vlan brief --verbose
```

Example

```
admin@sonic:~$ show vlan brief
```

VLAN ID	IP Address	Ports	Port Tagging	DHCP Helper	DHCP Source	DHCP Link
Proxy ARP				Address	Interface	Selection
100	1.1.2.2/16	Ethernet0	tagged	192.0.0.1	PortChannel0001	enabled
disabled		Ethernet4	tagged	192.0.0.2		
				192.0.0.3		

```
admin@sonic:~$ show vlan brief --verbose
```

VLAN ID	IP Address	Ports	Port Tagging	DHCP Helper	DHCP Source	DHCP Link
Proxy ARP				Address	Interface	Selection
100	1.1.2.2/16	Ethernet0	tagged	192.0.0.1	PortChannel0001	enabled
disabled		Ethernet4	tagged	192.0.0.2		
				192.0.0.3		

DHCP Relay config commands

config vlan dhcp_relay add

This command adds a DHCP Relay Destination IP address to a VLAN. Note that more than one DHCP Relay Destination IP address can be added on a VLAN interface.

Usage

```
config vlan dhcp_relay add <vlan_id> <dhcp_relay_destination_ip>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay add 1000 7.7.7.7
Added DHCP relay destination address 7.7.7.7 to Vlan1000
Restarting DHCP relay service..
```

config vlan dhcp_relay delete

This command deletes a configured DHCP Relay destination IP address from a VLAN interface.

Usage

```
config vlan dhcp_relay del <vlan-id> <dhcp_relay_destination_ip>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay del 1000 7.7.7.7
Removed DHCP relay destination address 7.7.7.7 from Vlan1000
Restarting DHCP relay service...
Running command: systemctl restart dhcp_relay
```

config vlan dhcp_relay link_selection add

This command is used to enable DHCP relay link selection on a VLAN. Note that users need to add the dhcp relay source interface first.

Usage

```
config vlan dhcp_relay link_selection add <vlan_id>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay link_selection add 1000
Enable DHCP relay link selection for Vlan1000
Restarting DHCP relay service...
```

config vlan dhcp_relay link_selection del

This command is used to disable DHCP relay link selection on a VLAN.

Usage

```
config vlan dhcp_relay link_selection del <vlan_id>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay link_selection del 1000
Removed DHCP relay link selection from Vlan1000
Restarting DHCP relay service...
```

config vlan dhcp_relay src_intf add

This command is used to set the DHCP relay source interface on a VLAN.

Usage

```
config vlan dhcp_relay src_intf add <vlan_id> <dhcp_relay_src_intf>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay src_intf add 1000 PortChannel0001
Added DHCP relay source interface PortChannel0001 for Vlan1000
Restarting DHCP relay service...
```

config vlan dhcp_relay src_intf del

This command is used to remove the configured DHCP relay source interface from a VLAN. Note that users need to disable the dhcp relay link selection first.

Usage

```
config vlan dhcp_relay src_intf del <vlan_id> <dhcp_relay_src_intf>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay src_intf del 1000 PortChannel0001
Removed DHCP relay source interface PortChannel0001 from Vlan1000
Restarting DHCP relay service...
```


(202111.2) DHCPv6 Relay

- DHCPv6 Relay show commands
 - `show dhcprelay_helper ipv6`
 - `show dhcpv6_replay counter`
 - `clear dhcpv6_replay counter`
- DHCPv6 Relay config commands
 - `config vlan dhcp_relay add`
 - `config vlan dhcp_relay delete`

These commands are used to add or remove DHCPv6 Relay destination IP addresses for VLAN interfaces.

When the DHCPv6 Relay service is enabled, and the switch sees a DHCPv6 client request from an attached host, it inserts its own IP address into the request so that the DHCPv6 server will know the subnet where the client is located. Then, the switch forwards the packet to a DHCPv6 server (destination IP) on another network. When the server receives the DHCPv6 request, it allocates a free IP address for the DHCPv6 client from its defined scope for the DHCPv6 client's subnet, and sends a DHCPv6 response back to the DHCPv6 Relay agent (the switch). The switch then passes the DHCPv6 response received from the server to the client.

DHCPv6 Relay show commands

`show dhcprelay_helper ipv6`

This command is used to show the config of DHCPv6 Relay.

Usage

```
show dhcprelay_helper ipv6
```

Example

```
admin@sonic:~$ show dhcprelay_helper ipv6
-----
Vlan1000  fc02:2000::5
-----
```

`show dhcpv6_replay counter`

This command is used to show the counter of DHCPv6 relay packets

Usage

```
show dhcp6relay_counters counts
```

Example

```
admin@sonic:~$ show dhcp6relay_counters counts
Message Type      Vlan1000
-----
Unknown           0
Solicit           2
Advertise          2
Request            2
Confirm            0
Renew              0
Rebind             0
Reply              2
Release            0
Decline            0
Relay-Forward      16
Relay-Reply        4
```

clear dhcpv6_replay counter

This command is used to clear counter of DHCPv6 relay packets

Usage

```
sudo sonic-clear dhcp6relay_counters
```

Example

```
admin@sonic:~$ sudo sonic-clear dhcp6relay_counters

admin@sonic:~$ show dhcp6relay_counters counts
Message Type      Vlan1000
-----
Unknown           0
Solicit            0
Advertise          0
Request            0
Confirm            0
Renew              0
Rebind             0
Reply              0
Release            0
Decline            0
Relay-Forward      0
Relay-Reply        0
```

DHCPv6 Relay config commands

config vlan dhcp_relay add

1. This command is used to add a DHCPv6 Relay Destination IPv6 address to a VLAN.

Usage

```
config vlan dhcp_relay add <vlan_id> <dhcpv6_relay_destination_ips>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay add 1000 fc02:2000::5
Added DHCP relay destination addresses ['fc02:2000::5'] to Vlan1000
Restarting DHCP relay service...
```

2. add the following config to config_db.json. (after images Edgecore-SONiC_20220720_083059_ec202111_159, this step can be skipped. After config in step1, the config in step2 would be also added into config_db in redis.)

config

```
"DHCP_RELAY": {
  "Vlan1000": {
    "dhcpv6_servers": [
      "fc02:2000::5"
    ]
  }
},
```

config vlan dhcp_relay delete

1. This command is used to delete a DHCPv6 Relay Destination IPv6 address.

Usage

```
config vlan dhcp_relay del <vlan_id> <dhcpv6_relay_destination_ips>
```

Example

```
admin@sonic:~$ sudo config vlan dhcp_relay del 1000 fc02:2000::5
Removed DHCP relay destination addresses ['fc02:2000::5'] from Vlan1000
Restarting DHCP relay service...
```

2. Remove the following config from config_db.json (after images Edgecore-SONiC_20220720_083059_ec202111_159, this step can be skipped. After config in step1, the config in step2 would be also added into config_db in redis.)

config

```
"DHCP_RELAY": {  
  "vlan1000": {  
    "dhcpv6_servers": [  
      "fc02:2000::5"  
    ]  
  }  
},
```

(202111.2) Drop Counters

This section explains all the configurable drop counters show commands and configuration options that are supported in SONiC.

- [Drop Counters Show Commands](#)
 - [show dropcounters capabilities](#)
 - [show dropcounters configuration](#)
 - [show dropcounters counts](#)
- [Drop Counters Config Commands](#)
 - [config dropcounters install](#)
 - [config dropcounters add_reasons](#)
 - [config dropcounters remove_reasons](#)
 - [config dropcounters delete](#)
- [Drop Counters Clear Commands](#)
 - [sonic-clear dropcounters](#)

Drop Counters Show Commands

show dropcounters capabilities

This command is used to show the drop counter capabilities that are available on this device. It displays the total number of drop counters that can be configured on this device as well as the drop reasons that can be configured for the counters.

Usage

```
show dropcounters capabilities
```

Example

```
admin@sonic:~$ show dropcounters capabilities
Counter Type          Total
-----
PORT_INGRESS_DROPS    3
SWITCH_EGRESS_DROPS   2

PORT_INGRESS_DROPS:
  L2_ANY
  SMAC_MULTICAST
  SMAC_EQUALS_DMACE
  INGRESS_VLAN_FILTER
  EXCEEDS_L2_MTU
  SIP_CLASS_E
  SIP_LINK_LOCAL
  DIP_LINK_LOCAL
  UNRESOLVED_NEXT_HOP
  DECAP_ERROR

SWITCH_EGRESS_DROPS:
  L2_ANY
  L3_ANY
  A_CUSTOM_REASON
```

show dropcounters configuration

This command is used to show the current running configuration of the drop counters on this device.

Usage

```
show dropcounters configuration [-g <group name>]
```

Example

```
admin@sonic:~$ show dropcounters configuration
Counter  Alias  Group  Type  Reasons  Description
-----  -
DEBUG_0  RX_LEGIT  LEGIT  PORT_INGRESS_DROPS  SMAC_EQUALS_DMAC  Legitimate port-level RX pipeline drops
          INGRESS_VLAN_FILTER
DEBUG_1  TX_LEGIT  None  SWITCH_EGRESS_DROPS  EGRESS_VLAN_FILTER  Legitimate switch-level TX pipeline drops

admin@sonic:~$ show dropcounters configuration -g LEGIT
Counter  Alias  Group  Type  Reasons  Description
-----  -
DEBUG_0  RX_LEGIT  LEGIT  PORT_INGRESS_DROPS  SMAC_EQUALS_DMAC  Legitimate port-level RX pipeline drops
          INGRESS_VLAN_FILTER
```

show dropcounters counts

This command is used to show the current statistics for the configured drop counters. Standard drop counters are displayed as well for convenience. Because clearing counters (see below) is handled on a per-user basis, different users may see different drop counts.

Usage

```
show dropcounters counts \[-g <group name>\] \[-t <counter type>\]
```

Example

```
admin@sonic:~$ show dropcounters counts
  IFACE  STATE  RX_ERR  RX_DROPS  TX_ERR  TX_DROPS  RX_LEGIT
-----
Ethernet0  U      10      100       0         0         20
Ethernet4  U       0     1000       0         0        100
Ethernet8  U     100       10       0         0         0

DEVICE  TX_LEGIT
-----
sonic    1000

admin@sonic:~$ show dropcounters counts -g LEGIT
  IFACE  STATE  RX_ERR  RX_DROPS  TX_ERR  TX_DROPS  RX_LEGIT
-----
Ethernet0  U      10      100       0         0         20
Ethernet4  U       0     1000       0         0        100
Ethernet8  U     100       10       0         0         0

admin@sonic:~$ show dropcounters counts -t SWITCH_EGRESS_DROPS
DEVICE  TX_LEGIT
-----
sonic    1000
```

Drop Counters Config Commands

config dropcounters install

This command is used to initialize a new drop counter. The user must specify a name, type, and initial list of drop reasons. This command will fail if the given name is already in use, if the type of counter is not supported, or if any of the specified drop reasons are not supported. It will also fail if all available counters are already in use on the device.

Usage

```
config dropcounters install <counter name> <counter type> <reasons list> [-d <description>] [-g <group>] [-a <alias>]
```

Example

```
admin@sonic:~$ sudo config dropcounters install DEBUG_2 PORT_INGRESS_DROPS \[EXCEEDS_L2_MTU,DECAP_ERROR\] -d
"More port ingress drops" -g BAD -a BAD_DROPS
```

config dropcounters add_reasons

This command is used to add drop reasons to an already initialized counter. This command will fail if any of the specified drop reasons are not supported.

Usage

```
config dropcounters add_reasons <counter name> <reasons list>
```

Example

```
admin@sonic:~$ sudo config dropcounters add_reasons DEBUG_2 \[SIP_CLASS_E\]
```

config dropcounters remove_reasons

This command is used to remove drop reasons from an already initialized counter.

Usage

```
config dropcounters remove_reasons <counter name> <reasons list>
```

Example

```
admin@sonic:~$ sudo config dropcounters remove_reasons DEBUG_2 \[SIP_CLASS_E\]
```

config dropcounters delete

This command is used to delete a drop counter.

Usage

```
config dropcounters delete <counter name>
```

Example

```
admin@sonic:~$ sudo config dropcounters delete DEBUG_2
```

Drop Counters Clear Commands

sonic-clear dropcounters

This command is used to clear drop counters. This is done on a per-user basis.

Usage

```
sonic-clear dropcounters
```

Example

```
admin@sonic:~$ sonic-clear dropcounters  
Cleared drop counters
```

Note: Only supports the type `PORT_INGRESS_DROPS` with two reasons (`SMAC_EQUALS_DMAC` and `EXCEEDS_L3_MTU`) on the hardware models uses Broadcom ASIC.

(202111.2) Fast Reboot

- [Fast Reboot Configuration Commands](#)
 - [fast-reboot](#)

Fast Reboot Configuration Commands

fast-reboot

Fast-reboot feature enables a switch to reboot up quickly, with minimum disruption to the data plane. It is important for data center networking operation where infrastructure maintenance and upgrade are unavoidable. Fast-reboot is one of the features to minimize traffic impact. This command requires root privilege.

Usage

```
fast-reboot [options]
```

Parameters

```
-h,-? : get this help
-v : turn on verbose
-f : force execution - ignore Orchagent RESTARTCHECK failure
-i : force execution - ignore ASIC MD5-checksum-verification
-d : force execution - ignore database integrity check
-r : reboot with /sbin/reboot
-k : reboot with /sbin/kexec -e [default]
-x : execute script with -x flag
-c : specify control plane assistant IP list
-s : strict mode: do not proceed without:
    - control plane assistant IP list.
-t : Don't tag the current kube images as latest
-D : detached mode - closing terminal will not cause stopping reboot
```

Example

```
admin@sonic:~$ sudo fast-reboot
```

(202111.2) FDB

- [FDB Show Commands](#)
 - [show mac](#)
- [FDB Configuration Commands](#)
 - [sonic-clear fdb all](#)

Switches store the addresses for all known devices. This information is used to forward traffic directly between the inbound and outbound ports. All the addresses learned by monitoring traffic are stored in the dynamic address table, or forwarding database (FDB).

FDB Show Commands

show mac

This command displays the MAC (FDB) entries either in full or partial as given below.

1. show mac - displays the full table
2. show mac -v <vlanid> - displays the MACs learnt on the particular VLAN ID.
3. show mac -p <port> - displays the MACs learnt on the particular port.

Usage

```
show mac [-v <vlan_id>] [-p <port_name>]
```

Example

```
admin@sonic:~$ show mac
No.      Vlan      MacAddress      Port
-----
1        1000      E2:8C:56:85:4A:CD Ethernet192
2        1000      A0:1B:5E:47:C9:76 Ethernet192
3        1000      AA:54:EF:2C:EE:30 Ethernet192
4        1000      A4:3F:F2:17:A3:FC Ethernet192
5        1000      0C:FC:01:72:29:91 Ethernet192
6        1000      48:6D:01:7E:C9:FD Ethernet192
7        1000      1C:6B:7E:34:5F:A6 Ethernet192
8        1000      EE:81:D9:7B:93:A9 Ethernet192
9        1000      CC:F8:8D:BB:85:E2 Ethernet192
10       1000      0A:52:B3:9C:FB:6C Ethernet192
11       1000      C6:E2:72:02:D1:23 Ethernet192
12       1000      8A:C9:5C:25:E9:28 Ethernet192
13       1000      5E:CD:34:E4:94:18 Ethernet192
14       1000      7E:49:1F:B5:91:B5 Ethernet192
15       1000      AE:DD:67:F3:09:5A Ethernet192
16       1000      DC:2F:D1:08:4B:DE Ethernet192
17       1000      50:96:23:AD:F1:65 Ethernet192
18       1000      C6:C9:5E:AE:24:42 Ethernet192
Total number of entries 18
```

Optionally, you can specify a VLAN ID or interface name in order to display only those particular entries.

Examples

```
admin@sonic:~$ show mac -v 1000
No.      Vlan      MacAddress      Port
-----
1        1000      E2:8C:56:85:4A:CD Ethernet192
2        1000      A0:1B:5E:47:C9:76 Ethernet192
3        1000      AA:54:EF:2C:EE:30 Ethernet192
4        1000      A4:3F:F2:17:A3:FC Ethernet192
5        1000      0C:FC:01:72:29:91 Ethernet192
6        1000      48:6D:01:7E:C9:FD Ethernet192
7        1000      1C:6B:7E:34:5F:A6 Ethernet192
8        1000      EE:81:D9:7B:93:A9 Ethernet192
9        1000      CC:F8:8D:BB:85:E2 Ethernet192
10       1000      0A:52:B3:9C:FB:6C Ethernet192
11       1000      C6:E2:72:02:D1:23 Ethernet192
12       1000      8A:C9:5C:25:E9:28 Ethernet192
13       1000      5E:CD:34:E4:94:18 Ethernet192
14       1000      7E:49:1F:B5:91:B5 Ethernet192
15       1000      AE:DD:67:F3:09:5A Ethernet192
16       1000      DC:2F:D1:08:4B:DE Ethernet192
17       1000      50:96:23:AD:F1:65 Ethernet192
18       1000      C6:C9:5E:AE:24:42 Ethernet192
Total number of entries 18
```

```
admin@sonic:~$ show mac -p Ethernet192
No.      Vlan      MacAddress      Port
-----
1        1000      E2:8C:56:85:4A:CD Ethernet192
2        1000      A0:1B:5E:47:C9:76 Ethernet192
3        1000      AA:54:EF:2C:EE:30 Ethernet192
4        1000      A4:3F:F2:17:A3:FC Ethernet192
5        1000      0C:FC:01:72:29:91 Ethernet192
6        1000      48:6D:01:7E:C9:FD Ethernet192
7        1000      1C:6B:7E:34:5F:A6 Ethernet192
8        1000      EE:81:D9:7B:93:A9 Ethernet192
9        1000      CC:F8:8D:BB:85:E2 Ethernet192
10       1000      0A:52:B3:9C:FB:6C Ethernet192
11       1000      C6:E2:72:02:D1:23 Ethernet192
12       1000      8A:C9:5C:25:E9:28 Ethernet192
13       1000      5E:CD:34:E4:94:18 Ethernet192
14       1000      7E:49:1F:B5:91:B5 Ethernet192
15       1000      AE:DD:67:F3:09:5A Ethernet192
16       1000      DC:2F:D1:08:4B:DE Ethernet192
17       1000      50:96:23:AD:F1:65 Ethernet192
18       1000      C6:C9:5E:AE:24:42 Ethernet192
Total number of entries 18
```

FDB Configuration Commands

sonic-clear fdb all

Clear the FDB table.

Usage

```
sonic-clear fdb all
```

Example

```
admin@sonic:~$ sonic-clear fdb all
FDB entries are cleared.
```

(202111.2) Interfaces

- Interface Show Commands
 - show interfaces alias
 - show interfaces autoneg status
 - show interfaces breakout
 - show interfaces counters
 - show interfaces description
 - show interfaces diagnostic
 - show interfaces fec status(Updated)
 - show interfaces link-training status(Updated)
 - show interfaces neighbor
 - show interfaces portchannel
 - show interface status
 - intfutil status_ext(Not support)
 - show interfaces tpid
 - show interfaces transceiver
 - show interfaces transceiver eeprom
 - show interfaces transceiver error-status [-hw]
 - show interfaces transceiver frequency (Updated)
 - show interfaces transceiver lpmode
 - show interfaces transceiver presence
 - Combo-example (show interface and eeprom)
 - sfputil show eeprom
 - show storm-control
- Interface Clear Counters Commands
 - sonic-clear clounters
 - sonic-clear rifcounters
- Interface Configuration Commands
 - config interface ip add (Configure interface IP address)
 - config interface remove (Remove interface IP address)
 - config interface ip add/remove --secondary
 - config interface link-training(Updated)
 - config interface pfc priority
 - config interface pfc asymmetric
 - config interface shutdown
 - config interface startup
 - config interface speed
 - config interface transceiver frequency (Updated)
 - config interface transceiver tx_power (Updated)
 - config interface transceiver lpmode
 - config interface transceiver reset
 - config interface mtu
 - config interface breakout
 - config interface storm-control
 - config interface autoneg(Updated)
 - config interface advertised-speeds
 - config interface advertised-types(Updated)
 - config interface type(Not support)
 - config interface tpid
- Interface Naming Mode
 - show interfaces naming_mode
 - config interface_naming_mode
- Interface VRF Binding Commands
 - config interface vrf bind
 - config interface vrf unbind

Interface Show Commands

This sub-section lists all the possible show commands for the interfaces available in the device. The following example gives a list of possible show commands on interfaces. Subsequent sections explain each of these commands in detail.

Example

```
admin@sonic:~$ show interfaces -?
Usage: show interfaces [OPTIONS] COMMAND [ARGS]...

  Show details of the network interfaces

Options:
  -?, -h, --help  Show this message and exit.

Commands:
  alias          Show Interface Name/Alias Mapping
  autoneg        Show interface autoneg information
  breakout       Show Breakout Mode information by interfaces
  buffer         Show interfaces buffer ...
  counters       Show interface counters
  description    Show interface status, protocol and description
  diagnostic     Show diagnostic cable test information
  fec            Show interface fec information
  mpls           Show Interface MPLS status
  naming_mode    Show interface naming_mode status
  neighbor       Show neighbor related information
  portchannel    Show PortChannel information
  qos            Show interface qos information
  rate-limit     Show interface rate-limit information
  scheduler      Show interface scheduler information
  status         Show Interface status information
  tpid           Show Interface tpid information
  transceiver    Show SFP Transceiver information
  wred           Show interface wred information
```

show interfaces alias

Shows the interface name/alias mapping.

Example

Name	Alias
-----	-----
Ethernet0	hundredGigE1
Ethernet4	hundredGigE2
Ethernet8	hundredGigE3
Ethernet12	hundredGigE4
... continue	

show interfaces autoneg status

Shows the interface auto-negotiation status.

Usage

```
show interfaces autoneg status
show interfaces autoneg status <interface_name>
```

Example

```
admin@sonic:~$ show interfaces autoneg status
```

Interface	Auto-Neg Mode	Oper Speed	Adv Speeds	Type	Adv Types	Oper	Admin
Ethernet0	N/A	10G	1000M,10G	N/A	N/A	up	up
Ethernet1	N/A	10G	N/A	N/A	N/A	up	up
Ethernet2	N/A	10G	N/A	N/A	N/A	up	up
Ethernet3	N/A	10G	N/A	N/A	N/A	up	up
Ethernet4	N/A	10G	N/A	N/A	N/A	up	up
Ethernet5	N/A	10G	N/A	N/A	N/A	up	up
...							

show interfaces breakout

This command displays the port capability for all interfaces, i.e. index, lanes, default break mode (default_brkout_mode), all available breakout modes (breakout_modes), and current breakout mode (brkout_mode). To display the current breakout mode, the "current-mode" subcommand can be used. For a specific interface, provide the <interface name> in the options.

Usage

```
show interfaces breakout
show interfaces breakout current-mode
show interfaces breakout current-mode <interface_name>
```

Example

```
admin@lnos-x1-a-fab01:~$ show interfaces breakout
```

```
{
  "Ethernet0": {
    "index": "1,1,1,1",
    "default_brkout_mode": "1x100G[40G]",
    "child ports": "Ethernet0",
    "child port speed": "100G",
    "breakout_modes": "1x100G[40G],2x50G,4x25G[10G]",
    "Current Breakout Mode": "1x100G[40G]",
    "lanes": "65,66,67,68",
    "alias_at_lanes": "Eth1/1, Eth1/2, Eth1/3, Eth1/4"
  },... continue
}
```

The "current-mode" option is used to display the current breakout mode for all interfaces.

Example (all interfaces)

```
admin@lnos-x1-a-fab01:~$ show interfaces breakout current-mode
+-----+-----+
| Interface | Current Breakout Mode |
+=====+=====+
| Ethernet0 | 4x25G[10G]           |
+-----+-----+
| Ethernet4 | 4x25G[10G]           |
+-----+-----+
| Ethernet8 | 4x25G[10G]           |
+-----+-----+
| Ethernet12| 4x25G[10G]           |
+-----+-----+
```

Example (specific interface)

```
admin@lnos-x1-a-fab01:~$ show interfaces breakout current-mode Ethernet0
+-----+-----+
| Interface | Current Breakout Mode |
+=====+=====+
| Ethernet0 | 4x25G[10G]           |
+-----+-----+
```

Note:

- breakout_mode: `number x speed1 [speed2, speed3]`
 - The parent port lanes are split equally to the number of ports denoted by the number. And, the speed for each breakout port defaults to speed1, but is changeable to speed2, speed3 etc.
 - For example: 4x25G[10G] means that all lanes are assigned to one port and are running at 25G by default, but the speed can be changed to 10G later.

show interfaces counters

This show command displays packet counters for all interfaces since the last time the counters were cleared.

Usage

```
show interfaces counters [-a|--printall] [-p|--period ]
show interfaces counters errors
show interfaces counters rates
show interfaces counters rif [-p|--period ] [-i <interface_name>]
```

Options:

- "-a" provides two additional columns - RX_PPS and TX_PPS.
- "-p" specifies the `period` (in seconds) in which to gather counters.
- "errors" displays the interface error counters.
- "rates" displays only the interface rates.
- "rif" displays Layer 3 counters for a `period` of time or a specific `<interface_name>`, which can include router interface, portchannel, and vlan.

Note:

- There is no facility to display counters for one specific Layer 2 interface.

Example (all interfaces)

```
admin@sonic:~$ show interfaces counters
```

IFACE	STATE	RX_OK	RX_BPS	RX_UTIL	RX_ERR	RX_DRP	RX_OVR
TX_OK	TX_BPS	TX_UTIL	TX_ERR	TX_DRP	TX_OVR		
Ethernet0	U	471,729,839,997	653.87 MB/s	12.77%	0	18,682	0
556.84 MB/s	10.88%	0	0	0			
Ethernet4	U	453,838,006,636	632.97 MB/s	12.36%	0	1,636	0
529.34 MB/s	10.34%	0	0	0			
Ethernet8	U	549,034,764,539	761.15 MB/s	14.87%	0	18,274	0
615.20 MB/s	12.02%	0	0	0			
Ethernet12	U	458,052,204,029	636.84 MB/s	12.44%	0	17,614	0
527.37 MB/s	10.30%	0	0	0			
Ethernet16	U	16,679,692,972	13.83 MB/s	0.27%	0	17,605	0
17.51 MB/s	0.34%	0	0	0			
Ethernet20	U	47,983,339,172	35.89 MB/s	0.70%	0	2,174	0
51.83 MB/s	1.01%	0	0	0			
Ethernet24	U	33,543,533,441	36.59 MB/s	0.71%	0	1,613	0
49.92 MB/s	0.97%	0	0	0			

Display the error counters only.

Example (error counters)

```
admin@str-s6000-ac11:~$ show interface counters errors
```

IFACE	STATE	RX_ERR	RX_DRP	RX_OVR	TX_ERR	TX_DRP	TX_OVR
Ethernet0	U	0	4	0	0	0	0
Ethernet4	U	0	0	0	0	0	0
Ethernet8	U	0	1	0	0	0	0
Ethernet12	U	0	0	0	0	0	0

Display the counters of rates.

Example (rate)

```
admin@str-s6000-ac11:/usr/bin$ show int counters rates
```

IFACE	STATE	RX_OK	RX_BPS	RX_PPS	RX_UTIL	TX_OK	TX_BPS	TX_PPS	TX_UTIL
Ethernet0	U	467510	N/A	N/A	N/A	466488	N/A	N/A	N/A
Ethernet4	U	469679	N/A	N/A	N/A	469245	N/A	N/A	N/A
Ethernet8	U	466660	N/A	N/A	N/A	465982	N/A	N/A	N/A
Ethernet12	U	466579	N/A	N/A	N/A	466318	N/A	N/A	N/A

Display the counters of specified interfaces

Example (specific interfaces)

```
admin@sonic:~$ show interfaces counters -i Ethernet4,Ethernet12-16
```

IFACE	STATE	RX_OK	RX_BPS	RX_UTIL	RX_ERR	RX_DRP	RX_OVR
TX_OK	TX_BPS	TX_UTIL	TX_ERR	TX_DRP	TX_OVR		
Ethernet4	U	453,838,006,636	632.97 MB/s	12.36%	0	1,636	0 388,299,875,056
529.34 MB/s	10.34%	0	0	0			
Ethernet12	U	458,052,204,029	636.84 MB/s	12.44%	0	17,614	0 388,341,776,615
527.37 MB/s	10.30%	0	0	0			
Ethernet16	U	16,679,692,972	13.83 MB/s	0.27%	0	17,605	0 18,206,586,265
17.51 MB/s	0.34%	0	0	0			

Note: The "-i" option is required when using `show interface counters` and it needs to specify the interface.

Example (rif)

```
admin@sonic:~$ show interfaces counters rif
```

IFACE	RX_OK	RX_BPS	RX_PPS	RX_ERR	TX_OK	TX_BPS	TX_PPS	TX_ERR
PortChannel0001	62,668	107.81 B/s	1.34/s	3	6	0.02 B/s	0.00/s	0
PortChannel0002	62,645	107.77 B/s	1.34/s	3	2	0.01 B/s	0.00/s	0
PortChannel0003	62,481	107.56 B/s	1.34/s	3	3	0.01 B/s	0.00/s	0
PortChannel0004	62,732	107.88 B/s	1.34/s	2	3	0.01 B/s	0.00/s	0
Vlan1000	0	0.00 B/s	0.00/s	0	0	0.00 B/s	0.00/s	0

Display the counters of a rif interface.

Example (one interface)

```
admin@sonic:~$ show interfaces counters rif PortChannel0001
PortChannel0001
  RX:
    3269 packets
    778494 bytes
    3 error packets
    292 error bytes
  TX:
    0 packets
    0 bytes
    0 error packets
    0 error bytes
```

Note: When using `show interface counters rif`, it needs to specify the interface.

Display the counters for a period of time in seconds.

Example (a period of time)

```
admin@sonic:~$ show interfaces counters -p 5
```

IFACE	TX_BPS	TX_UTIL	STATE	TX_ERR	RX_OK	RX_BPS	RX_UTIL	RX_ERR	RX_DRP	RX_OVR	TX_OK
Ethernet0	0.00%	0	U	0	515	59.14 KB/s	0.00%	0	0	0	1,305
/s	0.00%	0		0		0		0	0		127.60 KB
Ethernet4	0.00%	0	U	0	305	26.54 KB/s	0.00%	0	0	0	279
/s	0.00%	0		0		0		0	0		39.12 KB
Ethernet8	0.00%	0	U	0	437	42.96 KB/s	0.00%	0	0	0	182
/s	0.00%	0		0		0		0	0		18.37 KB
Ethernet12	0.00%	0	U	0	284	40.79 KB/s	0.00%	0	0	0	160
/s	0.00%	0		0		0		0	0		13.03 KB
Ethernet16	0.00%	0	U	0	377	32.64 KB/s	0.00%	0	0	0	214
/s	0.00%	0		0		0		0	0		18.01 KB
Ethernet20	0.00%	0	U	0	284	36.81 KB/s	0.00%	0	0	0	138
/s	0.00%	0		0		0		0	0		8758.25 B
Ethernet24	0.00%	0	U	0	173	16.09 KB/s	0.00%	0	0	0	169
/s	0.00%	0		0		0		0	0		11.39 KB

show interfaces description

This command displays the key fields of interfaces, such as Operational Status, Administrative Status, Alias, and Description.

Usage

```
show interfaces description [<interface_name>]
```

Example (all interfaces)

```
admin@sonic:~$ show interfaces description
```

Interface	Oper	Admin	Alias	Description
Ethernet0	down	up	hundredGigE1/1	T0-1:hundredGigE1/30
Ethernet4	down	up	hundredGigE1/2	T0-2:hundredGigE1/30
Ethernet8	down	down	hundredGigE1/3	hundredGigE1/3
Ethernet12	down	down	hundredGigE1/4	hundredGigE1/4

Example (a specific interface)

```
admin@sonic:~$ show interfaces description Ethernet4
```

Interface	Oper	Admin	Alias	Description
Ethernet4	down	up	hundredGigE1/2	T0-2:hundredGigE1/30

show interfaces diagnostic

This command displays information for all the interfaces for the cable TDR(Time Domain Reflectometer) data requested or a specific interface if the optional "interface_name" is specified.

Usage

```
show interfaces diagnostic cable-test [<interface_name>]
```

Example (all interfaces)

```
admin@sonic:~$ show interfaces diagnostic cable-test
Ethernet0: cable (4 pairs, length +/- 10 meters)
    pair A Open, length 0 meters
    pair B Open, length 0 meters
    pair C Open, length 0 meters
    pair D Open, length 0 meters

Ethernet1: cable (4 pairs, length +/- 10 meters)
    pair A Ok, length 9 meters
    pair B Ok, length 9 meters
    pair C Ok, length 10 meters
    pair D Ok, length 9 meters

Ethernet2: cable (4 pairs, length +/- 10 meters)
    pair A Ok, length 0 meters
    pair B Ok, length 10 meters
    pair C Ok, length 0 meters
    pair D Ok, length 0 meters

Ethernet3: cable (4 pairs, length +/- 10 meters)
    pair A Open, length 0 meters
    pair B Open, length 0 meters
    pair C Open, length 0 meters
    pair D Open, length 0 meters
```

Example (a specific interface)

```
admin@sonic:~$ show interfaces diagnostic cable-test Ethernet12
Ethernet12: cable (4 pairs, length +/- 10 meters)
    pair A Ok, length 9 meters
    pair B Ok, length 9 meters
    pair C Ok, length 9 meters
    pair D Ok, length 5 meters
```

Example (unsupported interface)

```
admin@sonic:~$ show interfaces diagnostic cable-test Ethernet51
Error! could not perform TDR test on Ethernet51
```

show interfaces fec status(Updated)

This command displays administrative and operational FEC mode mainly and along with the fields such as the operational and administrative status and operational Speed of the interfaces.

Usage

```
show interfaces fec status
show interfaces fec status <interface_name>
```

Example

```
admin@sonic:~$ show interface fec status
Interface      Oper      Admin      FEC      Oper FEC      Oper Speed
-----
Ethernet0      up        up         rs        rs        100G
Ethernet8      up        up         rs        rs        100G
Ethernet16     up        up         rs        rs        100G
Ethernet24     up        up         rs        rs        100G
Ethernet32     up        up         rs        rs        100G
Ethernet40     up        up         rs        rs        100G
...
```

show interfaces link-training status(Updated)

Use the following command to display the link-training configuration and operational status.

Usage

```
show interfaces link-training status
show interfaces link-training status <interface_name>
```

Example

```
admin@sonic:~$ show interfaces link-training status Ethernet0
Interface      LT Oper      LT Admin      Oper      Admin
-----
Ethernet0      trained      on          up          up
```

show interfaces neighbor

This command is used to display the list of expected neighbors for all interfaces (or for a particular interface) that is configured.

Usage

```
show interfaces neighbor expected [<interface_name>]
```

Example

```
admin@sonic:~$ show interfaces neighbor expected
LocalPort      Neighbor      NeighborPort      NeighborLoopback      NeighborMgmt      NeighborType
-----
Ethernet112    ARISTA01T1    Ethernet1          None                  10.16.205.100     ToRRouter
Ethernet116    ARISTA02T1    Ethernet1          None                  10.16.205.101     SpineRouter
Ethernet120    ARISTA03T1    Ethernet1          None                  10.16.205.102     LeafRouter
Ethernet124    ARISTA04T1    Ethernet1          None                  10.16.205.103     LeafRouter
```

show interfaces portchannel

This command displays information regarding port-channel interfaces.

Usage

```
show interfaces portchannel
```

Example

```
admin@sonic:~$ show interfaces portchannel
Flags: A - active, I - inactive, Up - up, Dw - Down, N/A - not available, S - selected, D - deselected
No.  Team Dev          Protocol  Ports
---  -
24   PortChannel24  LACP(A)(Up)  Ethernet28(S) Ethernet24(S)
48   PortChannel48  LACP(A)(Up)  Ethernet52(S) Ethernet48(S)
40   PortChannel40  LACP(A)(Up)  Ethernet44(S) Ethernet40(S)
0    PortChannel0   LACP(A)(Up)  Ethernet0(S)  Ethernet4(S)
8    PortChannel8   LACP(A)(Up)  Ethernet8(S)  Ethernet12(S)
```

show interface status

This command displays some more fields such as Lanes, Speed, MTU, Type, Asymmetric PFC status and also the operational and administrative status of the interfaces.

Usage

```
show interfaces status [<interface_name>]
```

Show the status of all interfaces.

Example (all interfaces)

```
admin@sonic:~$ show interfaces status
Interface      Lanes      Speed      MTU      FEC      Alias      Oper      Admin      Type      Asym PFC
-----
Ethernet0      49,50,51,52  100G      9100      N/A      hundredGigE1/1  down      up      N/A      off
Ethernet4      53,54,55,56  100G      9100      N/A      hundredGigE1/2  down      up      N/A      off
Ethernet8      57,58,59,60  100G      9100      N/A      hundredGigE1/3  down      down     N/A      off
```

Show the status of a specific interface.

Example (a specific interface)

```
admin@sonic:~$ show interface status Ethernet0
Interface      Lanes      Speed      MTU      FEC      Alias      Oper      Admin
-----
Ethernet0      101,102     40G      9100      rs      fortyGigE1/1/1  up      up
```

Show the status for a range of interfaces.

Example (a range of interfaces)

```
admin@sonic:~$ show interfaces status Ethernet8,Ethernet168-180
```

Interface	Lanes	Speed	MTU	FEC	Alias	Oper	Admin	Type	Asym	PFC
Ethernet8	49,50,51,52	100G	9100	N/A	hundredGigE3	down	down	N/A		N/A
Ethernet168	9,10,11,12	100G	9100	N/A	hundredGigE43	down	down	N/A		N/A
Ethernet172	13,14,15,16	100G	9100	N/A	hundredGigE44	down	down	N/A		N/A
Ethernet176	109,110,111,112	100G	9100	N/A	hundredGigE45	down	down	N/A		N/A
Ethernet180	105,106,107,108	100G	9100	N/A	hundredGigE46	down	down	N/A		N/A

intfutil status_ext(Not support)

This show command displays like **show interface status**, but it also shows extended fields for the specific platforms.

For the AS8000, which supports PIM hotswap, the extended fields are "PIM slot" and "PIM status".

Example (all interfaces)

```
admin@sonic:~$ intfutil status_ext
```

Interface	Lanes	Speed	MTU	FEC	Alias	Vlan	Oper	Admin	Type	Asym
Ethernet0	5,6	100G	9100	rs	onehundredGigE0	routed	up	up	N/A	
off	2	up								
Ethernet1	7,8	100G	9100	rs	onehundredGigE1	routed	up	up	N/A	
off	2	up								
Ethernet2	1,2	100G	9100	rs	onehundredGigE2	routed	up	up	N/A	
off	2	up								
Ethernet3	3,4	100G	9100	rs	onehundredGigE3	routed	up	up	N/A	
off	2	up								
Ethernet4	37,38	100G	9100	rs	onehundredGigE4	routed	up	up	N/A	
off	2	up								
Ethernet5	39,40	100G	9100	rs	onehundredGigE5	routed	up	up	N/A	
off	2	up								
Ethernet6	33,34	100G	9100	rs	onehundredGigE6	routed	up	up	N/A	
off	2	up								
Ethernet7	35,36	100G	9100	rs	onehundredGigE7	routed	up	up	N/A	
off	2	up								

show interfaces tpid

This command displays the key fields of the interfaces such as Operational Status, Administrative Status, Alias and TPID.

Usage

Usage: show interfaces tpid [<interface_name>]

Display the TPID for all interfaces.

Example (a specific interface)

```
admin@sonic:~$ show interfaces tpid
Interface      Alias      Oper      Admin      TPID
-----
Ethernet0      fortyGigE1/1/1      up      up      0x8100
Ethernet1      fortyGigE1/1/2      up      up      0x8100
Ethernet2      fortyGigE1/1/3      down    down    0x8100
Ethernet3      fortyGigE1/1/4      down    down    0x8100
Ethernet4      fortyGigE1/1/5      up      up      0x8100
Ethernet5      fortyGigE1/1/6      up      up      0x8100
Ethernet6      fortyGigE1/1/7      up      up      0x9200
Ethernet7      fortyGigE1/1/8      up      up      0x88A8
Ethernet8      fortyGigE1/1/9      up      up      0x8100
...
Ethernet63     fortyGigE1/4/16     down    down    0x8100
PortChannel0001      N/A      up      up      0x8100
PortChannel0002      N/A      up      up      0x8100
PortChannel0003      N/A      up      up      0x8100
PortChannel0004      N/A      up      up      0x8100
admin@sonic:~$
```

Display the TPID for a specific interface.

Example (a specific interface)

```
admin@sonic:~$ show interfaces tpid Ethernet6
Interface      Alias      Oper      Admin      TPID
-----
Ethernet6      fortyGigE1/1/7      up      up      0x9200
admin@sonic:~$
```

show interfaces transceiver

Usage

Usage: show interfaces transceiver [OPTIONS] COMMAND [ARGS]...

Show SFP Transceiver information

Options:

-, -h, --help Show this message and exit.

Commands:

eeprom Show interface transceiver EEPROM information
 error-status Show transceiver error-status
 frequency Show interface transceiver frequency
 lpmode Show interface transceiver low-power mode status
 presence Show interface transceiver presence

show interfaces transceiver eeprom

Show interface transceiver EEPROM information.

Usage (eeprom)

```
show interfaces transceiver eeprom [-d|--dom] [<interface_name>]
```

Parameters:

- "-d" or "-- dom" display DOM (Digital Optics Monitoring) information
- <interface_name> display specific interface transceiver information

Note: Without "-d", this command shows DOM capacity.

Example

```
Ethernet0: SFP EEPROM detected
  Connector: No separable connector
  DOM Capability:
    Rx_power_support: no
    Temp_support: no
    Tx_power_support: no
    Voltage_support: no
  Encoding: 64B66B
  Extended Identifier: Power Class 1(1.5W max)
  Extended RateSelect Compliance: QSFP+ Rate Select Version 1
  Identifier: QSFP28 or later
  Length Cable Assembly(m): 1
  Nominal Bit Rate(100Mbs): 255
  Specification compliance:
    10/40G Ethernet Compliance Code: 40GBASE-CR4
    Fibre Channel Speed: 1200 Mbytes/Sec
  Vendor Date Code(YYYY-MM-DD Lot): 2020-03-31
  Vendor Name: EdgecoreNetworks
  Vendor OUI: 70-72-cf
  Vendor PN: ET7402-DAC-1M
  Vendor Rev: A0
  Vendor SN: J22012100062

Ethernet4: SFP EEPROM detected
  Connector: No separable connector
... continue
```

Example (ceeprom info)

```
admin@sonic:~$ show interfaces transceiver eeprom
Ethernet0: SFP EEPROM detected
  Application Advertisement: N/A
  Connector: MPOx12
  DOM Capability:
    Rx_power_support: True
    Temp_support: False
    Tx_power_support: True
    Voltage_support: False
  Encoding: 256B257B
  Extended Identifier: Power Class 3(2.5W max), CDR present in Rx Tx
  Extended RateSelect Compliance: QSFP+ Rate Select Version 1
  Identifier: QSFP28 or later
  Length Cable Assembly(m): 50
  Nominal Bit Rate(100Mbs): 255
  Specification compliance:
    Extended Specification compliance: 100GBASE-SR4 or 25GBASE-SR
    Fibre Channel Speed: 100 Mbytes/Sec
  Vendor Date Code(YYYY-MM-DD Lot): 2018-09-14
  Vendor Name: Edgecore
  Vendor OUI: 70-72-cf
  Vendor PN: ET7402-SR4
  Vendor Rev: 01
  Vendor SN: J11837006041

Ethernet240: SFP EEPROM detected
  Application Advertisement: N/A
  Connector: MPOx12
  DOM Capability:
    Rx_power_support: True
    Temp_support: False
    Tx_power_support: True
    Voltage_support: False
  Encoding: 256B257B
  Extended Identifier: Power Class 3(2.5W max), CDR present in Rx Tx
  Extended RateSelect Compliance: QSFP+ Rate Select Version 1
  Identifier: QSFP28 or later
  Length Cable Assembly(m): 50
  Nominal Bit Rate(100Mbs): 255
  Specification compliance:
    Extended Specification compliance: 100GBASE-SR4 or 25GBASE-SR
    Fibre Channel Speed: 100 Mbytes/Sec
  Vendor Date Code(YYYY-MM-DD Lot): 2018-09-14
  Vendor Name: Edgecore
  Vendor OUI: 70-72-cf
  Vendor PN: ET7402-SR4
  Vendor Rev: 01
  Vendor SN: J11837006040

Ethernet248: SFP EEPROM Not detected
admin@sonic:~
```

Example (ZR transceiver with dom option)

```
root@sonic:/home/admin# show interfaces transceiver eeprom --dom
Ethernet160: SFP EEPROM detected
  Active App Selection Host Lane 1: 1
  Active App Selection Host Lane 2: 1
```

```
Active App Selection Host Lane 3: 1
Active App Selection Host Lane 4: 1
Active App Selection Host Lane 5: 1
Active App Selection Host Lane 6: 1
Active App Selection Host Lane 7: 1
Active App Selection Host Lane 8: 1
Active Firmware Version: 1.3
Application Advertisement:
    1: 400GAUI-8 C2M (Annex 120E) | 400ZR, DWDM, amplified
    2: 100GAUI-2 C2M (Annex 135G) | 400ZR, DWDM, amplified
CMIS Revision: 4.1
Connector: LC
Encoding: N/A
Extended Identifier: Power Class 8 (19.0W Max)
Extended RateSelect Compliance: N/A
Hardware Revision: 1.1
Host Electrical Interface: 400GAUI-8 C2M (Annex 120E)
Host Lane Assignment Options: 1
Host Lane Count: 8
Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
Inactive Firmware Version: 1.3
Length Cable Assembly(m): 0.0
Media Interface Code: 400ZR, DWDM, amplified
Media Interface Technology: 1550 nm DFB
Media Lane Assignment Options: 1
Media Lane Count: 1
Nominal Bit Rate(100Mbs): 0
Specification compliance: sm_media_interface
Supported Max Laser Frequency: 196100GHz
Supported Max TX Power: -9.0dBm
Supported Min Laser Frequency: 191300GHz
Supported Min TX Power: -14.0dBm
Vendor Date Code(YYYY-MM-DD Lot): 2022-03-28
Vendor Name: MOLEX
Vendor OUI: 00-09-3a
Vendor PN: 1837240006
Vendor Rev: 10
Vendor SN: Z22120BU5
ChannelMonitorValues:
    RX1Power: -40.0dBm
    RX2Power: -40.0dBm
    RX3Power: -40.0dBm
    RX4Power: -40.0dBm
    RX5Power: -40.0dBm
    RX6Power: -40.0dBm
    RX7Power: -40.0dBm
    RX8Power: -40.0dBm
    TX1Power: -40.0dBm
    TX2Power: -40.0dBm
    TX3Power: -40.0dBm
    TX4Power: -40.0dBm
    TX5Power: -40.0dBm
    TX6Power: -40.0dBm
    TX7Power: -40.0dBm
    TX8Power: -40.0dBm
ChannelThresholdValues:
    RxPowerHighAlarm : 6.0dBm
    RxPowerHighWarning: 3.0dBm
    RxPowerLowAlarm : -16.003dBm
    RxPowerLowWarning : -14.001dBm
    TxBiasHighAlarm : 0.0mA
    TxBiasHighWarning : 0.0mA
    TxBiasLowAlarm : 0.0mA
    TxBiasLowWarning : 0.0mA
    TxPowerHighAlarm : -2.0dBm
    TxPowerHighWarning: -4.0dBm
    TxPowerLowAlarm : -13.002dBm
```

```

    TxPowerLowWarning : -12.007dBm
ModuleMonitorValues:
    Temperature: 31.941C
    Vcc: 3.353Volts
ModuleThresholdValues:
    TempHighAlarm : 80.0C
    TempHighWarning: 75.0C
    TempLowAlarm : -6.0C
    TempLowWarning : -3.0C
    VccHighAlarm : 3.7Volts
    VccHighWarning : 3.6Volts
    VccLowAlarm : 3.0Volts
    VccLowWarning : 3.05Volts
Ethernet192: SFP EEPROM detected
    Application Advertisement: N/A
    Connector: MPO 1x12
    DOM Capability:
        Rx_power_support: True
        Temp_support: True
        Tx_power_support: True
        Voltage_support: True
    Encoding: 256B/257B (transcoded FEC-enabled data)
    Extended Identifier: Power Class 3 Module (2.5W max.), No CLEI code present in Page 02h, CDR present
in TX, CDR present in RX
    Extended RateSelect Compliance: Unknown
    Identifier: QSFP28 or later
    Length Cable Assembly(m): 50.0
    Nominal Bit Rate(100Mbs): 255
    Specification compliance:
        10/40G Ethernet Compliance Code: Extended
        Extended Specification Compliance: 100GBASE-SR4 or 25GBASE-SR
        Fibre Channel Link Length: Unknown
        Fibre Channel Speed: Extended
        Fibre Channel Transmission Media: Multi-mode 50 um (OM3)
        Fibre Channel Transmitter Technology: Unknown
        Gigabit Ethernet Compliant Codes: Unknown
        SAS/SATA Compliance Codes: Unknown
        SONET Compliance Codes: Unknown
    Vendor Date Code(YYYY-MM-DD Lot): 2020-06-16
    Vendor Name: EdgecoreNetworks
    Vendor OUI: 70-72-cf
    Vendor PN: ET7402-SR4
    Vendor Rev: A0
    Vendor SN: J22025400592
ChannelMonitorValues:
    RX1Power: 0.0dBm
    RX2Power: 0.0dBm
    RX3Power: 0.0dBm
    RX4Power: 0.0dBm
    TX1Bias: 6.718mA
    TX1Power: 1.058dBm
    TX2Bias: 6.712mA
    TX2Power: 1.057dBm
    TX3Bias: 6.704mA
    TX3Power: 1.055dBm
    TX4Bias: 6.698mA
    TX4Power: 1.054dBm
ChannelThresholdValues:
    RxPowerHighAlarm : 4.0dBm
    RxPowerHighWarning: 2.4dBm
    RxPowerLowAlarm : -18.013dBm
    RxPowerLowWarning : -15.003dBm
    TxBiasHighAlarm : 15.0mA
    TxBiasHighWarning : 12.0mA
    TxBiasLowAlarm : 0.0mA
    TxBiasLowWarning : 2.0mA
ModuleMonitorValues:

```

```

    Temperature: 27.0C
    Vcc: 3.326Volts
ModuleThresholdValues:
    TempHighAlarm   : 85.0C
    TempHighWarning : 70.0C
    TempLowAlarm    : -10.0C
    TempLowWarning  : 0.0C
    VccHighAlarm    : 3.6Volts
    VccHighWarning  : 3.5Volts
    VccLowAlarm     : 2.9Volts
    VccLowWarning   : 3.1Volts

```

show interfaces transceiver error-status [-hw]

Show interface transceiver error-status

Example

Port	Error Status
Ethernet0	OK
Ethernet8	OK
Ethernet16	Unplugged
Ethernet24	Unplugged
Ethernet32	OK
...	continue

show interfaces transceiver frequency (Updated)

Show interface transceiver frequency and space-grid for QSFP-DD. If the transceiver is not QSFP-DD, it would show 'N/A'.

Example

Port	Frequency	Grid space
Ethernet0	N/A	N/A
Ethernet8	N/A	N/A
Ethernet16	193100GHz	75GHz
Ethernet24	N/A	N/A
Ethernet32	N/A	N/A
...	continue	

show interfaces transceiver lpmode

Show interface transceiver low-power mode status.

Example

Port	Low-power Mode
Ethernet0	On
Ethernet4	On
Ethernet8	On
Ethernet12	On
...	continue

show interfaces transceiver presence

Show interface transceiver presence.

Example

```
Port          Presence
-----
Ethernet0     Present
Ethernet4     Not present
Ethernet8     Not present
Ethernet12    Not present
... continue
```

Combo-example (show interface and eeprom)

show interface with 400G transceiver (Edgecore ET7502-AOC-10M & Credo CAC47X281D1DC0HW)

Example (combo, interface and eeprom)

```
admin@sonic:~$ show interfaces status
Interface      Lanes      Speed      MTU      FEC      Alias      Vlan      Oper
Admin          Type      Asym PFC      Oper Speed
-----
Ethernet0      73, 74, 75, 76, 77, 78, 79, 80      400G      9100      rs      Eth1/1(Port1)      routed      up
up  QSFP-DD Double Density 8X Pluggable Transceiver      N/A      400G
Ethernet8      65, 66, 67, 68, 69, 70, 71, 72      400G      9100      rs      Eth2/1(Port2)      routed      up
up  QSFP-DD Double Density 8X Pluggable Transceiver      N/A      400G
Ethernet16     81, 82, 83, 84, 85, 86, 87, 88      400G      9100      rs      Eth3/1(Port3)      routed      up
up  QSFP-DD Double Density 8X Pluggable Transceiver      N/A      400G
Ethernet24     89, 90, 91, 92, 93, 94, 95, 96      400G      9100      rs      Eth4/1(Port4)      routed      up
up  QSFP-DD Double Density 8X Pluggable Transceiver      N/A      400G
Ethernet32     97, 98, 99, 100, 101, 102, 103, 104      400G      9100      rs      Eth5/1(Port5)      routed      down
up                                N/A      N/A      400G
Ethernet40     105, 106, 107, 108, 109, 110, 111, 112      400G      9100      rs      Eth6/1(Port6)      routed      down
up                                N/A      N/A      400G
Ethernet48     113, 114, 115, 116, 117, 118, 119, 120      400G      9100      rs      Eth7/1(Port7)      routed      down
up                                N/A      N/A      400G
Ethernet56     121, 122, 123, 124, 125, 126, 127, 128      400G      9100      rs      Eth8/1(Port8)      routed      down
up                                N/A      N/A      400G
Ethernet64     41, 42, 43, 44, 45, 46, 47, 48      400G      9100      rs      Eth9/1(Port9)      routed      down
up                                N/A      N/A      400G
Ethernet72     33, 34, 35, 36, 37, 38, 39, 40      400G      9100      rs      Eth10/1(Port10)      routed      down
up                                N/A      N/A      400G
Ethernet80     49, 50, 51, 52, 53, 54, 55, 56      400G      9100      rs      Eth11/1(Port11)      routed      down
up                                N/A      N/A      400G
Ethernet88     57, 58, 59, 60, 61, 62, 63, 64      400G      9100      rs      Eth12/1(Port12)      routed      down
up                                N/A      N/A      400G
Ethernet96     129, 130, 131, 132, 133, 134, 135, 136      400G      9100      rs      Eth13/1(Port13)      routed      down
up                                N/A      N/A      400G
Ethernet104    137, 138, 139, 140, 141, 142, 143, 144      400G      9100      rs      Eth14/1(Port14)      routed      down
up                                N/A      N/A      400G
Ethernet112    145, 146, 147, 148, 149, 150, 151, 152      400G      9100      rs      Eth15/1(Port15)      routed      down
up                                N/A      N/A      400G
Ethernet120    153, 154, 155, 156, 157, 158, 159, 160      400G      9100      rs      Eth16/1(Port16)      routed      down
up                                N/A      N/A      400G
Ethernet128    169, 170, 171, 172, 173, 174, 175, 176      400G      9100      rs      Eth17/1(Port17)      routed      down
up                                N/A      N/A      400G
Ethernet136    161, 162, 163, 164, 165, 166, 167, 168      400G      9100      rs      Eth18/1(Port18)      routed      down
up                                N/A      N/A      400G
Ethernet144    177, 178, 179, 180, 181, 182, 183, 184      400G      9100      rs      Eth19/1(Port19)      routed      down
```

```

up
Ethernet152 185,186,187,188,189,190,191,192 N/A N/A 400G
up Ethernet160 1,2,3,4,5,6,7,8 N/A N/A 400G
up Ethernet168 9,10,11,12,13,14,15,16 N/A N/A 400G
up Ethernet176 17,18,19,20,21,22,23,24 N/A N/A 400G
up Ethernet184 25,26,27,28,29,30,31,32 N/A N/A 400G
up Ethernet192 201,202,203,204,205,206,207,208 N/A N/A 400G
up Ethernet200 193,194,195,196,197,198,199,200 N/A N/A 400G
up Ethernet208 217,218,219,220,221,222,223,224 N/A N/A 400G
up Ethernet216 209,210,211,212,213,214,215,216 N/A N/A 400G
up Ethernet224 233,234,235,236,237,238,239,240 N/A N/A 400G
up Ethernet232 225,226,227,228,229,230,231,232 N/A N/A 400G
up Ethernet240 249,250,251,252,253,254,255,256 N/A N/A 400G
up Ethernet248 241,242,243,244,245,246,247,248 N/A N/A 400G
up
admin@sonic:~$
admin@sonic:~$ show interfaces transceiver eeprom Ethernet0
Ethernet0: SFP EEPROM detected
    Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 2.6x10^-4

    Connector: No separable connector
    DOM Capability:
        Rx_power_support: True
        Temp_support: True
        Tx_power_support: True
        Voltage_support: True
    Encoding: Not supported for CMIS cables
    Extended Identifier: Power Class 2(12.0W Max)
    Extended RateSelect Compliance: Not supported for CMIS cables
    Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
    Length Cable Assembly(m): 10
    Nominal Bit Rate(100Mbs): Not supported for CMIS cables
    Specification compliance: Not supported for CMIS cables
    Vendor Date Code(YYYY-MM-DD Lot): 2021-04-07
    Vendor Name: Edgecore
    Vendor OUI: 68-e8-eb
    Vendor PN: ET7502-AOC-10M
    Vendor Rev: 10
    Vendor SN: L12115000001
admin@sonic:~$ show interfaces transceiver eeprom Ethernet08
Ethernet8: SFP EEPROM detected
    Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 2.6x10^-4

    Connector: No separable connector
    DOM Capability:
        Rx_power_support: True
        Temp_support: True
        Tx_power_support: True
        Voltage_support: True
    Encoding: Not supported for CMIS cables
    Extended Identifier: Power Class 2(12.0W Max)
    Extended RateSelect Compliance: Not supported for CMIS cables
    Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
    Length Cable Assembly(m): 10
    Nominal Bit Rate(100Mbs): Not supported for CMIS cables

```



```
Specification compliance: Not supported for CMIS cables
Vendor Date Code(YYYY-MM-DD Lot): 2021-04-07
Vendor Name: Edgecore
Vendor OUI: 68-e8-eb
Vendor PN: ET7502-A0C-10M
Vendor Rev: 10
Vendor SN: L12115000001
admin@sonic:~$ show interfaces transceiver eeprom Ethernet816
Ethernet16: SFP EEPROM detected
  Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                           200GAUI-4 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                           100GAUI-2 C2M (Annex 135G) - Undefined
                           50GAUI-1 C2M (Annex 135G) - Active Cable assembly with BER < 5x10^-5

  Connector: No separable connector
  DOM Capability:
    Rx_power_support: True
    Temp_support: True
    Tx_power_support: True
    Voltage_support: True
  Encoding: Not supported for CMIS cables
  Extended Identifier: Power Class 3(4.5W Max)
  Extended RateSelect Compliance: Not supported for CMIS cables
  Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
  Length Cable Assembly(m): 7
  Nominal Bit Rate(100Mbs): Not supported for CMIS cables
  Specification compliance: Not supported for CMIS cables
  Vendor Date Code(YYYY-MM-DD Lot): 2021-01-06
  Vendor Name: Credo
  Vendor OUI: 9a-ad-ca
  Vendor PN: CAC47X281D1DC0HW
  Vendor Rev:
  Vendor SN: FN4N47X10100002
admin@sonic:~$
admin@sonic:~$ show interfaces transceiver eeprom Ethernet1624
Ethernet24: SFP EEPROM detected
  Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                           200GAUI-4 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                           100GAUI-2 C2M (Annex 135G) - Undefined
                           50GAUI-1 C2M (Annex 135G) - Active Cable assembly with BER < 5x10^-5

  Connector: No separable connector
  DOM Capability:
    Rx_power_support: True
    Temp_support: True
    Tx_power_support: True
    Voltage_support: True
  Encoding: Not supported for CMIS cables
  Extended Identifier: Power Class 3(4.5W Max)
  Extended RateSelect Compliance: Not supported for CMIS cables
  Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
  Length Cable Assembly(m): 7
  Nominal Bit Rate(100Mbs): Not supported for CMIS cables
  Specification compliance: Not supported for CMIS cables
  Vendor Date Code(YYYY-MM-DD Lot): 2021-01-06
  Vendor Name: Credo
  Vendor OUI: 9a-ad-ca
  Vendor PN: CAC47X281D1DC0HW
  Vendor Rev:
  Vendor SN: FN4N47X10100002
```

sfputil show eeprom

This command uses sfp utility to retrieve transceiver's DOM information. It can support transceiver maker to make sure DOM information can be read and correct in EEPROM.

Example (get transceiver eeprom DOM information)

```
admin@sonic:~$ show interfaces status
```

Interface	Lanes	Speed	MTU	FEC	Alias	Vlan	Oper
Admin		Type	Asym	PFC	Oper Speed		
Ethernet0	73,74,75,76,77,78,79,80	400G	9100	rs	Eth1/1(Port1)	routed	up
up	QSFP-DD Double Density 8X Pluggable Transceiver		N/A		400G		
Ethernet8	65,66,67,68,69,70,71,72	400G	9100	rs	Eth2/1(Port2)	routed	up
up	QSFP-DD Double Density 8X Pluggable Transceiver		N/A		400G		
Ethernet16	81,82,83,84,85,86,87,88	400G	9100	rs	Eth3/1(Port3)	routed	up
up	QSFP-DD Double Density 8X Pluggable Transceiver		N/A		400G		
Ethernet24	89,90,91,92,93,94,95,96	400G	9100	rs	Eth4/1(Port4)	routed	up
up	QSFP-DD Double Density 8X Pluggable Transceiver		N/A		400G		
Ethernet32	97,98,99,100,101,102,103,104	400G	9100	rs	Eth5/1(Port5)	routed	down
up		N/A	N/A		400G		

```
...
admin@sonic:~$ sudo sfputil show eeprom --dom
Ethernet0: SFP EEPROM detected
  Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 2.6x10^-4

  Connector: No separable connector
  Encoding: Not supported for CMIS cables
  Extended Identifier: Power Class 2(12.0W Max)
  Extended RateSelect Compliance: Not supported for CMIS cables
  Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
  Length Cable Assembly(m): 10
  Nominal Bit Rate(100Mbs): Not supported for CMIS cables
  Specification compliance: Not supported for CMIS cables
  Vendor Date Code(YYYY-MM-DD Lot): 2021-04-07
  Vendor Name: Edgecore
  Vendor OUI: 68-e8-eb
  Vendor PN: ET7502-AOC-10M
  Vendor Rev: 10
  Vendor SN: L12115000001
  ChannelMonitorValues:
    RX1Power: 0.4591dBm
    RX2Power: 1.1311dBm
    RX3Power: 1.3440dBm
    RX4Power: 1.1883dBm
    RX5Power: -0.0463dBm
    RX6Power: 0.7918dBm
    RX7Power: 0.8052dBm
    RX8Power: 0.7972dBm
    TX1Bias: 6.6000mA
    TX1Power: 0.6483dBm
    TX2Bias: 6.6000mA
    TX2Power: 1.1428dBm
    TX3Bias: 6.6000mA
    TX3Power: 1.1388dBm
    TX4Bias: 6.6000mA
    TX4Power: 1.3348dBm
    TX5Bias: 6.6000mA
    TX5Power: 0.7547dBm
    TX6Bias: 6.6000mA
    TX6Power: 1.0673dBm
    TX7Bias: 6.6000mA
    TX7Power: 0.9989dBm
    TX8Bias: 6.6000mA
    TX8Power: 1.1348dBm
```

```
ChannelThresholdValues:
  RxPowerHighAlarm : 5.0dBm
  RxPowerHighWarning: 4.0dBm
  RxPowerLowAlarm : -9.3968dBm
  RxPowerLowWarning : -8.3983dBm
  TxBiasHighAlarm : 15.0mA
  TxBiasHighWarning : 14.0mA
  TxBiasLowAlarm : 2.0mA
  TxBiasLowWarning : 3.0mA
  TxPowerHighAlarm : 5.0dBm
  TxPowerHighWarning: 4.0dBm
  TxPowerLowAlarm : -7.5007dBm
  TxPowerLowWarning : -6.4995dBm
```

```
ModuleMonitorValues:
  Temperature: 36.7773C
  Vcc: 3.2323Volts
```

```
ModuleThresholdValues:
  TempHighAlarm : 75.0C
  TempHighWarning: 70.0C
  TempLowAlarm : -5.0C
  TempLowWarning : 0.0C
  VccHighAlarm : 3.6Volts
  VccHighWarning : 3.5Volts
  VccLowAlarm : 2.9Volts
  VccLowWarning : 3.0Volts
```

Ethernet8: SFP EEPROM detected

Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 2.6x10^-4

```
Connector: No separable connector
Encoding: Not supported for CMIS cables
Extended Identifier: Power Class 2(12.0W Max)
Extended RateSelect Compliance: Not supported for CMIS cables
Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
Length Cable Assembly(m): 10
Nominal Bit Rate(100Mbs): Not supported for CMIS cables
Specification compliance: Not supported for CMIS cables
Vendor Date Code(YYYY-MM-DD Lot): 2021-04-07
Vendor Name: Edgecore
Vendor OUI: 68-e8-eb
Vendor PN: ET7502-A0C-10M
Vendor Rev: 10
Vendor SN: L12115000001
```

```
ChannelMonitorValues:
  RX1Power: 0.2857dBm
  RX2Power: 0.6315dBm
  RX3Power: 0.6115dBm
  RX4Power: 0.8296dBm
  RX5Power: -0.2581dBm
  RX6Power: -0.0537dBm
  RX7Power: 0.2144dBm
  RX8Power: 0.0715dBm
  TX1Bias: 6.6000mA
  TX1Power: 0.8689dBm
  TX2Bias: 6.6000mA
  TX2Power: 1.4348dBm
  TX3Bias: 6.6000mA
  TX3Power: 1.4610dBm
  TX4Bias: 6.6000mA
  TX4Power: 1.4298dBm
  TX5Bias: 6.6000mA
  TX5Power: 0.6815dBm
  TX6Bias: 6.6000mA
  TX6Power: 1.2655dBm
  TX7Bias: 6.6000mA
  TX7Power: 1.2330dBm
  TX8Bias: 6.6000mA
```

```

    TX8Power: 1.3618dBm
ChannelThresholdValues:
    RxPowerHighAlarm : 5.0dBm
    RxPowerHighWarning: 4.0dBm
    RxPowerLowAlarm : -9.3968dBm
    RxPowerLowWarning : -8.3983dBm
    TxBiasHighAlarm : 15.0mA
    TxBiasHighWarning : 14.0mA
    TxBiasLowAlarm : 2.0mA
    TxBiasLowWarning : 3.0mA
    TxPowerHighAlarm : 5.0dBm
    TxPowerHighWarning: 4.0dBm
    TxPowerLowAlarm : -7.5007dBm
    TxPowerLowWarning : -6.4995dBm
ModuleMonitorValues:
    Temperature: 39.0938C
    Vcc: 3.2222Volts
ModuleThresholdValues:
    TempHighAlarm : 75.0C
    TempHighWarning: 70.0C
    TempLowAlarm : -5.0C
    TempLowWarning : 0.0C
    VccHighAlarm : 3.6Volts
    VccHighWarning : 3.5Volts
    VccLowAlarm : 2.9Volts
    VccLowWarning : 3.0Volts

Ethernet16: SFP EEPROM detected
  Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                           200GAUI-4 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                           100GAUI-2 C2M (Annex 135G) - Undefined
                           50GAUI-1 C2M (Annex 135G) - Active Cable assembly with BER < 5x10^-5

  Connector: No separable connector
  Encoding: Not supported for CMIS cables
  Extended Identifier: Power Class 3(4.5W Max)
  Extended RateSelect Compliance: Not supported for CMIS cables
  Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
  Length Cable Assembly(m): 7
  Nominal Bit Rate(100Mbs): Not supported for CMIS cables
  Specification compliance: Not supported for CMIS cables
  Vendor Date Code(YYYY-MM-DD Lot): 2021-01-06
  Vendor Name: Credo
  Vendor OUI: 9a-ad-ca
  Vendor PN: CAC47X281D1DC0HW
  Vendor Rev:
  Vendor SN: FN4N47X10100002
ChannelMonitorValues:
    RX1Power: -infdBm
    RX2Power: -infdBm
    RX3Power: -infdBm
    RX4Power: -infdBm
    RX5Power: -infdBm
    RX6Power: -infdBm
    RX7Power: -infdBm
    RX8Power: -infdBm
    TX1Bias: 0.0000mA
    TX1Power: -infdBm
    TX2Bias: 0.0000mA
    TX2Power: -infdBm
    TX3Bias: 0.0000mA
    TX3Power: -infdBm
    TX4Bias: 0.0000mA
    TX4Power: -infdBm
    TX5Bias: 0.0000mA
    TX5Power: -infdBm
    TX6Bias: 0.0000mA

```

```

TX6Power: -infdBm
TX7Bias: 0.0000mA
TX7Power: -infdBm
TX8Bias: 0.0000mA
TX8Power: -infdBm
ChannelThresholdValues:
  TxBiasHighAlarm : 0.0mA
  TxBiasHighWarning : 0.0mA
  TxBiasLowAlarm : 0.0mA
  TxBiasLowWarning : 0.0mA
ModuleMonitorValues:
  Temperature: 29.8008C
  Vcc: 3.3070Volts
ModuleThresholdValues:
  TempHighAlarm : 75.0C
  TempHighWarning: 70.0C
  TempLowAlarm : -5.0C
  TempLowWarning : 0.0C
  VccHighAlarm : 3.565Volts
  VccHighWarning : 3.465Volts
  VccLowAlarm : 3.035Volts
  VccLowWarning : 3.135Volts

```

Ethernet24: SFP EEPROM detected

```

Application Advertisement: 400GAUI-8 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                          200GAUI-4 C2M (Annex 120E) - Active Cable assembly with BER < 5x10^-5
                          100GAUI-2 C2M (Annex 135G) - Undefined
                          50GAUI-1 C2M (Annex 135G) - Active Cable assembly with BER < 5x10^-5

```

```

Connector: No separable connector
Encoding: Not supported for CMIS cables
Extended Identifier: Power Class 3(4.5W Max)
Extended RateSelect Compliance: Not supported for CMIS cables
Identifier: QSFP-DD Double Density 8X Pluggable Transceiver
Length Cable Assembly(m): 7
Nominal Bit Rate(100Mbs): Not supported for CMIS cables
Specification compliance: Not supported for CMIS cables
Vendor Date Code(YYYY-MM-DD Lot): 2021-01-06
Vendor Name: Credo
Vendor OUI: 9a-ad-ca
Vendor PN: CAC47X281D1DC0HW
Vendor Rev:
Vendor SN: FN4N47X10100002
ChannelMonitorValues:

```

```

RX1Power: -infdBm
RX2Power: -infdBm
RX3Power: -infdBm
RX4Power: -infdBm
RX5Power: -infdBm
RX6Power: -infdBm
RX7Power: -infdBm
RX8Power: -infdBm
TX1Bias: 0.0000mA
TX1Power: -infdBm
TX2Bias: 0.0000mA
TX2Power: -infdBm
TX3Bias: 0.0000mA
TX3Power: -infdBm
TX4Bias: 0.0000mA
TX4Power: -infdBm
TX5Bias: 0.0000mA
TX5Power: -infdBm
TX6Bias: 0.0000mA
TX6Power: -infdBm
TX7Bias: 0.0000mA
TX7Power: -infdBm
TX8Bias: 0.0000mA

```

```

TX8Power: -infdBm
ChannelThresholdValues:
  TxBiasHighAlarm   : 0.0mA
  TxBiasHighWarning : 0.0mA
  TxBiasLowAlarm    : 0.0mA
  TxBiasLowWarning  : 0.0mA
ModuleMonitorValues:
  Temperature: 33.5547C
  Vcc: 3.3180Volts
ModuleThresholdValues:
  TempHighAlarm   : 75.0C
  TempHighWarning : 70.0C
  TempLowAlarm    : -5.0C
  TempLowWarning  : 0.0C
  VccHighAlarm    : 3.565Volts
  VccHighWarning  : 3.465Volts
  VccLowAlarm     : 3.035Volts
  VccLowWarning   : 3.135Volts

```

show storm-control

This command is used to display the current threshold rate and burst size for BUM (broadcast/unknown-multicast/unknown-unicast) traffic on the physical interface.

Usage

```
show storm-control <command> <args>
```

Commands:

- all , Show storm-control settings for all interfaces
- interface , Show storm-control settings for a given interface

Example

```

admin@sonic:~$ show storm-control all
+-----+-----+-----+-----+
| Interface Name | Storm Type      | Rate (kbps) | Burst Size(kbits) |
+=====+=====+=====+=====+
| Ethernet0      | broadcast       | 1000        | 150                |
+-----+-----+-----+-----+
| Ethernet1      | unknown-multicast | 1000        | 150                |
+-----+-----+-----+-----+
| Ethernet2      | unknown-unicast  | 1000        | 150                |
+-----+-----+-----+-----+

admin@sonic:~$ show storm-control interface Ethernet1
+-----+-----+-----+-----+
| Interface Name | Storm Type      | Rate (kbps) | Burst Size(kbits) |
+=====+=====+=====+=====+
| Ethernet1      | unknown-multicast | 1000        | 150                |
+-----+-----+-----+-----+

```

Interface Clear Counters Commands

sonic-clear clounters

Clear interface counters.

Usage

```
sonic-clear counters
```

Example (clear all counters)

```
admin@sonic:~$ sonic-clear counters
```

sonic-clear rifcounters

Clear Layer 3 interface counters.

Usage

```
sonic-clear rifcounters
```

Example (clear rif counters)

```
admin@sonic:~$ sonic-clear rifcounters
```

Interface Configuration Commands

Interface configuration commands cover these sub-topics:

1. ip - To add or remove an IP address for an interface
2. pfc - to set the PFC configuration for an interface
3. shutdown - to administratively shut down an interface
4. speed - to set the interface speed
5. startup - to bring up an administratively shutdown interface
6. breakout - to set an interface breakout mode
7. tpid - to set interface tpid

config interface ip add (Configure interface IP address)

Add the IP address for an interface. The command syntax is changed after SONiC version 201904, which swaps the "ip add" and "<interface_name>".

Usage (SONiC version after 201904)

```
config interface ip add <interface_name> <ip_addr> [default_gw]
```

or

Usage (SONiC version before 201811)

```
config interface <interface_name> ip add <ip_addr>
```

Note:

- The IP address for a physical interface, portchannel, VLAN interface, or Loopback interface, can all be configured with this command.
- While configuring the IP address for the **management interface "eth0"**, users can provide the **default gateway IP address** as an optional parameter from SONiC release 201911.
- <interface_name>
 - interface VLAN: use "vlan<vlan-id>". For example, "vlan100: for VLAN-100.

Configure interface on port**Example (after 201904)**

```
admin@sonic:~$ sudo config interface ip add Ethernet63 10.11.12.13/24
admin@sonic:~$ sudo config interface ip add eth0 20.11.12.13/24 20.11.12.254
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface Ethernet63 ip add 10.11.12.13/24
```

Configure interface on vlan**Example (after 201904)**

```
admin@sonic:~$ sudo config interface ip add vlan100 10.11.12.13/24
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface vlan100 ip add 10.11.12.13/24
```

config interface remove (Remove interface IP address)

Remove an IP address from an interface. The command syntax is changed after 201904.

Usage (after 201904)

```
config interface ip remove <interface_name> <ip_addr>
```

or

Usage (before 201811)

```
config interface <interface_name> ip remove <ip_addr>
```

Remove IP address from an interface - port**Example (after 201904)**

```
admin@sonic:~$ sudo config interface ip remove Ethernet63 10.11.12.13/24
admin@sonic:~$ sudo config interface ip remove eth0 20.11.12.13/24
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface Ethernet63 ip remove 10.11.12.13/24
```

Remove IP address from an interface - vlan**Example (after 201904)**

```
admin@sonic:~$ sudo config interface ip remove vlan100 10.11.12.13/24
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface vlan100 ip remove 10.11.12.13/24
```

config interface ip add/remove --secondary

Since SONiC 202006 Dec. version, the IPv4 addresses for an interface are classified as the primary address and secondary. There is no primary/secondary classification for IPv6 addresses.

The principles are:

- There is only one primary address for an interface. Other addresses should be specified as secondary.
- The first added address SHOULD be the primary address. The secondary address SHOULD be added after the primary one.
- The command syntax for the addition of a primary address is like the original syntax. The syntax for addition of a secondary address is as follows:

Usage (Add a secondary IPv4 address)

```
config interface <interface_name> ip add <ip_addr> --secondary
```

- The primary address SHOULD be the last address to be removed from an interface. In order to remove the primary address, you must remove all secondary addresses first.
- The command syntax for the removal of a primary address is like the original syntax. The syntax for the removal of a secondary address is as follows:

Usage (Remove a secondary IPv4 address)

```
config interface <interface_name> ip remove <ip_addr> --secondary
```

config interface link-training(Updated)

Use the following command to enable or disable link-training at a interface when auto-negotiation is disabled. Once configured, use **show interfaces link-training status** to check the status.

Usage

Format:

```
config interface link-training <interface_name> <mode>
```

Arguments:

interface_name: [mandatory] name of the interface to be configured. e.g: Ethernet0
mode: [mandatory] link training mode, can be either "on" or "off"

Return:

- error message if interface_name or mode is invalid
- empty upon success

Example

```
admin@sonic:~$ sudo config interface link-training Ethernet0 on
admin@sonic:~$
admin@sonic:~$ sudo config interface link-training Ethernet8 off
admin@sonic:~$
```

config interface pfc priority

This command is used to set PFC to a given priority on a given interface to either "on" or "off". Once successfully configured, it will show current loss priorities on the given interface. Otherwise, it will show error information.

Usage

```
config interface pfc priority <interface_name> <priority> (on | off)
```

Parameters:

- <priority> - range in (0..7)

Example

```
admin@sonic:~$ sudo config interface pfc priority Ethernet0 3 off
Interface      Lossless priorities
-----
Ethernet0      4

admin@sonic:~$ sudo config interface pfc priority Ethernet0 8 off
Usage: pfc config priority [OPTIONS] STATUS INTERFACE PRIORITY
Error: Invalid value for "priority": invalid choice: 8. (choose from 0, 1, 2, 3, 4, 5, 6, 7)

admin@sonic:~$ sudo config interface pfc priority Ethernet101 3 off
Cannot find interface Ethernet101

admin@sonic:~$ sudo config interface pfc priority Ethernet0 3 on
Interface      Lossless priorities
-----
Ethernet0      3,4
```

config interface pfc asymmetric

This command is used for setting the asymmetric PFC for an interface to either "on" or "off". Once it is configured, use **show interfaces status** to check the status.

Usage (after 201904)

```
config interface pfc asymmetric <interface_name> (on | off)
```

or

Usage (before 201811)

```
config interface <interface_name> pfc asymmetric (on | off)
```

Example (after 201904)

```
admin@sonic:~$ sudo config interface pfc asymmetric Ethernet60 on
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface Ethernet60 pfc asymmetric on
```

config interface shutdown

This command is used to administratively shut down either a physical interface or a port-channel interface. Once configured, use **show interfaces status** to check the status.

Usage (after 201904)

```
config interface shutdown <interface_name>
```

or

Usage (before 201811)

```
config interface <interface_name> shutdown
```

Example (after 201904)

```
admin@sonic:~$ sudo config interface shutdown Ethernet63
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface Ethernet63 shutdown
```

Shutdown multi-interfaces**Example (multi-interfaces)**

```
admin@sonic:~$ sudo config interface shutdown Ethernet8,Ethernet16-20,Ethernet32
```

config interface startup

This command is used for administratively bringing up a physical interface or a port-channel interface. Once configured, use **show interfaces status** to check the status.

Usage (after 201904)

```
config interface startup <interface_name>
```

or

Usage (before 201811)

```
config interface <interface_name> startup
```

Example (after 201904)

```
admin@sonic:~$ sudo config interface startup Ethernet63
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface Ethernet63 startup
```

Startup mulit-interfaces**Example (startup multi-interfaces)**

```
admin@sonic:~$ sudo config interface startup Ethernet8,Ethernet16-20,Ethernet32
```

config interface speed

This command is used to configure the speed for a physical interface.

The supported speed list can be changed by breakout-mode. Use **show interfaces breakout** to see the current breakout-mode for the supported speed.

Usage (after 201904)

```
config interface speed <interface_name> <speed_value>
```

or

Usage (before 201811)

```
config interface <interface_name> speed <speed_value>
```

Parameters:

- <speed_value>, user needs to know the interface configurable speeds
 - 40000 for 40G
 - 100000 for 100G

Example (after 201904)

```
admin@sonic:~$ sudo config interface speed Ethernet63 40000
```

or

Example (before 201811)

```
admin@sonic:~$ sudo config interface Ethernet63 speed 40000
```

Note:

- breakout_mode: `number` x speed1 [speed2, speed3]
 - The parent port lanes are split equally to the number of ports denoted by the number. And, the speed for each breakout port defaults to speed1, but is changeable to speed2, speed3 etc.
 - For example: 1x100G[40G] means all lanes are assigned to one port and are running at 100G by default, but the speed can be changed to 40G later.

config interface transceiver frequency (Updated)

This command is used to set frequency and space-grid for ZR transceiver.

Usage

```
config interface transceiver frequency <interface_name> <frequency> <grid_space:75|100>
```

Example

```
user@sonic~$ sudo config interface transceiver frequency Ethernet0 193100 75
Set frequency for port Ethernet0... OK
user@sonic~$ sudo config interface transceiver frequency Ethernet0 192900 100
Set frequency for port Ethernet0... OK
```

config interface transceiver tx_power (Updated)

This command is used to set tx power for ZR transceiver.

Usage

```
config interface transceiver tx_power <interface_name> <txpower>
```

Example

```
user@sonic~$ sudo config interface transceiver tx_power Ethernet0 -- -9
Setting target Tx output power to -9.0 dBm on port Ethernet0
```

config interface transceiver lpmode

This command is used to enable or disable the low-power mode of an SFP transceiver.

Usage

```
config interface transceiver lpmode <interface_name> (enable | disable)
```

Example

```
user@sonic~$ sudo config interface transceiver lpmode Ethernet0 enable
Enabling low-power mode for port Ethernet0... OK
user@sonic~$ sudo config interface transceiver lpmode Ethernet0 disable
Disabling low-power mode for port Ethernet0... OK
```

config interface transceiver reset

Resets an SFP transceiver module.

Usage

```
config interface transceiver reset <interface_name>
```

Example

```
user@sonic~$ sudo config interface transceiver reset Ethernet0
Resetting port Ethernet0... OK
```

config interface mtu

This command is used to configure the MTU for a physical interface. Use the value 1500 for setting the max transfer unit size to 1500 bytes.

Usage

```
config interface mtu <interface_name> <mtu_value>
```

Example

```
admin@sonic:~$ sudo config interface mtu Ethernet64 1500
```

config interface breakout

This command is used to set an available breakout mode for a user-specified interface. Use a double tab (i.e. <tab><tab>) to see the available breakout options customized for each interface specified by the user.

Usage

```
config interface breakout [OPTIONS] <interface_name> <breakout_mode>
```

Options:

- -f, --force-remove-dependencies

- Clear all dependencies internally first.
- -l, --load-predefined-config
 - load predefined user configuration (alias, lanes, speed etc) first.
- -y, --yes
- -v, --verbose
 - Enable verbose output
- -?, -h, --help
 - Show this message and exit.

Note:

- breakout_mode: `number x speed1 [speed2, speed3]`
 - The parent port lanes are split equally to the number of ports denoted by the number. And, the speed for each breakout port defaults to speed1, but is changeable to speed2, speed3 etc.
 - For example: 1x100G[40G] means that all lanes are assigned to one port and running at 100G by default, but the speed can be changed to 40G later.
 - Use **show interface breakout** to see the supported breakout mode.
- The new breakout ports are created by the default configuration, but you can predefine the port configuration by the option "--load-predefined-config". The file name of the predefined configuration must be "port_breakout_config_db.json" and the file must be placed in the directory /etc/sonic/.
- If the breakout_mode includes parentheses (), please add single quotes ' ' for the bash parser. For example: config interface breakout Ethernet0 '1x100G[40G](4)'

Example

```
admin@sonic:~$ sudo config interface breakout Ethernet0 <tab><tab>
<tab provides option for breakout mode>
1x100G[40G] 2x50G 4x25G[10G]

admin@sonic:~$ sudo config interface breakout Ethernet48 4x25G -f -l -v -y

Running Breakout Mode : 1x100G[40G]
Target Breakout Mode : 4x25G

Ports to be deleted :
{
  "Ethernet48": "100000"
}
Ports to be added :
{
  "Ethernet50": "25000",
  "Ethernet51": "25000",
  "Ethernet49": "25000",
  "Ethernet48": "25000"
}

After running Logic to limit the impact

Final list of ports to be deleted :
{
  "Ethernet48": "100000"
}
Final list of ports to be added :
{
  "Ethernet50": "25000",
  "Ethernet51": "25000",
  "Ethernet49": "25000",
  "Ethernet48": "25000"
}

Loaded below Yang Models
['sonic-acl', 'sonic-extension', 'sonic-interface', 'sonic-loopback-interface', 'sonic-port', 'sonic-
portchannel', 'sonic-types', 'sonic-vlan']
Note: Below table(s) have no YANG models:
TACPLUS_SERVER, DEVICE_METADATA, BGP_PEER_RANGE, PORTCHANNEL_MEMBER, FLEX_COUNTER_TABLE, SYSLOG_SERVER, CRM,
CONTAINER_FEATURE, BGP_NEIGHBOR, PORTCHANNEL_INTERFACE, MGMT_INTERFACE, NTP_SERVER,
DEVICE_NEIGHBOR_METADATA, TELEMETRY, DEVICE_NEIGHBOR, FEATURE, BREAKOUT_CFG, DHCP_SERVER, MGMT_PORT,
RESTAPI, VERSIONS, BUFFER_POOL, BUFFER_PROFILE,
Breakout process got successfully completed.
Please note loaded setting will be lost after system reboot. To preserve setting, run `config save`.
```

config interface storm-control

This command is used to add the threshold rate and burst size for BUM (broadcast/unknown-multicast/unknown-unicast) traffic, or to delete the previous setting on the physical interface. All types of storm traffic can be configured independently on the interface at the same time.

Usage

```
config interface storm-control <storm_type> add <interface_name> <rate> <burst size>
config interface storm-control <storm_type> del <interface_name>
```

Note: The parameter <burst size> is optional. A burst size of 10 times the MTU of the interface is applied by default if <burst size> has not been specified.

Example

```
admin@sonic:~$ sudo config interface storm-control broadcast add Ethernet0 1000
add broadcast storm-control
admin@sonic:~$ sudo config interface storm-control unknown-multicast add Ethernet0 100000 750
add unknown-multicast storm-control
admin@sonic:~$ sudo config interface storm-control unknown-unicast add Ethernet0 50000 150
add unknown-unicast storm-control
admin@sonic:~$ show storm-control all
+-----+-----+-----+-----+
| Interface Name | Storm Type      | Rate (kbps) | Burst Size(kbits) |
+-----+-----+-----+-----+
| Ethernet0      | broadcast       | 1000        | default            |
+-----+-----+-----+-----+
| Ethernet0      | unknown-multicast | 100000      | 750                |
+-----+-----+-----+-----+
| Ethernet0      | unknown-unicast  | 50000       | 150                |
+-----+-----+-----+-----+
admin@sonic:~$ sudo config interface storm-control broadcast del Ethernet0
del broadcast storm-control
admin@sonic:~$ show storm-control all
+-----+-----+-----+-----+
| Interface Name | Storm Type      | Rate (kbps) | Burst Size(kbits) |
+-----+-----+-----+-----+
| Ethernet0      | unknown-multicast | 100000      | 750                |
+-----+-----+-----+-----+
| Ethernet0      | unknown-unicast  | 50000       | 150                |
+-----+-----+-----+-----+
```

config interface autoneg(Updated)

This command enables (or disables) the interface auto-negotiation. The referred speed in negotiation is specified in the **config interface advertised-speeds** command. The referred interface type in negotiation is specified in the **config interface advertised-types** command.

Usage

```
config interface autonego <interface> { enabled | disabled }
```

Note,

- There are 3 kinds of IEEE standard to support autonego. The first one is IEEE 802.3 Clause 28, which is applicable to RJ-45 port to support BASE-T PHYs like 1000Base-T. The second is IEEE 802.3 Clause 37, which is applicable to 1000Base-X. The other one is IEEE 802.3 Clause 73, which is applicable to the port that plugged a DAC cable. Note that IEEE 802.3 Clause 73 is not applicable to the port that plugged an optical transceiver. Enable the autonego on the port which is not applicable to support autonego will cause the unpredictable result and the port may not work correctly.
- When auto-negotiation is enabled, the interface speed is negotiated, which refers to the speed list specified by the **config interface advertised-speed** command.
- When auto-negotiation is enabled, the interface type is negotiated, which refers to the speed list specified by the **config interface advertised-types** command.
- When auto-negotiation is disabled, the interface is forced to the speed specified in by the **config interface speed** command.

This is an example to enable (or disable) auto-negotiation on interface Ethernet0.

Example

```
admin@sonic:~$ sudo config interface autonego Ethernet0 enabled
admin@sonic:~$ sudo config interface autonego Ethernet0 disabled
```

config interface advertised-speeds

This command specifies a collection of speeds that are available as a linked speed after auto-negotiation. The configuration does not take effect if the interface has not been enabled for auto-negotiation using the command **config interface autonego**. The change to the speed list takes effect immediately if the interface has auto-negotiation enabled.

Usage

```
config interface advertised-speeds <interface> { all | <speed-list> }
```

Parameters:

- <interface>
 - the interface that applies the auto-negotiation speeds.
- option, user needs to specify one of the following:
 - **all**, set all the advertised-speeds that the interface can support. (default)
 - **<speed list>**, speed collection that the user wants to use in auto-negotiation.

Note:

- A user-configured speed may not be the linked speed if the peer cannot support it. For example, a user specifies 100M, 1G, and 10G in <speed list>, and then mis-configures the peer interface for 1G and 10G when it only supports 100M. The port will enter a link down state since there is no proper speed for both ends.
- The whole speed list is not accepted if it contains an invalid speed in the list.
- All advertised speeds are replaced if a user specifies a new set of speeds, even if auto-negotiation is enabled.

The following example configures an interface (Ethernet0) with a speed list (100M, 1G, 2.5G) or all. After this command, you need to use the **config interface autonego Ethernet0 enable** command to enable port auto-negotiation.

Example

```
admin@sonic:~$ sudo config interface advertised-speeds Ethernet0 100,1000,2500
admin@sonic:~$ sudo config interface advertised-speeds Ethernet0 all
```

The following example shows the result of configuring an invalid speed into the list.

Example

```
admin@sonic:~$ sudo config interface advertised-speeds Ethernet0 100,500,1000,2500
Invalid speed specified: 500
Valid speeds:100,1000,2500
```

config interface advertised-types(Updated)

This command specifies a collection of interface types that are available as a linked type after auto-negotiation. The configuration does not take effect if the interface has not been enabled for auto-negotiation using the command **config interface autonego**. The change to the type list takes effect immediately if the interface has auto-negotiation enabled.

Usage

```
config interface advertised-types <interface> { all | <interface_type> }
```

Parameters:

- <interface>
 - the interface that applies the auto-negotiation interface types
- option, user needs to specify one of the following:
 - **all**, set all the advertised-types that the interface can support. (default)
 - **<interface type>**, interface type collection that the user wants to use in auto-negotiation(KR, KR2, KR4, KR8, CR, CR2, CR4, CR8, and so on).

Note:

- The setting is only applicable to IEEE 802.3by Clause73, which means, only applicable to the port that plugged a DAC cable.

The following example configures an interface (Ethernet0) with a interface type list (CR4, KR4) or all. After this command, you need to use the **config interface autonego Ethernet0 enable** command to enable port auto-negotiation.

Example

```
admin@sonic:~$ sudo config interface advertised-types Ethernet0 CR4,KR4
admin@sonic:~$ sudo config interface advertised-types Ethernet0 all
```

config interface type(Not support)

Configuring interface type takes effect only if auto negotiation is disabled. If auto negotiation is enabled, this command still saves interface type value to CONFIG_DB.

Usage

```
config interface type <interface> <interface_type>
```

Parameters:

- <interface>
 - the interface that applies the interface type configuration
- <interface_type>
 - interface type, valid value include CR4, KR4, and son on

Note:

- This configuration is only applicable to the port, which is a SFP or QSFP form factor port.

The following example configures an interface (Ethernet0) with a interface type CR4.

Example

```
admin@sonic:~$ sudo config interface type Ethernet0 CR4
admin@sonic:~$ sudo config interface type Ethernet4 KR4
```

config interface tpid

This command is used to configure the TPID for the Physical/PortChannel interface. default is 0x8100. Other allowed values if supported by HW SKU (0x9100, 0x9200, 0x88A8).

Usage

```
config interface tpid <interface_name> <tpid_value>
```

Example

```
admin@sonic:~$ sudo config interface tpid Ethernet64 0x9200
```

Interface Naming Mode

show interfaces naming_mode

This command displays the current interface naming mode.

Usage

```
show interfaces naming_mode
```

Note:

- default, display all SONiC interface names in show commands and accept SONiC interface names as parameters in config commands.
- alias, display all hardware vendor interface aliases in show commands and accept hardware vendor interface aliases as parameters in config commands.
- Interface naming mode originally set to 'default'. Interfaces are referenced by default SONiC interface names. Users can change the naming_mode using the **config interface_naming_mode** command.

Example (default)

```
admin@sonic:~$ show interfaces naming_mode
default
```

or

Example (alisa)

```
admin@sonic:~$ show interfaces naming_mode
alias
```

config interface_naming_mode

This command is used to change the interface naming mode.

Users can select between the default mode (SONiC interface names) or alias mode (Hardware vendor names).

The user must log out and log back in for the changes to take effect. Note that the newly-applied interface mode will affect all interface-related show/config commands.

Note: Some platforms do not support alias mapping. In such cases, this command is not applicable. Such platforms always use the same SONiC interface names.

Usage

```
config interface_naming_mode (default | alias)
```

Note: Interface naming mode is originally set to 'default'. Interfaces are referenced by default SONiC interface names.

Example

```
admin@sonic:~$ show interfaces naming_mode
default
admin@sonic:~$ show interface status Ethernet0
Interface      Lanes      Speed      MTU      Alias      Oper      Admin
-----
Ethernet0      101,102      40G      9100      fortyGigE1/1/1      up      up
admin@sonic:~$ sudo config interface_naming_mode alias
Please logout and log back in for changes take effect.
```

After the user logs out and logs in again, the interfaces are referred to by the hardware vendor alias.

Example

```
admin@sonic:~$ show interfaces naming_mode
alias
admin@sonic:~$ sudo config interface fortyGigE1/1/1 shutdown
admin@sonic:~$ show interface status fortyGigE1/1/1
Interface      Lanes      Speed      MTU      Alias      Oper      Admin
-----
Ethernet0      101,102      40G      9100      fortyGigE1/1/1      down      down
```

Interface VRF Binding Commands

By default, all L3 interfaces are bound to the default VRF. Use the **config interface vrf bind** command to bind to different VRFs.

config interface vrf bind

This command is used to bind an interface to a non default VRF.

Usage

```
config interface vrf bind <interface_name> <vrf_name>
```

Example

```
admin@sonic:~$ show vrf
VRF    Interfaces
-----
admin@sonic:~$ config vrf add Vrf1
admin@sonic:~$
admin@sonic:~$
admin@sonic:~$
admin@sonic:~$ show vrf
VRF    Interfaces
-----
Vrf1
admin@sonic:~$ config interface vrf bind PortChannel0001 Vrf1
admin@sonic:~$
admin@sonic:~$
admin@sonic:~$ show vrf
VRF    Interfaces
-----
Vrf1   PortChannel0001
```

config interface vrf unbind

Unbind an interface from a non-default VRF.

Usage

```
config interface vrf unbind <interface_name>
```

Example

```
admin@sonic:~$ config interface vrf unbind PortChannel0001
admin@sonic:~$
admin@sonic:~$ show vrf
VRF    Interfaces
-----
Vrf1
```

(202111.2) IP/IPv6

- [IP Show Commands](#)
 - [show ip route](#)
 - [show ip interfaces](#)
 - [show ip prefix-list](#)
 - [show ip protocol](#)
- [IPv6 Show Commands](#)
 - [show ipv6 route](#)
 - [show ipv6 interfaces](#)
 - [show ipv6 prefix-list](#)
 - [show ipv6 protocol](#)

IP Show Commands

This sub-section explains the various IP protocol-specific show commands that are used to display the following.

1. routes
2. bgp details - Explained in the [bgp section](#)
3. IP interfaces
4. prefix-list
5. protocol

show ip route

This command displays either all the route entries from the routing table or a specific route.

Usage

```
show ip route [vrf <vrf-name>] [<ip_address>]
```

Example

```
admin@sonic:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

S>* 0.0.0.0/0 [200/0] via 10.11.162.254, eth0
C>* 1.1.0.0/16 is directly connected, Vlan100
C>* 10.1.1.0/31 is directly connected, Ethernet112
C>* 10.1.1.2/31 is directly connected, Ethernet116
C>* 10.11.162.0/24 is directly connected, eth0
```

Optionally, specify an IP address to display routes to the particular IP address.

Example

```
admin@sonic:~$ show ip route 10.1.1.0
Routing entry for 10.1.1.0/31
  Known via "connected", distance 0, metric 0, best
  Last update 00:07:49 ago
  * directly connected, Ethernet112
```


Specify <vrf_name> to get IPv4 routes programmed in the <vrf_name>.

Example

```
admin@sonic:~$ show ip route vrf Vrf-red
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

VRF Vrf-red:
C>* 11.1.1.1/32 is directly connected, Loopback11, 21:50:47
C>* 100.1.1.0/24 is directly connected, Vlan100, 03wid06h
admin@sonic:~$ show ip route vrf Vrf-red 11.1.1.1/32
Routing entry for 11.1.1.1/32
  Known via "connected", distance 0, metric 0, vrf Vrf-red, best
  Last update 21:57:53 ago
  * directly connected, Loopback11
```

show ip interfaces

This command displays the details about all the Layer3 IP interfaces in the device for which IP address have been assigned. The type of interfaces include the following.

1. Front panel physical ports
2. PortChannel interfaces
3. VLAN interfaces
4. Loopback interfaces
5. docker interface
6. management interface

Usage

```
show ip interfaces
```

Example

```
admin@sonic:~$ show ip interfaces
```

Interface	Master	IPv4 address/mask	Admin/Oper	BGP Neighbor	Neighbor IP	Flags
Loopback0		1.0.0.1/32	up/up	N/A	N/A	
Loopback11	Vrf-red	11.1.1.1/32	up/up	N/A	N/A	
Loopback100	Vrf-blue	100.0.0.1/32	up/up	N/A	N/A	
PortChannel01		10.0.0.56/31	up/down	DEVICE1	10.0.0.57	
PortChannel02		10.0.0.58/31	up/down	DEVICE2	10.0.0.59	
PortChannel03		10.0.0.60/31	up/down	DEVICE3	10.0.0.61	
PortChannel04		10.0.0.62/31	up/down	DEVICE4	10.0.0.63	
Vlan100	Vrf-red	1001.1.1/24	up/up	N/A	N/A	
Vlan1000		192.168.0.1/27	up/up	N/A	N/A	
docker0		240.127.1.1/24	up/down	N/A	N/A	
eth0		10.3.147.252/23	up/up	N/A	N/A	
lo		127.0.0.1/8	up/up	N/A	N/A	

show ip prefix-list

This command displays the ip prefix-list.

Usage

```
show ip prefix-list
```

Example

```
admin@sonic:~$ show ip prefix-list
BGP: ip prefix-list PL_LoopbackV4: 1 entries
    seq 5 permit 10.1.0.32/32
```

show ip protocol

This command displays the route-map that is configured for the routing protocol. Refer to the routing stack [Quagga Command Reference](#) or [FRR Command Reference](#) to know more about this command.

Usage

```
show ip protocol
```

Example

```
admin@sonic:~$ show ip protocol
VRF: default
Protocol      : route-map
-----
system       : none
kernel       : none
connected    : none
static       : none
rip          : none
ripng        : none
ospf         : none
ospf6        : none
isis         : none
bgp          : RM_SET_SRC
pim          : none
eigrp        : none
nhrp         : none
hsls         : none
olsr         : none
table        : none
ldp          : none
vnc          : none
vnc-direct   : none
vnc-rn       : none
bgp-direct   : none
bgp-direct-to-nve-groups : none
babel        : none
sharp        : none
pbr          : none
bfd          : none
openfabric   : none
vrrp         : none
wildcard     : none
any          : none
```

IPv6 Show Commands

This sub-section explains the various IPv6 protocol-specific show commands that are used to display the following.

1. routes
2. IPv6 bgp details - Explained in the [bgp section](#)
3. IP interfaces
4. prefix-list
5. protocol

show ipv6 route

This command displays either all the IPv6 route entries from the routing table or a specific IPv6 route.

Usage

```
show ipv6 route [vrf <vrf-name>] [<ipv6_address>]
```

Example

```
admin@sonic:~$ show ipv6 route
Codes: K - kernel route, C - connected, S - static, R - RIPng,
       O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
       v - VNC, V - VNC-Direct, A - Babel, D - SHARP, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

C>* 2018:2001::/126 is directly connected, Ethernet112
C>* 2018:2002::/126 is directly connected, Ethernet116
C>* fc00:2::102/128 is directly connected, eth0
C * fe80::/64 is directly connected, Vlan100
C * fe80::/64 is directly connected, Ethernet112
C * fe80::/64 is directly connected, Ethernet116
C * fe80::/64 is directly connected, Bridge
C * fe80::/64 is directly connected, PortChannel0011
C>* fe80::/64 is directly connected, eth0
```

Optionally, you can specify an IPv6 address in order to display only routes to that particular IPv6 address.

Example

```
admin@sonic:~$ show ipv6 route fc00:1::32
Routing entry for fc00:1::32/128
  Known via "connected", distance 0, metric 0, best
  Last update 00:26:02 ago
  * directly connected, lo
```

A VRF name can also be specified to see IPv6 routes configured in the VRF.

Example

```
admin@sonic:~$ show ipv6 route vrf Vrf-red
Codes: K - kernel route, C - connected, S - static, R - RIPng,
       O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
       v - VNC, V - VNC-Direct, A - Babel, D - SHARP, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

VRF Vrf-red:
C>* 1100::1/128 is directly connected, Loopback11, 21:50:47
C>* 100::/112 is directly connected, Vlan100, 03w1d06h
C>* fe80::/64 is directly connected, Loopback11, 21:50:47
C>* fe80::/64 is directly connected, Vlan100, 03w1d06h
admin@sonic:~$ show ipv6 route vrf Vrf-red 1100::1/128
Routing entry for 1100::1/128
  Known via "connected", distance 0, metric 0, vrf Vrf-red, best
  Last update 21:57:53 ago
  * directly connected, Loopback11
```

show ipv6 interfaces

This command displays the details about all the Layer3 IPv6 interfaces in the device for which IPv6 address have been assigned. The type of interfaces include the following.

1. Front panel physical ports
2. PortChannel interfaces
3. VLAN interfaces
4. Loopback interfaces
5. management interface

Usage

```
show ipv6 interfaces
```

Example

```
admin@sonic:~$ show ipv6 interfaces
```

Interface	Master	IPv6 address/mask	Admin/Oper	BGP Neighbor	
Neighbor IP					
-----	-----	-----	-----	-----	-----
Bridge		fe80::7c45:1dff:fe08:cdd%Bridge/64	up/up	N/A	N/A
Loopback11	Vrf-red	1100::1/128	up/up		
PortChannel01		fc00::71/126	up/down	DEVICE1	fc00::72
PortChannel02		fc00::75/126	up/down	DEVICE2	fc00::76
PortChannel03		fc00::79/126	up/down	DEVICE3	fc00::7a
PortChannel04		fc00::7d/126	up/down	DEVICE4	fc00::7e
Vlan100	Vrf-red	100::1/112	up/up	N/A	N/A
		fe80::eef4:bbff:fe08:880a%Vlan100/64			
eth0		fe80::eef4:bbff:fe08:880a%eth0/64	up/up	N/A	N/A
lo		fc00:1::32/128	up/up	N/A	N/A

show ipv6 prefix-list

This command displays the IPv6 prefix-list.

Usage

```
show ipv6 prefix-list
```

Example

```
admin@sonic:~$ show ipv6 prefix-list
BGP: ipv6 prefix-list PL_LoopbackV6: 1 entries
seq 5 permit fc00:1::/64
```

show ipv6 protocol

This command displays the route-map that is configured for the IPv6 routing protocol. Refer to the routing stack [Quagga Command Reference](#) or [FRR Command Reference](#) to know more about this command.

Usage

```
show ipv6 protocol
```

Example

```
admin@sonic:~$ show ipv6 protocol
VRF: default
Protocol    : route-map
-----
system      : none
kernel      : none
connected   : none
static      : none
rip         : none
ripng       : none
ospf        : none
ospf6       : none
isis        : none
bgp         : none
pim         : none
eigrp       : none
nhrp        : none
hsls        : none
olsr        : none
table       : none
ldp         : none
vnc         : none
vnc-direct  : none
vnc-rn      : none
bgp-direct  : none
bgp-direct-to-nve-groups : none
babel       : none
sharp       : none
pbr         : none
bfd         : none
openfabric  : none
vrrp        : none
wildcard    : none
any         : none
```

(202111.2) Kubernetes Configuration

- [Kubernetes Server Setting](#)
 - [Show the Kubernetes Server Config](#)
 - [Show the Kubernetes Link Status](#)
 - [Configure the Kubernetes Server IP](#)
 - [Enable/Disable the Kubernetes Connection](#)
- [SONiC Feature Setting](#)
 - [Configure the Feature Owner to Kubernetes](#)
 - [Show the feature owner](#)
 - [Show the feature status](#)

Kubernetes Server Setting

Show the Kubernetes Server Config

This command displays the Kubernetes server configuration if any, otherwise it will report as not configured.

- Usage:

```
show kubernetes server config
```

- Example:

```
admin@sonic:~$ show kubernetes server config
ip          port    insecure  disable
-----
10.250.4.167 6443    True      False
```

Show the Kubernetes Link Status

This command displays the Kubernetes server status.

- Usage:

```
show kubernetes server status
```

- Example:

```
admin@sonic:~$ show kubernetes server status
ip          port    connected  update-time
-----
10.250.4.167 6443    true       2022-4-15 18:25:05
```

Configure the Kubernetes Server IP

This command is used to configure the kubernetes server address

- Usage:

```
sudo config kubernetes server ip [<server_ip>]
```

- Example:

```
admin@sonic:~$ sudo config kubernetes server ip 10.250.4.167
```

Enable/Disable the Kubernetes Connection

To enable or disable the connection between the kubernetes server.

- Usage:

```
sudo config kubernetes server disable [on/off]
```

- Example: Enable the connection

```
admin@sonic:~$ sudo config kubernetes server disable off
```

- Example: Disable the connection

```
admin@sonic:~$ sudo config kubernetes server disable on
```

SONiC Feature Setting

Configure the Feature Owner to Kubernetes

Configures the owner of a feature as "local" or "kube". The "local" implies starting the feature container from local image. The "kube" implies that kubernetes server is made eligible to deploy the feature. The deployment of a feature by kubernetes is conditional based on many factors like, whether the kube server is configured or not, connected-to-kube-server or not and if that master has manifest for this feature for this switch or not and more. At some time in the future, the deployment *could* happen and beyond that time point the feature can run from the local image, called "fallback". The fallback is allowed by default and it could be toggled to "not allowed". When fallback is not allowed, the feature would run only upon deployment by kubernetes master.

- Usage:

```
config feature owner [<feature name>] [local/kube]
```

- Example:

```
admin@sonic:~$ sudo config feature owner snmp kube
```

Show the feature owner

By executing the command below, the the owner configuration of the corresponding feature can be displayed.

- Usage:

```
show feature config [<feature name>]
```

- Example:


```
admin@sonic:~$ show feature config
Feature      State      AutoRestart  Owner      fallback
-----
bgp           enabled    enabled      local
database      enabled    disabled     local
dhcp_relay    enabled    enabled      kube
lldp          enabled    enabled      kube      true
mgmt-framework enabled    enabled      local
nat           disabled   enabled      local
pmon          enabled    enabled      kube
radv          enabled    enabled      kube
sflow         disabled   enabled      local
snmp          enabled    enabled      kube
swss          enabled    enabled      local
syncd         enabled    enabled      local
teamd         enabled    enabled      local
telemetry     enabled    enabled      kube
```

Show the feature status

To see some detailed status like the current owner or the remote state of the feature.

- Usage:

```
show feature status [<feature name>]
```

- Example:

```
admin@sonic:~$ show feature status
Feature          State    AutoRestart  SystemState  UpdateTime          ContainerId
ContainerVersion  SetOwner  CurrentOwner  RemoteState
-----
bgp              enabled  enabled
up              none
database        enabled
disabled
dhcp_relay      enabled  enabled      up           2020-11-15 18:21:09 249e70102f55
20201230.100    kube      local
lldp            enabled  enabled      up           2020-11-15 18:21:09 779c2d55ee12
20201230.100    kube      local
mgmt-framework  enabled  enabled
up              none
nat             disabled
enabled
pmon            enabled  enabled      up           2020-11-15 18:20:27 a2b9ffa8aba3
20201230.100    kube      local
radv            enabled  enabled      up           2020-11-15 18:21:05 d8ff27dcfe46
20201230.100    kube      local
sflow           disabled
enabled
snmp            enabled  enabled      up           2020-11-15 18:25:51 8b7d5529e306
20201230.111    kube      kube      running
swss            enabled  enabled
up              none
syncd           enabled  enabled
up              none
teamd           enabled  enabled
up              none
telemetry       enabled  enabled      down        2020-11-15 18:24:59
20201230.100    kube      none
```

(202111.2) LLDP

- LLDP Show Commands
 - show lldp table
 - show lldp neighbors
 - show lldp med location
 - show lldp med policy
 - show lldp med power
 - show lldp dot3
 - show lldp global
 - show lldp interfaces
- LLDP Config Commands
 - config lldp med location
 - config lldp med policy voice
 - config lldp med power
 - config lldp dot3 power
 - config lldp global hello_timer
 - config lldp global management_ip
 - config lldp global system_description
 - config lldp global system_name

The Link Layer Discovery Protocol (LLDP) is used to discover basic information about neighboring devices in the local broadcast domain. LLDP is a Layer 2 protocol that uses periodic broadcasts to advertise information about the sending device. Advertised information is represented in Type Length Value (TLV) format according to the IEEE 802.1AB standard, and can include details such as device identification, capabilities, and configuration settings. LLDP also defines how to store and maintain information gathered about the neighboring network nodes it discovers.

LLDP Show Commands

show lldp table

This command displays a brief summary of all LLDP neighbors.

Usage

```
show lldp table
```

Example

```
admin@sonic:~$ show lldp table
Capability codes: (R) Router, (B) Bridge, (O) Other
LocalPort    RemoteDevice  RemotePortID  Capability  RemotePortDescr
-----
Ethernet112  ARISTA01T1   Ethernet1     BR
Ethernet116  ARISTA02T1   Ethernet1     BR
eth0         00:e0:0c:00:01:05  BR           Ethernet Port on unit 1, port 8
-----
Total entries displayed:  3
```

show lldp neighbors

This command displays more details about all LLDP neighbors or only the neighbors connected to a specific interface.

Usage

```
show lldp neighbors <interface_name>
```

Example 1: Display all neighbors on all interfaces

```
admin@sonic:~$ show lldp neighbors
-----
LLDP neighbors:
-----
Interface:    eth0, via: LLDP, RID: 5, Time: 0 day, 00:04:12
Chassis:
  ChassisID:   mac 00:e0:0c:00:00:fd
  SysDescr:    ECS4120-28T
  Capability:   Bridge, on
  Capability:   Router, on
Port:
  PortID:      mac 00:e0:0c:00:01:05
  PortDescr:    Ethernet Port on unit 1, port 8
  TTL:         120
  MFS:         1522
  PMD autoneg: supported: yes, enabled: yes
    Adv:        10Base-T, HD: yes, FD: yes
    Adv:        100Base-TX, HD: yes, FD: yes
    Adv:        1000Base-T, HD: no, FD: yes
  MAU oper type: 1000BaseTFD - Four-pair Category 5 UTP, full duplex mode
VLAN:         20, pvid: yes
-----
Interface:    Ethernet112, via: LLDP, RID: 1, Time: 0 day, 00:04:32
Chassis:
  ChassisID:   mac 52:54:00:02:b9:09
  SysName:      ARISTA01T1
  SysDescr:     Arista Networks EOS version 4.20.15M running on an Arista Networks vEOS
  MgmtIP:       10.250.2.100
  Capability:   Bridge, on
  Capability:   Router, on
Port:
  PortID:      ifname Ethernet1
  TTL:         120
  MFS:         9236
  Port is aggregated. PortAggregID: 1000001
-----
Interface:    Ethernet116, via: LLDP, RID: 3, Time: 0 day, 00:04:32
Chassis:
  ChassisID:   mac 52:54:00:25:75:8e
  SysName:      ARISTA02T1
  SysDescr:     Arista Networks EOS version 4.20.15M running on an Arista Networks vEOS
  MgmtIP:       10.250.2.101
  Capability:   Bridge, on
  Capability:   Router, on
Port:
  PortID:      ifname Ethernet1
  TTL:         120
  MFS:         9236
  Port is aggregated. PortAggregID: 1000001
-----
```

Optionally, you can specify an interface name in order to display only that particular interface.

Example 2

```
admin@sonic:~$ show lldp neighbors Ethernet112
-----
LLDP neighbors:
-----
Interface:    Ethernet112, via: LLDP, RID: 5, Time: 0 day, 00:05:08
Chassis:
  ChassisID:   mac 52:54:00:02:b9:09
  SysName:     ARISTA01T1
  SysDescr:    Arista Networks EOS version 4.20.15M running on an Arista Networks vEOS
  MgmtIP:      10.250.2.100
  Capability:   Bridge, on
  Capability:   Router, on
Port:
  PortID:      ifname Ethernet1
  TTL:         120
  MFS:         9236
  Port is aggregated. PortAggregID: 1000001
-----
```

show lldp med location

This command displays the summary of LLDP-MED location configuration of specified port or all ports.

Usage

```
show lldp med location [port]
```

Example

```
admin@sonic:~$ show lldp med location

INTERFACE NAME      ELIN
-----
Global              1234556
```

show lldp med policy

This command displays the summary of LLDP-MED policy configuration of specified port or all ports.

Usage

```
show lldp med policy [port]
```

Example

```
admin@sonic:~$ show lldp med policy

INTERFACE NAME  APPLICATION  UNKNOWN  TAGGED  VLAN  PRIORITY  DSCP
-----
Global          voice        True     True    -----
background      63
```

show lldp med power

This command displays the summary of LLDP-MED PoE-MDI configuration of a specified port or all ports.

Usage

```
show lldp med power [port]
```

Example

```
admin@sonic:~$ show lldp med power
```

INTERFACE NAME	TYPE	SOURCE	PRIORITY	VALUE
Global	pse	primary	unknown	100

show lldp dot3

This command displays the summary of LLDP Dot3 PoE-MDI configuration of a specified port or all ports.

Usage

```
show lldp dot3 [port]
```

Example

```
admin@sonic:~$ show lldp dot3
```

INTERFACE NAME	TYPE	POWER PAIRS	SUPPORTED	ENABLED	PAIR CONTROL	CLASS	802.3
TYPE SOURCE	PRIORITY	REQUESTED	ALLOCATED				
Global	pse	signal	False	False	False		

show lldp global

This command displays the summary of LLDP global configuration.

Usage

```
show lldp global
```

Example

```
admin@sonic:~$ show lldp global
-----
LLDP Global configuration:
-----
HELLO TIMER          30
SYSTEM NAME          as7326-56x-3
SYSTEM DESCRIPTION   SONiC Software Version: SONiC.Edgecore-SONiC_20220803_092523_ec202111_172 - HwSku:
Accton-AS7326-56X - Distribution: Debian 11.4 - Kernel: 5.10.0-8-2-amd64
MANAGEMENT IP        10.250.0.182
-----
```

show lldp interfaces

This command displays the LLDP interfaces status.

Usage

```
show lldp interfaces [port]
```

Example

```
admin@sonic:~$ show lldp interfaces Ethernet0
INTERFACE NAME    LLDP ENABLED
-----
Ethernet0        true
```

LLDP Config Commands**config lldp med location**

This command is used to set the LLDP-MED location.

Usage

```
config lldp med location <--elin <elin>> [--port <port-name>]
```

Parameters:

- **elin**: Emergency Location Identification Number.
- **port**: Port name. Default is set to all ports.

Note: The configuration cannot be removed after it is configured.

The example below sets the LLDP-MED location to port Ethernet27 and emergency number to 911.

Example

```
admin@sonic:~$ sudo config lldp med location --elin 911 --port Ethernet27
```

config lldp med policy voice

This command is used to configure the specific network policy for voice applications advertised by LLDP for a given port or for all ports.

Usage

```
config lldp med policy voice [--port <port-name>] [--vlan <1-4094>] [--tagged] [--unknown] [--priority (best-effort | background | excellent-effort | critical-applications | video | voice | internetwork-control | network-control)] [--dscp <0-63>]
```

Parameters:

- **port**: Port name. Default is set to all ports.
- **unknown**: Means the network policy for voice application is currently unknown. Default is known.
- **tagged**: Means the IEEE 802.1Q tagged frame format is used by voice application. Default is untagged.
- **vlan**: The VLAN ID (the range is from 1 to 4094). Default value is 0.
- **priority**: IEEE 802.1p Layer 2 Priority, also known as Class of Service (CoS). Default value is best-effort.
 - **best-effort** (0).
 - **background** (1).
 - **excellent-effort** (2).
 - **critical-applications** (3).
 - **video** (4).
 - **voice** (5).
 - **internetwork-control** (6).
 - **network-control** (7).
- **dscp**: DSCP value to be advertised for the given network policy (the range is from 0 to 63). Default value is 0.

Note: The configuration cannot be removed after it is configured.

The example below sets the LLDP-MED policy for the voice VLAN to port Ethernet27, vlan 100 tagged, network policy unknown, priority voice, and dscp 46.

Example

```
sudo config lldp med policy voice --port Ethernet27 --vlan 100 --tagged --unknown --priority voice --dscp 46
```

config lldp med power

These commands are used to configure the LLDP-MED PoE-MDI TLV advertised on a given port or all ports.

The following command is used to configure LLDP-MED PoE-MDI TLV of power providers.

Usage

```
config lldp med power --power-type pse --source (primary | backup) --priority (unknown | critical | high | low) --value <milliwatts> [--port <port-name>]
```

The following command is used to configure LLDP-MED PoE-MDI TLV of power consumers.

Usage

```
config lldp med power --power-type pd --source (pse | local | both) --priority (unknown | critical | high | low) --value <milliwatts> [--port <port-name>]
```

Parameters:

- **source**: Power source.

- **primary**: Primary power source.
- **backup**: Backup power source.
- **pse**: power source equipment.
- **local**: Local power.
- **both**: Both **pse** and **local**.
- **priority**: Power priority.
 - **unknown**
 - **critical**
 - **high**
 - **low**
- **value**: Total power in milliwatts required by the PD device or available by the PSE device. It shall be divisible by 100.
- **port**: Port name. Default is set to all ports.

Note: The configuration cannot be removed after it is configured.

The example below sets LLDP-MED PoE-MDI to port Ethernet27, power-type pse, source primary, priority high, and available power 4000 milliwatts.

Example

```
sudo config lldp med power --port Ethernet27 --power-type pse --source primary --priority high --value 4000
```

config lldp dot3 power

These commands are used to configure LLDP Dot3 PoE-MDI TLV for a given port or for all ports.

The following command is used to configure LLDP Dot3 PoE-MDI TLV of power providers.

Usage

```
config lldp dot3 power --power-type pse <--power-pairs (signal | spare)> [--port <port-name>] [--supported]
[--enabled] [--pair-control] [--power-class <0-4>]
    [--dot3-type (1 | 2) --source (primary | backup) --priority (unknown | critical | high | low) --
requested <milliwatts> --allocated <milliwatts> ]
```

The following command is used to configure LLDP Dot3 PoE-MDI TLV of power consumers.

Usage

```
config lldp dot3 power --power-type pd <--power-pairs (signal | spare)> [--port <port-name>] [--supported]
[--enabled] [--pair-control] [--power-class <0-4>]
    [--dot3-type (1 | 2) --source (pse | local | both) --priority (unknown | critical | high | low) --
requested <milliwatts> <--allocated <milliwatts> ]
```

Parameters:

- **power-pairs**: Power pairs.
 - **signal**: Signal pair.
 - **spare**: Spare pair.
- **port**: Port name. Default is set to all ports.
- **supported**: Means MDI power is supported on the given port, otherwise default is not supported.
- **enabled**: Means MDI power is enabled, otherwise the default is disabled.
- **pair-control**: Means pair selection can be controlled, otherwise the default is cannot be controlled.
- **power-class**: Power class (the range is from 0 to 4).
- **dot3-type**: Type of 802.3.
 - **1**: Type 1, IEEE802.3af.
 - **2**: Type 2, IEEE802.3at.
- **source**: Power source.
 - **primary**: Primary power source.
 - **backup**: Backup power source.
 - **pse**: Power source equipment.
 - **local**: Local power.
 - **both**: Both **pse** and **local**.
- **priority**: Power priority.

- unknown
- critical
- high
- low
- **requested**: Requested power in milliwatts. It shall be divisible by 100.
- **allocated**: Allocated power in milliwatts. It shall be divisible by 100.

Note: The configuration cannot be removed after it is configured.

The example below sets LLDP Dot3 PoE-MDI to port Ethernet27, power-type pse, power pairs signal, power class 3, MDI power supported and enabled, 802.3at type 1, source backup, priority critical, requested 10000 milliwatts, and allocated 10000 milliwatts.

Example

```
sudo config lldp dot3 power --port Ethernet27 --power-type pse --power-pairs signal --power-class 3 --supported --enabled --dot3-type 1 --source backup --priority critical --requested 10000 --allocated 10000
```

config lldp global hello_timer

The command is used to configure the transmission frequency of LLDP PDU updates in seconds. The range is 5 to 65534 seconds.

Usage

```
config lldp global hello_timer <seconds>
```

Example

```
sudo config lldp global hello_timer 10
```

config lldp global management_ip

The command is used to configure the management address to be advertised in LLDP PDUs.

Usage

```
config lldp global management_ip <ip_str>
```

Example

```
sudo config lldp global management_ip 192.168.1.1
```

config lldp global system_description

The command is used to configure the system description to be advertised in LLDP PDUs.

Usage

```
config lldp global system_description <desc>
```

Example

```
sudo config lldp global system_description "Edgecore Networks"
```

config lldp global system_name

The command is used to configure the system name to be advertised in LLDP PDUs.

Usage

```
config lldp global system_name <name>
```

Example

```
sudo config lldp global system_name "Edgecore SONiC DUT"
```

(202111.2) Management VRF

- Management VRF Show Commands
 - `show mgmt-vrf`
 - `show mgmt-vrf routes`
 - `show management_interface address`
- Management VRF Config Commands
 - `config vrf add mgmt`
 - `config vrf del mgmt`
 - `config in-band-mgmt`

Management Virtual Routing and Forwarding (VRF) supports a separate routing table for management traffic that is forwarded by out-of-band management ports on a switch. This enables management traffic to be routed across a management network and still remain secure and separate from network data traffic.

Management VRF Show Commands

`show mgmt-vrf`

This command displays whether the management VRF and in-band management VRF are enabled or disabled. It also displays the details about the links (eth0, mgmt, lo-m, L3 interface) that are related to the management VRF.

Usage:

```
show mgmt-vrf
```

Example:

```
admin@sonic:~$ show mgmt-vrf
```

```
ManagementVRF : Enabled
```

```
In-band ManagementVRF : Enabled
```

```
Management VRF interfaces in Linux:
```

```
348: mgmt: <NOARP,MASTER,UP,LOWER_UP> mtu 65536 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether f2:2a:d9:bc:e8:f0 brd ff:ff:ff:ff:ff:ff
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq master mgmt state UP mode DEFAULT group
default qlen 1000
    link/ether 4c:76:25:f4:f9:f3 brd ff:ff:ff:ff:ff:ff
66: Ethernet104: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9100 qdisc pfifo_fast master mgmt state UP mode
DEFAULT group default qlen 1000
    link/ether 4c:76:25:f4:f9:f3 brd ff:ff:ff:ff:ff:ff
350: lo-m: <BROADCAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc noqueue master mgmt state UNKNOWN mode DEFAULT
group default qlen 1000
    link/ether b2:4c:c6:f3:e9:92 brd ff:ff:ff:ff:ff:ff
```

NOTE: The management interface "eth0" shows the "master" as "mgmt" since it is part of management VRF.

`show mgmt-vrf routes`

This command displays the routes that are present in the management VRF routing table.

Usage:

```
show mgmt-vrf routes
```

Example:

```
admin@sonic:~$ show mgmt-vrf routes
```

```
Routes in Management VRF Routing Table:
default via 10.16.210.254 dev eth0 metric 201
broadcast 10.16.210.0 dev eth0 proto kernel scope link src 10.16.210.75
10.16.210.0/24 dev eth0 proto kernel scope link src 10.16.210.75
local 10.16.210.75 dev eth0 proto kernel scope host src 10.16.210.75
broadcast 10.16.210.255 dev eth0 proto kernel scope link src 10.16.210.75
broadcast 127.0.0.0 dev lo-m proto kernel scope link src 127.0.0.1
127.0.0.0/8 dev lo-m proto kernel scope link src 127.0.0.1
local 127.0.0.1 dev lo-m proto kernel scope host src 127.0.0.1
broadcast 127.255.255.255 dev lo-m proto kernel scope link src 127.0.0.1
```

show management_interface address

This command displays the IP addresses configured for the management interface "eth0" and the management network default gateway.

Usage:

```
show management_interface address
```

Example:

```
admin@sonic:~$ show management_interface address
Management IP address = 10.16.210.75/24
Management Network Default Gateway = 10.16.210.254
Management IP address = FC00:2::32/64
Management Network Default Gateway = fc00:2::1
```

Management VRF Config Commands**config vrf add mgmt**

This command enables the management VRF in the system. This command restarts the "interfaces-config" service, which in turn regenerates the /etc/network/interfaces file and restarts the "networking" service. This creates a new interface and l3mdev CGROUP with the name as "mgmt" and enslaves the management interface "eth0" into this master interface "mgmt". Note that the VRFName "mgmt" (or "management") is reserved for the management VRF. That is, data VRFs should not use these reserved VRF names.

Usage:

```
config vrf add mgmt
```

Example:

```
admin@sonic:~$ sudo config vrf add mgmt
```

config vrf del mgmt

This command disables the management VRF in the system. This command restarts the "interfaces-config" service, which in turn regenerates the /etc/network/interfaces file and restarts the "networking" service. This deletes the interface "mgmt" and deletes the l3mdev CGROUP named "mgmt" and puts back the management interface "eth0" into the default VRF. Note that the VRFName "mgmt" (or "management") is reserved for the management VRF. That is, data VRFs should not use these reserved VRF names.

Usage:

```
config vrf del mgmt
```

Example:

```
admin@sonic:~$ sudo config vrf del mgmt
```

config in-band-mgmt

This command is used to enable or disable the in-band management VRF in the system. Globally, in-band management VRF is disabled by default.

Usage:

```
config in-band-mgmt <enable|disable>
```

Example:

```
admin@sonic:~$ sudo config in-band-mgmt enable
```

(202111.2) MC LAG

- MCLAG show commands
 - show mclag brief
 - show mclag interface
 - show mclag unique ip
- MCLAG Configuration Commands
 - config mclag add/del
 - config mclag member add/del
 - config mclag unique-ip add/del
 - config mclag keepalive-interval
 - config mclag session-timeout

This section explains all the MCLAG (Multi-Chassis Link Aggregation Group) commands that are supported in SONiC.

MCLAG show commands

show mclag brief

This command displays brief information about an MCLAG domain.

Usage

```
show mclag brief
```

Example

```
admin@sonic:~$ show mclag brief
Domain ID           : 5
Role                : Active
Session Status      : Up
Peer Link Status    : Up
Source Address      : 192.168.1.1
Peer Address        : 192.168.1.2
Peer Link           : PortChannel30
Keepalive Interval  : 1 secs
Session Timeout     : 15 secs
System MAC          : b8:6a:97:73:6c:96
Number of MCLAG Interfaces : 2
MCLAG Interface     Local/Remote Status
PortChannel50       Up/Up
PortChannel60       Up/Up
```

show mclag interface

This command displays interface information in an MCLAG domain.

Usage

```
show mclag interface <domain-id> <portchannel-name>
```

Example

```
admin@sonic:~$ show mclag interface 5 PortChannel50
Local/Remote Status      : Up/Up
IsolateWithPeerLink      : Yes
```

show mclag unique ip

This command displays the unique IP status.

Usage

```
show mclag unique-ip
```

Example

```
admin@sonic:~$ show mclag unique-ip
Unique IP:           : Vlan200, Vlan300
```

MCLAG Configuration Commands

In this section, it describes how to configure a MCLAG domain, add MCLAG member, enable/disable unique IP, and configure timer for keep-alive and session timeout.

config mclag add/del

Use this command to configure MCLAG domain attributes.

Usage

```
config mclag {add | del} <domain-id> <local-ip-addr> <peer-ip-addr> [<peer-ifname>]
```

Description:

- **domain-id** - MCLAG node's unique domain ID; this domain-id should be the same as peer node.
- **local-ip-addr** - MCLAG node's local IPv4 address
- **peer-ip-addr** - MCLAG node's peer IPv4 address
- **peer-ifname** - MCLAG peer interface name; optional for L3 MCLAG, mandatory for L2 MCLAG config

Example

```
admin@sonic:~$ sudo config mclag add 5 192.168.1.1 192.168.1.2 PortChannel30
```

config mclag member add/del

Use this command to configure a domain's MCLAG interfaces.

Usage

```
config mclag member {add | del} <domain-id> <portchannel-name>
```

Description:

- MCLAG interfaces can be only portchannel.

Example

```
admin@sonic:~$ sudo config mclag member add 5 PortChannel50
```

config mclag unique-ip add/del

Use this command to configure the unique IP status on a specified VLAN. User may enable (or disable) unique IP on different VLANs but this configuration should be the same in both active node and standby node. When the unique IP of a VLAN is enabled, active node and standby node should use different IP addresses, that is, the assigned IP address of active node and standby node on the VLAN cannot be the same.

Usage

```
config mclag unique-ip {add | del} <Vlan-interface>n
```

Parameter:

- **add** - enable unique IP on the VLAN. After enabled, the active node and standby node should not assign the same IP address on the VLAN.
- **del** - disable unique IP on the VLAN. After disabled, the active node and standby node share the same IP address on the VLAN.
- **Vlan-interface** - VLAN interface for which a unique IP can be configured.

Before applying this command, please make sure

- The MCLAG Domain should be created.
- VLAN interface should not be bound to VRF.
- There are not any IP address on VLAN interface.

Note, this restriction is derived from following user scenario

- when enabling (or disabling) the unique IP on a VLAN, the IP address should be re-assigned after the command executed.
- the IP address re-assignment should be manually configured by user. This command would not automatically clean (or assign) the IP address
- as the IP address is re-assigned, the routing tables of VRF should be not referred any more.
- Due to the re-assignment is manually, the command will check the MCLAG domain, IP assignment, and bound on VRF before execution.

Example

```
admin@sonic:~$ sudo config mclag unique-ip add Vlan200
```

config mclag keepalive-interval

Use this command to configure keep-alive timer.

Usage

```
config mclag keepalive-interval <domain-id> <keepalive-timer-val>
```

Description:

- **keepalive-timer-val** - keepalive-timer in the range of 1 to 60 seconds with default value of 1 seconds.
- session-timeout should be at least greater than 3 times keepalive-timer value.

Example

```
admin@sonic:~$ sudo config mclag keepalive-interval 5 5
```

config mclag session-timeout

Use this command to configure session timeout.

Usage

```
config mclag session-timeout <domain_id> <time_in_secs>
```

Description:

- **time_in_secs** - session-timeout in the range of 3 to 3600 seconds with default value of 15 seconds
- session-timeout should be at least greater than 3 times keepalive-timer value.

Example

```
admin@sonic:~$ sudo config mclag session-timeout 5 20
```

(202111.2) Mirroring

- [Mirroring Show Commands](#)
 - `show mirror_session`
- [Mirroring Configuration Commands](#)
 - `config mirror_session add`
 - `config mirror_session remove`

Mirroring Show Commands

`show mirror_session`

This command displays all the mirror sessions that are configured.

Usage

```
show mirror_session
```

Example

```
admin@sonic:~$ show mirror_session
```

```
ERSPAN Sessions
Name      Status  SRC IP    DST IP      GRE      DSCP      TTL  Queue  Policer  Monitor Port  SRC
Port      Direction
-----
mrr_legacy inactive 1.2.3.4    20.21.22.23 0x88be    8         100  UC0

SPAN Sessions
Name      Status  DST Port  SRC Port    Direction  Queue  Policer
-----
mrr_port0 active  Ethernet0 PortChannel10 rx
```

Mirroring Configuration Commands

`config mirror_session add`

This command is used to add mirroring sessions. A mirror session is identified by a "session_name". This command supports configuring both SPAN and ERSPAN sessions. For a SPAN session, you can configure mirroring for a list of source ports/LAG to a destination port in ingress, egress, or both directions. For an ERSPAN session, you can configure mirroring for a list of source ports/LAG to a destination IP. Both SPAN/ERSPAN support ACL-based mirroring and can be used in ACL configurations.

The command below is used to create an ERSPAN mirror session:

Usage

```
config mirror_session erspan add <session-name> <src_ip> <dst_ip> <dscp> <ttl> [gre_type] [queue] [src_port]
[direction] [--policer <policer_name>]
```

or

Usage (backward compatible, to be deprecated)

```
config mirror_session add <session_name> <src_ip> <dst_ip> <dscp> <ttl> [gre_type] [queue]
```

Parameters:

1. Session name
2. Source IP address
3. Destination IP address
4. DSCP (QoS) value with which mirrored packets are forwarded
5. TTL value
6. Optional - GRE type in a GRE tunnel packet. The default value is 0x88be. Valid values are:
 - a. 0x88be: ERSPAN Type I.
 - b. 0x22eb: ERSPAN Type III (Only support on Intel devices).
7. Optional - Queue in which packets shall be sent out of the device. The default value is 0. Only support to change the value on Intel devices. Valid values are 0..7.
8. Optional - List of source ports that can have both Ethernet and LAG ports.
9. Optional - Direction - Mirror session direction when configured along with source port. (Supported rx/tx/both. default direction is both.)
10. Optional - Policer that will be used to control the rate at which frames are mirrored. (refer [\(202111.2\) Policer](#))

The example below shows how to create an ERSPAN session with name **mrr_legacy**, source address **1.2.3.4**, destination address **20.21.2.23**, DSCP **8**, TTL **100**, and GRE-type **0x88be**.

Example1

```
admin@sonic:~# config mirror_session add mrr_legacy 1.2.3.4 20.21.22.23 8 100 0x88be 0
admin@sonic:~# show mirror_session
```

ERSPAN Sessions									
Name	Status	SRC IP	DST IP	GRE	DSCP	TTL	Queue	Policer	Monitor Port
mrr_legacy	inactive	1.2.3.4	20.21.22.23	0x88be	8	100	UC0		

```

SPAN Sessions
Name      Status   DST Port   SRC Port   Direction   Queue   Policer
-----
admin@sonic:~#
```

The example below shows how to create an ERSPAN session with name **mrr_abcd**, source address **1.2.3.4**, destination address **20.21.22.23**, DSCP **8**, TTL **100**, and GRE-type **0x88be**.

Example2

```
admin@sonic:~# config mirror_session erspan add mrr_abcd 1.2.3.4 20.21.22.23 8 100 0x88be 0
admin@sonic:~# show mirror_session
```

ERSPAN Sessions										
Name	Status	SRC IP	DST IP	GRE	DSCP	TTL	Queue	Policer	Monitor Port	SRC
mrr_abcd	inactive	1.2.3.4	20.21.22.23	0x88be	8	100	UC0			

```

SPAN Sessions
Name      Status   DST Port   SRC Port   Direction   Queue   Policer
-----
admin@sonic:~#
```

The example below shows how to create an ERSPAN session with name **mrr_port**, source address **1.2.3.4**, destination address **20.21.22.23**, DSCP **8**, TTL **100**, GRE-type **0x88be**, and source port **Ethernet0**.

Example3

```
admin@sonic:~# config mirror_session erspan add mrr_port 1.2.3.4 20.21.22.23 8 100 0x88be 0 Ethernet0
admin@sonic:~# show mirror_session
```

ERSPAN Sessions										
Name	Status	SRC IP	DST IP	GRE	DSCP	TTL	Queue	Policer	Monitor Port	SRC
Port	Direction									
mrr_port	inactive	1.2.3.4	20.21.22.23	0x88be	8	100	UC0			
Ethernet0	both									

```

SPAN Sessions
Name      Status   DST Port   SRC Port   Direction   Queue   Policer
-----
admin@sonic:~#
```

The command below is used to create a SPAN mirror session:

Usage

```
config mirror_session span add <session_name> <dst_port> [src_port] [direction] [queue] [--policer
<policer_name>]
```

Parameters:

1. Session name
2. Destination port
3. Optional - List of source ports- List of source ports that can have both Ethernet and LAG ports.
4. Optional - Direction - Mirror session direction when configured along with source port. (Supported rx/tx/both. default direction is both.)
5. Optional - Queue in which packets shall be sent out of the device. The default value is 0. Only support to change the value on Intel devices. Valid values are 0..7.
6. Optional - Policer that will be used to control the rate at which frames are mirrored.

The example below shows how to create a SPAN session with name **mrr_port0**, destination port **Ethernet0** and source ports **Ethernet4**, **PortChannel10**, **Ethernet8**.

Example

```
admin@sonic:~# config mirror_session span add mrr_port0 Ethernet0 Ethernet4,PortChannel10,Ethernet8
admin@sonic:~# show mirror_session
```

ERSPAN Sessions										
Name	Status	SRC IP	DST IP	GRE	DSCP	TTL	Queue	Policer	Monitor Port	SRC Port
Direction										

```

SPAN Sessions
Name      Status   DST Port   SRC Port   Direction   Queue   Policer
-----
mrr_port0 active   Ethernet0   Ethernet4,PortChannel10,Ethernet8 both       UC0
admin@sonic:~#
```

config mirror_session remove

This command is used to remove mirroring sessions.

Usage

```
config mirror_session remove <session-name>
```

Parameter:

1. Session name

The example below shows how to remove a mirror session with session name `mrr_port0`.

Example

```
admin@sonic:~# show mirror_session
ERSPAN Sessions
Name      Status    SRC IP    DST IP    GRE    DSCP    TTL    Queue    Policer    Monitor Port    SRC Port
Direction
-----
-----

SPAN Sessions
Name      Status    DST Port    SRC Port                                Direction    Queue    Policer
-----
mrr_port0 active    Ethernet0    Ethernet4,PortChannel10,Ethernet8    both        UC0
admin@sonic:~# config mirror_session remove mrr_port0
admin@sonic:~# show mirror_session
ERSPAN Sessions
Name      Status    SRC IP    DST IP    GRE    DSCP    TTL    Queue    Policer    Monitor Port    SRC Port
Direction
-----
-----

SPAN Sessions
Name      Status    DST Port    SRC Port                                Direction    Queue    Policer
-----
-----
```

(202111.2) NAT

- NAT Show Commands
 - `show nat config`
 - `show nat statistics`
 - `show nat translations`
- NAT Configuration Commands
 - `config nat add static`
 - `config nat add pool`
 - `config nat add binding`
 - `config nat add interface`
 - `config nat set`
 - `config nat feature`
- NAT Clear Commands
 - `sonic-clear nat translations`
 - `sonic-clear nat statistics`

Network Address Translation (NAT) is a standard method of mapping “internal” (private) network IP addresses to “external” (public) IP address on devices at the edge of a network. Typically, NAT not only translates the inbound and outbound IP addresses of network traffic, but also Layer 4 ports. This allows multiple internal IP addresses to “share” one public external IP address and also provides a level of security for private network hosts that remain “hidden” from the public network.

NAT Show Commands

`show nat config`

This command displays the NAT configuration.

Usage:

```
show nat config [static | pool | bindings | globalvalues | zones]
```

With no optional arguments, the whole NAT configuration is displayed.

Example:

```
admin@sonic:~$ show nat config static
```

Nat Type	IP Protocol	Global IP	Global L4 Port	Local IP	Local L4 Port	Twice-Nat Id
dnat	all	65.55.45.5	---	10.0.0.1	---	---
dnat	all	65.55.45.6	---	10.0.0.2	---	---
dnat	tcp	65.55.45.7	2000	20.0.0.1	4500	1
snat	tcp	20.0.0.2	4000	65.55.45.8	1030	1

```
admin@sonic:~$ show nat config pool
```

Pool Name	Global IP Range	Global L4 Port Range
Pool1	65.55.45.5	1024-65535
Pool2	65.55.45.6-65.55.45.8	---
Pool3	65.55.45.10-65.55.45.15	500-1000

```
admin@sonic:~$ show nat config bindings
```

Binding Name	Pool Name	Access-List	Nat Type	Twice-Nat Id
Bind1	Pool1	---	snat	---
Bind2	Pool2	1	snat	1
Bind3	Pool3	2	snat	--

```
admin@sonic:~$ show nat config globalvalues
```

```
Admin Mode      : enabled
Global Timeout  : 600 secs
TCP Timeout     : 86400 secs
UDP Timeout     : 300 secs
```

```
admin@sonic:~$ show nat config zones
```

Port	Zone
Ethernet2	0
Vlan100	1

show nat statistics

This command displays the NAT translation statistics for each entry.

Usage:

```
show nat statistics
```


Example:

```
admin@sonic:~$ show nat statistics
```

Protocol	Source	Destination	Packets	Bytes
-----	-----	-----	-----	-----
all	10.0.0.1	---	802	1009280
all	10.0.0.2	---	23	5590
tcp	20.0.0.1:4500	---	110	12460
udp	20.0.0.1:4000	---	1156	789028
tcp	20.0.0.1:6000	---	30	34800
tcp	20.0.0.1:5000	65.55.42.1:2000	128	110204
tcp	20.0.0.1:5500	65.55.42.1:2000	8	3806

show nat translations

This command displays the NAT translation entries.

Usage:

```
show nat translations [count]
```

Giving the optional count argument displays only the details about the number of translation entries.

Example:

```
admin@sonic:~$ show nat translations
```

```
Static NAT Entries      ..... 4
Static NAPT Entries     ..... 2
Dynamic NAT Entries     ..... 0
Dynamic NAPT Entries    ..... 4
Static Twice NAT Entries ..... 0
Static Twice NAPT Entries ..... 4
Dynamic Twice NAT Entries ..... 0
Dynamic Twice NAPT Entries ..... 0
Total SNAT/SNAPT Entries ..... 9
Total DNAT/DNAPT Entries ..... 9
Total Entries           ..... 14
```

Protocol	Source	Destination	Translated Source	Translated Destination
all	10.0.0.1	---	65.55.42.2	---
all	---	65.55.42.2	---	10.0.0.1
all	10.0.0.2	---	65.55.42.3	---
all	---	65.55.42.3	---	10.0.0.2
tcp	20.0.0.1:4500	---	65.55.42.1:2000	---
tcp	---	65.55.42.1:2000	---	20.0.0.1:4500
udp	20.0.0.1:4000	---	65.55.42.1:1030	---
udp	---	65.55.42.1:1030	---	20.0.0.1:4000
tcp	20.0.0.1:6000	---	65.55.42.1:1024	---
tcp	---	65.55.42.1:1024	---	20.0.0.1:6000
tcp	20.0.0.1:5000	65.55.42.1:2000	65.55.42.1:1025	20.0.0.1:4500
tcp	20.0.0.1:4500	65.55.42.1:1025	65.55.42.1:2000	20.0.0.1:5000
tcp	20.0.0.1:5500	65.55.42.1:2000	65.55.42.1:1026	20.0.0.1:4500
tcp	20.0.0.1:4500	65.55.42.1:1026	65.55.42.1:2000	20.0.0.1:5500

```
admin@sonic:~$ show nat translations count
```

```
Static NAT Entries      ..... 4
Static NAPT Entries     ..... 2
Dynamic NAT Entries     ..... 0
Dynamic NAPT Entries    ..... 4
Static Twice NAT Entries ..... 0
Static Twice NAPT Entries ..... 4
Dynamic Twice NAT Entries ..... 0
Dynamic Twice NAPT Entries ..... 0
Total SNAT/SNAPT Entries ..... 9
Total DNAT/DNAPT Entries ..... 9
Total Entries           ..... 14
```

NAT Configuration Commands

config nat add static

This command adds a static NAT or NAPT entry. When configuring a static NAT entry, you have to specify the following fields with these "basic" keywords.

1. Global IP address,
2. Local IP address,
3. NAT type (snat / dnat) to be applied on the Global IP address. Default value is dnat. This is an optional argument.
4. Twice NAT Id. This is an optional argument used in the case of twice NAT configuration.

When configuring a static NAPT entry, you have to specify the following fields.

1. IP protocol type (tcp / udp)
2. Global IP address + Port
3. Local IP address + Port

4. NAT type (snat / dnat) to be applied on the Global IP address + Port. Default value is dnat. This is an optional argument.
5. Twicw NAT Id. This is an optional argument used in the case of twice NAT configuration.

Usage:

```
config nat add static (basic <global-ip> <local-ip> | (tcp | udp) <global-ip> <global-port> <local-ip>
<local-port>) [-nat_type (snat | dnat)] [-twice_nat_id <value>]
```

To delete a static NAT or NAPT entry, use the command below. Giving the **all** argument deletes all the configured static NAT and NAPT entries.

Usage:

```
config nat remove static (basic <global-ip> <local-ip> | (tcp | udp) <global-ip> <global-port> <local-ip>
<local-port> | all)
```

Example:

```
admin@sonic:~$ sudo config nat add static basic 65.55.45.1 12.12.12.14 -nat_type dnat
admin@sonic:~$ sudo config nat add static tcp 65.55.45.2 100 12.12.12.15 200 -nat_type dnat
```

```
admin@sonic:~$ show nat translations
```

```
Static NAT Entries           ..... 2
Static NAPT Entries          ..... 2
Dynamic NAT Entries           ..... 0
Dynamic NAPT Entries          ..... 0
Static Twice NAT Entries      ..... 0
Static Twice NAPT Entries     ..... 0
Dynamic Twice NAT Entries     ..... 0
Dynamic Twice NAPT Entries    ..... 0
Total SNAT/SNAPT Entries      ..... 2
Total DNAT/DNAPT Entries      ..... 2
Total Entries                 ..... 4
```

Protocol	Source	Destination	Translated Source	Translated Destination
all	12.12.12.14	---	65.55.42.1	---
all	---	65.55.42.1	---	12.12.12.14
tcp	12.12.12.15:200	---	65.55.42.2:100	---
tcp	---	65.55.42.2:100	---	12.12.12.15:200

config nat add pool

This command creates a NAT pool used for dynamic source NAT or NAPT translations. The pool can be configured in one of the following combinations.

1. Global IP address range (or)
2. Global IP address + L4 port range (or)
3. Global IP address range + L4 port range.

Usage:

```
config nat add pool <pool-name> <global-ip-range> <global-port-range>
```

To delete a NAT pool, use the following command. A pool cannot be removed if it is referenced by a NAT binding. Using the **pools** argument removes all the configured pools.

Usage:

```
config nat remove (pool <pool-name> | pools)
```

Example:

```
admin@sonic:~$ sudo config nat add pool pool1 65.55.45.2-65.55.45.10
admin@sonic:~$ sudo config nat add pool pool2 65.55.45.3 100-1024
```

```
admin@sonic:~$ show nat config pool
```

Pool Name	Global IP Range	Global Port Range
pool1	65.55.45.2-65.55.45.10	---
pool2	65.55.45.3	100-1024

config nat add binding

This command creates a NAT binding between a pool and an ACL. The following fields are needed for configuring the binding.

1. ACL is an optional argument. If an ACL argument is not given, the NAT binding is applicable to match all traffic.
2. NAT type is an optional argument. Only DNAT type is supported for binding.
3. Twice NAT Id is an optional argument. This ID is used to form a twice NAT grouping with the static NAT/NAPT entry configured with the same ID.

Usage:

```
config nat add binding <binding-name> [<pool-name>] [<acl-name>] [-nat_type (snat | dnat)] [-twice_nat_id <value>]
```

To delete a NAT pool, use the following command. A pool cannot be removed if it is referenced by a NAT binding. Using the **pools** argument removes all the configured pools.

Usage:

```
config nat remove (binding <binding-name> | bindings)
```

Example:

```
admin@sonic:~$ sudo config nat add binding bind1 pool1 acl1
admin@sonic:~$ sudo config nat add binding bind2 pool2
```

```
admin@sonic:~$ show nat config bindings
```

Binding Name	Pool Name	Access-List	Nat Type	Twice-NAT Id
bind1	pool1	acl1	snat	---
bind2	pool2		snat	---

config nat add interface

This command is used to configure a NAT zone on an L3 interface. The default value of a NAT zone on an L3 interface is 0. The valid range of zone values is 0-3 (On Intel devices, the valid values are: 0, 1).

Usage:

```
config nat add interface <interface-name> -nat_zone <value>
```

To reset the NAT zone on an interface, use the command below. Using the **interfaces** argument resets the NAT zone on all L3 interfaces to 0.

Usage:

```
config nat remove (interface <interface-name> | interfaces)
```

Example:

```
admin@sonic:~$ sudo config nat add interface Ethernet28 -nat_zone 1
```

```
admin@sonic:~$ show nat config zones
```

Port	Zone
-----	-----
Ethernet0	0
Ethernet28	1
Ethernet22	0
Vlan2091	0

config nat set

This command is used to set NAT timeout values. Different timeout values can be configured for the NAT entry timeout, NAPT TCP entry timeout, and NAPT UDP entry timeout. The range for the Global NAT entry timeout is 300 sec to 432000 sec, the default value is 600 sec. The range for the TCP NAT/NAPT entry timeout is 300 sec to 432000 sec, the default value is 86400 sec. The range for the UDP NAT/NAPT entry timeout is 120 sec to 600 sec, the default value is 300 sec.

Usage:

```
config nat set (tcp-timeout <value> | timeout <value> | udp-timeout <value>)
```

To reset the timeout values to the default values, use the following command.

Usage:

```
config nat reset (tcp-timeout | timeout | udp-timeout)
```

Example:

```
admin@sonic:~$ sudo config nat add set tcp-timeout 3600

admin@sonic:~$ show nat config globalvalues

Admin Mode      : enabled
Global Timeout  : 600 secs
TCP Timeout     : 600 secs
UDP Timeout     : 300 secs
```

config nat feature

This command is used to enable or disable the NAT feature.

Usage:

```
config nat feature (enable | disable)
```

Example:

```
admin@sonic:~$ sudo config nat feature enable
admin@sonic:~$ sudo config nat feature disable
```

NAT Clear Commands

sonic-clear nat translations

This command is used to clear the dynamic NAT and NAPT translation entries.

Usage:

```
sonic-clear nat translations
```

sonic-clear nat statistics

This command is used to clear the statistics of all NAT and NAPT entries.

Usage:

```
sonic-clear nat statistics
```

(202111.2) NDP

- [NDP Show Commands](#)
 - [show ndp](#)

NDP Show Commands

show ndp

This command displays the IPv6 neighbors.

Usage

```
show ndp [-if|--iface <interface_name>] <ipv6_address>
```

Example (all IPv6 neighbors)

```
admin@sonic:~$ show ndp
Address                MacAddress            Iface    Vlan    Status
-----
fe80::20c:29ff:feb8:b11e 00:0c:29:b8:b1:1e eth0     -      REACHABLE
fe80::20c:29ff:feb8:cff0 00:0c:29:b8:cf:f0 eth0     -      REACHABLE
fe80::20c:29ff:feb9:324  00:0c:29:f9:03:24 eth0     -      REACHABLE
Total number of entries 3
```

Example (a specific IPv6 neighbor)

```
admin@sonic:~$ show ndp fe80::20c:29ff:feb8:b11e
Address                MacAddress            Iface    Vlan    Status
-----
fe80::20c:29ff:feb8:b11e 00:0c:29:b8:b1:1e eth0     -      REACHABLE
Total number of entries 1
```

Example (a specific interface)

```
admin@sonic:~$ show ndp -if eth0
Address                MacAddress            Iface    Vlan    Status
-----
fe80::20c:29ff:feb8:b11e 00:0c:29:b8:b1:1e eth0     -      REACHABLE
fe80::20c:29ff:feb8:cff0 00:0c:29:b8:cf:f0 eth0     -      REACHABLE
fe80::20c:29ff:feb9:324  00:0c:29:f9:03:24 eth0     -      REACHABLE
Total number of entries 3
```

(202111.2) NTP

- [NTP Show Commands](#)
 - [show ntp](#)
- [NTP Config Commands](#)
 - [config ntp add](#)
 - [config ntp delete](#)

NTP Show Commands

show ntp

This command displays a list of NTP peers known to the server as well as a summary of their state.

Usage

```
show ntp
```

Example

```
admin@sonic:~$ show ntp
synchronised to NTP server (204.2.134.164) at stratum 3
time correct to within 326797 ms
polling server every 1024 s
```

remote	refid	st	t	when	poll	reach	delay	offset	jitter
23.92.29.245	.XFAC.	16	u	-	1024	0	0.000	0.000	0.000
*204.2.134.164	46.233.231.73	2	u	916	1024	377	3.079	0.394	0.128

NTP Config Commands

This sub-section of commands is used to add or remove the configured NTP servers.

config ntp add

This command is used to add an NTP server IP address to the NTP server list. Note that more than one NTP server IP address can be added in the device.

Usage

```
config ntp add <ip_address>
```

The following example shows how to add a NTP server.

Example

```
admin@sonic:~$ sudo config ntp add 9.9.9.9
NTP server 9.9.9.9 added to configuration
Restarting ntp-config service...
admin@sonic:~$ sudo config ntp add 2000::2
NTP server 2000::2 added to configuration
Restarting ntp-config service...
admin@sonic:~$
```

Notice: If your NTP connect to NTP server through network interface eth0 which IP is get from DHCP, please indicated NTP listen interface as eth0

Please add following section into /etc/sonic/config_db.json

Example

```
{
  ...
  "NTP": {
    "global": {
      "src_intf": "eth0"
    }
  }
  ...
}
```

config ntp delete

This command is used to delete a configured NTP server IP address.

Usage

```
config ntp del <ip_address>
```

The following example shows how to remove an NTP server.

Example

```
admin@sonic:~$ sudo config ntp del 9.9.9.9
NTP server 9.9.9.9 removed from configuration
Restarting ntp-config service...
```

(202111.2) PFC

- PFC Show commands
 - show pfc asymmetric
 - show pfc priority
 - show pfc counters
 - show qos pfc-priority-queue
 - show interface pfc config
- PFC Configuration Commands
 - config interface pfc asymmetric
 - config interface pfc priority
 - config qos pfc-priority-queue add
 - config qos pfc-priority-queue update
 - config qos pfc-priority-queue del
 - config interface qos pfc-priority-queue bind
 - config interface qos pfc-priority-queue unbind
- PFC Clear Commands
 - sonic-clear pfccounters

PFC Show commands

show pfc asymmetric

This command displays the status of asymmetric PFC for all interfaces or a given interface.

Usage

```
show pfc asymmetric [<interface>]
```

Example: Show on factory default

```
admin@sonic:~$ show pfc asymmetric
```

Interface	Asymmetric
Ethernet0	N/A
Ethernet4	N/A

Note: The factory default of asymmetric PFC is disabled and show as N/A.

Example: Show after configured

```
admin@sonic:~$ show pfc asymmetric
```

Interface	Asymmetric
Ethernet0	on
Ethernet2	off
Ethernet4	off
Ethernet6	off
Ethernet8	off
Ethernet10	off
Ethernet12	off
Ethernet14	off

Example: Show on specify interface

```
admin@sonic:~$ show pfc asymmetric Ethernet0
```

Interface	Asymmetric
Ethernet0	off

show pfc priority

This command displays the lossless priorities for all interfaces or a given interface.

Usage

```
show pfc priority [<interface>]
```

Example

```
admin@sonic:~$ show pfc priority
```

Interface	Lossless priorities
Ethernet0	3,4
Ethernet2	3,4
Ethernet8	3,4
Ethernet10	
Ethernet16	3,4

Example

```
admin@sonic:~$ show pfc priority Ethernet0
```

Interface	Lossless priorities
Ethernet0	3,4

show pfc counters

This command displays the details of Rx and Tx priority-flow-control (PFC) for all ports.

Usage

```
show pfc counters
```

Example

```
admin@sonic:~$ show pfc counters
```

Port Rx	PFC0	PFC1	PFC2	PFC3	PFC4	PFC5	PFC6	PFC7
Ethernet0	0	0	0	0	0	0	0	0
Ethernet4	0	0	0	0	0	0	0	0
Ethernet8	0	0	0	0	0	0	0	0
Ethernet12	0	0	0	0	0	0	0	0

Port Tx	PFC0	PFC1	PFC2	PFC3	PFC4	PFC5	PFC6	PFC7
Ethernet0	0	0	0	0	0	0	0	0
Ethernet4	0	0	0	0	0	0	0	0
Ethernet8	0	0	0	0	0	0	0	0
Ethernet12	0	0	0	0	0	0	0	0

show qos pfc-priority-queue

This command displays the PFC-priority-to-queue mapping.

Usage

```
show qos pfc-priority-queue [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show qos pfc-priority-queue
```

```
pfc-queue policy: pfc_queue_profile
```

PFC Priority	Queue
0	0
1	2
2	2

show interface pfc config

This command displays the PFC configuration for all interfaces or a given interface.

Usage

```
show interface pfc config [<interface_name>]
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
admin@sonic:~$ show interfaces pfc config
Ethernet88
  PFC mode: Symmetric
  PFC Lossless Priorities: 0,1

  PFC to Queue: pfc_q_prof
    PFC      Queue
  ----
    0        2
    1        2
```

PFC Configuration Commands

config interface pfc asymmetric

This command is used to set PFC asymmetric of an interface to "on"(enable) or "off"(disable).

Usage

```
config interface pfc asymmetric <interface_name> (on | off)
```

To disable PFC asymmetric on Ethernet0, use the following command:

Example

```
admin@sonic:~$ sudo config interface pfc asymmetric Ethernet0 on
```

config interface pfc priority

This command is used to enable ("on") or disable ("off") a PFC priority on an interface.

Usage

```
config interface pfc priority <interface_name> {0 | 1 | 2 | 3 | 4 | 5 | 6 | 7} {on | off}
```

To enable PFC priority 7 on Ethernet0, use the following command.

Example

```
admin@sonic:~$ sudo config interface pfc priority Ethernet0 7 on
```

config qos pfc-priority-queue add

This command is used to create a profile of PFC-priority-to-queue mapping.

Usage

```
config qos pfc-priority-queue add <profile_name> --pfc-priority <pfc-priority> --queue <queue>
```

Parameters:

- <profile_name>: Profile name.
- <pfc-priority>: List of PFC priority. Minimum: 0. Maximum: 7. For example: "1", "1,2-3".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).

Note:

- This command is only supported on the devices using Broadcom ASIC.

Example

```
sudo config qos pfc-priority-queue add pfc_queue_profile --pfc-priority 0 --queue 0
```

config qos pfc-priority-queue update

This command is used to modify an existing profile of PFC-priority-to-queue mapping.

Usage

```
config qos pfc-priority-queue update <profile_name> --pfc-priority <pfc-priority> [--queue <queue>] [--remove]
```

Parameters:

- <profile_name>: Profile name.
- <pfc-priority>: List of PFC priority. Minimum: 0. Maximum: 7. For example: "1", "1,2-3".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).
- remove: Remove the mapping. For example: "--pfc-priority 1 --remove".

Note:

- This command is only supported on the devices using Broadcom ASIC.

Example

```
sudo config qos pfc-priority-queue update pfc_queue_profile --pfc-priority 1,2 --queue 2
```

config qos pfc-priority-queue del

This command is used to delete a profile of PFC-priority-to-queue mapping.

Usage

```
config qos pfc-priority-queue del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Note:

- This command is only supported on the devices using Broadcom ASIC.

Example

```
sudo config qos pfc-priority-queue del pfc_queue_profile
```

config interface qos pfc-priority-queue bind

This command is used to configure the PFC-priority-to-queue mapping on the specified interface.

Usage

```
config interface qos pfc-priority-queue bind <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Default mapping:

PFC Priority	Queue
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7

Any packet field value that is not specified in the configuration is assigned to a default value.

Note:

- This command is only supported on the devices using Broadcom ASIC.

Example

```
sudo config interface qos pfc-priority-queue bind Ethernet0 pfc_queue_profile
```

config interface qos pfc-priority-queue unbind

This command is used to reset the PFC-priority-to-queue mapping on the specified interface.

Usage

```
config interface qos pfc-priority-queue unbind <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Note:

- This command is only supported on the devices using Broadcom ASIC.

Example

```
sudo config interface qos pfc-priority-queue unbind Ethernet0
```

PFC Clear Commands

sonic-clear pfccounters

This command is used to clear the counters of Rx and Tx priority-flow-control for all ports.

Usage

```
sonic-clear pfccounters
```

Example

```
admin@sonic:~$ sonic-clear pfccounters  
Clear saved counters
```


(202111.2) PFC Watchdog

- [PFC Watchdog Command](#)
 - `config pfcwd start`
 - `config pfcwd stop`
 - `config pfcwd interval`
 - `config pfcwd counter_poll`
 - `config pfcwd big_red_switch`
 - (Not supported) `config pfcwd start_default`
 - `show pfcwd config`

PFC Watchdog Command

A detailed description of the PFC Watchdog can be found on [this wiki page](#).

`config pfcwd start`

This command starts the PFC Watchdog.

Usage

```
config pfcwd start [--action (drop | forward | alert)] [--restoration-time <ms_value>] ports (all | <PORTS>)
detection-time <ms_value>
```

all : All ethernet ports

<PORTS> : <ethernet_interface_name> separate by space ' ', e.g Ethernet0 Ethernet1 Ethernet2

Example: Set pfcwd on all ethernet port

```
admin@sonic:~$ config pfcwd start --action drop --restoration-time 400 ports all detection-time 400
```

Example: Set pfcwd on Ethernet0

```
admin@sonic:~$ config pfcwd start --action drop --restoration-time 400 ports Ethernet0 detection-time 400
```

Example: Set pfcwd on Ethernet0 and Ethernet4

```
admin@sonic:~$ config pfcwd start --action drop --restoration-time 400 ports Ethernet0 Ethernet4 detection-time 400
```

`config pfcwd stop`

This command stops the PFC Watchdog.

Usage

```
config pfcwd stop
```

Example: Stop pfcwd on all ports

```
admin@sonic:~$ config pfcwd stop
```

config pfcwd interval

This command sets the PFC Watchdog counter polling interval (ms).

Usage

```
config pfcwd interval <interval_in_ms>
```

Example

```
admin@sonic:~$ config pfcwd interval 200
```

config pfcwd counter_poll

This command enables or disables the PFC Watchdog related counters polling.

Usage

```
config pfcwd counter_poll (enable | disable)
```

Example

```
admin@sonic:~$ config pfcwd counter_poll disable
```

config pfcwd big_red_switch

This command enables or disables the PFC Watchdog's "BIG RED SWITCH"(BRS). When enabled, the BRS PFC Watchdog will be disabled on all ports/queues it is configured for no matter whether a storm was detected or not.

Usage

```
config pfcwd big_red_switch (enable | disable)
```

Example

```
admin@sonic:~$ config pfcwd big_red_switch enable
```

(Not supported) config pfcwd start_default

This command starts the PFC Watchdog with the default settings.

Usage

```
config pfcwd start_default
```

Default values are the following:

```
detection time - 200ms
restoration time - 200ms
polling interval - 200ms
action - 'drop'
```

Additionally, if the number of ports in the system exceeds 32, all times will be multiplied by roughly $\text{num_ports} / 32 + 1$.

Example

```
admin@sonic:~$ config pfcwd start_default
```

Note: Due to the incomplete implementation of `config pfcwd start_default`, this command malfunctions.

show pfcwd config

This command shows the current PFC Watchdog configuration.

Usage

```
show pfcwd config
```

Example

```
admin@sonic:~$ show pfcwd config
  PORT      ACTION      DETECTION TIME      RESTORATION TIME
  -----
Ethernet0    drop              400                400
```

show pfcwd stats

This command shows the current PFC Watchdog statistics (storms detected, packets dropped, etc).

Usage

show pfcwd stats

Example

```
admin@sonic:~$ show pfcwd stats
      QUEUE      STATUS  STORM DETECTED/RESTORED  TX OK/DROP  RX OK/DROP  TX LAST OK/DROP  RX
LAST OK/DROP
-----
Ethernet0:3    stormed           1/0      100/300      100/300           0
/200           0/200
Ethernet0:4  operational           2/2      100/100      100/100           0
/0             0/0
```

(202111.2) Policer

- [Policer Show Commands](#)
 - [show policer](#)
- [Policer Config Commands](#)
 - [config policer add](#)
 - [config policer update](#)
 - [config policer remove](#)

This section explains various show and configuration commands available for users.

Policer Show Commands

show policer

This command displays a summary of the policer configuration.

Usage

```
show policer
```

Example

```
admin@sonic:~$ show policer
Name      Type      Mode      Color Source      CIR      CBS      PIR      PBS      Green Packet Action      Yellow
Packet Action      Red Packet Action
-----
POLICER_TR_TCM bytes    tr_tcm    blind          1500    3000    2000    4000    forward
forward                                drop
```

Policer Config Commands

config policer add

This command is used to add a policer.

Usage

```
config policer add <policer_name> sr_tcm <meter_type> <color_source> [--cir <cir>] [--cbs <cbs>] [--pbs <pbs>]
[--green_packet_action <green_packet_action>] [--yellow_packet_action <yellow_packet_action>] [--red_packet_action <red_packet_action>]
```

```
config policer add <policer_name> tr_tcm <meter_type> <color_source> [--cir <cir>] [--cbs <cbs>] [--pir <pir>]
[--pbs <pbs>] [--green_packet_action <green_packet_action>] [--yellow_packet_action <yellow_packet_action>] [--red_packet_action <red_packet_action>]
```

```
config policer add <policer_name> storm_control <meter_type> <color_source> [--cir <cir>] [--cbs <cbs>] [--green_packet_action <green_packet_action>]
[--red_packet_action <red_packet_action>]
```

Parameters:

- `policer_name`: Policer name.
- `meter_type`: The unit of metering. Valid options are the following:
 - `bytes`
 - `packets`, packets mode is only supported on Broadcom switches.
- `color_source`: Be aware of or ignore the [pre-coloring](#) of packets.
 - `aware`
 - `blind`

- **cir**: Bandwidth limit for guaranteed traffics. It ranges from:
 - 1000 to 12500000000 in bytes mode,
 - 1 to 148809523 in packets mode.
- **cbs**: Maximum packet size permitted for bursts of data that exceed the cir. It ranges from:
 - 2000 to 256000000 in bytes mode,
 - 1 to 536576 in packets mode.
- **pir**: Bandwidth limit for peak traffics. It ranges from:
 - 1000 to 12500000000 in bytes mode,
 - 1 to 148809523 in packets mode.
- **pbs**: Maximum packet size permitted for bursts of data that exceed the pir. It ranges from:
 - 2000 to 256000000 in bytes mode,
 - 1 to 536576 in packets mode.
- **green_packet_action**: Specifies the action for packets marked green by the meter.
- **yellow_packet_action**: Specifies the action for packets marked yellow by the meter.
- **red_packet_action**: Specifies the action for packets marked red by the meter.

The following example shows how to create a policer named POLICER_TR_TCM with tr_tcm mode, meter type "bytes", color source "blind", cir = 1000, cbs = 2000, pir = 2000, pbs = 4000, which drops packets marked red by the meter.

Example

```
admin@sonic:~$ sudo config policer add POLICER_TR_TCM tr_tcm bytes blind --cir 1000 --cbs 2000 --pir 2000 --pbs 4000 --green_packet_action forward --yellow_packet_action forward --red_packet_action drop
```

config policer update

This command is used to update an existing policer.

Usage

```
config policer update <policer_name> [--cir <cir>] [--cbs <cbs>] [--pir <pir>] [--pbs <pbs>]
```

Parameters:

- **cir**: Bandwidth limit for guaranteed traffics. It ranges from:
 - 1000 to 12500000000 in bytes mode,
 - 1 to 148809523 in packets mode.
- **cbs**: Maximum packet size permitted for bursts of data that exceed the cir. It ranges from:
 - 2000 to 256000000 in bytes mode,
 - 1 to 536576 in packets mode.
- **pir**: Bandwidth limit for peak traffics. It ranges from:
 - 1000 to 12500000000 in bytes mode,
 - 1 to 148809523 in packets mode.
- **pbs**: Maximum packet size permitted for bursts of data that exceed the pir. It ranges from:
 - 2000 to 256000000 in bytes mode,
 - 1 to 536576 in packets mode.

The following example updates a policer named POLICER_TR_TCM with cir = 1500, cbs = 3000.

Example

```
admin@sonic:~$ sudo config policer update POLICER_TR_TCM --cir 1500 --cbs 3000
```

config policer remove

This command is used to remove a policer.

Usage

```
config policer remove <policer_name>
```

The following example removes a policer named POLICER_TR_TCM.

Example

```
admin@sonic:~$ sudo config policer remove POLICER_TR_TCM
```

(202111.2) Port Channels

- [Port Channel Show Commands](#)
 - [show interfaces portchannel](#)
- [Port Channel Config Commands](#)
 - [config portchannel](#)
 - [config portchannel member](#)

This section includes port channel creation, adding (or removing) members, and deletion.

In creating a port channel, user may specify the port channel is static (manually configured) or dynamic (running LACP protocol) which should be consistent with peer switch.

Port Channel Show Commands

show interfaces portchannel

This command displays all the port channels that are configured in the device and its current status.

Usage

```
show interfaces portchannel
```

Example

```
admin@sonic:~$ show interfaces portchannel
Flags: A - active, I - inactive, Up - up, Dw - Down, N/A - not available,
       S - selected, D - deselected, * - not synced
```

No.	Team Dev	Protocol	Ports	Oper Key	Admin Key	Fast Rate
0	PortChannel0	LACP(A)(Up)	Ethernet0(S) Ethernet4(S)	1	1	false
2	PortChannel2	NONE(-)(Up)	Ethernet2(S) Ethernet1(S)	N/A	N/A	N/A
5	PortChannel5	LACP(A)(Dw)	N/A	N/A	auto	false
8	PortChannel8	LACP(A)(Up)	Ethernet8(S) Ethernet12(S)	8	8	true
24	PortChannel24	LACP(A)(Up)	Ethernet28(S) Ethernet24(S)	124	auto	false
40	PortChannel40	LACP(A)(Up)	Ethernet44(S) Ethernet40(S)	40	40	false
48	PortChannel48	LACP(A)(Up)	Ethernet52(S) Ethernet48(S)	48		

Parameter

- **Protocol**
 - LACP protocol mode, active (A) or inactive(I) on the port channel.
 - LACP(A) means LACPDU frames are sent along the configured links periodically.
 - LACP(I) means the port channel is not in an active negotiating state. It acts as "speak when spoken to", does respond to incoming LACP packets.
 - The 'NONE' keyword means static port channel.
 - The port channel state, (Up) or (Dw).
- **Ports**
 - S - selected, the port is already attached to the port channel and ready to transmit and receive.
 - D - deselected, the port is not attached to the port channel yet.
 - * - not synced, the port is not synchronized. It is currently not in the right aggregation.
- **Oper Key and Admin Key**
 - The Admin Key is specified value in "--lACP-key" option of **config portchannel add** command.
 - The Oper Key is value of actually participating in the LACP protocol.
 - The Admin Key and Oper Key are always N/A for static port channel.
- **Fast Rate**
 - It is used to set the rate at which the LACP control packets are sent from partner.
 - true means LACP fast rate mode, request partner to transmit LACPDUs every 1 second.
 - false means LACP slow rate mode, request partner to transmit LACPDUs every 30 seconds.
 - N/A, it is meaningless for static port channel.

Note:

If user created a LACP port channel and set lacp-key as "auto", the "Oper Key" value will be generated automatically after first port channel member is added to port channel.

Port Channel Config Commands

This sub-section explains how to configure a port channel and its member ports.

config portchannel

This command is used to add or delete a port channel. User can specify some attributes of the port channel when adding the port channel name.

Usage

```
config portchannel add <portchannel_name> [--min-links <num_min_links>] [--fallback {true | false}] [--lacp-key {auto | <num_lacp_key>}] [--static {true | false}] [--fast-rate {true | false}]
config portchannel del <portchannel_name>
```

Parameter:

- **add**, add a new port-channel name in the system.
 - default attributes: LACP if do not specify any option or do not specify "-static" option.
- **del**, delete the port-channel name from the system. User doesn't need to specify any options.
- **<portchannel_name>**, the port channel name to be added or deleted.
 - The syntax of port channel name is "PortChannel"+<suffix> where <suffix> range is 0..9999.
 - For example, <suffix> uses "200", the name could be "PortChannel200".
- **--min-links <num_min_links>**, minimum number of links in the LACP port channel before the LACP port channel can be brought up and active.
 - The default value is 1.
 - The range from 1 to 1024.
- **--fallback option**, LACP fallback feature can be enabled/disabled.
 - **true**, enable LACP fallback, which keep the port channel stays in "up" before receiving any LACP PDUs from peer DUT.
 - **false**, disable LACP fallback, turn the port channel state to down when it does not receive the LACP PDUs from peer DUT.
 - The default value is **false**.
- **--lacp-key**, specify the exchanged lacp-key in LACP protocol, either **auto** or **<num_lacp_key>**.
 - **auto**, it concatenates "1"+<suffix> as the key. For example, <suffix> uses "200", the lacp_key would be "1200".
 - **<num_lacp_key>**, the range of given number from 1 to 65535.
 - The default value is **auto**.
- **--static**, specify the port channel type, either **true** for static port channel or **false** for dynamic LACP port channel.
- **--fast-rate**, specify LACP rate mode.
 - **true**, LACP fast rate mode, request partner to transmit LACPDUs every 1 second. LACP timeout is 3 seconds.
 - **false**, LACP slow rate mode, request partner to transmit LACPDUs every 30 seconds. LACP timeout is 90 seconds.
 - the default value is **false**.

Note:

1. The member of a portchannel cannot be added into another portchannel.
 - If any port is already a member of a port channel and a user tries to add the same port to another port channel (without first deleting it from the current port channel), the command fails and error message is printed. In such cases, remove the member port from the current port channel and then add it to new port channel.
2. Portchannel cannot be deleted if there is any configured member.
 - A port channel can be deleted only if it does not have any members or the members are already deleted. When a user tries to delete a port channel and the port channel still has one or more members that exist, the deletion of port channel is blocked.
3. Refer to [https://github.com/Azure/SONiC/blob/master/doc/lag/LACP Fallback Feature for SONiC_v0.5.md](https://github.com/Azure/SONiC/blob/master/doc/lag/LACP%20Fallback%20Feature%20for%20SONiC_v0.5.md) for more details about the fallback feature.
4. Only **--static** option is required when adding static port channel. Extra option, such as **lacp_key**, **min_link**, **fast_rate** or **fallback**, would cause an error.
5. The attributes of created port channel cannot be modified or changed.
 - If port channel is already created and user tries to specify options to change port channel attributes, the command fails and error message is printed.

Example of creating a LACP port channel: create a port-channel with name: "PortChannel200".

Example

```
admin@sonic:~$ sudo config portchannel add PortChannel200
```

Example of creating a static port channel: create a port-channel with name: "PortChannel300" and specify "--static true".

Example

```
admin@sonic:~$ sudo config portchannel add PortChannel300 --static true
```

Example of deleting a port channel: delete a port-channel with name: "PortChannel200"

Example

```
admin@sonic:~$ sudo config portchannel del PortChannel200
```

config portchannel member

This command adds or deletes a member port to/from the existed port channel.

Usage

```
config portchannel member {add | del} <portchannel_name> <member_portname>
```

Example: add interface Ethernet4 as a member of port channel "PortChannel200"

Example

```
admin@sonic:~$ sudo config portchannel member add PortChannel200 Ethernet4
```

(202111.2) Port Rate Limit

Port Rate Limit

This section explains the commands of limiting the rate of transmission. It could be counted in bytes or packets.

Port Rate Limit Show Commands

show interface rate-limit

To display the rate limit configuration on an interface, use **show interface rate-limit** command.

Usage

```
show interface rate-limit [<interface_name>]
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
admin@sonic:~$ show interface rate-limit
```

Interface	Meter Type	Rate	Burst Size
Ethernet0	bytes	100 kbps	1 Kib

Port Rate Limit Configuration Commands

config interface rate-limit add

To enable to limit the incoming traffic rate on a port, use **config interface rate-limit add** command.

Usage

```
config interface rate-limit add <interface_name> --meter-type {bytes|packets} --rate <rate> [--burst-size <burst_size>]
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- meter-type: Meter type. Valid values are:
 - **bytes**: Meter type as bytes mode.
 - **packets**: Meter type as packets mode.
- <rate>: Maximum rate of ingress traffic in kilo-bits/second (kbps) or packet/second (pps). Valid values are:
 - If meter-type is **bytes**:
 - The format is a decimal number or a decimal number followed by the abbreviation k (1,000), m (1,000,000), or g (1,000,000,000). Minimum: 8 kbps (32 kbps for AS8000 and AS9716-32D). Maximum: 400,000,000 kbps. For example: "10", "10k".
 - If meter-type is **packets**:
 - Minimum: 1 pps. Maximum: 148,809,523 pps.
- <burst_size>: The burst size in bytes or packets. If burst size is not given, the default value of bytes mode and packet mode are $\text{<rate>} \times 10$ and $\text{<rate>} \div 8 \times 10$ milliseconds respectively. Valid values are:
 - If meter-type is **bytes**:
 - The format is a decimal number of a decimal number followed by the abbreviation k (2^{10}), or m (2^{20}). Minimum: 2,000 bytes. Maximum: 256,000,000 bytes. For example: "100", "100k".
 - If meter-type is **packets**:
 - Minimum: 1 pps. Maximum: 536,576 pps.

Example

```
sudo config interface rate-limit add Ethernet0 --meter-type bytes --rate 10g --burst-size 9100
```

config interface rate-limit del

Disable to limit the incoming traffic rate on a port, use **config interface rate-limit del** command.

Usage

```
config interface rate-limit del <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
sudo config interface rate-limit del Ethernet0
```

(202111.2) PVST (STP)

- PVST show commands
 - show spanning-tree
- PVST configuration commands
 - config spanning-tree
 - config spanning-tree forward_delay
 - config spanning-tree hello
 - config spanning-tree max_age
 - config spanning-tree priority
 - config spanning-tree root_guard_timeout
 - config spanning-tree vlan
 - config spanning-tree vlan interface cost
 - config spanning-tree vlan interface priority
 - config spanning-tree interface
 - config spanning-tree interface root_guard
 - config spanning-tree interface bpdu_guard
 - config spanning-tree interface portfast
 - config spanning-tree interface uplink_fast
 - config spanning-tree interface priority
 - configure spanning-tree interface cost
- PVST debug commands
 - debug spanning-tree
- PVST clear commands
 - sonic-clear spanning-tree

This section describes PVST (STP) feature and commands which are supported in EC SONiC.

PVST (Per VLAN Spanning Tree) is a feature in SONiC, [SONiC_PVST_HLD](#), which planned in 202106 branch. It supports STP (802.1D Spanning Tree) for those ports, assigned to untagged VLAN1, but there is an issue - dual root bridge when interacts with 802.1D switch.

To resolve this, EC SONiC adding **stp** mode in [config spanning-tree](#) command, which runs a single spanning tree instance for all ports which is a member of any VLAN, and disables other instances in PVST.

PVST show commands

show spanning-tree

This command displays spanning tree information .

Note:

- VLAN related settings cannot be displayed when spanning-tree mode is stp.

Usage

```
show spanning-tree
show spanning-tree vlan <vlanid>
show spanning-tree vlan <vlanid> interface <ifname>
```

Example

```
admin@sonic:~$ show spanning-tree vlan 100
Spanning-tree Mode: PVST
```

```
VLAN 100 - STP instance 3
```

```
-----
STP Bridge Parameters:
```

Bridge Identifier	Bridge MaxAge	Bridge Hello	Bridge FwdDly	Bridge Hold Time	LastTopology Change	Topology Change
hex	sec	sec	sec	sec	sec	cnt
8000002438eefbc3	20	2	15	1	0	0

RootBridge Identifier	RootPath Cost	DesignatedBridge Identifier	RootPort	Max Age	Hel lo	Fwd Dly
hex	hex	hex	sec	sec	sec	sec
8000002438eefbc3	0	8000002438eefbc3	Root	20	2	15

```
STP Port Parameters:
```

Port Name	Prio	Path Cost	Port Uplink	State	Designated Cost	Designated Root	Designated Bridge
Ethernet13	128	4	Y	N	FORWARDING	0	8000002438eefbc3 8000002438eefbc3

- show spanning-tree bpduguard: This command displays the interfaces which are BPDU guard enabled and also the state if the interface is disabled due to BPDU guard.

Usage

```
show spanning-tree bpduguard
```

Example

```
admin@sonic:~$ show spanning-tree bpduguard
PortNum      Shutdown      Port shut
              Configured    due to BPDU guard
-----
Ethernet1     Yes           Yes
Ethernet2     Yes           No
Port-Channel2 No            NA
```

- show spanning-tree rootguard: This command displays the interfaces where root guard is active and the pending time for root guard timer expiry

Usage

```
show spanning-tree rootguard
```

Example

```
admin@sonic:~$ show spanning-tree root_guard
Root guard timeout: 120 secs
```

Port	VLAN	Current State
Ethernet1	1	Inconsistent state (102 seconds left on timer)
Ethernet8	100	Consistent state

- `show spanning-tree statistics`
- `show spanning-tree statistics vlan <vlanid>`: This command displays the spanning-tree bpdv statistics. Statistics will be synced to APP DB every 10 seconds.

Usage

```
show spanning-tree statistics
show spanning-tree statistics vlan <vlanid>
```

Example

```
admin@sonic:~$ show spanning-tree statistics vlan 100
VLAN 100 - STP instance 3
```

PortNum	BPDU Tx	BPDU Rx	TCN Tx	TCN Rx
Ethernet13	10	4	3	4
PortChannel15	20	6	4	1

PVST configuration commands

config spanning-tree

This command enables/disables the spanning tree mode for the device. **pvst** and **stp** are exclusive. User's configuration should be one of (pvst, stp, disable).

Usage

```
config spanning-tree {enable|disable} {pvst | stp}
```

option

- **pvst**: Per VLAN Spanning Tree mode. When enabled, it enables multi-spanning tree instances for first 255 VLANs. The first instance is for VLAN 1 and delivers BPDU for untagged member interface.
- **stp**: (802.1D) Spanning Tree mode. When enabled, it enables 1st spanning tree instance with system id (0) and disable other instance. All ports which is member of VLAN, either tagged or untagged, are joined the instance.

Note:

- Only one mode can be enabled at any given time. That is, the configuration should be one of (disable, pvst, stp).
- When switching the mode, eg pvst to stp, it will stop **pvst** and then start **stp**.

Example

```
admin@sonic:~$ sudo config spanning-tree enable pvst
```

config spanning-tree forward_delay

This command allows configuring the forward delay time in seconds (default = 15), range 4-30.

Note:

- $2 * (\text{forward_delay} - 1) \geq \text{max_age} \geq 2 * (\text{hello_time} + 1)$.
- The specified value on the global level command will also apply to value on these VLAN if original per-VLAN value is same as the global one.

Usage

```
config spanning-tree forward_delay <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree forward_delay 30
```

config spanning-tree hello

This command allow configuring the hello interval in secs for transmission of bpdus (default = 2), range 1-10.

Note:

- $2 * (\text{forward_delay} - 1) \geq \text{max_age} \geq 2 * (\text{hello_time} + 1)$.
- The specified value on the global level command will also apply to value on these VLAN if original per-VLAN value is same as the global one.

Usage

```
config spanning-tree hello <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree hello 10
```

config spanning-tree max_age

This command allows configuring the maximum time to listen for root bridge in seconds (default = 20), range 6-40.

Note:

- $2 * (\text{forward_delay} - 1) \geq \text{max_age} \geq 2 * (\text{hello_time} + 1)$.
- The specified value on the global level command will also apply to value on these VLAN if original per-VLAN value is same as the global one.

Usage

```
config spanning-tree max_age <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree max_age 40
```

config spanning-tree priority

This command allows configuring the bridge priority in increments of 4096 (default = 32768), range 0-61440.

Note:

- The specified value on the global level command will also apply to value on these VLAN if original per-VLAN value is same as the global one.

Usage

```
config spanning-tree priority <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree priority 61440
```

config spanning-tree root_guard_timeout

This command allows configuring a root guard timeout value. Once superior BPDUs stop coming on the port, device will wait for a period until root guard timeout before moving the port to forwarding state(default = 30 secs), range 5-600.

Usage

```
config spanning-tree root_guard_timeout <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree root_guard_timeout 600
```

config spanning-tree vlan

Note:

- VLAN related settings cannot be configured when spanning-tree mode is stp.

This command allows enabling or disabling spanning-tree on a VLAN.

Usage

```
config spanning-tree vlan {enable|disable} <vlan>
```

Example

```
admin@sonic:~$ sudo config spanning-tree vlan enable 1
```

Below commands are similar to the global level commands but allow configuring on per VLAN basis.

Usage

```
config spanning-tree vlan forward_delay <vlan> <fwd-delay-value>
config spanning-tree vlan hello <vlan> <hello-value>
config spanning-tree vlan max_age <vlan> <max-age-value>
config spanning-tree vlan priority <vlan> <priority-value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree vlan priority 1 61440
```

config spanning-tree vlan interface cost

This command allows to configure the port level cost value for a VLAN, range 1 - 200000000

Usage

```
config spanning-tree vlan interface cost <vlan> <ifname> <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree vlan interface cost 1 Ethernet0 10
```

config spanning-tree vlan interface priority

This command allows to configure the port level priority value for a VLAN, range 0 - 240 (default 128)

Usage

```
config spanning-tree vlan interface priority <vlan> <ifname> <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree vlan interface priority 1 Ethernet0 240
```

config spanning-tree interface

This command allows enabling or disabling of STP on an interface, by default STP will be enabled on the interface if global STP mode is configured.

Usage

```
config spanning-tree interface {enable|disable} <ifname>
```

Example

```
admin@sonic:~$ sudo config spanning-tree interface enable Ethernet0
```

config spanning-tree interface root_guard

This command allow enabling or disabling of root guard on an interface.

Usage

```
config spanning-tree interface root_guard {enable|disable} <ifname>
```

Example

```
admin@sonic:~$ sudo config spanning-tree interface root_guard enable Ethernet0
```

config spanning-tree interface bpdu_guard

Below command allows enabling or disabling of bpdu guard on an interface.

Usage

```
config spanning-tree interface bpdu_guard {enable|disable} <ifname>
```

By default, BPDU guard will only generate a syslog indicating the condition, for taking an action like disabling the port below command can be used with shutdown option

Usage

```
config spanning-tree interface bpdu_guard {enable|disable} <ifname> [--shutdown | -s]
```

Example

```
admin@sonic:~$ sudo config spanning-tree interface bpdu_guard enable Ethernet0
```

config spanning-tree interface portfast

Below command allows enabling or disabling of portfast on an interface.

Usage

```
config spanning-tree interface portfast {enable|disable} <ifname>
```

Example

```
admin@sonic:~$ sudo config spanning-tree interface portfast enable Ethernet0
```

config spanning-tree interface uplink_fast

Usage

```
config spanning-tree interface uplink_fast {enable|disable} <ifname>
```

Example

```
admin@sonic:~$ sudo config spanning-tree interface uplink_fast enable Ethernet0
```

config spanning-tree interface priority

This command allows to configure the port level priority value, range 0 - 240 (default 128)

Usage

```
config spanning-tree interface priority <ifname> <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree interface priority Ethernet0 240
```

configure spanning-tree interface cost

This command allows to configure the port level cost value, range 1 - 200000000

Usage

```
config spanning-tree interface cost <ifname> <value>
```

Example

```
admin@sonic:~$ sudo config spanning-tree interface cost Ethernet0 1000
```

PVST debug commands**debug spanning-tree**

Following debug commands will be supported for enabling additional logging which can be viewed in /var/log/stdp.log, orchangent related logs can be viewed in /var/log/syslog.

Usage

```
debug spanning-tree
debug spanning-tree bpd [tx|rx]
debug spanning-tree event
debug spanning-tree interface <ifname>
debug spanning-tree verbose
debug spanning-tree vlan <id>
```

Note: This "debug spanning-tree" command must be enabled for logs to be written to log file, after enabling any of the below commands.

Example

```
admin@sonic:~$ debug spanning-tree
Command: sudo stpctl dbg enable

admin@sonic:~$ debug spanning-tree bpdu tx
Command: sudo stpctl dbg bpdu tx-on

admin@sonic:~$ debug spanning-tree event
Command: sudo stpctl dbg event on

admin@sonic:~$ debug spanning-tree verbose
Command: sudo stpctl dbg verbose on

admin@sonic:~$ debug spanning-tree vlan 1
Command: sudo stpctl dbg vlan 1 on

admin@sonic:~$ debug spanning-tree interface Ethernet3
Command: sudo stpctl dbg port Ethernet3 on
```

To disable the debugging controls enabled '-d' or '--disable' option can be used, an example of disabling bpdu debugging is shown below -

Usage

```
debug spanning-tree bpdu -d
```

Example

```
admin@sonic:~$ debug spanning-tree bpdu -d
Command: sudo stpctl dbg bpdu off
```

Follow commands can be used to reset and display the debugging controls enabled respectively

Usage

```
debug spanning-tree reset
debug spanning-tree show
```

Example

```
admin@sonic:~$ debug spanning-tree bpdu -d
Command: sudo stpctl dbg bpdu off
```

```
admin@sonic:~$ debug spanning-tree show
Command: sudo stpctl dbg show
```

```
STP Debug Parameters
-----
STP debugging is : ON
  Verbose       : ON
  Event         : ON
  BPDU-RX       : OFF
  BPDU-TX       : OFF
Ports: 3
VLANs: 1
```

Following debug commands will be supported for displaying internal data structures

Usage

```
debug spanning-tree dump global
debug spanning-tree dump vlan <vid>
debug spanning-tree dump interface <vid> <ifname>
```

Example

```
admin@sonic:~$ debug spanning-tree dump interface 2 Ethernet3
Command: sudo stpctl port 2 Ethernet3
PORT CLASS - VLAN 2 PORT 3(Ethernet3)
=====
port_id           = 8 3
state             = FORWARDING
path_cost         = 2000
designated_root    = 0x800268215f8101b6
designated_cost    = 0
designated_bridge  = 0x800268215f8101b6
designated_port    = Pri-8, Num-3
topology_change_acknowledge = 0
config_pending    = 0
change_detection_enabled = 1
self_loop         = 0
auto_config       = 1
message_age_timer = INACTIVE 0
forward_delay_timer = INACTIVE 0
hold timer        = INACTIVE 0
root_protect_timer = INACTIVE 0
forward_transitions = 1
rx_config_bpdu    = 0
tx_config_bpdu    = 266
rx_tcn_bpdu       = 0
tx_tcn_bpdu       = 0
```

PVST clear commands

`sonic-clear spanning-tree`

Following clear commands will be supported

Usage

```
sonic-clear spanning-tree statistics
sonic-clear spanning-tree statistics vlan <vid>
sonic-clear spanning-tree statistics vlan-interface <vid> <ifname>
sonic-clear spanning-tree statistics interface <ifname>
```

Example

```
admin@sonic:~$ sonic-clear spanning-tree statistics
All stats cleared

admin@sonic:~$ sonic-clear spanning-tree statistics vlan 1
Stats cleared for VLAN 1

admin@sonic:~$ sonic-clear spanning-tree statistics vlan-interface 1 Ethernet0
Stats cleared for VLAN 1 Ethernet0

admin@sonic:~$ sonic-clear spanning-tree statistics interface Ethernet0
Stats cleared for Ethernet0
```

(202111.2) QoS

- Qos Classification
 - Classification Show Commands
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 - QoS Classification Configuration Commands
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 - config qos tc-pg del
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 - config qos remark dscp update
 - config qos remark dscp del
 - config interface qos remark dscp bind
 - config interface qos remark dscp unbind
- Qos Configuration Commands
 - config qos clear
 - config qos reload

Qos Classification

Classification Show Commands

show qos dot1p-tc

This command is used to display the Dot1p-to-TC profile(s).

Usage

```
show qos dot1p-tc [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show qos dot1p-tc
dot1p-tc policy: dot1p_to_tc_profile
  Dot1p    TC
  -----
    0      2
    1      3
    2      4

dot1p-tc policy: dot1p_to_tc_profile2
  Dot1p    TC
  -----
    2      3
    3      3
    4      3
    5      3
```

show qos dscp-tc

This command is used to display the DSCP-to-TC profile(s).

Usage

```
show qos dscp-tc [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show qos dscp-tc
dscp-tc policy: dscp_to_tc_profile
DSCP    TC
-----
0        0
8        1
16 24    2
```

show qos tc-pg

This command is used to display the TC-to-PG profile(s).

Usage

```
show qos tc-pg [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show qos tc-pg
tc-pg policy: tc_pg_prof1
  TC    PG
  ----  --
    2    3
    3    3
    4    3
    5    3
```

show qos tc-queue

This command is used to display the Tc-to-queue profile(s).

Usage

```
show qos tc-queue [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show qos tc-queue
tc-queue policy: tc_to_queue_profile
  TC    Queue
  ----  -
    0    0
    1    2
    2    2
```

show interface qos

This command is used to display the QoS configuration for all interfaces or a given interface.

Usage

```
show interface qos [<interface_name>]
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

```
admin@sonic:~$ show interface qos
Ethernet0:
  TC to Queue: tc_to_queue_profile

Ethernet16:
  Dot1p to TC: AZURE
  PFC to Queue: AZURE
  TC to PG: AZURE
  TC to Queue: AZURE

Ethernet18:
  pfc-priority: 0
  PFC to Queue: AZURE
```

Example

```
admin@sonic:~$ show interface qos Ethernet0
  TC to Queue: tc_to_queue_profile
```

QoS Classification Configuration Commands

config qos dot1p-tc add

This command is used to create a Dot1p-to-TC profile.

Usage

```
config qos dot1p-tc add <profile_name> --dot1p <dot1p> --tc <tc>
```

Parameters:

- <profile_name>: Profile name.
- <dot1p>: List of dot1p. Minimum: 0. Maximum: 7. For example: "1", "1,2-3".
- <tc>: Traffic Class. Minimum: 0. Maximum: 7.

Example

```
sudo config qos dot1p-tc add dot1p_to_tc_profile --dot1p 0,1,2,3 --tc 0
```

config qos dot1p-tc update

This command is used to modify an existing Dot1p-to-TC profile.

Usage

```
config qos dot1p-tc update <profile_name> --dot1p <dot1p> [--tc <tc>] [--remove]
```

Parameters:

- <profile_name>: Profile name.
- <dot1p>: List of dot1p. Minimum: 0. Maximum: 7. For example: "1", "1,2-3".
- <tc>: Traffic Class. Minimum: 0. Maximum: 7.

- remove: Remove the mapping. For example: "--dot1p 1 --remove".

Example

```
sudo config qos dot1p-tc update dot1p_to_tc_profile --dot1p 0 --tc 1
sudo config qos dot1p-tc update dot1p_to_tc_profile --dot1p 4 --tc 0
sudo config qos dot1p-tc update dot1p_to_tc_profile --dot1p 1 --tc 0 --remove
```

config qos dot1p-tc del

This command is used to delete a Dot1p-to-TC profile.

Usage

```
config qos dot1p-tc del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config qos dot1p-tc del dot1p_to_tc_profile
```

config interface qos dot1p-tc bind

This command is used to configure Dot1p-to-TC mapping on the specified interface.

Usage

```
config interface qos dot1p-tc bind <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Default mapping:

The following table shows the default mapping on Broadcom platform.

DOT1P	TC
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7

The following table shows the default mapping on Intel platform.

DOT1P	TC
0 ~ 7	0

Any packet field value that is not specified in the configuration is assigned to a default value.

Example

```
sudo config interface qos dot1p-tc bind Ethernet0 dot1p_to_tc_profile
```

config interface qos dot1p-tc unbind

This command is used to reset Dot1p-to-TC mapping on the specified interface.

Usage

```
config interface qos dot1p-tc unbind <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
sudo config interface qos dot1p-tc unbind Ethernet0
```

config qos dscp-tc add

This command is used to create a DSCP-to-TC profile.

Usage

```
config qos dscp-tc add <profile_name> --dscp <dscp> --tc <tc>
```

Parameters:

- <profile_name>: Profile name.
- <dscp>: List of DSCP. Minimum: 0. Maximum: 63. For example: "1", "1,2-3".
- <tc>: Traffic Class. Minimum: 0. Maximum: 7.

Example

```
sudo config qos dscp-tc add dscp_to_tc_profile --dscp 0,1,2,3 --tc 0
```

config qos dscp-tc update

This command is used to modify an existing DSCP-to-TC profile.

Usage

```
config qos dscp-tc update <profile_name> --dscp <dscp> [--tc <tc>] [--remove]
```

Parameters:

- <profile_name>: Profile name.
- <dscp>: List of DSCP. Minimum: 0. Maximum: 63. For example: "1", "1,2-3".
- <tc>: Traffic Class. Minimum: 0. Maximum: 7.
- remove: Remove the mapping. For example: "--dscp 1 --remove".

Example

```
sudo config qos dscp-tc update dscp_to_tc_profile --dscp 0 --tc 1
sudo config qos dscp-tc update dscp_to_tc_profile --dscp 16,24 --tc 2
sudo config qos dscp-tc update dscp_to_tc_profile --dscp 1 --tc 2 --remove
```

config qos dscp-tc del

This command is used to delete a DSCP-to-TC profile.

Usage

```
config qos dscp-tc del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config qos dscp-tc del dscp_to_tc_profile
```

config interface qos dscp-tc bind

This command is used to configure the DSCP-to-TC mapping on the specified interface.

Usage

```
config interface qos dscp-tc bind <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Default mapping:

DSCP	TC
0 ~ 63	0

Any packet field value that is not specified in the configuration is assigned to a default value.

Example

```
sudo config interface qos dscp-tc bind Ethernet0 dscp_to_tc_profile
```

config interface qos dscp-tc unbind

This command is used to reset the DSCP-to-TC mapping on the specified interface.

Usage

```
config interface qos dscp-tc unbind <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
sudo config interface qos dscp-tc unbind Ethernet0
```

config qos tc-pg add

This command is used to create a TC-to-PG profile.

Usage

```
config qos tc-pg add <profile_name> --tc <tc> --pg <pg>
```

Parameters:

- <profile_name>: Profile name.
- <tc>: the TC (Traffic Class), or switch-priority, the value could be a list that is comma separated, valid range is 0 to 7.
- <pg>: the priority group value, valid range is 0 to 7 for Broadcom platform, 1 to 5 for Intel platform.

Note:

- On the devices using Intel ASIC, it only supports one profile per system. The configuration is applied immediately after profile is added. Any packet field value that is not specified in the configuration is assigned to a default value.

Example

```
sudo config qos tc-pg add tc_to_pg_profile --tc 0,1,2,3 --pg 0
```

config qos tc-pg update

This command is used to modify an existing TC-to-PG profile.

Usage

```
config qos tc-pg update <profile_name> --tc <tc> --pg <pg> [--remove]
```

Parameters:

- <profile_name>: Profile name.
- <tc>: the TC (Traffic Class), or switch-priority, the value could be a list that is comma separated, valid range is 0 to 7.
- <pg>: the priority group value, valid range is 0 to 7 for Broadcom platform, 1 to 5 for Intel platform.
- remove: Remove the mapping. For example: "--tc 1 --remove".

Example

```
sudo config qos tc-pg update tc_to_pg_profile --tc 0 --pg 1
sudo config qos tc-pg update tc_to_pg_profile --tc 4 --pg 0
sudo config qos tc-pg update tc_to_pg_profile --tc 1 --pg 0 --remove
```

config qos tc-pg del

This command is used to delete a TC-to-PG profile.

Usage

```
config qos tc-pg del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config qos tc-pg del tc_to_pg_profile
```

config interface qos tc-pg bind

This command is used to configure the TC-to-PG mapping on the specified interface.

Usage

```
config interface qos tc-pg bind <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Default mapping:

The following table shows the default mapping on Broadcom platform.

TC	PG
0 ~ 7	7

The following table shows the default mapping on Intel platform.

TC	PG
0 ~ 7	0

Any packet field value that is not specified in the configuration is assigned to a default value.

Example

```
sudo config interface qos tc-pg bind Ethernet0 tc_to_pg_profile
```

config interface qos tc-pg unbind

This command is used to reset the TC-to-PG mapping on the specified interface.

Usage

```
config interface qos tc-pg unbind <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
sudo config interface qos tc-pg unbind Ethernet0
```

config qos tc-queue add

This command is used to create a TC-to-queue profile.

Usage

```
config qos tc-queue add <profile_name> --tc <tc> --queue <queue>
```

Parameters:

- <profile_name>: Profile name.
- <tc>: the TC (Traffic Class), or switch-priority, the value could be a list that is comma separated, valid range is 0 to 7.
- <queue>: the queue value, valid range is 0 to 7.

Note:

- On the devices using Intel ASIC, it only supports one profile per system. The configuration is applied immediately after profile is added. Any packet field value that is not specified in the configuration is assigned to a default value.

Example

```
sudo config qos tc-queue add tc_to_queue_profile --tc 0,1,2,3 --queue 0
```

config qos tc-queue update

This command is used to modify an existing TC-to-queue profile. The configuration is applied immediately after profile is updated on Intel platform.

Usage

```
config qos tc-queue update <profile_name> --tc <tc> --queue <queue> [--remove]
```

Parameters:

- <profile_name>: Profile name.

- <tc>: the TC (Traffic Class), or switch-priority, the value could be a list that is comma separated, valid range is 0 to 7.
- <queue>: the queue value, valid range is 0 to 7.
- remove: Remove the mapping. For example: "--tc 1 --remove".

Note:

- On the devices using Intel ASIC, it only supports one profile per system. The configuration is applied immediately after profile is added. Any packet field value that is not specified in the configuration is assigned to a default value.

Example

```
sudo config qos tc-queue update tc_to_queue_profile --tc 0 --queue 1
sudo config qos tc-queue update tc_to_queue_profile --tc 4 --queue 2
sudo config qos tc-queue update tc_to_queue_profile --tc 1 --queue 2 --remove
```

config qos tc-queue del

This command is used to delete a TC-to-queue profile.

Usage

```
config qos tc-queue del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Note:

- On the devices using Intel ASIC, this command is used to restore the TC-to-queue mapping. All TC maps to queue 0. The following table shows the default mapping.

TC	Queue
0 ~ 7	0

Example

```
sudo config qos tc-queue del tc_to_queue_profile
```

config interface qos tc-queue bind

This command is used to configure the TC-queue mapping on the specified interface. This command is not supported on Intel platform because configuration is applied immediately after profile is added or updated.

Usage

```
config interface qos tc-queue bind <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Default mapping:

The following table shows the default mapping on Broadcom platform.

TC	Queue
0	0

1	1
2	2
3	3
4	4
5	5
6	6
7	7

Any packet field value that is not specified in the configuration is assigned to a default value.

Note:

- This command is only supported on the devices using Broadcom ASIC.

Example

```
sudo config interface qos tc-queue bind Ethernet0 tc_to_queue_profile
```

config interface qos tc-queue unbind

This command is used to reset the TC-to-queue mapping on the specified interface.

Usage

```
config interface qos tc-queue unbind <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Note:

- This command is only supported on the devices using Broadcom ASIC.

Example

```
sudo config interface qos tc-queue unbind Ethernet0
```

Qos Marking

QoS Marking Show Commands

show qos remark dot1p

This command is used to display the dot1p remarking profile(s).

Usage

```
show qos remark dot1p [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show qos remark dot1p
dot1p policy: remark_dot1p_profile
  TC      Dot1p
  ----      -
  0         0
  1         1
  2         2
```

show qos remark dscp

This command is used to display the DSCP remarking profile(s).

Usage

```
show qos remark dscp [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show qos remark dscp
dscp policy: remark_dscp_profile
  TC      DSCP
  ----      -
  0         2
  1         9
  2        17
```

QoS Marking Configuration Commands

config qos remark dot1p add

This command is used to create a Dot1p remarking profile.

Usage

```
config qos remark dot1p add <profile_name> --tc <tc> --dot1p <dot1p>
```

Parameters:

- <profile_name>: Profile name.
- <tc>: List of Traffic Class. Minimum: 0. Maximum: 7. For example: "1", "1,2-3"
- <dot1p>: Dot1p. Minimum: 0. Maximum: 7.

Example

```
sudo config qos remark dot1p add remark_dot1p_profile --tc 0 --dot1p 0
```

config qos remark dot1p update

This command is used to modify an existing Dot1p remarking profile.

Usage

```
config qos remark dot1p update <profile_name> --tc <tc> [--dot1p <dot1p>] [--remove]
```

Parameters:

- <profile_name>: Profile name.
- <tc>: List of Traffic Class. Minimum: 0. Maximum: 7. For example: "1", "1,2-3".
- <dot1p>: Dot1p. Minimum: 0. Maximum: 7
- remove: Remove the mapping. For example: "--tc 1 --remove".

Example

```
sudo config qos remark dot1p update remark_dot1p_profile --tc 1 --dot1p 1
```

config qos remark dot1p del

This command is used to delete a Dot1p remarking profile.

Usage

```
config qos remark dot1p del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config qos remark dot1p del remark_dot1p_profile
```

config interface qos remark dot1p bind

This command is used to enable Dot1p remarking on the specified interface.

Usage

```
config interface qos remark dot1p bind <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Default mapping:

TC	Dot1p
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7

Any TC value that is not specified in the configuration is assigned to a default Dot1p value for rewriting.

Example

```
sudo config interface qos remark dot1p bind Ethernet0 remark_dot1p_profile
```

config interface qos remark dot1p unbind

This command is used to disable Dot1p remarking on the specified interface.

Usage

```
config interface qos remark dot1p unbind <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
sudo config interface qos remark dot1p unbind Ethernet0
```

config qos remark dscp add

This command is used to create a DSCP remarking profile.

Usage

```
config qos remark dscp add <profile_name> --tc <tc> --dscp <dscp>
```

Parameters:

- <profile_name>: Profile name.
- <tc>: List of Traffic Class. Minimum: 0. Maximum: 7. For example: "1", "1,2-3".
- <dscp>: DSCP. Minimum: 0. Maximum: 63.

Example

```
sudo config qos remark dscp add remark_dscp_profile --tc 0 --dscp 2
```

config qos remark dscp update

This command is used to modify an existing DSCP remarking profile.

Usage

```
config qos remark dscp update <profile_name> --tc <tc> [--dscp <dscp>] [--remove]
```

Parameters:

- <profile_name>: Profile name.
- <tc>: List of Traffic Class. Minimum: 0. Maximum: 7. For example: "1", "1,2-3"
- <dscp>: DSCP. Minimum: 0. Maximum: 63.
- remove: Remove the mapping. For example: "--tc 1 --remove".

Example

```
sudo config qos remark dscp update remark_dscp_profile --tc 1 --dscp 9
sudo config qos remark dscp update remark_dscp_profile --tc 2 --dscp 17
```

config qos remark dscp del

This command is used to delete a DSCP remarking profile.

Usage

```
config qos remark dscp del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config qos remark dscp del remark_dscp_profile
```

config interface qos remark dscp bind

This command is used to enable DSCP remarking on the specified interface.

Usage

```
config interface qos remark dscp bind <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

- <profile_name>: Profile name.

Default mapping:

TC	DSCP
0 ~ 7	0

Any TC value that is not specified in the configuration is assigned to a default DSCP value for rewriting.

Example

```
sudo config interface qos remark dscp bind Ethernet0 remark_dscp_profile
```

config interface qos remark dscp unbind

This command is used to disable DSCP remarking on the specified interface.

Usage

```
config interface qos remark dscp unbind <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
sudo config interface qos remark dscp unbind Ethernet0
```

Qos Configuration Commands

config qos clear

This command is used to clear all the QoS configuration from all the following QOS Tables in ConfigDB.

1. TC_TO_PRIORITY_GROUP_MAP
2. MAP_PFC_PRIORITY_TO_QUEUE
3. TC_TO_QUEUE_MAP
4. DSCP_TO_TC_MAP
5. DOT1P_TO_TC_MAP
6. MPLS_TC_TO_TC_MAP
7. SCHEDULER
8. PFC_PRIORITY_TO_PRIORITY_GROUP_MAP
9. PORT_QOS_MAP
10. WRED_PROFILE
11. QUEUE
12. CABLE_LENGTH
13. BUFFER_POOL
14. BUFFER_PROFILE
15. BUFFER_PG
16. BUFFER_QUEUE

Usage

```
config qos clear
```


Example

```
admin@sonic:~$ sudo config qos clear
```

config qos reload

This command is used to clear and load the predefined lossless buffer configuration. The **config qos clear** command is issued by this command to clear all the QoS configuration. After this command, use **config save** command to save the configuration file, then use **reboot** command to restart the switch to take the change effect.

Usage

```
config qos reload
```

Example

```
admin@sonic:~$ sudo config qos reload
Running command: /usr/local/bin/sonic-cfggen -d -t /usr/share/sonic/device/x86_64-dell_z9100_c2538-r0
/Force10-Z9100-C32/buffers.json.j2 >/tmp/buffers.json
Running command: /usr/local/bin/sonic-cfggen -d -t /usr/share/sonic/device/x86_64-dell_z9100_c2538-r0
/Force10-Z9100-C32/qos.json.j2 -y /etc/sonic/sonic_version.yml >/tmp/qos.json
Running command: /usr/local/bin/sonic-cfggen -j /tmp/buffers.json --write-to-db
Running command: /usr/local/bin/sonic-cfggen -j /tmp/qos.json --write-to-db
```

In this example, it uses the buffers.json.j2 file and qos.json.j2 file from platform specific folders. When there are no changes in the platform specific configuration files, they internally use the file "/usr/share/sonic/templates/buffers_config.j2" and "/usr/share/sonic/templates/qos_config.j2" to generate the configuration.

(202111.2) Rate

- [Rate Config Commands](#)
 - `config rate smoothing-interval`

Rate Config Commands

`config rate smoothing-interval`

This command is used to change the window size of exponential moving average to make the rates and utilization values more smooth in the output of "show interface counters"

Usage

```
config rate smoothing-interval <interval>
```

Parameter:

- `<interval>`: The window size of exponential moving average.(The range is from 1 to 1000)

Example

```
admin@sonic:~$ sudo config rate smoothing-interval 20
```

(202111.2) Routing Stack

The SONiC software is agnostic of the routing software that is being used in the device. For example, users can use either the Quagga or FRR routing stack as per their requirement. A separate shell (**vtysh**) is provided to configure such routing stacks. Once users enter **vtysh**, they can use the routing stack specific commands as given in the following example.

Example (FRR Routing Stack)

```
admin@sonic:~$ vtysh

Hello, this is FRRouting (version 7.2.1-sonic).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

sonic# show route-map
ZEBRA:
route-map: RM_SET_SRC Invoked: 82595
  permit, sequence 10 Invoked 82595
    Match clauses:
    Set clauses:
      src 10.1.0.32
    Call clause:
    Action:
      Exit routemap
```

Refer to the routing stack [Quagga Command Reference](#) or [FRR Command Reference](#) to know more about about the routing stack configuration.

(202111.2) SAG

- SAG Show Commands
 - show sag
 - show sag ip
 - show sag ipv6
- SAG Configuration Commands
 - config sag mac_address add/del
 - config sag ipv4 enable/disable
 - config sag ipv6 enable/disable
 - config sag interface sag ip add/del

SAG Show Commands

show sag

This command displays the global SAG configuration.

Usage

```
show sag
```

Example

```
admin@sonic:~#show sag
Static Anycast Gateway Information

MacAddress      IPv4    IPv6
-----
00:11:22:33:44:55  enable  enable
```

show sag ip

This command displays the configured SAG IPv4 addresses.

Usage

```
show sag ip
```

Example

```
admin@sonic~#show sag ip
Vlan Interface Name    IPv4 address/mask
-----
Vlan100                1.1.1.254/24
```

show sag ipv6

This command displays the configured SAG IPv6 addresses.

Usage

```
show sag ipv6
```

Example

```
admin@sonic~#show sag ipv6
Vlan Interface Name   IPv6 address/mask
-----
Vlan100               2001:1000::FE/64
```

SAG Configuration Commands

config sag mac_address add/del

This command is used to add/delete the SAG MAC address.
This setting is global and only a single MAC address is allowed.

Usage

```
config sag mac_address <add|del> <mac-address>
```

Parameters:

- mac-address: Specifies the MAC address used for SAG interface.

Example

```
admin@sonic:~# sudo config sag mac_address add 00:11:22:33:44:55
admin@sonic:~# sudo config sag mac_address del 00:11:22:33:44:55
```

config sag ipv4 enable/disable

This command is used to enable/disable the IPv4 address on the SAG interface.

Usage

```
config sag ipv4 <enable|disable>
```

Example

```
admin@sonic:~# sudo config sag ipv4 enable
admin@sonic:~# sudo config sag ipv4 disable
```

config sag ipv6 enable/disable

This command is used to enable/disable the IPv6 address on the SAG interface.

Usage

```
config sag ipv6 <enable|disable>
```

Example

```
admin@sonic:~# sudo config sag ipv6 enable
admin@sonic:~# sudo config sag ipv6 disable
```

config sag interface sag ip add/del

This command is used to add/del the IPv4/IPv6 address on the SAG interface, and the VLAN interface that the SAG interface is based on.

Usage

```
config interface sag ip <add|del> <vlan-name> <ip-address>
```

Parameters:

- vlan-name: Specifies the SAG's parent VLAN interface name.
- ip-address: Specifies the IPv4/IPv6 address used on the SAG interface.

Example

```
admin@sonic:~# sudo config interface sag ip add Vlan1 192.168.1.1/24
admin@sonic:~# sudo config interface sag ip del Vlan1 192.168.1.1/24
```

(202111.2) Scheduling & Shaping

- [Scheduling & Shaping](#)
 - [Scheduling & Shaping Show Commands](#)
 - [show scheduler](#)
 - [show interface scheduler](#)
 - [Scheduling & Shaping Configuration Commands](#)
 - [config scheduler add](#)
 - [config scheduler update](#)
 - [config scheduler del](#)
 - [config interface scheduler bind queue](#)
 - [config interface scheduler unbind queue](#)
 - [config interface scheduler bind port](#)
 - [config interface scheduler unbind port](#)

Scheduling & Shaping

This section explain the commands for queue scheduling and shaping, either in bytes or packets.

Scheduling & Shaping Show Commands

show scheduler

This command is used to display the scheduler profile(s).

Usage

```
show scheduler [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@as5835-54t-2:~$ show scheduler
Name                               Scheduling Type   Weight  Shaper Type   Bandwidth
-----
scheduler_profile_queue0          WRR                20     bytes         10 Gbps
```

show interface scheduler

This command is used to display the scheduling and shaping for all interfaces or a given interface.

Usage

```
show interface scheduler [<interface_name>]
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
admin@sonic:~$ show interface scheduler
Ethernet0
Type      Queue    Profile                               Scheduling Mode  Weight  Shaper Type  Bandwidth
-----
Port      N/A      egress_port_shaper_profile          DWRR             N/A     bytes        10 Gbps
Queue     0        scheduler_profile_queue0            WRR              20      bytes        20 Gbps
```

Scheduling & Shaping Configuration Commands

User may create a schedule profile and bind/unbind the profile with port or egress queue of a port.

config scheduler add

This command is used to create a scheduler profile.

Usage

```
config scheduler add <profile_name> [--sched_type {WRR | DWRR | STRICT}] [--weight <weight>] [--shaper_type {bytes | packets}] [--bandwidth <bandwidth>]
```

Parameters:

- <profile_name>: Profile name
- sched-type: Scheduling mode. Valid values are:
 - **WRR**: Weighted Round Robin.
 - **note**: mode WRR is not supported in Intel ASIC; for example Wedge100BF-32X/32QS/65X or AS9516-32D.
 - **DWRR**: Deficit Weighted Round Robin.
 - Default value for Intel ASIC.
 - **STRICT**: Strict mode.
- <weight>: Weight of the mode; in the range (1..100)
 - Default: 1.
- shaper-type: Shaper type of shaping. Valid values are:
 - **bytes**: Bytes mode.
 - **packets**: Packets mode.
- <bandwidth>: Maximum rate of traffic can pass through in kilo-bits/second (kbps) or packet/second (pps). Valid values are:
 - If meter-type is **bytes**:
 - The format is a decimal number or a decimal number followed by the abbreviation k (1000), m (1,000,000), or g (1,000,000,000). Minimum: 8 kbps (32 kbps for AS8000 and AS9716-32D). Maximum: 400,000,000 kbps. For example: "10", "10k".
 - If meter-type is **packets**:
 - Minimum: 1 pps (2 pps for AS8000 and AS9716-32D). Maximum: 148,809,523 pps (67,108,608 pps for AS8000 and AS9716-32D).

Example

```
sudo config scheduler add scheduler_profile_queue0 --sched_type DWRR --weight 10 --meter_type bytes --bandwidth 1250000
```

config scheduler update

This command is used to modify an existing scheduler profile.

Usage

```
config scheduler update <profile_name> [--sched-type {WRR | DWRR | STRICT}] [--weight <weight>] [--shaper-type {bytes | packets}] [--bandwidth <bandwidth>] [--no-shaping]
```

Parameters:

- <profile_name>: Profile name
- sched-type: Scheduling mode. (This parameter is required for Intel platform) Valid values are:
 - **WRR**: Weighted Round Robin.
 - **note**: mode WRR is not supported in Intel ASIC; for example Wedge100BF-32X/32QS/65X or AS9516-32D.
 - **DWRR**: Deficit Weighted Round Robin.
 - Default value for Intel ASIC.
 - **STRICT**: Strict mode.
- <weight>: Weight of the mode; in the range (1..100)
- Default: 1..
- shaper-type: Shaper type of shaping. Valid values are:
 - **bytes**: Bytes mode.
 - **packets**: Packets mode.
- <bandwidth>: Maximum rate of traffic can pass through in kilo-bits/second (kbps) or packet/second (pps). Valid values are:
 - If meter-type is **bytes**:
 - The format is a decimal number or a decimal number followed by the abbreviation k (1000), m (1,000,000), or g (1,000,000,000). Minimum: 8 kbps (32 kbps for AS8000 and AS9716-32D). Maximum: 400,000,000 kbps. For example: "10", "10k".
 - If meter-type is **packets**:
 - Minimum: 1 pps (2 pps for AS8000 and AS9716-32D). Maximum: 148,809,523 pps (67,108,608 pps for AS8000 and AS9716-32D).
- no-shaping: Disable the traffic shaping.

Example

```
sudo config scheduler update scheduler_profile_queue0 --sched_type WRR --weight 20 --meter_type bytes --bandwidth 2500000
```

config scheduler del

This command is used to delete a scheduler profile.

Usage

```
config scheduler del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config scheduler del scheduler_profile_queue0
```

config interface scheduler bind queue

This command is used to configure the scheduling and shaping setting on the specified queue.

Usage

```
config interface scheduler bind queue <interface_name> <queue> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).
- <profile_name>: Profile name.

Example

```
sudo config interface scheduler bind queue Ethernet0 0 scheduler_profile_queue0
```

config interface scheduler unbind queue

This command is used to reset the scheduling and shaping setting on the specified queue.

Usage

```
config interface scheduler unbind queue <interface_name> <queue>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).

Example

```
sudo config interface scheduler unbind queue Ethernet0 0
```

config interface scheduler bind port

This command is used to configure the shaping setting on the specified interface.

Usage

```
config interface scheduler bind port <interface_name> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Note:

- This command is only supported on Intel switches.

Example

```
sudo config interface scheduler bind port Ethernet0 egress_port_shaper_profile
```

config interface scheduler unbind port

This command is used to reset the shaping setting on the specified interface.

Usage

```
config interface scheduler unbind port <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Note:

- This command is only supported on Intel switches.

Example

```
sudo config interface scheduler unbind port Ethernet0
```

(202111.2) sFlow

- [sFlow Show Commands](#)
 - `show sflow`
 - `show sflow interface`
- [sFlow Config Commands](#)
 - `config sflow collector add`
 - `config sflow collector del`
 - `config sflow agent-id add`
 - `config sflow agent-id del`
 - `config sflow enable`
 - `config sflow interface`
 - `config sflow interface sample-rate`
 - `config sflow polling-interval`

Sampled flow, or sFlow, collects sampled packets and counters from traffic at a predefined sampling rate, encapsulates the data in sFlow packets, and then delivers them to remote collectors. This section describes the sFlow command syntax and usage.

sFlow Show Commands

`show sflow`

This command displays the global sFlow configuration that includes the admin state, collectors, the Agent ID, and counter polling interval.

Usage

```
show sflow
```

Example

```
admin@sonic:~# show sflow
sFlow Global Information:
sFlow Admin State: up
sFlow Polling Interval: default
sFlow AgentID: lo
2 Collectors configured:
Name: collector_A IP addr: 10.11.46.2 UDP port: 6343
Name: collector_lo IP addr: 127.0.0.1 UDP port: 6343
```

`show sflow interface`

This command displays each interface's sFlow admin status and the sampling rate.

Usage

```
show sflow interface
```

Example

```
admin@sonic:~# show sflow interface
```

```
sFlow interface configurations
```

Interface	Admin State	Sampling Rate
Ethernet0	up	4000
Ethernet1	up	4000
...		
Ethernet61	up	4000
Ethernet62	up	4000
Ethernet63	up	4000

sFlow Config Commands

config sflow collector add

This command adds an sFlow collector.

Usage

```
config sflow collector add <collector-name> {<ipv4-address | ipv6-address>} [--port <number>] [--vrf {default | mgmt}]
```

Parameters:

- <collector-name>, a unique name for the collector, max length is 16. The maximum number of collectors that can be configured is 2.
- <ipv4-address>, IPv4 address of the collector in dotted decimal format.
- <ipv6-address>, IPv6 address of the collector in x: x: x: x::x format, where :: notation specifies successive hexadecimal fields of zeros.
- port (OPTIONAL), specifies the UDP port of the collector.
 - valid range is from 0 to 65535. (default is 6343)
- vrf (OPTIONAL), specifies the VRF of the collector. (default is default)
 - default, connects using the default VRF.
 - mgmt, connects using the mgmt VRF. If the mgmt VRF doesn't be created yet, the default VRF will be used to connect to server.

The following example configures a collector "collect_A" whose IP address is 10.11.46.2.

Example

```
admin@sonic:~# sudo config sflow collector add collect_A 10.11.46.2
```

config sflow collector del

This command deletes an sFlow collector whose name is specified.

Usage

```
config sflow collector del <collector-name>
```

Parameters:

- <collector-name>, the collector to be removed.

This example deletes a collector "collector_A".

Example

```
admin@sonic:~# sudo config sflow collector del collector_A
```

config sflow agent-id add

This command configures an interface as the sampling agent ID.

Usage

```
config sflow agent-id add <interface-name>
```

Parameters:

- <interface-name>, the interface whose IPv4 or IPv6 address is to be used as the agent address in sFlow packets.

Note:

- This setting is global and applied to configured collector(s).
- If there is a configured agent ID, you need to delete it before adding a new one.

This example specifies interface **lo** as the agent ID.

Example

```
admin@sonic:~# sudo config sflow agent-id add lo
```

config sflow agent-id del

This command removes the current agent ID.

Usage

```
config sflow agent-id del
```

Example

```
admin@sonic:~# sudo config sflow agent-id del
```

config sflow enable

Enables (or disables) sFlow sampling, which starts (or stops) the sFlow daemon monitoring the traffic on the interface(s).

Usage

```
config sflow {enable | disable}
```

This example starts sFlow monitoring.

Example

```
admin@sonic:~# sudo config sflow enable
```

config sflow interface

Enables/disables sFlow monitoring for sampling on an interface or all interfaces.

Usage

```
config sflow interface {enable | disable} {<interface-name> | all}
```

Parameters:

- <interface-name>, the interface on which the flow is monitored. It can be enabled or disabled individually.
- all, apply this configuration to all interfaces.

Note:

- To enable sFlow sampling correctly, you need to configure COPP. Here is a sample configuration that is appended to **/usr/share/sonic/templates/copp_cfg.j2**.

copp_cfg.j2

```
{
  "COPP_GROUP": {
    "queue2_group1": {
      "cbs": "1000",
      "cir": "1000",
      "genetlink_mcgrp_name": "packets",
      "genetlink_name": "psample",
      "meter_type": "packets",
      "mode": "sr_tcm",
      "queue": "2",
      "red_action": "drop",
      "trap_action": "trap",
      "trap_priority": "1"
    }
  },
  "COPP_TRAP": {
    "sflow": {
      "trap_group": "queue2_group1",
      "trap_ids": "sample_packet"
    }
  }
}
```

The following example shows how to disable flow sampling on Ethernet40.

Example

```
admin@sonic:~# sudo config sflow interface disable Ethernet40
```

config sflow interface sample-rate

Configure the sampling rate of an interface.

Usage

```
config sflow interface sample-rate <interface-name> <value>
```

Parameters:

- <interface-name>, the interface on which the sampling rate is applied.
- <value>, sampling rate. Valid range: (100..8388608)
 - The value is the average number of packets skipped before the sample is taken. "The sampling rate specifies random sampling probability as the ratio of packets observed to samples generated. For example, a sampling rate of 256 specifies that, on average, 1 sample will be generated for every 256 packets observed."
- default sampling rates for link speeds

	link speed	sampling rate
1	1G	100
2	10G	1,000
3	40G	4,000
4	50G	5,000
5	100G	10,000

6	400G	40,000
---	------	--------

Note:

- The Broadcom MAC chip supports a maximum of 4 different sampling rates per switch. It is recommended to use the default sampling rate unless it is a MUST.

The following example shows to how set a sampling rate of 1000 on interface Ethernet32.

Example

```
admin@sonic:~# sudo config sflow interface sample-rate Ethernet32 1000
```

config sflow polling-interval

This command is used to set the counter polling interval.

Usage

```
config sflow polling-interval <value>
```

Parameters:

- <value>, polling interval
 - valid range: 0, 5 to 300 seconds.
 - 0: disable counter polling
 - default: 20 seconds.

The following example shows how to set the polling interval to 30 seconds.

Example

```
admin@sonic:~# sudo config sflow polling-interval 30
```

(202111.2) Shared Buffer

- Shared Buffer
 - Shared Buffer Show Commands
 - `show buffer pool`
 - `show buffer profile`
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 - Shared Buffer Configuration Commands
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- Queue and Priority-Group
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Shared Buffer

Shared buffer is used to buffer the ingress packets, process, and insert into egress queue before transmitting out from the device. In some applications, it needs trigger notification to send (back pressure) pause frame to inform transmitting device slow down, user needs to adjust the threshold on ingress or egress.

Shared Buffer Show Commands

`show buffer pool`

This command is used to display the buffer pool configuration.

Usage

```
show buffer pool <[pool_name]>
```

Parameters:

- `<pool_name>`: Pool name.

Example

```
admin@sonic:~$ show buffer pool
Pool                Type      Size                Xoff
-----
egress_lossless_pool egress    6,383,104 Bytes
egress_lossy_pool    egress    6,383,104 Bytes
ingress_lossless_pool ingress    1,148,959 Bytes   5,744,794 Bytes
ingress_lossy_pool    ingress    5,872,456 Bytes
```

`show buffer profile`

This command is used to display the buffer profile configuration.

Usage

```
show buffer profile <[profile_name]>
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show buffer profile
Profile      Xon      Xon-offset      Pool      Size      Shared Mode      Shared Size      Xoff
-----
egress_lossless_profile  egress_lossless_pool  0 Bytes  dynamic      66%
egress_lossy_profile     egress_lossy_pool     0 Bytes  dynamic      66%
ingress_lossless_profile  ingress_lossless_pool  0 Bytes  dynamic      66%      5,744,794 Bytes  18
KiB
ingress_lossy_profile     ingress_lossy_pool     0 Bytes  dynamic      66%
```

show interface buffer priority-group

This command is used to display the buffer configuration of the priority-group (PG) for all interfaces or a given interface.

Usage

```
show interface buffer priority-group <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
admin@sonic:~$ show interface buffer priority-group
Interface  PG      Profile
-----
Ethernet0  0-2     buffer_prof1
```

show interface buffer queue

This command is used to display the buffer configuration for the queue for all interfaces or a given interface.

Usage

```
show interface buffer queue <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
admin@sonic:~$ show interface buffer queue
Interface      Queue  Profile
-----
Ethernet0      0      buffer_prof2
```

show buffer_pool

This command is used to display the watermark of shared buffer for buffer pools.

Usage

```
show buffer_pool {watermark | persistent-watermark}
```

Parameters:

- watermark: show how user watermark for buffer pools.
- persistent-watermark: show persistent watermark for buffer pools.

Example

```
admin@sonic:~$ show buffer_pool watermark
Shared pool maximum occupancy:
      Pool      Bytes
-----
egress_lossless_pool  982320
egress_lossy_pool      0
ingress_lossless_pool  919680
ingress_lossy_pool      0
admin@sonic:~$
admin@sonic:~$ show buffer_pool persistent-watermark
Shared pool maximum occupancy:
      Pool      Bytes
-----
egress_lossless_pool  982320
egress_lossy_pool      0
ingress_lossless_pool  919680
ingress_lossy_pool      0
```

Shared Buffer Configuration Commands

config buffer pool add

This command is used to create a buffer pool profile.

Usage

```
config buffer pool add {pool_name} --type {ingress | egress} --size <size> [--xoff-size <xoff-size>]
```

Parameters:

- <pool_name>: Pool name.
- type: Stage. Valid values are:
 - **ingress**: Ingress stage.
 - **egress**: Egress stage.

- <size>: The size of the shared buffer in bytes.
- <xoff-size>: The size of the shared headroom in bytes. Available only for ingress stage.

Note: If changed the buffer pool setting, it is needed to restart the switch for the setting to take effect.

Example

```
sudo config buffer pool add ingress_pool --type ingress --size 9600000 --xoff-size 300000
```

Example

```
sudo config buffer pool add egress_pool --type egress --size 8400000
```

config buffer pool del

This command is used to delete a buffer pool profile.

Usage

```
config buffer pool del {pool_name}
```

Parameters:

- <pool_name>: Pool name.

Note: If changed the buffer pool setting, it is needed to restart the switch for the setting to take effect.

Example

```
sudo config buffer pool del ingress_pool
sudo config buffer pool del egress_pool
```

config buffer profile add

This command is used to create a buffer profile.

Usage

```
config buffer profile add <profile_name> --pool <buffer_pool_name> [--size <size>] [--shared-size <shared_size> | --shared-percent <shared_percent>] [--xoff <xoff_size> --xon <xon_size> --xon-offset <xon_offset>]
```

Parameters:

- <profile_name>: Profile name.
- <buffer_pool_name>: Pool name.
- <size>: Reserved size in bytes.
- <shared_size>: Shared size in bytes.
- <shared_percent>: Shared size in percentage. The shared size is "the free size of buffer pool × shared percentage". Valid values are:
 - 0.78: 0.78% (only available on Broadcom platform).
 - 1.5: 1.5%.
 - 3: 3%.
 - 6: 6%.
 - 11: 11%.
 - 20: 20%.

- 33: 33%.
- 66: 66%.
- 80: 80%.
- 89: 89% (only available on Broadcom platform).
- <xoff_size>: Headroom size in bytes. Only for ingress lossless profile.
- <xon_size>: Resume size in bytes. Only for ingress lossless profile.
- <xon_offset_size>: Resume offset size in bytes. Only for ingress lossless profile. This size doesn't support on Intel platform.

Example

```
sudo config buffer profile add ingress_profile_lossy --pool ingress_pool --shared_percent 1 --size 0
```

config buffer profile update

This command is used to modify an existing buffer profile.

Usage

```
config buffer profile update <profile_name> [--shared-percent <shared_percent>] [--xon <xon_size>]
```

Parameters:

- <profile_name>: Profile name.
- <shared_percent>: Shared size in percentage. The shared size is "the free size of buffer pool × shared percentage". Valid values are:
 - 0.78: 0.78% (only available on Broadcom platform).
 - 1.5: 1.5%.
 - 3: 3%.
 - 6: 6%.
 - 11: 11%.
 - 20: 20%.
 - 33: 33%.
 - 66: 66%.
 - 80: 80%.
 - 89: 89% (only available on Broadcom platform).
- <xon_size>: Resume size in bytes. Only for ingress lossless profile.

Example

```
sudo config buffer profile update ingress_profile_lossy --pool ingress_pool --shared-dynamic-alpha 2 --size 100
```

config buffer profile del

This command is used to delete a buffer profile.

Usage

```
config buffer profile del <profile_name>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config buffer profile del ingress_profile_lossy
```

config interface buffer bind priority-group

This command is used to bind the buffer profile to the ingress priority group.

Usage

```
config interface buffer bind priority-group <interface_name> <pg> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <pg>: Priority Group. Valid values are:
 - Minimum: 0. Maximum: 7. (Broadcom platform)
 - Minimum: 1. Maximum: 5. (Intel platform)
- <profile_name>: Profile name.

Example

```
sudo config interface buffer bind priority-group Ethernet0 0 ingress_profile_lossy
```

config interface buffer unbind priority-group

This command is used to unbind the buffer profile from the ingress priority group.

Usage

```
config interface buffer unbind priority-group <interface_name> <pg>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <profile_name>: Profile name.

Here is an example for unbinding profile "ingress_profile_lossy" from interface "Ethernet0" pg 0.

Example

```
sudo config interface buffer unbind priority-group Ethernet0 0
```

config interface buffer bind queue

This command is used to bind the buffer profile to the egress queue.

Usage

```
config interface buffer bind queue <interface_name> <queue> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).
- <profile_name>: Profile name.

Example

```
sudo config interface buffer bind queue Ethernet0 0 egress_profile
```

config interface buffer unbind queue

This command is used to unbind the buffer profile from the egress queue.

Usage

```
config interface buffer unbind queue <interface_name> <queue>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).

Example

```
sudo config interface buffer unbind queue Ethernet0 0
```

Queue and Priority-Group

This sub-section describes some queue and priority-group parameters. It can be displayed in **show queue** commands or **show priority-group** commands.

- queue counters
- queue watermark
- priority-group

Queue and Priority-Group Show Commands

show queue counters

This command displays packet and byte counters for all queues of all ports or one specific-port specified as an argument.

Display packet and byte counters of interface(s).

Usage

```
show queue counters [<interface_name>]
```

Parameters:

- <interface name>: The interface of associated counter. Without this, it shows all ports.

Example

```
admin@sonic:~$ show queue counters
```

Port	TxQ	Counter/pkts	Counter/bytes	Drop/pkts	Drop/bytes
Ethernet0	UC0	0	0	0	0
Ethernet0	UC1	0	0	0	0
Ethernet0	UC2	0	0	0	0
Ethernet0	UC3	0	0	0	0
Ethernet0	UC4	0	0	0	0
Ethernet0	UC5	0	0	0	0
Ethernet0	UC6	0	0	0	0
Ethernet0	UC7	0	0	0	0
Ethernet0	UC8	0	0	0	0
Ethernet0	UC9	0	0	0	0
Ethernet0	MC0	0	0	0	0
Ethernet0	MC1	0	0	0	0
Ethernet0	MC2	0	0	0	0
Ethernet0	MC3	0	0	0	0
Ethernet0	MC4	0	0	0	0
Ethernet0	MC5	0	0	0	0
Ethernet0	MC6	0	0	0	0
Ethernet0	MC7	0	0	0	0
Ethernet0	MC8	0	0	0	0
Ethernet0	MC9	0	0	0	0

Port	TxQ	Counter/pkts	Counter/bytes	Drop/pkts	Drop/bytes
Ethernet4	UC0	0	0	0	0
Ethernet4	UC1	0	0	0	0
Ethernet4	UC2	0	0	0	0
Ethernet4	UC3	0	0	0	0
Ethernet4	UC4	0	0	0	0
Ethernet4	UC5	0	0	0	0
Ethernet4	UC6	0	0	0	0
Ethernet4	UC7	0	0	0	0
Ethernet4	UC8	0	0	0	0
Ethernet4	UC9	0	0	0	0
Ethernet4	MC0	0	0	0	0
Ethernet4	MC1	0	0	0	0
Ethernet4	MC2	0	0	0	0
Ethernet4	MC3	0	0	0	0
Ethernet4	MC4	0	0	0	0
Ethernet4	MC5	0	0	0	0
Ethernet4	MC6	0	0	0	0
Ethernet4	MC7	0	0	0	0
Ethernet4	MC8	0	0	0	0
Ethernet4	MC9	0	0	0	0

...

Optionally, you can specify an interface name in order to only display the particular interface.

Example

```
admin@sonic:~$ show queue counters Ethernet72
```

Port	TxQ	Counter/pkts	Counter/bytes	Drop/pkts	Drop/bytes
Ethernet72	UC0	0	0	0	0
Ethernet72	UC1	0	0	0	0
Ethernet72	UC2	0	0	0	0
Ethernet72	UC3	0	0	0	0
Ethernet72	UC4	0	0	0	0
Ethernet72	UC5	0	0	0	0
Ethernet72	UC6	0	0	0	0
Ethernet72	UC7	0	0	0	0
Ethernet72	UC8	0	0	0	0
Ethernet72	UC9	0	0	0	0
Ethernet72	MC0	0	0	0	0
Ethernet72	MC1	0	0	0	0
Ethernet72	MC2	0	0	0	0
Ethernet72	MC3	0	0	0	0
Ethernet72	MC4	0	0	0	0
Ethernet72	MC5	0	0	0	0
Ethernet72	MC6	0	0	0	0
Ethernet72	MC7	0	0	0	0
Ethernet72	MC8	0	0	0	0
Ethernet72	MC9	0	0	0	0

Note: Queue counters can be cleared by using the following command: **sonic-clear queuecounters**

show queue watermark

This command displays the user watermark or persistent-watermark for unicast or multicast queues (egress shared pool occupancy per queue) for all ports.

Usage

```
show queue {watermark | persistent-watermark} {multicast | unicast}
```

Parameters:

- **watermark**, the highest watermark after the last clean command.
- **persistent-watermark**, the highest watermark after system bring up.

Note: Both "user watermark" and "persistent watermark" can be cleared by the user.

Example: show for unitcast of watermark

```
admin@sonic:~$ show queue watermark unicast
```

Egress shared pool occupancy per unicast queue:

Port	UC0	UC1	UC2	UC3	UC4	UC5	UC6	UC7	UC8	UC9
Ethernet0	0	0	0	0	0	0	0	0	0	0
Ethernet4	416	0	0	0	0	0	0	0	0	0
Ethernet8	416	0	0	0	0	0	0	0	0	0
Ethernet12	416	0	0	0	0	0	0	0	0	0

Example: show for multicast of watermark

```
admin@sonic:~$ show queue watermark multicast
Egress shared pool occupancy per multicast queue:
  Port    MC10    MC11    MC12    MC13    MC14    MC15    MC16    MC17    MC18    MC19
-----
Ethernet0      0      0      0      0      0      0      0      0      0      0
Ethernet4    416      0      0      0      0      0      0      0      0      0
Ethernet8    416      0      0      0      0      0      0      0      0      0
Ethernet12   416      0      0      0      0      0      0      0      0      0
Ethernet16   416      0      0      0      0      0      0      0      0      0
```

Example: show for unicast of persistent-watermark

```
admin@sonic:~$ show queue persistent-watermark unicast
Egress shared pool occupancy per unicast queue:
  Port    UC0     UC1     UC2     UC3     UC4     UC5     UC6     UC7     UC8     UC9
-----
Ethernet0      0      0      0      0      0      0      0      0      0      0
Ethernet4    416      0      0      0      0      0      0      0      0      0
Ethernet8    416      0      0      0      0      0      0      0      0      0
```

Example: show for multicast of persistent-watermark

```
admin@sonic:~$ show queue persistent-watermark multicast
Egress shared pool occupancy per multicast queue:
  Port    MC10    MC11    MC12    MC13    MC14    MC15    MC16    MC17    MC18    MC19
-----
Ethernet0      0      0      0      0      0      0      0      0      0      0
Ethernet4      0      0      0      0      0      0      0      0      0      0
Ethernet8      0      0      0      0      0      0      0      0      0      0
```

show priority-group watermark

This command displays the user watermark or persistent-watermark for the Ingress "headroom" or "shared pool occupancy" per priority-group for all ports.

Usage

```
show priority-group {watermark | persistent-watermark} {headroom | shared}
```

Parameters:

- **watermark**, the highest watermark after the last clean command.
- **persistent-watermark**, the highest watermark after system bring up.
- **headroom**, the max size of headroom has been used.
- **shared**, the max size has been used in shared pool.

Note: Both "user watermark" and "persistent watermark" can be cleared by the user.

Example

```
admin@sonic:~$ show priority-group watermark shared
Ingress shared pool occupancy per PG:
  Port    PG0    PG1    PG2    PG3    PG4    PG5    PG6    PG7
-----
Ethernet0 0      0      0      0      0      0      0      0
Ethernet4 0      0      0      0      0      0      0      0
Ethernet8 0      0      0      0      0      0      0      0
Ethernet12 0      0      0      0      0      0      0      0
```

Example (Ingress headroom per PG)

```
admin@sonic:~$ show priority-group watermark headroom
Ingress headroom per PG:
  Port    PG0    PG1    PG2    PG3    PG4    PG5    PG6    PG7
-----
Ethernet0 0      0      0      0      0      0      0      0
Ethernet4 0      0      0      0      0      0      0      0
Ethernet8 0      0      0      0      0      0      0      0
```

Example (Ingress shared pool occupancy per PG)

```
admin@sonic:~$ show priority-group persistent-watermark shared
Ingress shared pool occupancy per PG:
  Port    PG0    PG1    PG2    PG3    PG4    PG5    PG6    PG7
-----
Ethernet0 0      0      0      0      0      0      0      0
Ethernet4 0      0      0      0      0      0      0      0
Ethernet8 0      0      0      0      0      0      0      0
```

Example (Ingress headroom per PG)

```
admin@sonic:~$ show priority-group persistent-watermark headroom
Ingress headroom per PG:
  Port    PG0    PG1    PG2    PG3    PG4    PG5    PG6    PG7
-----
Ethernet0 0      0      0      0      0      0      0      0
Ethernet4 0      0      0      0      0      0      0      0
Ethernet8 0      0      0      0      0      0      0      0
```

Queue and Priority-Group Clear Commands

sonic-clear queuecounters

This command will clear the packet and byte counters for all queues of all ports.

Usage

```
sonic-clear queuecounters
```

Example

```
admin@sonic:~$ sonic-clear queuecounters
Clear and update saved counters for Ethernet0
Clear and update saved counters for Ethernet1
Clear and update saved counters for Ethernet2
Clear and update saved counters for Ethernet3
```

sonic-clear queue watermark

This command will clear the user watermark or persistent-watermark for unicast or multicast queues (egress shared pool occupancy per queue) for all ports.

Usage

```
sonic-clear queue {watermark | persistent-watermark} {unicast | multicast}
```

Example: clear unicast of user watermark

```
admin@sonic:~$ sudo sonic-clear queue watermark unicast
```

Example: clear multicast of user watermark

```
admin@sonic:~$ sudo sonic-clear queue watermark multicast
```

Example: clear unicast of persistent-watermark

```
admin@sonic:~$ sudo sonic-clear queue persistent-watermark unicast
```

Example: clear multicast of persistent-watermark

```
admin@sonic:~$ sudo sonic-clear queue persistent-watermark multicast
```

sonic-clear priority-group watermark

This command will clear the user watermark or persistent-watermark for the Ingress "headroom" or "shared pool occupancy" per priority-group for all ports.

Usage

```
sonic-clear priority-group {watermark | persistent-watermark} {headroom | shared}
```

Parameters:

- **watermark**, the highest watermark after the last clean command.
- **persistent-watermark**, the highest watermark after system bring up.

- **headroom**, the max size of headroom has been used.
- **shared**, the max size has been used in shared pool.

Example: clear headroom of user watermark

```
admin@sonic:~$ sudo sonic-clear priority-group watermark headroom
```

Example: clear shared pool occupancy of user watermark

```
admin@sonic:~$ sudo sonic-clear priority-group watermark shared
```

Example: clear headroom of persistent-watermark

```
admin@sonic:~$ sudo sonic-clear priority-group persistent-watermark headroom
```

Example: clear shared pool occupancy of persistent-watermark

```
admin@sonic:~$ sudo sonic-clear priority-group persistent-watermark shared
```

(202111.2) SNMP

- **SNMP Show commands**
 - `show runningconfiguration snmp`
 - `show runningconfiguration snmp location`
 - `show runningconfiguration snmp contact`
 - `show runningconfiguration snmp community`
- **SNMP Config commands**
 - `config snmp location modify`
 - `config snmp location del`
 - `config snmp contact add`
 - `config snmp contact modify`
 - `config snmp contact del`
 - `config snmp community add`
 - `config snmp community replace`
 - `config snmp community del`

SNMP Show commands

`show runningconfiguration snmp`

This command displays the global SNMP configuration that includes the location, contact, and community settings.

Usage

```
show runningconfiguration snmp
```

Example

```
admin@sonic:~$ show runningconfiguration snmp
Location
-----
Emerald City

SNMP_CONTACT      SNMP_CONTACT_EMAIL
-----
joe                joe@contoso.com

Community String   Community Type
-----
Jack              RW
```

`show runningconfiguration snmp location`

This command displays the SNMP location setting.

Usage

```
show runningconfiguration snmp location [--json]
```

Parameters:

- **json:** Display the output in JSON format.

Example

```
admin@sonic:~$ show runningconfiguration snmp location
Location
-----
Emerald City
```

Example with json format

```
admin@sonic:~$ show runningconfiguration snmp location --json
{'Location': 'Emerald City'}
```

show runningconfiguration snmp contact

This command displays the SNMP contact setting.

Usage

```
show runningconfiguration snmp contact [--json]
```

Parameters:

- **json:** Display the output in JSON format.

Example

```
admin@sonic:~$ show runningconfiguration snmp contact
Contact      Contact Email
-----
joe           joe@contoso.com
```

Example with json format

```
admin@sonic:~$ show runningconfiguration snmp contact --json
{'joe': 'joe@contoso.com'}
```

show runningconfiguration snmp community

This command displays the SNMP community setting.

Usage

```
show runningconfiguration snmp community [--json]
```

Parameters:

- **json:** Display the output in JSON format.

Example

```
admin@sonic:~$ show runningconfiguration snmp community
Community String      Community Type
-----
Jack                  RW
```

Example with json format

```
admin@sonic:~$ show runningconfiguration snmp community --json
{'Jack': {'TYPE': 'RW'}}
```

SNMP Config commands

config snmp location modify

This command is used to modify the SNMP location.

Usage

```
config snmp location modify <location>
```

Parameters:

- <location>: SNMP location, default value is "public".

Example

```
admin@as7326-56x:~$ sudo config snmp location modify USA
```

config snmp location del

This command is used to restore the SNMP location.

Usage

```
config snmp location del <location>
```

Parameters:

- <location>: SNMP location, default value is "public".

Example

```
admin@as7326-56x:~$ sudo config snmp location del USA
```

config snmp contact add

This command is used to add the SNMP contact.

Usage

```
config snmp contact add <contact_name> <contact_email>
```

Parameters:

- <contact_name>: SNMP contact.
- <contact_email>: SNMP contact email.

Example

```
admin@sonic:~$ sudo config snmp contact add joe joe@contoso.com
```

config snmp contact modify

This command is used to modify the SNMP contact.

Usage

```
config snmp contact modify <contact> <contact_email>
```

Parameters:

- <contact>: SNMP contact.
- <contact_email>: SNMP contact email.

Example

```
admin@sonic:~$ sudo config snmp contact modify test test@contoso.com
```

config snmp contact del

This command is used to delete the SNMP contact.

Usage

```
config snmp contact del <contact>
```

Parameters:

- <contact>: SNMP contact.

Example

```
admin@sonic:~$ sudo config snmp contact del test
```

config snmp community add

This command is used to add the SNMP community.

Usage

```
config snmp community add <snmp_community> <access_type>
```

Parameters:

- <snmp_community>: SNMP community.
- <access_type>: one of the following:
 - **RO**: Read-only community.
 - **RW**: Read-write community.

Example

```
admin@sonic:~$ sudo config snmp community add testcomm ro
```

config snmp community replace

This command is used to replace the SNMP community.

Usage

```
config snmp community replace <current_community_string> <new_community_string>
```

Parameters:

- <current_community_string>: current SNMP community.
- <new_community_string>: new SNMP community.

Example

```
admin@sonic:~$ sudo config snmp community replace testcomm newtestcomm
```

config snmp community del

This command is used to delete the SNMP community.

Usage

```
config snmp community del <snmp_community>
```

Parameters:

- <snmp_community>: SNMP community.

Example

```
admin@sonic:~$ sudo config snmp community del newtestcomm
```

(202111.2) SNMP Traps

- [SNMP Trap Show Commands](#)
 - [show snmptrap](#)
- [SNMP Trap Config Commands](#)
 - [config snmptrap modify](#)
 - [config snmptrap del](#)

SNMP Trap Show Commands

show snmptrap

This command displays the configured SNMP trap server IP addresses.

Usage

```
show snmptrap
```

Example

```
admin@sonic:~$ show snmptrap
Version  TrapReceiverIP      Port  VRF    Community
-----  -
          2    31.31.31.31      456  mgmt   public
```

SNMP Trap Config Commands

config snmptrap modify

This command configures the trap server IP address and some options for the traps.

Usage

```
config snmptrap modify <snmp version> [-p <port>] [-v <vrf name>] [-c <community>] <trap server ip>
```

Parameters:

- <snmp version>: SNMP version. Each SNMP version can be configured with one trap server IP address.
 - 1
 - 2
- <port>: the port to receive the SNMP traps. (default: 162)
- <vrf name>: VRF name; using this VRF name, the SNMP agent sends the trap through the <vrf name> domain. The option can be one of following VRF types:
 - **mgmt**, the keyword to specify management VRF
 - **none**, the keyword to clear the VRF specified before
 - **<name>**, specifies the name of one data VRF
- <community>
- <trap server ip>: The server IP address that receives the trap

Note:

- The configuration is stored in **snmpd.conf**.
- The SNMP service is restarted to take the configuration effect.

Example

```
admin@sonic:~$ sudo config snmptrap modify 2 -p 456 -v mgmt 21.21.21.21

trap2sink 21.21.21.21:456%mgmt public
```

In this example, a user configures a trap server address (21.21.21.21) for SNMP version 2, changes the port (456), and specifies the management VRF domain. This configuration is stored in **/etc/snmp/snmpd.conf** inside of the SNMP docker.

In the output information, the "%mgmt" would not be present if the "-v" option is not specified.

In the case of SNMPv1, "trapsink" will be updated, and in the case of v2, "trap2sink" will be updated.

config snmptrap del

This command deletes the trap server configuration of the specified SNMP version. This configuration is removed from **snmpd.conf** and the SNMP service restarted to make this configuration effective in the SNMP agent.

Usage

```
config snmptrap del <snmp_version>
```

Parameters:

- <snmp version>, the version to be deleted.

Example

```
admin@sonic:~$ sudo config snmptrap del 2
```

(202111.2) Software Installation and Management

- SONiC Installer
 - sonic-installer list
 - sonic-installer install(Updated)
 - sonic-installer set-default
 - sonic-installer set-nex-boot
 - sonic-installer remove
 - sonic-installer cleanup
 - sonic_installer binary_version (Not Supported)
 - sonic_installer rollback_docker (Not Supported)
 - sonic_installer upgrade_docker (Not Supported)

The SONiC software can be installed by two methods; using the "sonic_installer" tool or "ONIE Installer".

SONiC Installer

This is a command line tool available as part of the SONiC software; If the device is already running the SONiC software, this tool can be used to install an alternate image in the partition. This tool has the facility to install an alternate image, list the available images, and set the next reboot image. This command requires elevated (root) privileges to run.

sonic-installer list

This command displays information about currently installed images. It displays a list of installed images, the currently running image, and the image set to be loaded at the next reboot.

Usage:

```
sonic-installer list
```

Example:

```
admin@tor:~$ sudo sonic-installer list
Current: Edgecore-SONiC_20220819_084543_ec202111_184
Next: Edgecore-SONiC_20220819_084543_ec202111_184
Available:
Edgecore-SONiC_20220819_084543_ec202111_184
Edgecore-SONiC_20220526_081320_ec202111_117
```



This output can be obtained without root privileges by running the show boot command.

sonic-installer install(Updated)

This command is used to install a new image in the alternate image partition. This command takes a path to an installable SONiC image or URL and installs the image.

Usage:

```
sonic-installer install [OPTIONS] <image_file_path>
```

Options:

- -f, --force Forces installation of an image of a type that differs from that of the currently running image
- --skip-platform-check Force installation of an image of a type which is not of the same platform

- `--skip_migration` image Do not migrate the current configuration to the newly installed image
- `--skip-package-migration` Do not migrate current packages to the newly installed image
- `--skip-setup-swap` Skip setup temporary SWAP memory used for installation
- `--swap-mem-size` INTEGER SWAP memory space size [default: (1024 MiB)] NOTE: this argument is mutually exclusive with arguments: `skip_setup_swap`
- `--available-mem-threshold` INTEGER If system available memory is lower than threshold, setup SWAP memory [default: (1200 MiB)] NOTE: this argument is mutually exclusive with arguments: `skip_setup_swap`
- `--help` Show this message and exit

Note

- By default, the current configuration will be migrated to the newly installed image. The option "`--skip_migration`" can be applied to omit the migration of configuration.
- If a kernel panic has ever occurred on the running of the current image, there will be `dmesg` files in `/sys/fs/pstore/`, which can be used for debugging. It is strongly recommended to have a backup before installing a new image. Also, remember to note down the current image version so the `dmesg` files can be correlated to the correct image version.
- The format of files in the `/sys/fs/pstore/` may not be compatible with all Linux kernel versions, which could result in error logs that read "pstore: crypto comp decompress failed, ret = -22!" when the system boots. These files will be deleted after the first boot with the freshly loaded image to prevent them from showing harmless error logs that can confuse users.

Example:

```
admin@sonic:~$ sudo sonic-installer install https://sonic-jenkins.westus.cloudapp.azure.com/job/xxxx/job/buildimage-xxxx-all/xxx/artifact/target/sonic-xxxx.bin
New image will be installed, continue? [y/N]: y
Downloading image...
...100%, 480 MB, 3357 KB/s, 146 seconds passed
Command: /tmp/sonic_image
Verifying image checksum ... OK.
Preparing image archive ... OK.
ONIE Installer: platform: XXXX
onie_platform:
Installing SONiC in SONiC
Installing SONiC to /host/image-xxxx
Directory /host/image-xxxx/ already exists. Cleaning up...
Archive: fs.zip
creating: /host/image-xxxx/boot/
inflating: /host/image-xxxx/boot/vmlinuz-3.16.0-4-amd64
inflating: /host/image-xxxx/boot/config-3.16.0-4-amd64
inflating: /host/image-xxxx/boot/System.map-3.16.0-4-amd64
inflating: /host/image-xxxx/boot/initrd.img-3.16.0-4-amd64
creating: /host/image-xxxx/platform/
extracting: /host/image-xxxx/platform/firsttime
inflating: /host/image-xxxx/fs.squashfs
inflating: /host/image-xxxx/dockerfs.tar.gz
Log file system already exists. Size: 4096MB
Installed SONiC base image SONiC-OS successfully
Command: cp /etc/sonic/minigraph.xml /host/
Command: grub-set-default --boot-directory=/host 0
Done
```

sonic-installer set-default

This command is used to change the image that can be loaded by default in all the subsequent reboots.

Usage:

```
sonic-installer set-default <image_name>
```

Example:

```
admin@sonic:~$ sudo sonic-installer set-default SONiC-OS-HEAD.XXXX
```

sonic-installer set-nex-boot

This command is used to change the image that can be loaded in the *next* reboot only. Note that following the *next* reboot, it will always fall back to the current image.

Usage:

```
sonic-installer set-next-boot <image_name>
```

Example:

```
admin@sonic:~$ sudo sonic-installer set-next-boot SONiC-OS-HEAD.XXXX
```

sonic-installer remove

This command is used to remove the unused SONiC image from the disk. Note that it's *not* allowed to remove the currently running image.

Usage:

```
sonic-installer remove [y|-yes] <image_name>
```

Example:

```
admin@sonic:~$ sudo sonic-installer remove SONiC-OS-HEAD.YYYY
Image will be removed, continue? [y/N]: y
Updating GRUB...
Done
Removing image root filesystem...
Done
Command: grub-set-default --boot-directory=/host 0
Image removed
```

sonic-installer cleanup

This command removes all unused images from the device, leaving only the currently active image and the image that will be booted next (if different). If there are no images that can be removed, the command will output "No image(s) to remove".

Usage:

```
sonic-installer cleanup [y|-yes]
```

Example:

```
admin@sonic:~$ sudo sonic-installer cleanup
Remove images which are not current and next, continue? [y/N]: y
No image(s) to remove
```

sonic_installer binary_version (Not Supported)

This command is used to get the version from the local binary image file.

sonic_installer rollback_docker (Not Supported)

This command is used to rollback the docker image to a previous version.

sonic_installer upgrade_docker (Not Supported)

This command is used to upgrade a docker image from a local binary or URL.

(202111.2) Startup & Running Configuration

- [Startup Configuration](#)
 - `show startupconfiguration bgp`
- [Running Configuration](#)
 - `show runningconfiguration bgp`
 - `show runningconfiguration interfaces`
 - `show runningconfiguration ntp`
 - `show runningconfiguration snmp`
 - `show runningconfiguration all`
 - `show runningconfiguration acl`
 - `show runningconfiguration ports`
 - `show runningconfiguration syslog`

Startup Configuration

`show startupconfiguration bgp`

This command is used to display the startup configuration for the BGP module.

Usage
<code>show startupconfiguration bgp</code>

Example

```
admin@sonic:~$ show startupconfiguration bgp
Routing-Stack is: frr
!
! template: bgpd/bgpd.conf.j2
!
!
! ===== Managed by sonic-cfggen DO NOT edit manually! =====
! generated by templates/quagga/bgpd.conf.j2 with config DB data
! file: bgpd.conf
!
!
! template: common/daemons.common.conf.j2
!
hostname as5812-54x
password zebra
enable password zebra
!
log syslog informational
log facility local4
!
! end of template: common/daemons.common.conf.j2!
agentx
!
!
! template: bgpd/bgpd.main.conf.j2
!
! bgp multiple-instance
!
! BGP configuration
!
! TSA configuration
!
ip prefix-list PL_LoopbackV4 permit 10.1.0.32/32
!
ipv6 prefix-list PL_LoopbackV6 permit fc00:1::/64
!
!
!
router bgp 65100
!
    bgp log-neighbor-changes
    no bgp default ipv4-unicast
!
    bgp bestpath as-path multipath-relax
!
    bgp graceful-restart restart-time 240
    bgp graceful-restart
    bgp graceful-restart preserve-fw-state
...

```

Running Configuration

This sub-section explains the show commands for displaying the running configuration for the following modules.

1. BGP
2. Interfaces
3. NTP
4. SNMP
5. ALL
6. ACL
7. Ports

8. Syslog

show runningconfiguration bgp

This command displays the running configuration of the BGP module.

Usage

```
show runningconfiguration bgp
```

Example

```
admin@sonic:~$ show runningconfiguration bgp
Building configuration...

Current configuration:
!
frr version 7.2.1-sonic
frr defaults traditional
hostname as5812-54x
log syslog informational
log facility local4
agentx
no service integrated-vtysh-config
!
enable password zebra
password zebra
!
ip route 0.0.0.0/0 10.250.0.1 200
!
router bgp 65100
  bgp router-id 10.1.0.32
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
  bgp graceful-restart restart-time 240
  bgp graceful-restart
  bgp graceful-restart preserve-fw-state
  bgp bestpath as-path multipath-relax
  neighbor PEER_V4 peer-group
...

```

show runningconfiguration interfaces

This command displays the running configuration for the interfaces.

Usage

```
show runningconfiguration interfaces
```

Example

```
admin@sonic:~$ show runningconfiguration interfaces
```

```
{  
  "Ethernet0": {},  
  "Ethernet1": {},  
  "Ethernet2": {},  
  "Ethernet3": {},  
  "Ethernet4": {},  
  "Ethernet5": {},  
  "Ethernet6": {},  
  "Ethernet7": {},  
  "Ethernet8": {},  
  "Ethernet9": {},  
  "Ethernet10": {},  
  "Ethernet11": {},  
  "Ethernet12": {},  
  "Ethernet13": {},  
  "Ethernet14": {},  
  "Ethernet15": {},  
  "Ethernet16": {},  
  "Ethernet17": {},  
  "Ethernet18": {},  
  "Ethernet19": {},  
  "Ethernet20": {},  
  "Ethernet21": {},  
  "Ethernet22": {},  
  "Ethernet23": {},  
  "Ethernet24": {},  
  "Ethernet25": {},  
  "Ethernet26": {},  
  "Ethernet27": {},  
  "Ethernet28": {},  
  "Ethernet29": {},  
  "Ethernet30": {},  
  "Ethernet31": {},  
  "Ethernet0|10.0.0.0/31": {},  
  "Ethernet0|FC00::1/126": {},  
  "Ethernet1|10.0.0.2/31": {},  
  "Ethernet1|FC00::5/126": {},  
  "Ethernet2|10.0.0.4/31": {},  
  ...  
}
```

show runningconfiguration ntp

This command displays the running configuration of the NTP module.

Usage

```
show runningconfiguration ntp
```

Example

```
admin@sonic:~$ show runningconfiguration ntp
NTP Servers
-----
1.1.1.1
2.2.2.2
```

show runningconfiguration snmp

This command displays the running configuration of the SNMP module.

Usage

```
show runningconfiguration snmp
```

Example

```
admin@sonic:~$ show runningconfiguration snmp
#####
# Managed by sonic-config-engine
#####
#
# EXAMPLE.conf:
#   An example configuration file for configuring the Net-SNMP agent ('snmpd')
#   See the 'snmpd.conf(5)' man page for details
#
# Some entries are deliberately commented out, and will need to be explicitly activated
#
#####
#
# AGENT BEHAVIOUR
#

# Listen for connections on all ip addresses, including eth0, ipv4 lo
#
agentAddress udp:161
agentAddress udp6:161

#####
#
# ACCESS CONTROL
#

view systemonly included .1.3.6.1.2.1.1          # system + hrSystem groups only
view systemonly included .1.3.6.1.2.1.25.1

rocommunity public          # Default access to basic system info
rocommunity6 public

#####
#
# SYSTEM INFORMATION
#

# Note that setting these values here, results in the corresponding MIB objects being 'read-only'
# See snmpd.conf(5) for more details
sysLocation    public
sysContact     Azure Cloud Switch vteam <linuxnetdev@microsoft.com>
               # Application + End-to-End layers
sysServices    72

...
```

show runningconfiguration all

This command displays the entire running configuration.

Usage

```
show runningconfiguration all
```


Example

```
admin@sonic:~$ show runningconfiguration all
```

```
{
  "ACL_TABLE": {
    "DATAACL": {
      "policy_desc": "DATAACL",
      "ports": [
        "Ethernet16",
        "Ethernet0",
        "Ethernet17",
        "Ethernet1",
        "Ethernet18",
        "Ethernet2",
        "Ethernet19",
        "Ethernet3",
        "Ethernet20",
        "Ethernet4",
        "Ethernet21",
        "Ethernet5",
        "Ethernet22",
        "Ethernet6",
        "Ethernet23",
        "Ethernet7",
        "Ethernet24",
        "Ethernet8",
        "Ethernet25",
        "Ethernet9",
        "Ethernet26",
        "Ethernet10",
        "Ethernet27",
        "Ethernet11",
        "Ethernet28",
        "Ethernet12",
        "Ethernet29",
        "Ethernet13",
        "Ethernet30",
        "Ethernet14",
        "Ethernet31",
        "Ethernet15"
      ],
      "stage": "ingress",
      "type": "L3"
    },
    "EVERFLOW": {
      "policy_desc": "EVERFLOW",
      "ports": [
        "Ethernet10",
        "Ethernet9",
        "Ethernet16",
        "Ethernet15",
        "Ethernet14",
        "Ethernet13",
        "Ethernet12",
        "Ethernet11",
        "Ethernet18",
        "Ethernet17",
        "Ethernet2",
        "Ethernet1",
        "Ethernet0",
        "Ethernet6",
        "Ethernet5",
        "Ethernet4",
        "Ethernet3",
        "Ethernet8",
        "Ethernet7",
```

```

        "Ethernet21",
        "Ethernet22",
        "Ethernet19",
        "Ethernet20",
        "Ethernet25",
        "Ethernet26",
        "Ethernet23",
        "Ethernet24",
        "Ethernet27",
        "Ethernet28",
        "Ethernet30",
        "Ethernet29",
        "Ethernet31"
    ],
    "stage": "ingress",
    "type": "MIRROR"
},
...

```

show runningconfiguration acl

This command displays the running configuration of the ACLs.

Usage

```
show runningconfiguration acl
```

Example

```

root@sonic:~# show runningconfiguration acl
{
  "TEST1|DEFAULT_RULE": {
    "ETHER_TYPE": "2048",
    "PACKET_ACTION": "DROP",
    "PRIORITY": "1"
  },
  "TEST1|RULE_1": {
    "PACKET_ACTION": "FORWARD",
    "PRIORITY": "9999",
    "SRC_IP": "20.0.0.2/32"
  }
}

```

show runningconfiguration ports

This command displays the running configuration of the Ports.

Usage

```
show runningconfiguration ports [<portname>]
```

Example

```
admin@sonic:~$ show runningconfiguration ports
{
  "Ethernet0": {
    "admin_status": "up",
    "alias": "tenGigE1",
    "description": "ARISTA01T2:Ethernet1",
    "index": "1",
    "lanes": "13",
    "mtu": "9100",
    "pfc_asym": "off",
    "speed": "10000"
  },
  "Ethernet1": {
    "admin_status": "up",
    "alias": "tenGigE2",
    "description": "ARISTA02T2:Ethernet1",
    "index": "2",
    "lanes": "14",
    "mtu": "9100",
    "pfc_asym": "off",
    "speed": "10000"
  },
  "Ethernet2": {
    ...
  }
}

admin@sonic:~$ show runningconfiguration ports Ethernet0
{
  "Ethernet0": {
    "index": "1",
    "lanes": "13",
    "description": "ARISTA01T2:Ethernet1",
    "admin_status": "up",
    "mtu": "9100",
    "alias": "tenGigE1",
    "pfc_asym": "off",
    "speed": "10000"
  }
}
```

show runningconfiguration syslog

This command displays the running configuration of the Syslog module.

Usage

```
show runningconfiguration syslog
```

Example

```
admin@sonic:~$ show runningconfiguration syslog
Syslog Servers
-----
[10.250.1.44]
```

**Note**

For all "show startupconfiguration" and "show runningconfiguration" commands, only the partial output is shown here. In the realtime, all the configuration information is displayed.

(202111.2) Syslog

- [Syslog Configuration Commands](#)
 - [config syslog add](#)
 - [config syslog delete](#)

Syslog Configuration Commands

The **config syslog** command is used to add or remove the configured syslog servers.

config syslog add

This command is used to add a new syslog server to the Syslog server list. Note that more than one syslog server can be added to the list.

Usage

```
config syslog add <ipv4-address | ipv6-address>
```

Parameters:

- **ipv4-address** : IP address of the collector in dotted decimal format for IPv4
- **ipv6-address** : x:x:x:x::x format for IPv6 address of the collector (where :: notation specifies successive hexadecimal fields of zeros)

The following example shows how to add a remote syslog server with the ip address 1.1.1.1.

Example

```
admin@sonic:~$ sudo config syslog add 1.1.1.1
Syslog server 1.1.1.1 added to configuration
Restarting rsyslog-config service...
```

config syslog delete

This command is used to delete a configured syslog server.

Usage

```
config syslog del <ipv4-address | ipv6-address>
```

Parameters:

- **ipv4-address** : IP address of the collector in dotted decimal format for IPv4
- **ipv6-address** : x:x:x:x::x format for IPv6 address of the collector (where :: notation specifies successive hexadecimal fields of zeros)

The following example shows how to delete a remote syslog server with the ip address 1.1.1.1.

Example

```
admin@sonic:~$ sudo config syslog del 1.1.1.1
Syslog server 1.1.1.1 removed from configuration
Restarting rsyslog-config service...
```

Note

The format of the syslog messages sent to configured syslog servers is defined in RFC3164. The following packet opened in Wireshark is an example. At the header of the syslog message, the facility and severity levels are encoded and placed between angle brackets ("**<**" and "**>**").

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.250.0.168	10.250.2.44	Syslog	307	USER.INFO: Mar 9 10:48:33.458161 as7726-32x-4 INFO syncd#/su

> Frame 1: 307 bytes on wire (2456 bits), 307 bytes captured (2456 bits)
 > Ethernet II, Src: Edgecore_21:ac:0f (90:3c:b3:21:ac:0f), Dst: ae:41:e1:45:ff:95 (ae:41:e1:45:ff:95)
 > Internet Protocol Version 4, Src: 10.250.0.168, Dst: 10.250.2.44
 > User Datagram Protocol, Src Port: 60998, Dst Port: 514
 ✓ [truncated]Syslog message: USER.INFO: Mar 9 10:48:33.458161 as7726-32x-4 INFO syncd#/supervisord: syncd 0:_bcm_esw_linkscan_thread: LINK
 0000 1... = Facility: USER - random user-level messages (1)
 110 = Level: INFO - informational (6)
 Message [truncated]: Mar 9 10:48:33.458161 as7726-32x-4 INFO syncd#/supervisord: syncd 0:_bcm_esw_linkscan_thread: LINKSCAN interval mc

0020	02 2c ee 46 02 02 01 11 2d e5 3c 31 34 3e 4d 61	...Facility: <14>Ma
0030	72 20 20 39 20 31 30 3a 34 38 3a 33 33 2e 34 35	... 9 10: 48:33.45
0040	38 31 36 31 20 61 73 37 37 32 36 2d 33 32 78 2d	8161 as7 726-32x-

(202111.2) Subport

- [Subport Show commands](#)
 - [show subinterfaces status](#)
- [Subport Configuration commands](#)
 - [config interface ip add/remove](#)
 - [config interface vrf bind/unbind](#)
 - [Different command sequence](#)
 - [config interface startup/shutdown](#)

Subport Show commands

show subinterfaces status

This command display the Subport configuration and status.

Usage

```
show subinterfaces status
```

Example

```
admin@sonic:~# show subinterfaces status
```

Sub port interface	Speed	MTU	Vlan	Admin	Parent admin	Type
Ethernet4.10	100G	9100	10	up	up	802.1q-encapsulation
Ethernet4.4094	100G	9100	4094	up	up	802.1q-encapsulation

Subport Configuration commands

A subport interface is added (or removed) when it's configured with IP address ("[config interface ip](#)") or bound on a VRF ("[config interface vrf](#)"). If user does not specify the <vrf name>, the L3 configuration on subport interface is bound on default VRF.

User should be careful in configuring sequence of "config interface vrf" and "config interface ip". (please refer example)

config interface ip add/remove

Configuring IP address on subport is the same as [config interface ip add/remove command](#).

This command will automatically add (or remove) subport interface if the subport interface does not exist (or already existed). The VLAN ID in subport interface name specifies the 802.1q Vlan tagging. The ingress and egress packets of subport interface should be tagged with the VLAN ID. For example, for sub port interface Ethernet4.20, Vlan 20 is used for 802.1q Vlan tagging.

Usage

```
config interface ip {add|remove} <sub_port_interface_name> <ip_addr>
```

Parameters

- sub_port_interface_name: specify subport to add (or remove) subport interface

Example

```
admin@sonic:~# sudo config interface ip add Ethernet4.20 192.169.1.1/24
admin@sonic:~# sudo config interface ip remove Ethernet4.20 192.169.1.1/24
```

config interface vrf bind/unbind

This command is used to bind (or unbind) subport interface to a VRF name space. If a subport interface is not explicitly specified a VRF name, it would be bound on default VRF.

Usage

```
config interface vrf bind <sub_port_interface_name> <vrf_name>
config interface vrf unbind <sub_port_interface_name>
```

Note,

- when binding a subport interface to a <named vrf>, it will clear IP address assigned to the subport interface if there is.

Example

```
admin@sonic:~# sudo config vrf add Vrf1
admin@sonic:~# sudo config interface vrf bind Ethernet1.10 Vrf1
admin@sonic:~# sudo config interface vrf unbind Ethernet1.10
```

Different command sequence

Here are two examples, which has different sequence of commands which introduced different result.

Following example, the IP address (192.169.1.1) and a static route are bound on Vrf1

Example: L3 rif on Vrf1

```
admin@sonic:~# sudo config vrf add Vrf1
admin@sonic:~# sudo config interface vrf bind Ethernet1.10 Vrf1
admin@sonic:~# sudo config interface ip add Ethernet1.10 192.169.1.1/24
admin@sonic:~# sudo config route add prefix vrf Vrf1 12.12.12.0/24 nexthop 192.168.1.20
```

Following example, the IP address is cleared when the subport is bound on Vrf1, and the static route is inactive.

Example: IP address is clean

```
admin@sonic:~# sudo config interface ip add Ethernet1.10 192.169.1.1/24
admin@sonic:~# sudo config route add prefix 12.12.12.0/24 nexthop 192.169.1.20
admin@sonic:~# sudo config vrf add Vrf1
admin@sonic:~# sudo config interface vrf bind Ethernet1.10 Vrf1
```

config interface startup/shutdown

Change admin status on a subport is the same as [config interface startup/shutdown command](#).

Usage

```
config interface {startup|shutdown} <sub_port_interface_name>
```


Example

```
admin@sonic:~# sudo config interface startup Ethernet4.20
admin@sonic:~# sudo config interface shutdown Ethernet4.20
admin@sonic:~# sudo config interface startup PortChannel1.20
admin@sonic:~# sudo config interface shutdown PortChannel1.20
```

Note: In 202111, the sub interface admin status is dependent on the parent interface. ex: config interface shutdown Ethernet4. The admin status of Ethernet4.10 in 'show subinterfaces status' is also down.

(202111.2) System State

- [Processes](#)
 - [show processes cpu](#)
 - [show processes memory](#)
 - [show processes summary](#)
- [Services & Memory](#)
 - [show services](#)
 - [show system-memory](#)
 - [show mmu](#)
 - [show line \(Not Supported\)](#)
- [System-Health](#)
 - [show system-health detail](#)
 - [show system-health monitor-list](#)
 - [show system-health summary](#)

Processes

This command is used to determine the CPU utilization. It also lists the active processes along with their corresponding process ID and other relevant parameters.

This sub-section explains the various "processes" specific data that includes the following.

- [cpu](#) Show processes CPU info
- [memory](#) Show processes memory info
- [summary](#) Show processes info

The "show processes" commands provide a wrapper over Linux's "top" command.

The "show process cpu" command sorts the processes being displayed by cpu-utilization, whereas "show process memory" does it for processes' memory-utilization.

show processes cpu

This command displays the current CPU usage by process. This command uses Linux's "top -bn 1 -o %CPU" command to display the output.

Usage

```
show processes cpu
```

Tip: Users can pipe the output to "head" to display only the "n" number of lines (e.g., show processes cpu | head -n 10).

Example

```
admin@sonic:~$ show processes cpu
top - 23:50:08 up 1:18, 1 user, load average: 0.25, 0.29, 0.25
Tasks: 161 total, 1 running, 160 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.8 us, 1.0 sy, 0.0 ni, 95.1 id, 0.1 wa, 0.0 hi, 0.0 si, 0.0 st
KiB Mem: 8181216 total, 1161060 used, 7020156 free, 105656 buffers
KiB Swap: 0 total, 0 used, 0 free. 557560 cached Mem

  PID USER      PR  NI   VIRT   RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 2047 root        20   0 683772 109288 39652 S   23.8   1.3   7:44.79 syncd
 1351 root        20   0 43360   5616  2844 S   11.9   0.1   1:41.56 redis-server
10093 root        20   0 21944   2476  2088 R    5.9   0.0   0:00.03 top
   1 root        20   0 28992   5508  3236 S    0.0   0.1   0:06.42 systemd
   2 root        20   0      0      0      0 S    0.0   0.0   0:00.00 kthreadd
   3 root        20   0      0      0      0 S    0.0   0.0   0:00.56 ksoftirqd/0
   5 root         0 -20      0      0      0 S    0.0   0.0   0:00.00 kworker/0:0H
```

Tip: Advanced users can view individual processes using variations of the **ps** command (e.g., ps -ax | grep <process name>).

show processes memory

This command displays the current memory usage by processes. This command uses Linux's "top -bn 1 -o %MEM" command to display the output.

Usage

```
show processes memory
```

Note:

- the pipe option can be used using " | head -n" to display only the "n" number of lines.

Example

```
admin@sonic:~$ show processes memory
top - 23:41:24 up 7 days, 39 min, 2 users, load average: 1.21, 1.19, 1.18
Tasks: 191 total, 2 running, 189 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.8 us, 20.7 sy, 0.0 ni, 76.3 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
KiB Mem : 8162264 total, 5720412 free, 945516 used, 1496336 buff/cache
KiB Swap: 0 total, 0 free, 0 used. 6855632 avail Mem

  PID USER      PR  NI   VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
18051 root        20   0 851540 274784  8344 S   0.0   3.4   0:02.77 syncd
17760 root        20   0 1293428 259212 58732 S   5.9   3.2 96:46.22 syncd
  508 root        20   0 725364 76244 38220 S   0.0   0.9 4:54.49 dockerd
30853 root        20   0 96348 56824 7880 S   0.0   0.7 0:00.98 show
17266 root        20   0 509876 49772 30640 S   0.0   0.6 0:06.36 docker
24891 admin      20   0 515864 49560 30644 S   0.0   0.6 0:05.54 docker
17643 admin      20   0 575668 49428 30628 S   0.0   0.6 0:06.29 docker
23885 admin      20   0 369552 49344 30840 S   0.0   0.6 0:05.57 docker
18055 root        20   0 509076 49260 30296 S   0.0   0.6 0:06.36 docker
17268 root        20   0 371120 49052 30372 S   0.0   0.6 0:06.45 docker
 1227 root        20   0 443284 48640 30100 S   0.0   0.6 0:41.91 docker
23785 admin      20   0 443796 48552 30128 S   0.0   0.6 0:05.58 docker
17820 admin      20   0 435088 48144 29480 S   0.0   0.6 0:06.33 docker
  506 root        20   0 1151040 43140 23964 S   0.0   0.5 8:51.08 containerd
18437 root        20   0 84852 26388 7380 S   0.0   0.3 65:59.76 python3.6
```

show processes summary

This command displays the current summary information about all the processes.

Usage

```
show processes summary
```

Example

```
admin@sonic:~$ show processes summary
PID  PPID CMD                %MEM %CPU
1     0  /sbin/init             0.0  0.0
2     0  [kthreadd]             0.0  0.0
3     2  [ksoftirqd/0]          0.0  0.0
5     2  [kworker/0:0H]         0.0  0.0
```

Services & Memory

These commands are used to know the services that are running and the memory that is currently utilized.

show services

This command displays the state of all the SONiC processes running inside a docker container. This helps to identify the status of SONiC's critical processes.

Usage

```
show services
```

Example

```
admin@sonic:~$ show services
dhcp_relay      docker
-----
UID      PID  PPID  C  STIME TTY      TIME CMD
root      1    0    0  05:26 ?      00:00:12 /usr/bin/python /usr/bin/supervi
root     24    1    0  05:26 ?      00:00:00 /usr/sbin/rsyslogd -n

nat      docker
-----
USER      PID PPID  C  STIME TTY      TIME CMD
root      1    0    0  05:26 ?      00:00:12 /usr/bin/python /usr/bin/supervisord
root     18    1    0  05:26 ?      00:00:00 /usr/sbin/rsyslogd -n
root     23    1    0  05:26 ?      00:00:01 /usr/bin/natmgrd
root     34    1    0  05:26 ?      00:00:00 /usr/bin/natsyncd

snmp      docker
-----
UID      PID  PPID  C  STIME TTY      TIME CMD
root      1    0    0  05:26 ?      00:00:16 /usr/bin/python /usr/bin/supervi
root     24    1    0  05:26 ?      00:00:02 /usr/sbin/rsyslogd -n
Debian-+ 29    1    0  05:26 ?      00:00:04 /usr/sbin/snmpd -f -LS4d -u Debi
root     31    1    1  05:26 ?      00:15:10 python3.6 -m sonic_ax_impl

syncd      docker
-----
UID      PID  PPID  C  STIME TTY      TIME CMD
root      1    0    0  05:26 ?      00:00:13 /usr/bin/python /usr/bin/supervi
root     12    1    0  05:26 ?      00:00:00 /usr/sbin/rsyslogd -n
root     17    1    0  05:26 ?      00:00:00 /usr/bin/dsserve /usr/bin/syncd
root     27    17   22  05:26 ?      04:09:30 /usr/bin/syncd --diag -p /usr/sh
root     51    27    0  05:26 ?      00:00:01 /usr/bin/syncd --diag -p /usr/sh

swss      docker
-----
UID      PID  PPID  C  STIME TTY      TIME CMD
root      1    0    0  05:26 ?      00:00:29 /usr/bin/python /usr/bin/supervi
root     25    1    0  05:26 ?      00:00:00 /usr/sbin/rsyslogd -n
root     30    1    0  05:26 ?      00:00:13 /usr/bin/orchagent -d /var/log/s
root     42    1    1  05:26 ?      00:12:40 /usr/bin/portsyncd -p /usr/share
root     45    1    0  05:26 ?      00:00:00 /usr/bin/intfsyncd
root     48    1    0  05:26 ?      00:00:03 /usr/bin/neighsyncd
root     59    1    0  05:26 ?      00:00:01 /usr/bin/vlanmgrd
root     92    1    0  05:26 ?      00:00:01 /usr/bin/intfmgrd
root    3606    1    0  23:36 ?      00:00:00 bash -c /usr/bin/arp_update; sle
root    3621  3606    0  23:36 ?      00:00:00 sleep 300
...
```

show system-memory

This command displays the system-wide memory utilization information - just a wrapper over the Linux native "free" command.

Usage

```
show system-memory
```

Example

```
admin@sonic:~$ show system-memory
Command: free -m -h
```

	total	used	free	shared	buffers	cached
Mem:	3.9G	2.0G	1.8G	33M	324M	791M
-/+ buffers/cache:		951M	2.9G			
Swap:	0B	0B	0B			

show mmu

This command displays the virtual address to physical address translation status of the Memory Management Unit (MMU).

Usage

```
show mmu
```

Example

```
admin@sonic:~$ show mmu
Pool: ingress_lossless_pool
----
xoff 4194112
type ingress
mode dynamic
size 10875072
----

Pool: egress_lossless_pool
----
type egress
mode static
size 15982720
----

Pool: egress_lossy_pool
----
type egress
mode dynamic
size 9243812
----

Profile: egress_lossy_profile
-----
dynamic_th 3
pool [BUFFER_POOL|egress_lossy_pool]
size 1518
-----

Profile: pg_lossless_100000_300m_profile
-----
xon_offset 2288
dynamic_th -3
xon 2288
xoff 268736
pool [BUFFER_POOL|ingress_lossless_pool]
size 1248
```

```

-----
Profile: egress_lossless_profile
-----
static_th  3995680
pool       [BUFFER_POOL|egress_lossless_pool]
size       1518
-----

Profile: pg_lossless_100000_40m_profile
-----
xon_offset 2288
dynamic_th -3
xon        2288
xoff       177632
pool       [BUFFER_POOL|ingress_lossless_pool]
size       1248
-----

Profile: ingress_lossy_profile
-----
dynamic_th  3
pool       [BUFFER_POOL|ingress_lossless_pool]
size       0
-----

Profile: pg_lossless_40000_40m_profile
-----
xon_offset 2288
dynamic_th -3
xon        2288
xoff       71552
pool       [BUFFER_POOL|ingress_lossless_pool]
size       1248
-----

```

show line (Not Supported)

This command displays the serial port or a virtual network connection status. This command is used only when SONiC is used as a console switch. This command is not applicable when SONiC used as a regular switch.

Note: This command is not working. It crashes as follows. A bug ticket is opened for this issue.

Usage

```
show line
```

Example

```
admin@sonic:~$ show line
```

System-Health

System health monitor is intended to monitor both critical services/processes and peripheral device status and leverage system log, system status LED to and CLI command output to indicate the system status.

show system-health detail

This command displays the system-health detail information.

Usage

```
show system-health detail
```

Example

```
admin@wedge100bf-32x-1:~$ sudo show system-health detail
System status summary
```

System status LED

Services:

Status: OK

Hardware:

Status: Not OK

Reasons: Invalid temperature data for PSU 2, temperature=N/A, threshold=N/A
PSU 1 is out of power

System services and devices monitor list

Name	Status	Type
-----	-----	-----
wedge100bf-32x-1	OK	System
rsyslog	OK	Process
root-overlay	OK	Filesystem
var-log	OK	Filesystem
routeCheck	OK	Program
diskCheck	OK	Program
container_checker	OK	Program
vnetRouteCheck	OK	Program
container_memory_telemetry	OK	Program
telemetry:telemetry	OK	Process
telemetry:dialout	OK	Process
snmp:snmpd	OK	Process
snmp:snmp-subagent	OK	Process
sflow:sflowmgrd	OK	Process
lldp:lldpd	OK	Process
lldp:lldp-syncd	OK	Process
lldp:lldpmgrd	OK	Process
nat:natmgrd	OK	Process
nat:natsyncd	OK	Process
syncd:syncd	OK	Process
teamd:teammgrd	OK	Process
teamd:teamsyncd	OK	Process
teamd:tlm_teamd	OK	Process
p4rt:p4rt	OK	Process
swss:orchagent	OK	Process
swss:portsyncd	OK	Process
swss:neighsyncd	OK	Process
swss:fdbsyncd	OK	Process
swss:vlanmgrd	OK	Process
swss:intfmgrd	OK	Process
swss:portmgrd	OK	Process
swss:buffermgrd	OK	Process
swss:vrfmgrd	OK	Process
swss:nbrmgrd	OK	Process
swss:vxlanmgrd	OK	Process
swss:coppmgrd	OK	Process
swss:tunnelmgrd	OK	Process
swss:esmgrd	OK	Process
bgp:zebra	OK	Process


```

bgp:staticd          OK      Process
bgp:bgpd             OK      Process
bgp:fpmsyncd         OK      Process
bgp:essyncd          OK      Process
bgp:bgpcfgd          OK      Process
database:redis       OK      Process
PSU 1                Not OK  PSU
PSU 2                Not OK  PSU
counter-rotating-fan-1 OK      Fan
counter-rotating-fan-2 OK      Fan
counter-rotating-fan-3 OK      Fan
counter-rotating-fan-4 OK      Fan
counter-rotating-fan-5 OK      Fan

```

System services and devices ignore list

```

Name      Status    Type
-----

```

show system-health monitor-list

This command displays system-health monitored services and devices name list.

Usage

```
show system-health monitor-list
```

Example

```
admin@wedge100bf-32x-1:~$ sudo show system-health monitor-list
```

System services and devices monitor list

Name	Status	Type
-----	-----	-----
wedge100bf-32x-1	OK	System
rsyslog	OK	Process
root-overlay	OK	Filesystem
var-log	OK	Filesystem
routeCheck	OK	Program
diskCheck	OK	Program
container_checker	OK	Program
vnetRouteCheck	OK	Program
container_memory_telemetry	OK	Program
telemetry:telemetry	OK	Process
telemetry:dialout	OK	Process
snmp:snmpd	OK	Process
snmp:snmp-subagent	OK	Process
sflow:sflowmgrd	OK	Process
lldp:lldpd	OK	Process
lldp:lldp-syncd	OK	Process
lldp:lldpmgrd	OK	Process
nat:natmgrd	OK	Process
nat:natsyncd	OK	Process
syncd:syncd	OK	Process
teamd:teammgrd	OK	Process
teamd:teamsyncd	OK	Process
teamd:tlm_teamd	OK	Process
p4rt:p4rt	OK	Process
swss:orchagent	OK	Process
swss:portsyncd	OK	Process
swss:neighsyncd	OK	Process
swss:fdbsyncd	OK	Process
swss:vlanmgrd	OK	Process
swss:intfmgrd	OK	Process
swss:portmgrd	OK	Process
swss:buffermgrd	OK	Process
swss:vrfmgrd	OK	Process
swss:nbrmgrd	OK	Process
swss:vxlanmgrd	OK	Process
swss:coppmgrd	OK	Process
swss:tunnelmgrd	OK	Process
swss:esmgrd	OK	Process
bgp:zebra	OK	Process
bgp:staticd	OK	Process
bgp:bgpd	OK	Process
bgp:fpmsyncd	OK	Process
bgp:essyncd	OK	Process
bgp:bgpcfgd	OK	Process
database:redis	OK	Process
PSU 1	Not OK	PSU
PSU 2	Not OK	PSU
counter-rotating-fan-1	OK	Fan
counter-rotating-fan-2	OK	Fan
counter-rotating-fan-3	OK	Fan
counter-rotating-fan-4	OK	Fan
counter-rotating-fan-5	OK	Fan

show system-health summary

This command displays system-health summary information.

Usage

```
show system-health summary
```

Example

```
admin@wedge100bf-32x-1:~$ sudo show system-health summary
System status summary

System status LED
Services:
  Status: OK
Hardware:
  Status: Not OK
  Reasons: Invalid temperature data for PSU 2, temperature=N/A, threshold=N/A
           PSU 1 is out of power
```

(202111.2) Troubleshooting

- `show techsupport`
- Usage of "logger"
- Examples of collecting device information and log files, and sending them to a remote host using the "show techsupport" command
 - Example 2: Collect the device information and log files separately

show techsupport

In field service, there is limited time to gather information for troubleshooting and debugging. The **show techsupport** command gathers relevant information about the state of the device; information as diverse as syslog entries, database state, routing-stack state, etc. It then compresses the information into an archive file. This archived file can then be sent to the SONiC development team for examination.

Usage

```
show techsupport [--since=<time_specifier>]
show techsupport [--nolog]
```

Options:

- **--since** Collect logs and core files since a given date.
- **--silent** Run techsupport in silent mode.
- **--nolog** Collect system information without syslog to reduce the archive file size.



Note

- The archive file is stored in `/var/dump/<DEVICE_HOST_NAME>_YYYYMMDD_HHMMSS.tar.gz`
- The location and name of the archive file will be displayed at the end of the execution of the command "**show techsupport**"

show techsupport can generate a large dump file after the SONiC system has been running for a long time and there are exhaustive system resources. When the log files are specified for a certain period (by using the "--since" option) or are not been included (by using the "--nolog" option), the dump file size can be reduced to MB level and the time to collect the data is also reduced relatively.

Example

```
admin@sonic:~$ show techsupport --since=yesterday # Will collect syslog and core files for the last 24 hours
admin@sonic:~$ show techsupport --since='hour ago' # Will collect syslog and core files for the last one hour
admin@sonic:~$ show techsupport --nolog           # Will not collect syslog and core files
```

Usage of "logger"

Use the 'logger' command to add a marker to make it simple to find associated Syslog entries for a reported issue.

If there is a way to reproduce an issue by a specific procedure, it would be very helpful to execute a **logger** command to add a special mark into the syslog before and after reproducing the issue.

For example, execute the first **logger** command before recreating the issue, and then execute the second command after recreating the issue as shown below. Using this marker, you can **grep** the syslog to know which part of the syslog is related to the reported issue. The related syslog content will be the messages between "==== test start====" and "==== test finish====".

```
logger ==== test start====
logger ==== test finish====
```

Examples of collecting device information and log files, and sending them to a remote host using the "show techsupport" command

It is known that a large number of log files will be generated if the device has been running for a long time. Therefore, it will consume more time to process those files into an archive file.

Example 1: Collect the device information along with the log files

Step 1: Execute the **show techsupport** command with "--since" option.

Example

```
admin@as5835-54x-3:~$ show techsupport --since='hour ago'
main
mkdir: created directory '/var/dump/sonic_dump_as5835-54x_20210129_071517'
'/var/dump/sonic_dump_as5835-54x_20210129_071517/generate_dump' -> '/usr/bin/generate_dump'
sonic_dump_as5835-54x_20210129_071517/
sonic_dump_as5835-54x_20210129_071517/generate_dump
mkdir: created directory '/var/dump/sonic_dump_as5835-54x_20210129_071517/proc'
...
(skipped)
...
/var/dump/sonic_dump_as5835-54x_20210129_071517.tar.gz
```

Step 2: Use the **scp** command to copy the archive file from the local device (File generated from step 1) to a remote site.

Example

```
admin@as5835-54x:~$ sudo scp /var/dump/sonic_dump_as5835-54x_20210129_071517.tar.gz sonic@10.250.2.44:/home
/sonic/tmp_dennis/
sonic@10.250.2.44's password:
sonic_dump_as5835-54x_20210129_071517.tar.gz 100% 3111KB 86.4MB/s 00:00
```

Example 2: Collect the device information and log files separately

Step 1: Execute the **show techsupport** command with "--nolog" option to collect the device information.

Example

```
admin@as5835-54x-3:~$ show techsupport --nolog
main
mkdir: created directory '/var/dump/sonic_dump_as5835-54x-3_20210129_082320'
'/var/dump/sonic_dump_as5835-54x-3_20210129_082320/generate_dump' -> '/usr/bin/generate_dump'
sonic_dump_as5835-54x-3_20210129_082320/
sonic_dump_as5835-54x-3_20210129_082320/generate_dump
mkdir: created directory '/var/dump/sonic_dump_as5835-54x-3_20210129_082320/proc'
...
(skipped)
...
/var/dump/sonic_dump_as5835-54x-3_20210129_082320.tar.gz
```

Step 2: Use the **scp** command to copy the archive file from the local device to a remote site.

Example

```
admin@as5835-54x-3:~$ sudo scp /var/dump/sonic_dump_as5835-54x-3_20210129_082320.tar.gz sonic@10.250.2.44:
/home/sonic/tmp_dennis/
sonic@10.250.2.44's password:
sonic_dump_as5835-54x-3_20210129_082320.tar.g 100% 2007KB 81.7MB/s 00:00
```

Step 3: Use the **scp** to copy the log (ex:syslog) file from the local device to a remote site.

Example

```
admin@as5835-54x-3:~$ sudo scp /var/log/syslog* sonic@10.250.2.44:/home/sonic/tmp_dennis/sonic_log/
sonic@10.250.2.44's password:
syslog 100% 20KB 11.4MB/s 00:00
syslog.1 100% 2056KB 79.5MB/s 00:00
syslog.2.gz 100% 162KB 58.4MB/s 00:00
syslog.3.gz 100% 160KB 59.7MB/s 00:00
syslog.4.gz 100% 170KB 65.7MB/s 00:00
...
(skipped)
...
syslog.4.gz 100% 170KB 65.7MB/s 00:00
```

(202111.2) VLAN

- [VLAN Show Commands](#)
 - [show vlan brief](#)
 - [show vlan config](#)
- [VLAN Config Commands](#)
 - [config vlan add/del](#)
 - [config vlan member add/del](#)

VLAN Show Commands

show vlan brief

This command displays brief information about all the VLANs configured on the device. It displays the VLAN ID, IP address (if configured for the VLAN), list of VLAN member ports, whether the port is tagged or in untagged mode, and the DHCP Helper Address.

Usage

```
show vlan brief
```

Example

```
admin@sonic:~$ show vlan brief
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+
| VLAN ID | IP Address | Ports | Port Tagging | Proxy ARP | DHCP Helper | DHCP Source |
| DHCP Link | | | | | | |
| Selection | | | | | | |
+=====+=====+=====+=====+=====+=====+=====+=====+
=====+
|      100 | 1.1.2.2/16 | Ethernet0 | tagged | disabled | 192.0.0.1 | | |
|          | | | | | | |
|          | | Ethernet4 | tagged | | | 192.0.0.2 | |
|          | | | | | | |
|          | | | | | | |
|          | | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+
```

show vlan config

This command displays all the VLAN configuration.

Usage

```
show vlan config
```

Example

```
admin@sonic:~$ show vlan config
Name      VID  Member      Mode
-----
Vlan100   100  Ethernet0    tagged
Vlan100   100  Ethernet4    tagged
```

VLAN Config Commands

This sub-section explains how to configure a VLAN and its member ports.

config vlan add/del

This command is used to add or delete a VLAN.

Usage

```
config vlan (add | del) <vlan_id>
```

Example (Create the VLAN "Vlan100" if it does not already exist)

```
admin@sonic:~$ sudo config vlan add 100
```

config vlan member add/del

This command is used to add or delete a member port to/from an already created VLAN.

Usage

```
config vlan member add/del [-u|--untagged] <vlan_id> <member_portname>
```

Note:

- Adding the **-u** or **--untagged** flag will set the port member in "untagged" mode.
- It is not allowed to add an Ethernet port / port channel with a L3 configuration to a VLAN.
- An Ethernet port / port channel could only be added to one VLAN in untagged mode.

Example (Add Ethernet0 as member of the vlan 100)

```
admin@sonic:~$ sudo config vlan member add 100 Ethernet0
```


Example (Add Ethernet4 as member of the vlan 100)

```
admin@sonic:~$ sudo config vlan member add 100 Ethernet4
```

(202111.2) VLAN Stacking

- [VLAN Stacking Show Commands](#)
 - [show vlan-stacking](#)
- [VLAN Stacking Config Commands](#)
 - [config vlan-stacking add](#)
 - [config vlan-stacking del](#)

VLAN Stacking Show Commands

show vlan-stacking

This command displays the information about all the VLAN stacking rule configured on the device.

Usage

```
show vlan-stacking
```

Example

```
admin@sonic:~$ show vlan-stacking
Interface      Stage      Match VLAN  Action      Apply VLAN
-----
Ethernet0      egress      200        pop
Ethernet0      ingress     100        push         200
Ethernet4      egress      200        swap         100
Ethernet4      ingress     100        swap         200
```

VLAN Stacking Config Commands

This sub-section explains how to add(or delete) a VLAN stacking rule to(or from) the specified interface.

config vlan-stacking add

This command is used to add a VLAN stacking rule to interface.

Usage

```
config vlan-stacking add <interface_name> <stage> <match_vlan> <action> <apply_vlan>
```

Parameter:

- <interface_name>: The name of physical interface or PortChannel, e.g. "Ethernet0" or "PortChannel0001".
- <stage> : The traffic direction of the packet, e.g. "ingress" or "egress".
- <match_vlan> : The VLAN ID on the packet which expected to match the rule(The range is from 1 to 4094).
- <action> : The action applied to the packet when it matched the rule, e.g. "push", "pop" or "swap".
- <apply_vlan> : The VLAN ID applied to the packet according to the action field(The range is from 1 to 4094).

Example (Add vlan id 200 when ingress packet's vlan id is matched 100 on Ethernet0)

```
admin@sonic:~$ sudo config vlan-stacking add Ethernet0 ingress 100 push 200
```

config vlan-stacking del

This command is used to delete a VLAN stacking rule from interface.

Usage

```
config vlan-stacking del <interface_name> <stage> <match_vlan>
```

Parameter:

- <interface_name>: The name of physical interface or PortChannel, e.g. "Ethernet0" or "PortChannel0001".
- <stage> : The traffic direction of the packet, e.g. "ingress" or "egress".
- <match_vlan> : The VLAN ID on the packet which expected to match the rule(The range is from 1 to 4094).

Example (Delete the configuration applied to egress packet which matched vlan id 200 on Ethernet0)

```
admin@sonic:~$ sudo config vlan-stacking del Ethernet0 egress 200
```

(202111.2) VRF Configuration

- [VRF Show Commands](#)
 - [show vrf](#)
- [VRF Config Commands](#)
 - [config vrf add](#)
 - [config vrf del](#)
 - [config vrf add_vrf_vni_map](#)
 - [config vrf del_vrf_vni_map](#)

VRF Show Commands

show vrf

This command displays all VRFs configured on the system along with interface binding to the VRF. If **vrf-name** is also provided as part of the command, when the VRF is created it will display all interfaces binding to the VRF. If the VRF is not created, nothing will be displayed.

Usage

```
show vrf [<vrf_name>]
```

Example

```
admin@sonic:~$ show vrf
VRF          Interfaces
-----
Vrf-red      Vlan100
              Loopback11
Vrf-blue     Loopback100
              Loopback102
```

Note: You may refer to [\(202111.2\) Interfaces](#) for binding a VRF and interface.

VRF Config Commands

config vrf add

This command creates a VRF in the SONiC system with the provided VRF name.

Note: The **vrf-name** should always start with the keyword "Vrf".

Usage

```
config vrf add <vrf-name>
```

config vrf del

This command deletes the VRF with the name **vrf-name**.

Usage

```
config vrf del <vrf-name>
```

config vrf add_vrf_vni_map

This command creates VRF-VNI mapping in the SONiC system.

Note: The show command is in the VxLAN CLI guide.

Usage

```
config vrf add_vrf_vni_map <vrf-name> <vni>
```

Example

```
config vrf add Vrf1
config vxlan add vtep 10.1.0.32
config vxlan evpn_nvo add evpnvno vtep
config vrf add_vrf_vni_map Vrf1 20000
```

config vrf del_vrf_vni_map

This command deletes VRF-VNI mapping from the SONiC system.

Usage

```
config vrf del_vrf_vni_map <vrf-name>
```

(202111.2) VRF Route Leaking

- [Configuring VRF Route Leaking](#)
 - [Static VRF Route Leaking](#)
 - [Dynamic VRF Route Leaking](#)
 - [Basic Configuration](#)
 - [Combine with route-map and prefix-list](#)

Configuring VRF Route Leaking

The CLI configuration for VRF route leaking uses the **vttysh** command to configure static and dynamic route leaks. For dynamic route leaking, BGP is used as an example.

Static VRF Route Leaking

The following command creates a static route that leaks the next hop into a specified VRF.

Usage

```
ip route <A.B.C.D/M> <A.B.C.D> nexthop-vrf <vrf-name>
```

Example

```
admin@sonic:~# vtysh
sonic# configure
sonic(config)# vrf Vrf1
sonic(config-vrf)# ip route 2.2.2.2/32 10.0.0.59 nexthop-vrf Vrf2
sonic(config-vrf)# end
sonic# exit
admin@sonic:~#
admin@sonic:~# show ip route vrf Vrf1
Codes: K - kernel route, C - connected, S - static, R - RIP,
        O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
        T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
        F - PBR, f - OpenFabric,
        > - selected route, * - FIB route, q - queued route, r - rejected route
VRF Vrf1:
S>* 2.2.2.2/32 [1/0] via 10.0.0.59, PortChannel0002(vrf Vrf2), 01:34:15
C>* 10.0.0.56/31 is directly connected, PortChannel0001, 01:40:04
```

Dynamic VRF Route Leaking

Basic Configuration

The following command configures a dynamic VRF route leak that automatically imports routes between VRFs. In this example, the routes in the default VRF are imported to Vrf1.

Usage

```
import vrf <vrf-name>
```

Note:

- **import vrf <vrf-name>** is shortcut syntax for specifying automatic leaking from **vrf-name** to the current VRF. In the other words, you do not need to consider that the underlying mechanism relies on the BGP constructs of Route Distinguisher (RD) and Route Targets (RTs), because they are automatically derived when you enable route leaking.

Example

```
admin@sonic:~# vtysh
sonic# configure
sonic(config)# router bgp 65200 vrf Vrf1
sonic(config-router)# address-family ipv4 unicast
sonic(config-router-af)# import vrf default
sonic(config-router-af)# end
sonic# exit
admin@sonic:~#
admin@sonic:~# show ip route vrf Vrf1 193.11.248.128
Routing entry for 193.11.248.128/25
  Known via "bgp", distance 200, metric 0, vrf Vrf1, best
  Last update 00:03:47 ago
  * 10.0.0.59, via PortChannel0002(vrf default)
  * 10.0.0.61, via PortChannel0003(vrf default)
  * 10.0.0.63, via PortChannel0004(vrf default)
```

Combine with route-map and prefix-list

You can also decide to select certain prefixes from being imported. The match rule must be configured in a route map.

Usage

```
import vrf route-map <name of route-map>
```

Note: Our test environment is a t0 topology, so the default VRF will get 6440 routes from peers.

Example

```
admin@sonic:~# vtysh
sonic# configure
sonic(config)# ip prefix-list test_vrf_route_leak seq 5 deny 193.11.248.0/25
sonic(config)# ip prefix-list test_vrf_route_leak seq 10 permit 193.11.240.128/25
sonic(config)# route-map VRF_ROUTE_LEAK permit 5
sonic(config-route-map)# match ip address prefix-list test_vrf_route_leak
sonic(config-route-map)# exit
sonic(config)# router bgp 65200 vrf Vrf1
sonic(config-router)# address-family ipv4 unicast
sonic(config-router-af)# import vrf route-map VRF_ROUTE_LEAK
sonic(config-router-af)# import vrf default
sonic(config-router-af)# end
```

(202111.2) VxLAN (includes EVPN and EVPN-MH)

- VxLAN Show Commands
 - show vxlan tunnel
 - show vxlan name <vxlanname>
 - show vxlan interface
 - show vxlan vlanvni
 - show vxlan vrfvni
 - show vxlan remote_mac <remoteip/esi/all>
 - show vxlan remote_vni <remoteip/all>
 - show vxlan remotevtep
 - show vxlan evpn es
- VxLAN Configuration Commands
 - config vxlan add <vtepname> <src_ipv4>
 - config vxlan evpn_nvo add <nvo> <vtepname>
 - config vxlan map add <vtepname> <vlanid> <vnid>
 - config vxlan evpn_es add <portchannelname> <esid> [<sysmac>]
 - config vxlan del <vtepname>
 - config vxlan evpn_nvo del <nvo>
 - config vxlan map del <vtepname> <vlanid> <vnid>
 - config vxlan evpn_es del <portchannelname>

VxLAN Show Commands

show vxlan tunnel

This command displays brief information about VxLANs configured in the device. It displays VxLAN tunnel name, source IP address, destination IP address and VLAN VNI tunnel mappings.

Usage

```
show vxlan tunnel
```

Example

```
admin@sonic:~$ show vxlan tunnel
vxlan tunnel name  source ip  destination ip  tunnel map name  tunnel map mapping(vni -> vlan)
-----
vtep               1.1.1.1        map_3000_vlan30  3000 -> Vlan30
```

show vxlan name <vxlanname>

This command displays brief information for a specified VxLAN tunnel. It displays VxLAN tunnel name, source IP address, destination IP address and VLAN VNI tunnel mappings.

Usage

```
show vxlan name <vxlanname>
```

Example

```
admin@sonic:~$ show vxlan name vtep
vxlan tunnel name  source ip  destination ip  tunnel map name  tunnel map mapping(vni -> vlan)
-----
vtep               1.1.1.1        map_3000_vlan30  3000 -> Vlan30
```


show vxlan interface

This command displays brief VTEP information.

Usage

```
show vxlan interface
```

Example

```
admin@sonic:~$ show vxlan interface
VTEP Information:

    VTEP Name : vtep, SIP : 1.1.1.1
    NVO Name  : nvo,  VTEP : vtep
    Source interface : Loopback0
```

show vxlan vlanvni

This command displays all the VLAN VNI mappings.

Usage

```
show vxlan vlanvni
```

Example

```
admin@sonic:~$ show vxlan vlanvni
+-----+-----+
| VLAN  | VNI   |
+-----+-----+
| Vlan30 | 3000  |
+-----+-----+
Total count : 1
```

show vxlan vrfvni

This command displays all the VRF VNI mappings.

Note: The config commands are in the VRF CLI guide.

Usage

```
show vxlan vrfvni
```

Example

```
admin@sonic:~$ show vxlan vrfvni
+-----+-----+
| VRF   | VNI   |
+-----+-----+
| Vrf-1 | 104   |
+-----+-----+
Total count : 1
```

show vxlan remote_mac <remoteip/esi/all>

This command displays all MACs learnt from a specified remote VTEP IP, from a specified ESI, or from all of the remote VTEP IPs for all VNIs.

Usage

```
show vxlan remote_mac <remoteip/esi/all>
```

Example

```
admin@sonic:~$ show vxlan remotemac 2.2.2.2
+-----+-----+-----+-----+-----+-----+
| VLAN   | MAC           | RemoteVTEP | ESI   | VNI | Type   |
+-----+-----+-----+-----+-----+-----+
| Vlan1000 | 00:12:34:56:78:90 | 2.2.2.2   |       | 100 | dynamic |
+-----+-----+-----+-----+-----+-----+
Total count : 1

admin@sonic:~$ show vxlan remotemac 03:00:11:22:33:44:55:00:00:a8
+-----+-----+-----+-----+-----+-----+
| VLAN   | MAC           | RemoteVTEP | ESI   | VNI | Type   |
+-----+-----+-----+-----+-----+-----+
| Vlan1000 | 00:55:66:77:88:AA |           | 03:00:11:22:33:44:55:00:00:a8 | 100 | dynamic |
+-----+-----+-----+-----+-----+-----+
Total count : 1

admin@sonic:~$ show vxlan remotemac all
+-----+-----+-----+-----+-----+-----+
| VLAN   | MAC           | RemoteVTEP | ESI   | VNI | Type   |
+-----+-----+-----+-----+-----+-----+
| Vlan1000 | 00:12:34:56:78:90 | 2.2.2.2   |       | 100 | dynamic |
+-----+-----+-----+-----+-----+-----+
| Vlan1000 | 00:55:66:77:88:AA |           | 03:00:11:22:33:44:55:00:00:a8 | 100 | dynamic |
+-----+-----+-----+-----+-----+-----+
Total count : 2
```

show vxlan remote_vni <remoteip/all>

This command displays all the VNIs learnt from the specified remote VTEP, or all the VNIs learnt from all of the remote VTEPs.

Usage

```
show vxlan remote_vni <remoteip/all>
```

Example

```
admin@sonic:~$ show vxlan remote_vni 3.3.3.3
+-----+-----+-----+
| VLAN   | RemoteVTEP | VNI   |
+=====+=====+=====+
| Vlan101 | 3.3.3.3    | 1001  |
+-----+-----+-----+
Total count : 1
```

show vxlan remotevtep

This command displays all remote VTEPs (a.k.a. VxLAN tunnels) learnt by EVPN/VxLAN.

Usage

```
show vxlan remotevtep
```

Example

```
admin@sonic:~$ show vxlan remotevtep
+-----+-----+-----+-----+-----+
| SIP      | DIP      | Creation Source | OperStatus |
+=====+=====+=====+=====+=====+
| 11.11.11.11 | 22.22.22.22 | EVPN           | oper_up    |
+-----+-----+-----+-----+-----+
Total count : 1
```

show vxlan evpn es

This command display all Ethernet Segments. The 'status' field indicates the operation status of the corresponding PortChannel interface.

Usage

```
show vxlan evpn es
```

Example

```
admin@sonic:~$ show vxlan evpn es
Type: L local, R remote, N non-DF
```

ESI	Type	ES Interface	status	VTEPS
03:44:38:39:ab:cd:01:00:00:01	LRN	PortChannel0001	down	1.1.1.1 2.2.2.2 3.3.3.3 4.4.4.4 5.5.5.5
03:44:38:39:ab:cd:01:00:00:02	L	PortChannel0002	up	
03:44:38:39:ab:cd:01:00:00:03	RN			172.0.0.24 172.0.0.25 172.0.0.26

Total count : 3

VxLAN Configuration Commands

config vxlan add <vtepname> <src_ipv4>

This command is for VTEP Source IP configuration. The **vtepname** is a string, and **src_ipv4** is an IPV4 address in dotted notation A.B.C.D .

Usage

```
config vxlan add <vtepname> <src_ipv4>
```

Example

```
sudo config vxlan add vtep 1.1.1.1
```

config vxlan evpn_nvo add <nvname> <vtepname>

This command is for NVO configuration.

Usage

```
config vxlan evpn_nvo add <nvname> <vtepname>
```

Example

```
sudo config vxlan evpn_nvo add nvo vtep
```

config vxlan map add <vtepname> <vlanid> <vnid>

This command creates VLAN-VNI mapping on the VTEP with name **vtepname**.

Usage

```
config vxlan map add <vtepname> <vlanid> <vnid>
```

Example

```
sudo config vxlan map add vtep 30 3000
```

config vxlan evpn_es add <portchannelname> <esid> [<sysmac>]

This command creates an Ethernet Segment on specified PortChannel interface.

Usage

```
config vxlan evpn_es add <protchannelname> <esid> [<sysmac>]
```

Example

```
#type 0
sudo config vxlan evpn_es add PortChannel0001 00:11:22:33:44:55:66:77:88:99
#type 3
sudo config vxlan evpn_es add PortChannel0001 168 00:11:22:33:ab:cd
```

config vxlan del <vtepname>

This command deletes a VTEP with name **vtepname** .

Usage

```
config vxlan del <vtepname>
```

Example

```
sudo config vxlan del vtep
```

config vxlan evpn_nvo del <nvoname>

This command deletes NVO with name **nvoname**.

Usage

```
config vxlan evpn_nvo del <nvoname>
```

Example

```
sudo config vxlan evpn_nvo del nvo
```

config vxlan map del <vtepname> <vlanid> <vnid>

This command deletes specific VLAN-VNI mapping on the VTEP with name **vtepname**.

Usage

```
config vxlan map del <vtepname> <vlanid> <vnid>
```

Example

```
sudo config vxlan map del vtep 30 3000
```

config vxlan evpn_es del <portchannelname>

This command deletes the Ethernet Segment on specified PortChannel interface.

Usage

```
config vxlan evpn_es del <portchannelname>
```

Example

```
sudo config vxlan evpn_es del PortChannel0001
```

(202111.2) Warm Reboot

- [Warm Reboot Configuration Commands](#)
 - [warm-reboot](#)

Warm Reboot Configuration Commands

warm-reboot

This command initiates a warm reboot of the device. The command does not require setting a warm restart configuration, it sets up everything needed to perform a warm reboot. This command requires root privilege.

Usage

```
warm-reboot \[-h\|-?\|-v\|-f\|-r\|-k\|-x\|-c <control plane assistant IP list>\|-s\]
```

Parameters:

```
-h,-? : get this help
-v : turn on verbose mode
-f : force execution
-r : reboot with /sbin/reboot
-k : reboot with /sbin/kexec -e \[default\]
-x : execute script with -x flag
-c : specify control plane assistant IP list
-s : strict mode: do not proceed without:
```

```
- control plane assistant IP list.
```

Example

```
admin@sonic:~$ sudo warm-reboot -v
Tue Oct 22 23:20:53 UTC 2019 Pausing orchagent ...
Tue Oct 22 23:20:53 UTC 2019 Stopping radv ...
Tue Oct 22 23:20:54 UTC 2019 Stopping bgp ...
Tue Oct 22 23:20:54 UTC 2019 Stopped bgp ...
Tue Oct 22 23:20:57 UTC 2019 Initialize pre-shutdown ...
Tue Oct 22 23:20:58 UTC 2019 Requesting pre-shutdown ...
Tue Oct 22 23:20:58 UTC 2019 Waiting for pre-shutdown ...
Tue Oct 22 23:20:59 UTC 2019 Pre-shutdown succeeded ...
Tue Oct 22 23:20:59 UTC 2019 Backing up database ...
Tue Oct 22 23:21:00 UTC 2019 Stopping teamd ...
Tue Oct 22 23:21:00 UTC 2019 Stopped teamd ...
Tue Oct 22 23:21:00 UTC 2019 Stopping syncd ...
Tue Oct 22 23:21:11 UTC 2019 Stopped syncd ...
Tue Oct 22 23:21:11 UTC 2019 Stopping all remaining containers ...
Tue Oct 22 23:21:13 UTC 2019 Stopped all remaining containers ...
Tue Oct 22 23:21:15 UTC 2019 Rebooting with /sbin/kexec -e to SONiC-OS-20191021.01 ...
```

(202111.2) Warm Restart

- Overview
- Warm Restart Show Commands
 - `show warm_restart config`
 - `show warm_restart state`
- Warm Restart Config Commands
 - `config warm_restart bgp_timer` (Not Supported)
 - `config warm_restart enable/disable`
 - `config warm_restart neighsyncd_timer` (Not Supported)
 - `config warm_restart bgp_timer` (Not Supported)
 - `config warm_restart teamsyncd_timer` (Not Supported)

Overview

Besides a device-level warm reboot, SONiC also provides a docker-based warm restart. This feature is currently supported by the following dockers: BGP, teamD, and SWSS. A user can manage to restart a particular docker with no interruption on packet forwarding and no effect on other services. This helps to reduce operational costs as well as development efforts. For example, to fix a bug in the BGP routing stack, only the BGP docker image needs to be built, tested, and upgraded.

To achieve uninterrupted packet forwarding during the restarting stage and database reconciliation at the post restarting stage, warm restart enables dockers with adjacency state machine to facilitate standardized protocols. For example, a BGP restarting switch must have BGP "Graceful Restart" enabled, and its BGP neighbors must be "Graceful Restart Helper Capable", as specified in [IETF RFC4724](#).

Before executing a warm restart for the BGP docker, the following BGP commands should be enabled:

```
bgp graceful-restart
bgp graceful-restart preserve-fw-state
```

In the current SONiC release, the above two commands are enabled by default.

You should be aware that during a warm restart, certain BGP fast convergence features and black hole avoidance features should either be disabled or set to a lower preference to avoid conflicts with the BGP graceful restart.

For example, BGP BFD could be disabled via:

```
no neighbor <A.B.C.D|X:X::X:X|WORD> bfd
```

otherwise, the fast failure detection would cause packet drop during the warm reboot. Another commonly deployed blackhole avoidance feature: dynamic route priority adjustment, could be disabled to avoid large routes churn during BGP restart. Use the following command.

```
no bgp max-med on-peerup
```

Warm Restart Show Commands

`show warm_restart config`

This command displays all the configuration related to a warm restart.

Usage

```
show warm_restart config
```


Example

```
admin@sonic:~$ show warm_restart config
name enable timer_name timer_duration
-----
bgp true bgp_timer 100
teamd false teamsyncd_timer 300
swss false neighsyncd_timer 200
system true NULL NULL
```

show warm_restart state

This command displays the warm restart state.

Usage

```
show warm_restart state
```

Example

```
admin@sonic:~$ show warm_restart state
name restore_count state
-----
orchagent 0
vlanmgrd 0
bgp 1 reconciled
portsyncd 0
teammgrd 1
neighsyncd 0
teamsyncd 1
syncd 0
natsyncd 0
```

Warm Restart Config Commands

This sub-section explains the various configurations related to the warm restart feature. The following parameters can be configured using this command.

1. bgp_timer
2. disable
3. enable
4. neighsyncd_timer
5. teamsyncd_timer

Each of these sub-commands are explained in the following section.

Users can use an optional parameter **-s** to use the unix domain socket for communicating with the RedisDB which will be faster when compared to using the default network sockets. All these commands have the following option.

Options: -s, --redis-unix-socket-path TEXT unix socket path for redis connection.

config warm_restart bgp_timer (Not Supported)

This command is used to set the BGP timer value for warm restart of the BGP service. **bgp_timer** holds the time interval utilized by **fpm syncd** during warm-restart episodes. During this interval, **fpm syncd** will recover all the routing states previously pushed to AppDB, as well as all new states coming from zebra/bgpd. Upon expiration of this timer, **fpm syncd** will execute the reconciliation logic to eliminate all the stale entries from AppDB. This timer should match the BGP-GR restart-timer configured within the elected routing-stack.

Usage

```
config warm_restart [-s|--redis-unix-socket-path <socket_path>] bgp_timer <seconds>
```

- Parameters:
 - seconds: Range from 1 to 3600

Example

```
admin@sonic:~$ sudo config warm_restart bgp_timer 1000
```

config warm_restart enable/disable

This command is used to enable or disable the warm restart for a particular service that supports warm reboot. The following four services support warm reboot. When a user restarts the particular service using **systemctl restart <service_name>**, this configured value will be checked for whether it is enabled or disabled. If this configuration is enabled for that service, it will perform a warm reboot for that service. Otherwise, it will do a cold restart of the service.

Usage

```
config warm_restart [-s|--redis-unix-socket-path <socket_path>] enable [<module_name>]
```

- Parameters:
 - module_name: Can be either **system**, **swss**, **bgp**, or **teamd**. If the **module_name** argument is not specified, it will enable the **system** module.

Example (Set warm_restart as "enable" for the "system" service)

```
admin@sonic:~$ sudo config warm_restart enable
Example (Set warm_restart as "enable" for the "swss" service. When user does "systemctl restart swss", it
will perform warm reboot instead of cold reboot)
admin@sonic:~$ sudo config warm_restart enable swss
Example (Set warm_restart as "enable" for the "teamd" service. When user does "systemctl restart teamd", it
will perform warm reboot instead of cold reboot)
admin@sonic:~$ sudo config warm_restart enable teamd
```

config warm_restart neighsyncd_timer (Not Supported)

This command is used to set the **neighsyncd_timer** value for a warm restart of the **swss** service. The **neighsyncd_timer** is the timer used for **swss** (neighsyncd) service during the warm restart. The timer is started after the neighborTable is restored to internal data structures. Then, neighborsyncd starts to read all Linux kernel entries and mark the entries in the data structures accordingly. Once the timer is expired, reconciliation is done and the delta is pushed to appDB. Valid value is 1-9999. 0 is invalid.

Usage

```
config warm_restart [-s|--redis-unix-socket-path <socket_path>] neighsyncd_timer <seconds>
```

- Parameters:
 - seconds: Range from 1 to 9999

Example

```
admin@sonic:~$ sudo config warm_restart neighsyncd_timer 2000
```

config warm_restart bgp_timer (Not Supported)

This command is used to set the **bgp_timer** value for a warm restart of the **bgp** service. The **bgp_timer** is the timer used for the **bgp** service during the warm restart. The timer is started after the BGP table is restored to internal data structures. BGP services then start to read all Linux kernel entries and mark the entries in the data structures accordingly. Once the timer is expired, reconciliation is done and the delta is pushed to appDB. Valid value is 1-9999. 0 is invalid.

Usage

```
config warm_restart [-s|--redis-unix-socket-path <socket_path>] bgp_timer <seconds>
```

- Parameters:
 - seconds: Range from 1 to 9999

Example

```
admin@sonic:~$ sudo config warm_restart bgp_timer 2000
```

config warm_restart teamsyncd_timer (Not Supported)

This command is used to set the **teamsyncd_timer** value for a warm restart of the **teamd** service. The **teamsyncd_timer** holds the time interval utilized by teamsyncd during warm-restart episodes. The timer is started when teamsyncd starts. During the timer interval, teamsyncd will preserve all LAG interface changes, but it will not apply them. The changes will only be applied when the timer expires. When the changes are applied, the stale LAG entries will be removed, the new LAG entries will be created. Supported range: 1-9999. 0 is invalid.

Usage

```
config warm_restart teamsyncd_timer <seconds>
```

- Parameters:
 - seconds: Range from 1 to 9999

Example

```
admin@sonic:~$ sudo config warm_restart teamsyncd_timer 3000
```

(202111.2) Watermark

- [Watermark Show Commands](#)
 - `show watermark telemetry interval`
- [Watermark Config Commands](#)
 - `config watermark telemetry interval`

Watermark Show Commands

`show watermark telemetry interval`

This command displays the configured interval for the telemetry.

Usage

```
show watermark telemetry interval
```

Example

```
admin@sonic:~$ show watermark telemetry interval
Telemetry interval 120 second(s)
```

Watermark Config Commands

`config watermark telemetry interval`

This command is used to configure the interval for telemetry. The default interval is 120 seconds. There is no regulation on the valid range of values; it leverages the Linux timer.

Usage

```
config watermark telemetry interval <value>
```

Example

```
admin@sonic:~$ sudo config watermark telemetry interval 999
```

(202111.2) WRED & ECN

This section explains all the Explicit Congestion Notification (ECN) show commands and ECN configuration options that are supported in SONiC.

- [WRED & ECN Show Commands](#)
 - [show wred](#)
 - [show interface wred](#)
- [WRED & ECN Configuration Commands](#)
 - [config wred add](#)
 - [config wred update](#)
 - [config wred del](#)
 - [config interface wred bind queue](#)
 - [config interface wred unbind queue](#)
- [ECN Show Commands](#)
 - [show ecn](#)
- [ECN Config Commands](#)
 - [config ecn](#)

WRED & ECN Show Commands

show wred

This command is used to display the WRED profile.

Usage

```
show wred [<profile_name>]
```

Parameters:

- <profile_name>: Profile name.

Example

```
admin@sonic:~$ show wred
Profile: ecn_prof
Color    Mode      Min Threshold  Max Threshold  Drop Probability
-----
Green    ECN         100            10000          25
Yellow   ECN         200            20000          50
Red      ECN         400            40000          75
```

show interface wred

This command is used to display the WRED configuration for all interfaces or a given interface.

Usage

```
show interface wred <interface_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".

Example

```
admin@as9516-32d-2:~$ show interface wred
Ethernet0
Queue: 0
ECN/WRED: ecn_prof2
Color      Mode      Min Threshold  Max Threshold  Drop Probability
-----
Green      ECN           5000          10000          100
Yellow     ECN
Red        ECN
```

WRED & ECN Configuration Commands

config wred add

This command is used to create a WRED profile.

Usage

```
config wred add <profile> --mode={ecn|wred} [--gmin <gmin> --gmax <gmax> --gdrop <gdrop>] [--ymin <ymin> --ymax <ymax> --ydrop <ydrop>] [--rmin <rmin> --rmax <rmax> --rdrop <rdrop>]
```

Parameters:

- <profile_name>: Profile name.
- mode: ECN or WRED mode.
 - **ecn**: Enable ECN marking.
 - **wred**: Enable WRED dropping.
- <gmin>: Minimum threshold in bytes to start WRED dropping or ECN marking for green packet.
- <gmax>: Maximum threshold in bytes for WRED dropping or ECN marking for green packet.
- <gdrop>: Maximum dropping or marking probability for green packet. Minimum: 0. Maximum: 100.
- <ymin>: Minimum threshold in bytes to start WRED dropping or ECN marking for yellow packet.
- <ymax>: Maximum threshold in bytes for WRED dropping or ECN marking for yellow packet.
- <ydrop>: Maximum dropping or marking probability for yellow packet. Minimum: 0. Maximum: 100.
- <rmin>: Minimum threshold in bytes to start WRED dropping or ECN marking for red packet.
- <rmax>: Maximum threshold in bytes for WRED dropping or ECN marking for red packet.
- <rdrop>: Maximum dropping or marking probability for red packet. Minimum: 0. Maximum: 100.

Note:

The devices using the Broadcom ASIC, the maximum drop probability supports one of [0-10, 25, 50, 75, 100] in hardware.

- If the user configuration value is from 0 to 10, takes the same value as hardware configuration.
- If the user configuration value is from 10 to 24, takes 10 as hardware configuration.
- If the user configuration value is from 25 to 49, takes 25 as hardware configuration.
- If the user configuration value is from 50 to 74, takes 50 as hardware configuration.
- If the user configuration value is from 75 to 99, takes 75 as hardware configuration.
- If user configuration value exceeds 100, takes 100 is used for hardware configuration.

Example

```
sudo config wred add ecn_profile --mode ecn --gmin 5000 --gmax 10000 --gdrop 100
```

config wred update

This command is used to modify an existing WRED profile.

Usage

```
config wred update <profile> [--mode={ecn|wred}] [--gmin <gmin> --gmax <gmax> --gdrop <gdrop>] [--ymin
<ymin> --ymax <ymax> --ydrop <ydrop>] [--rmin <rmin> --rmax <rmax> --rdrop <rdrop>] [--no-green] [--no-
yellow] [--no-red]
```

Parameters:

- <profile_name>: Profile name.
- mode: ECN or WRED mode.
 - **ecn**: Enable ECN marking.
 - **wred**: Enable WRED dropping.
- <gmin>: Minimum threshold in bytes to start WRED dropping or ECN marking for green packet.
- <gmax>: Maximum threshold in bytes for WRED dropping or ECN marking for green packet.
- <gdrop>: Maximum dropping or marking probability for green packet. Minimum: 0. Maximum: 100.
- <ymin>: Minimum threshold in bytes to start WRED dropping or ECN marking for yellow packet.
- <ymax>: Maximum threshold in bytes for WRED dropping or ECN marking for yellow packet.
- <ydrop>: Maximum dropping or marking probability for yellow packet. Minimum: 0. Maximum: 100.
- <rmin>: Minimum threshold in bytes to start WRED dropping or ECN marking for red packet.
- <rmax>: Maximum threshold in bytes for WRED dropping or ECN marking for red packet.
- <rdrop>: Maximum dropping or marking probability for red packet. Minimum: 0. Maximum: 100.
- **no-green**: Remove the green setting from the profile.
- **no-yellow**: Remove the yellow setting from the profile.
- **no-red**: Remove the red setting from the profile.

Note:

The devices using the Broadcom ASIC, the maximum drop probability supports one of [0~10, 25, 50, 75, 100] in hardware.

- If the user configuration value is from 0 to 10, takes the same value as hardware configuration.
- If the user configuration value is from 10 to 24, takes 10 as hardware configuration.
- If the user configuration value is from 25 to 49, takes 25 as hardware configuration.
- If the user configuration value is from 50 to 74, takes 50 as hardware configuration.
- If the user configuration value is from 75 to 99, takes 75 as hardware configuration.
- If user configuration value exceeds 100, takes 100 is used for hardware configuration.

Example

```
sudo config wred update ecn_profile --mode ecn --gmin 10000 --gmax 20000 --gdrop 50
```

The following example shows how to remove the yellow setting from an existing profile.

Example

```
sudo config wred update ecn_profile --no-yellow
```

config wred del

This command is used to delete a WRED profile.

Usage

```
config wred del <profile>
```

Parameters:

- <profile_name>: Profile name.

Example

```
sudo config wred del ecn_profile
```

config interface wred bind queue

This command is used to configure the WRED thresholds on the specified queue.

Usage

```
config interface wred bind queue <interface_name> <queue> <profile_name>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).
- <profile_name>: Profile name.

Example

```
sudo config interface wred bind queue Ethernet0 0 ecn_profile
```

config interface wred unbind queue

This command is used to reset the WRED thresholds on the specified queue.

Usage

```
config interface wred unbind queue <interface_name> <queue>
```

Parameters:

- <interface_name>: Physical interface name. For example: "Ethernet0".
- <queue>: Queue. Minimum: 0. Maximum: 7 (9 for the devices using Broadcom Trident 3 ASIC).

Example

```
sudo config interface wred unbind queue Ethernet0 0
```

ECN Show Commands

This sub-section contains the ECN show commands that are supported.

show ecn

Note: this command can be replaced by "show wred" command.

This command displays all the WRED profiles that are configured in the device.

Usage

```
show ecn
```

Example: show the existed profile

```
admin@sonic:~$ show ecn
Profile: AZURE_LOSSLESS
-----
red_max_threshold      2097152
red_drop_probability   5
yellow_max_threshold   2097152
ecn                    ecn_all
green_min_threshold    1048576
red_min_threshold      1048576
wred_yellow_enable     true
yellow_min_threshold   1048576
green_max_threshold    2097152
green_drop_probability 5
wred_green_enable      true
yellow_drop_probability 5
wred_red_enable        true
-----

Profile: wredprofileabcd
-----
red_max_threshold 100
-----
```

ECN Config Commands

This sub-section contains the configuration commands that can configure the WRED profiles.

config ecn

Note: this command can be replaced by "config wred add" and "config wred update" commands.

This command configures the possible fields in a particular WRED profile that is specified using the "-profile <profilename>" argument. The list of the WRED profile fields that are configurable is listed below.

Usage

```
config ecn -profile <profile_name> [-rmax <red_threshold_max>] [-rmin <red_threshold_min>] [-ymax
<yellow_threshold_max>] [-ymin <yellow_threshold_min>] [-gmax <green_threshold_max>] [-gmin
<green_threshold_min>] [-v|--verbose]
```

Parameters:

- profile_name Profile name.
- red_threshold_max Set the red max threshold.
- red_threshold_min Set the red min threshold.
- yellow_threshold_max Set the yellow max threshold.
- yellow_threshold_min Set the yellow min threshold.
- green_threshold_max Set the green max threshold.
- green_threshold_min Set the green min threshold.

Note: The command can only be used to configure an existing WRED profile that is bound to a queue. It does not support creating a WRED profile and only limited fields can be configured.

Firstly, create a json file (here named ecn.json as an example) that includes a WRED profile (e.g., AZURE_LOSSLESS) and bind it to Ethernet0 queue 3 and Ethernet 4 queue4.

ecn.json

```
"WRED_PROFILE": {
  "AZURE_LOSSLESS": {
    "wred_green_enable": "true",
    "green_min_threshold": "1248",
    "green_max_threshold": "1872",
    "green_drop_probability": "25",
    "wred_yellow_enable": "true",
    "yellow_min_threshold": "1456",
    "yellow_max_threshold": "1872",
    "yellow_drop_probability": "50",
    "wred_red_enable": "true",
    "red_max_threshold": "2097152",
    "red_min_threshold": "2097152",
    "red_drop_probability": "75",
    "ecn": "ecn_all"
  }
},
"QUEUE": {
  "Ethernet0|3": {
    "wred_profile": "[WRED_PROFILE|AZURE_LOSSLESS]"
  },
  "Ethernet4|4": {
    "wred_profile": "[WRED_PROFILE|AZURE_LOSSLESS]"
  }
}
```

And then, use the **sonic-cfggen** command to load the json file that creates a WRED profile and applies it to queues.

Example

```
admin@sonic:~$ sudo sonic-cfggen -w -j ecn.json
```

The following example shows how to set a red max threshold 100 for the profile AZURE_LOSSLESS.

Example: Modify the red max threshold in profile 'AZURE_LOSSLESS'

```
admin@sonic:~$ sudo config ecn -profile AZURE_LOSSLESS -rmax 100
```

(202111.2) ZTP

- [ZTP Show Commands](#)
 - `show ztp status`
 - `show ztp status --verbose`
- [ZTP Configuration Commands](#)
 - `config ztp enable`
 - `config ztp disable`
 - `config ztp run`

This section explains all the Zero Touch Provisioning (ZTP) commands that are supported in SONiC.

ZTP Show Commands

`show ztp status`

`show ztp status --verbose`

This command displays the current ZTP configuration of the switch. It also displays detailed information about current state of a ZTP session. It displays information related to all configuration sections as defined in the switch provisioning information discovered in a particular ZTP session.

Usage

```
show ztp status
show ztp status --verbose
```

Example

```
root@B1-SP1-7712:/home/admin# show ztp status
ZTP Admin Mode : True
ZTP Service : Inactive
ZTP Status : SUCCESS
ZTP Source : dhcp-opt67 (eth0)
Runtime : 05m 31s
Timestamp : 2019-09-11 19:12:24 UTC
ZTP Service is not running
01-configdb-json: SUCCESS
02-connectivity-check: SUCCESS
```

Use the verbose option to display more detailed information.

Example

```

root@B1-SP1-7712:/home/admin# show ztp status --verbose
Command: ztp status --verbose
=====
ZTP
=====
ZTP Admin Mode : True
ZTP Service : Inactive
ZTP Status : SUCCESS
ZTP Source : dhcp-opt67 (eth0)
Runtime : 05m 31s
Timestamp : 2019-09-11 19:12:16 UTC
ZTP JSON Version : 1.0
ZTP Service is not running
-----
01-configdb-json
-----
Status : SUCCESS
Runtime : 02m 48s
Timestamp : 2019-09-11 19:11:55 UTC
Exit Code : 0
Ignore Result : False
-----
02-connectivity-check
-----
Status : SUCCESS
Runtime : 04s
Timestamp : 2019-09-11 19:12:16 UTC
Exit Code : 0
Ignore Result : False

```

Description:

- **ZTP Admin Mode** - Displays if the ZTP feature is administratively enabled or disabled. Possible values are True or False. This value is configurable using the config **ztp enabled** and **config ztp disable** commands.
- **ZTP Service** - Displays the ZTP service status. The following are possible values this field can display:
 - *Active Discovery*: ZTP service is operational and is performing DHCP discovery to learn switch provisioning information.
 - *Inactive*: ZTP service is not running.
 - *Processing*: ZTP service has discovered switch provisioning information and is processing it.
- **ZTP Status** - Displays the current state and result of ZTP session. The following are possible values this field can display:
 - *IN-PROGRESS*: ZTP session is currently in progress. ZTP service is processing switch provisioning information.
 - *SUCCESS*: ZTP service has successfully processed the switch provisioning information.
 - *FAILED*: ZTP service has failed to process the switch provisioning information.
 - *Not Started*: ZTP service has not started processing the discovered switch provisioning information.
- **ZTP Source** - Displays the DHCP option and then interface name from which switch provisioning information has been discovered.
- **Runtime** - Displays the time taken for ZTP process to complete from start to finish. For individual configuration sections it indicates the time taken to process the associated configuration section.
- **Timestamp** - Displays the date/time stamp when the status field has last changed.
- **ZTP JSON Version** - Version of ZTP JSON file used for describing switch provisioning information.
- **Status** - Displays the current state and result of a configuration section. The following are possible values this field can display:
 - *IN-PROGRESS*: Corresponding configuration section is currently being processed.
 - *SUCCESS*: Corresponding configuration section was processed successfully.
 - *FAILED*: Corresponding configuration section failed to execute successfully.
 - *Not Started*: ZTP service has not started processing the corresponding configuration section.
 - *DISABLED*: Corresponding configuration section has been marked as disabled and will not be processed.
- **Exit Code** - Displays the program exit code of the configuration section executed. Non-zero exit code indicates that the configuration section has failed to execute successfully.
- **Ignore Result** - If this value is True, the result of the corresponding configuration section is ignored and not used to evaluate the overall ZTP result.
- **Activity String** - In addition to above information an activity string is displayed indicating the current action being performed by the ZTP service and how much time it has been performing the mentioned activity. Below is an example.
 - (04m 12s) Discovering provisioning data

ZTP Configuration Commands

This sub-section explains the list of configuration options available for ZTP.

config ztp enable

Use this command to enable ZTP administrative mode.

Usage

```
config ztp enable
```

Example

```
root@sonic:/home/admin# config ztp enable
Running command: ztp enable
```

config ztp disable

Use this command to disable ZTP administrative mode. This command can also be used to abort a current ZTP session and load the factory default switch configuration.

Usage

```
config ztp disable
config ztp disable -y
```

Example

```
root@sonic:/home/admin# config ztp disable
Active ZTP session will be stopped and disabled, continue? [y/N]: y
Running command: ztp disable -y
```

config ztp run

Use this command to manually restart a new ZTP session. This command deletes the existing `/etc/sonic/config_db.json` file and starts the ZTP service. It also erases the previous ZTP session data. ZTP configuration is loaded on the switch and ZTP discovery is performed.

Usage

```
config ztp run
config ztp run -y
```

Example

```
root@sonic:/home/admin# config ztp run
ZTP will be restarted. You may lose switch data and connectivity, continue? [y/N]: y
Running command: ztp run -y
```